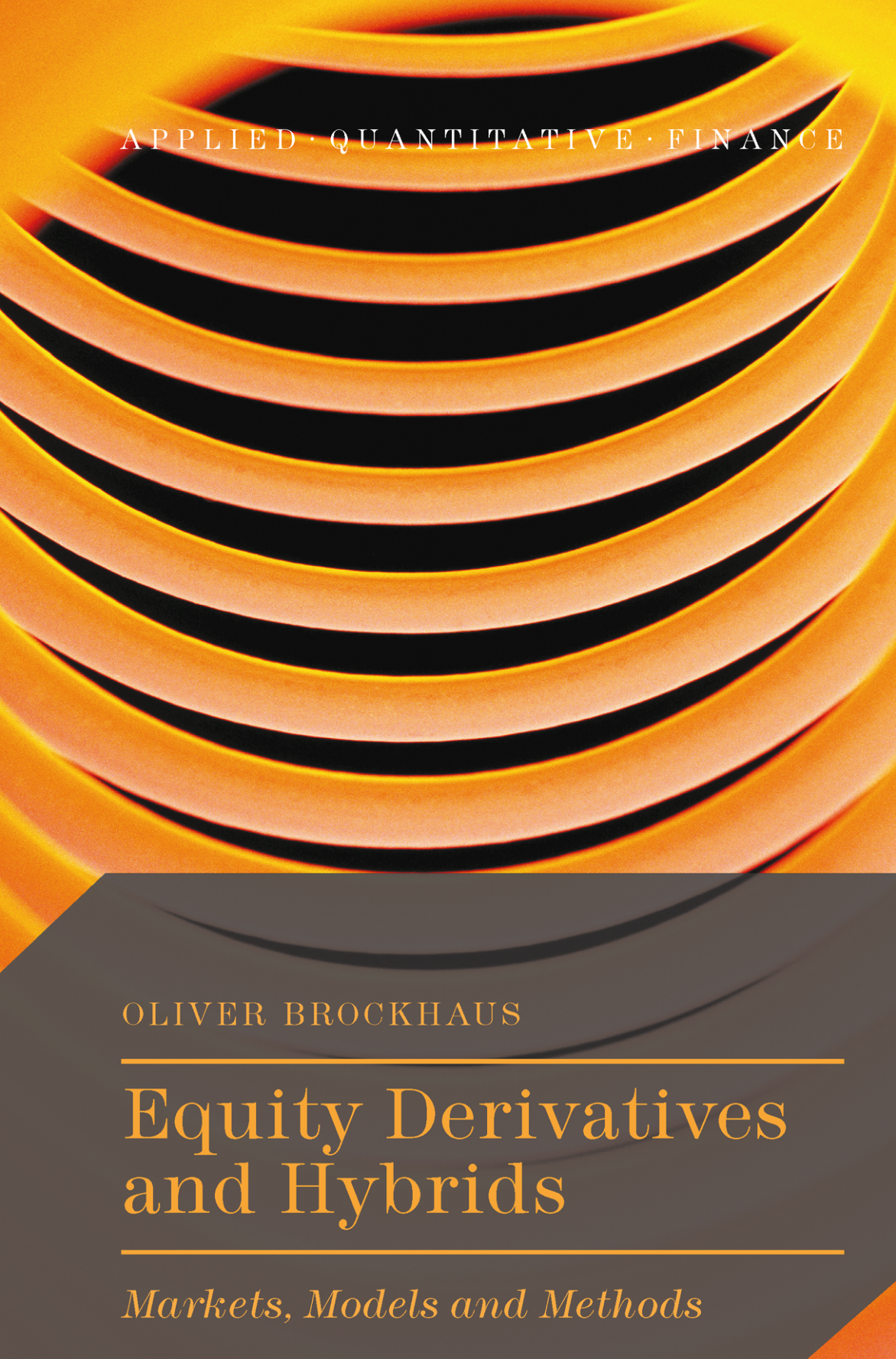




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OLIVER BROCKHAUS

Equity Derivatives and Hybrids

Markets, Models and Methods

Equity Derivatives and Hybrids

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To Miranda, Oskar, Lotte and Elise

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Preface

In a BSc course on Mathematical Finance I taught at King's College in 2003 a student asked me at the end of the first lecture what he needed to do to get my job. At the time I was an equity derivatives researcher, more commonly referred to as 'equity quant', at JP Morgan Chase. I might have mumbled that we only hired PhDs, being aware that this was at best part of the answer.

Around the same time I started giving practitioners' courses organized by World Business Strategies, which prompted me to collect ideas and concepts I had found relevant for equity and equity hybrid derivatives modelling. This book expands the material thus assembled over the years and makes it available to a wider audience. To achieve a self-contained and instructive presentation, derivations (or at least ideas of proof) are given throughout, avoiding unnecessary technical detail.

The choice of material is somewhat subjective. For example, a probabilistic approach is preferred to partial differential equations (PDEs) or Green's functions. Having seen too many quants getting into trouble trying to deliver sound PDE engines to hectic traders with a short fuse and attention span, Monte Carlo has become my favourite numerical method. Closed-form or semi-analytical formulae are also quite satisfying, both for the mathematician getting excited by more or less elegant formulae and for the quant trying to calibrate a model. The reader will find a few new ones for equity-credit hybrids in this book. On the models side I always stayed clear of jumps. Why add unhedgeable risk at the outset when delivering tools aimed at facilitating hedging? In my view academics should not be encouraged to waste time calibrating processes with independent returns to Vanillas.

My interest in correlation dates back to my time as credit quant, when I was briefly part of the dodgy business of arbitraging rating agency models against internal quantitative models for CDOs. The combination of cliquet models, copulas and factor correlation sparked some ideas for both stochastic volatility and multi-asset models, which have been presented elsewhere but are included in a textbook for the first time.

Every equity quant worth their salt will be interested in dividends. This is exemplified by Deutsche Bank's equity quant team and their four books (Brockhaus et al. (1999, 2000) and Overhaus et al. (2002, 2007)), the first two of which I was lucky enough to be involved in. The present book goes beyond the market standard approach for dividends modelling established in these publications.

On the hybrid front I attempt to bring together the available literature on asset models based on correlated stochastic (including local) processes representing

volatility, short interest rate and hazard rate. While scanning the literature I also felt that some gaps needed to be filled in the equity-credit space, in particular with respect to counterparty credit risk and structural models for credit.

The aforementioned BSc student was clearly not lacking self-confidence but needed theoretical and applied knowledge. Now my answer would be that he should read this book.

Oliver Brockhaus
Lorsbach, July 2015

Acknowledgements

During my years in London I was fortunate enough to benefit from discussions with many quants, traders, structurers, risk managers and academics. I am grateful to all of them for trying to teach me the trade. Needless to say, I am responsible for all misunderstandings and typos you will undoubtedly find in this book.

Neil Fowler from World Business Strategies motivated me to compile notes for the equity derivatives courses that ultimately led to this book. I also learnt a great deal from collaborating with Alexander Giese and Dr. Peter Jäckel on those courses. Their contribution is greatly appreciated.

Special thanks go to Prof. Dr. Uwe Wystup for supporting this project. I also wish to thank Andreas Weber for helpful comments.

This book would not have been possible without my wife's constant encouragement ("Get it out of your system!").

List of Notations

$x \wedge y, x \vee y$	$\min\{x, y\}, \max\{x, y\}$
i, j, m, n	discrete time index
k, l	space index
s, t, T	continuous time index
$\mathcal{T}$	time discretization
S_t, S_t^k	equity (stock or index)
X_t	exchange rate
r_t	instantaneous interest rate
$f_t(T)$	instantaneous forward rate
B_t	cash bond
N_t	numéraire
$P_t(s, T)$	discount factor
$F_t(T)$	forward
D_t^i	value of i -th dividend
L_t^i	Libor
$V_t(s, T)$	expected realized variance
I_t	index, basket
$Call_t(K, T), Put_t(K, T)$	call respectively put price
d_t	dividend yield
σ_t	volatility
$\hat{\sigma}_t(K, T)$	implied volatility
$g_t(S_t)$	local factor in lsv model
ρ_t	Pearson correlation
h_t, H_t	intensity, integrated intensity
$p_t(T)$	survival probability
τ_t	stopping time
$\mathbb{IP}, \mathbb{IE}$	probability, expectation
U, U^k	independent uniform random variates
X, X_i, X_i^k, Y, Z	independent standard Gaussians
W_t, W_t^k	Brownian motions
M_t	martingale
F^X, f^X	cumulative distribution and density of X with respect to $\mathbb{IP}$

$\Phi(x), \phi(x)$	cumulative normal distribution and density
$\Phi_2(x, y; \rho)$	bivariate cumulative normal distribution
$BSCall(F, K, \sigma, T)$	Black-Scholes formula without discounting
d_+, d_-	$BSCall(F, K, \sigma, T) = F\Phi(d_+) - K\Phi(d_-)$
Δ, Γ, Υ	Black-Scholes delta, gamma, vega
$S_t^*, S_{*,t}$	running maximum respectively minimum of S
R_i	equity return

1 [Empirical Evidence

When studying derivatives it is useful to be aware of the empirical properties of the underlying. Those properties should be taken into account when attempting to model the underlying for the risk management of derivatives. Derivatives often depend on one or more closing prices within a time period ranging from few months up to several years. Thus the focus in this chapter is a time series of closing prices, although the methods can also be applied to higher frequency data.

Let a time series $(S_i, i = 0, 1, 2, \dots, n)$ of closing prices for a stock or index be given where i ranges through all business days within a given time interval. Stock returns are defined for $i = 1, 2, \dots, n$ as

$$R_i = \log \frac{S_i}{S_{i-1}}$$

As an example consider the European stock index Stoxx 50, Figure 1.1.

The corresponding return sequence in Figure 1.2 shows periods of high volatility in 2002 (dot com bubble) and 2008 (Lehman default).

1.1 Distribution

When analyzing the first four moments of the return distribution one observes that the distribution of returns is negatively skewed (third moment) with heavier tails (fourth moment) than the Gaussian distribution. Figure 1.3 exhibits the empirical density of one business day's returns from January 1998 to March 2015, in comparison with the Gauss distribution with the same first and second moment. The observed return density motivates Lévy models, including models with jumps as introduced in Section 6.4, as an alternative to the log-normal process of the Black-Scholes model discussed in Section 18.2.

Negative skewness is often explained as being due to risk averse investors in connection with risky stocks. Both news and small negative price movements

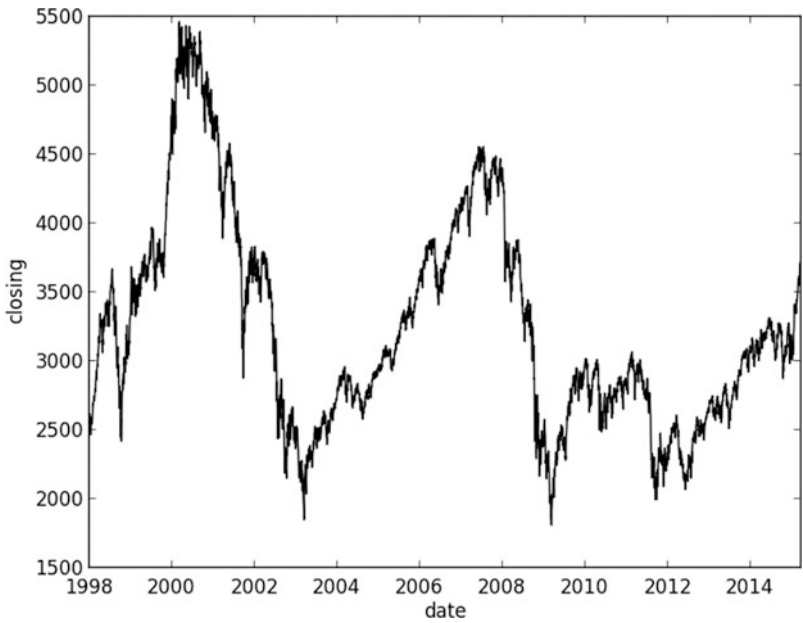


Figure 1.1 Stoxx 50 level

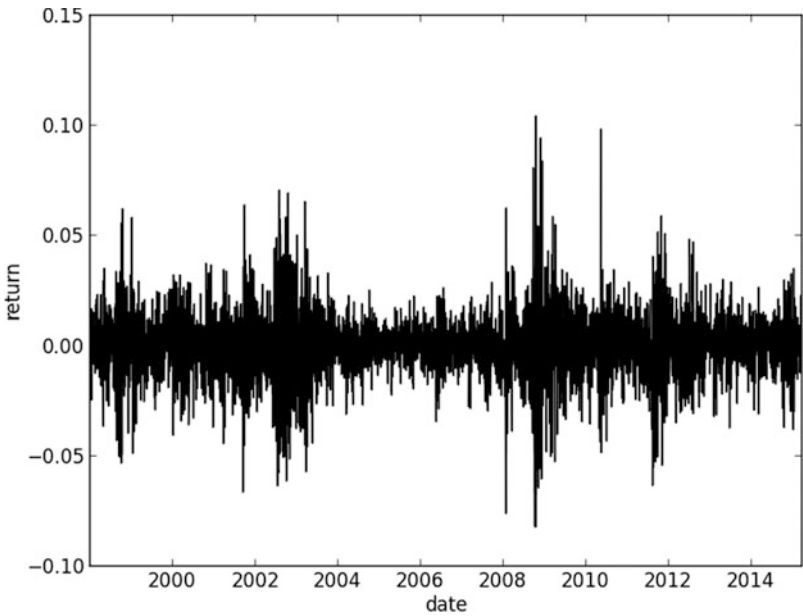


Figure 1.2 Stoxx 50 returns

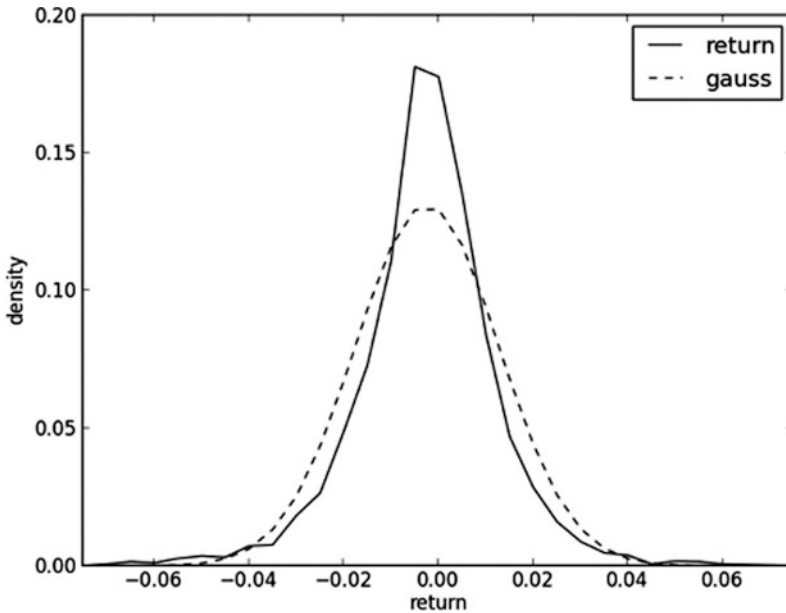


Figure 1.3 Stoxx 50 density

may trigger larger investors fearing default to cut losses and sell. This effect is not symmetric as investors do not apply the same urgency when reacting to positive price moves. Heavy tails (kurtosis) are a feature of processes with persistent volatility as discussed in Section 1.3.4.

1.2 Drift

1.2.1 Level

The observed drift level depends on the dividends and interest rate environment. Average returns often exceed the corresponding drift differential of interest rates and dividend yields as shareholders expect to be compensated for taking a higher risk than bond holders. In a risk neutral setting higher drift may also apply. The risk of default or dropping share value due to unpredictable events is compensated for by a drift exceeding the risk free rate. The assumption that equities are a better long term investment than fixed income products has been challenged by repeatedly collapsing equity markets since 2000.

1.2.2 Mean reversion

Equity processes are generally not mean reverting. This assertion can be tested by estimating parameters of the model

$$R_i = \alpha + \beta t + \gamma S_{i-1} + \delta R_{i-1}$$

using the Augmented Dickey-Fuller test. Vanishing autocorrelation indicates $\delta = 0$. If the null-hypothesis for γ against $\gamma < 0$ cannot be rejected then the process does not exhibit mean reversion. Note that mean reversion for an asset is not compatible with the no-arbitrage assumption that in a risk neutral setting the expected return of an asset has to be equal to the risk free interest rate.

1.2.3 Seasonality

Seasonality within equity processes is typically not considered, in contrast to commodities processes. Thus seasonality adjustments via a time-dependent periodic drift function are not made. Periodicity of trading activity and hence volatility can be observed intraday, as well as on a weekly basis and, presumably, to some extent with yearly periodicity. This can be modelled through a trader time function of physical time which grows slower during weekends, holidays and, possibly, closing times of relevant exchanges.

1.3 Autocorrelation

1.3.1 Return level

In order to assess whether returns are independent one can measure the autocorrelation, namely the correlation of $R_1, R_2, \dots, R_{n-\tau}$ and $R_{1+\tau}, R_{2+\tau}, \dots, R_n$. If returns are independent then the outcome should be 1 for $\tau = 0$ and numbers close to 0 for $\tau > 0$. This seems to be supported by the data: the autocorrelation curve labelled ‘returns’ in Figure 1.4 remains near zero, indicating that past returns are not an indicator for future returns. In the case of independent returns autocorrelation should also be low when replacing R_i by $f(R_i)$ for arbitrary functions f . With $f(x) = |x|$, however, one observes significant autocorrelation, see the curve labelled ‘absolute returns’ in Figure 1.4.

1.3.2 Absolute return

The absolute value $|R_i|$ serves as a proxy for return volatility. This can be justified since returns may be written as $R_i = \sigma_i X_i$ with $\sigma \geq 0$, such that the variance of X_i is 1. The law of large numbers then motivates replacing $|X_i|$ by its average. With this

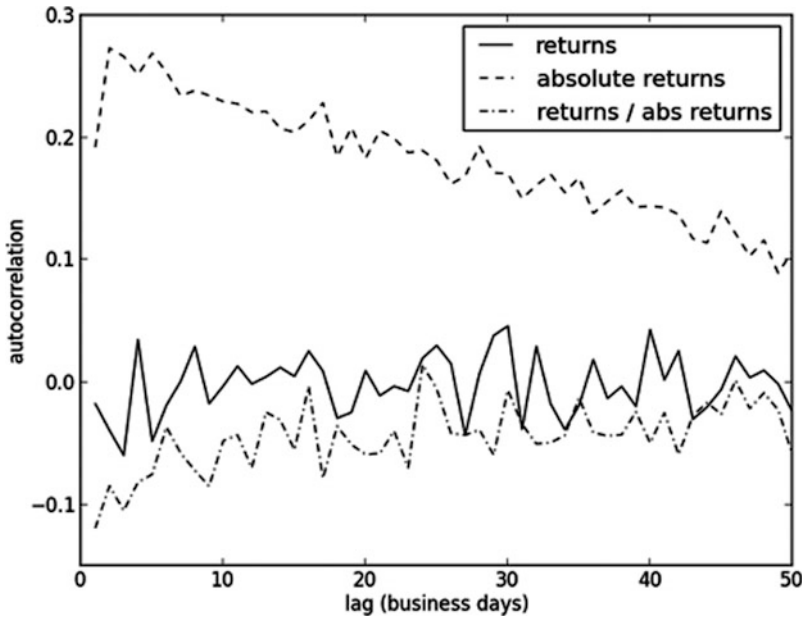


Figure 1.4 Stoxx 50 autocorrelation

interpretation the data indicate that future volatility can be predicted to some extent from past volatility levels. One can also observe this behaviour from the return series in Figure 1.2. When observing high volatility on any day in 2002 or 2008 it is likely that volatility will remain high on the following days since periods of high volatility cluster. This has led to equity models with continuous instantaneous volatility processes, such as local and stochastic volatility processes, which will be discussed in Chapter 6. Note that this observation rules out models with independent returns, such as Lévy models for $\log S$, including the Black-Scholes model.

1.3.3 Leverage

One can also study whether there is autocorrelation between stock returns and (proxy) volatility. The curve labelled ‘returns / abs returns’ in Figure 1.4 exhibits the correlation between $R_1, R_2, \dots, R_{n-\tau}$ and $|R_{1+\tau}|, |R_{2+\tau}|, \dots, |R_n|$. Negative correlation indicates that a decreasing stock level is often followed by increased (proxy) volatility. This has been modelled in stochastic volatility models through negative correlation between volatility increment and stock return, see Section 6.2. Alternatively local volatility models postulate a functional dependence of volatility on stock level, see Section 6.1.

1.3.4 Persistent volatility

Autocorrelation of absolute returns suggests models with persistent stochastic volatility, namely

$$R_i = \sigma_i X_i$$

with Gaussian noise X_i and some process σ_i .

An example for a time series exhibiting stochastic volatility is the autoregressive stochastic volatility, or ARSV(1), model introduced by Taylor (1986):

$$\log \sigma_i^2 = \phi + \alpha Y_i + \beta \log \sigma_{i-1}^2 \quad (1.1)$$

with Gaussian noise Y_i independent of X_i and persistence parameter $\beta \in [0, 1]$. It is easy to see that σ is log-normally distributed with log-variance

$$\sigma_v^2 = \frac{\alpha^2}{(1-\beta)^2}$$

Variance σ_R^2 , kurtosis κ_R and autocorrelation $\rho(\tau)$ of order $\tau \geq 1$ of R are given as

$$\begin{aligned} \sigma_R^2 &= e^{\phi/(1-\beta)} \left(e^{\sigma_v^2/2} - e^{\sigma_v^2/4} \right) \\ \kappa_R &= \kappa_X e^{\sigma_v^2} \\ \rho(\tau) &= \frac{e^{\sigma_v^2 \beta^\tau} - 1}{\kappa_R e^{\sigma_v^2} - 1} \approx \frac{e^{\sigma_v^2} - 1}{\kappa_R e^{\sigma_v^2} - 1} \beta^\tau \end{aligned}$$

where $\kappa_X = \mathbb{E}[X^4] = 3$ denotes the kurtosis of the standard Gaussian X . Thus autocorrelation decays exponentially. An asymmetric variant of ARSV(1) is obtained through correlation between X_i and Y_i in (1.1). The corresponding autocorrelation is

$$\rho(\tau) = \frac{(1 + \rho^2 \alpha^2) e^{\sigma_v^2 \beta^\tau} - 1}{\kappa_R e^{\sigma_v^2} - 1}$$

The generalized autoregressive conditional heteroscedasticity GARCH(1,1) model assumes

$$\sigma_i^2 = \phi + \alpha R_{i-1}^2 + \beta \sigma_{i-1}^2 \quad (1.2)$$

with $\phi > 0$ and $\alpha, \beta \geq 0$ as well as persistence $p = \alpha + \beta < 1$. Variance σ_R^2 , kurtosis κ_R and autocorrelation $\rho(\tau)$ are known to be

$$\begin{aligned}\sigma_R^2 &= \frac{\phi}{1-p} \\ \kappa_R &= \kappa_X \left(1 - \frac{\alpha^2 (\kappa_X - 1)}{1 - (\alpha + \beta)^2} \right)^{-1} \\ \rho(\tau) &= \frac{\alpha(1 - p^2 + p\alpha)}{1 - p^2 + \alpha^2} p^{\tau-1}\end{aligned}$$

Autocorrelation also decays exponentially. Asymmetric variants of GARCH include EGARCH and QGARCH. The simplest variant, EGARCH(1,0), evolves according to

$$\log \sigma_i^2 = \phi + \alpha (|X_{i-1}| - \mathbb{E}[|X_{i-1}|]) + \beta \log \sigma_{i-1}^2 + \gamma X_{i-1}$$

EGARCH(1,0) shares with ARSV(1) that log variance is modelled. Without asymmetry $\gamma = 0$ it behaves similarly to GARCH(1,1) since $|X_{i-1}| - \mathbb{E}[|X_{i-1}|]$ is a proxy for R_{i-1}^2 . QGARCH(1,1) extends GARCH by an asymmetry term:

$$\sigma_i^2 = \phi + \alpha R_{i-1}^2 + \beta \sigma_{i-1}^2 + \gamma R_{i-1}$$

Skewed conditional return distribution can be achieved through alternative distributions for the innovations X_i . Viable distributions include t -distribution, as well as generalized error distribution (GED).

Statistical packages written in languages such as R and Python allow for parameter estimation and thus volatility forecasting.

1.4 Jumps

The drop of stock prices at dividend dates can be explained by the loss of rights at ex-dividend dates. In order to assess whether equity processes exhibit jumps beyond dividend drops one can test whether there are absolute returns exceeding a threshold. This threshold can be defined as a multiple of empirical volatility.

Assume a stock process follows the dynamics

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_t dW_t + j_t dN_t$$

where μ_t is drift, σ_t is diffusive volatility, j_t is jump size and N_t is a counting process. A discrete-time realization at times t_i allows one to define log returns

$$R_i = \log \frac{S_{t_i}}{S_{t_{i-1}}}$$

A discrete test process is defined as

$$T_i = \frac{R_i}{\sigma_i}$$

where σ_i is n -step bipower variation, introduced by Barndorff-Nielsen and Shephard (2004) as

$$\sigma_i^2 = \frac{1}{n} \sum_{j=i-n}^{i-1} |R_j R_{j-1}|$$

Lee and Mykland (2008) discuss limiting behaviour for small time increments of the approximation

$$T_i \approx \frac{W_{t_i} - W_{t_{i-1}} + \frac{1}{\sigma_{t_{i-1}}} \sum_{\tau \in (t_{i-1}, t_i]} j_\tau}{\text{IE} [|W_{t_i} - W_{t_{i-1}}|]}$$

Here the number of observations n has to be chosen within $[\sqrt{N}, N]$ where N is the number of yearly observations.

Applying jump tests to equity data one finds that single stocks are more likely to have jumps than indices. As an example see Figure 1.5, exhibiting T for Stoxx 50 and SAP with $n = 130$. One also finds that stock jumps are typically negative as news tends to be bad news. Jumps also cluster in the sense that large moves do not arrive in isolation, compare Figure 1.2. This is an alternative interpretation of return data to persistent volatility. Jumps are often modelled using exponential distribution for arrival time of the next jump, as well as log-normal distribution for the jump size.

Modelling jumps also has an impact on hedging. Since jumps cannot be anticipated hedging strategies will be exposed to gap risk. While this risk will be better quantified with a realistic model that includes potential jumps it is not obvious that jump models lead to better hedging strategies.

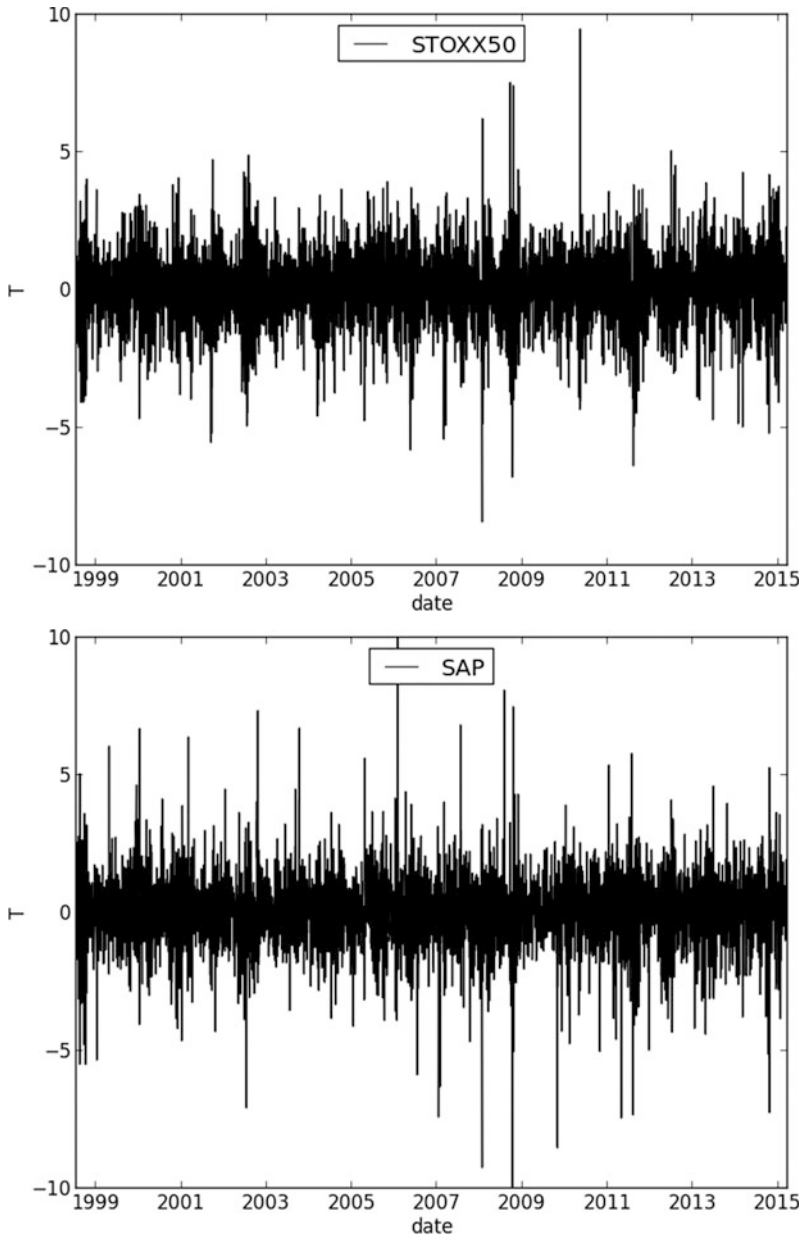


Figure 1.5 Jump test process for Stox 50 and SAP

2 [Equity Derivatives Market

2.1 Underlyings

In fixed income markets the relevant underlyings are yield curves and there is basically one of those per currency. With the end of risk neutral pricing the use of single curve modelling has become more complicated since effects such as liquidity, funding costs, credit risk and collateral agreements can no longer be ignored. However, arguably, this complicates the nature of fixed income underlyings rather than creates new ones.

To every pair of currencies there is a corresponding foreign exchange process. Interest rate processes are thus a prerequisite. The number of relevant underlyings in foreign exchange markets remains limited.

In equities, one has a multitude of single stocks, which are underlyings in their own right. Additionally, there are indices defined as baskets of single stocks adding to the amount of data required. Since those baskets may be denoted in different currencies both interest rates and foreign exchange processes need to be taken into consideration. Thus equity markets are characterized by a vast amount of underlyings. Their modelling requires taking into account interest and foreign exchange rates.

2.2 Dividends

Dividend amounts and schedules are not known in advance with any certainty. However, they are set by the board of directors for the near future and are only changed if necessary. For short maturities one can therefore use the latest dividend information to extrapolate into the future. The i -th dividend seen from today is defined through expected amount d_i , ex date t_i and payment date t'_i . The share price will typically drop at time t_i by an amount equal to the realized dividend at time t_i , discounted to the payment date t'_i . If stocks and dividends are treated differently for tax purposes, there may also be a tax factor impacting the share drop.

2.3 Repo rate

A stock repurchase agreement consists in selling stock with the promise to buy it back at a specified later date. The price difference between the stock price today and the repurchase price defines the repo rate. The repo rate represents the interest rate on a loan secured by the stock as collateral. It is therefore typically higher than the corresponding risk free rate.

2.4 Delta one products

Equity products which do not depend on stock volatility are denoted delta one products, although their delta may not be equal to one. Examples include the underlying itself, equity forward, future and swap contracts.

2.4.1 Spot

The spot price S_0 today is quoted by market data providers and brokers. As for any financial product this quote may include a discount factor due to premium settlement delay.

Deriving from the spot price there exist geared retail products with value $S_t - K$ at time t . Retail products cannot require the investor to be liable to the financial institution as a result of buying a contract. For that reason there is typically a barrier $B \geq K$ such that the product expires as worthless as soon as the barrier is hit:

$$(S_t - K) \mathbf{1}_{\{S_{*,t} > B\}}$$

where S_* is the running minimum process defined as

$$S_{*,t} = \min\{S_u, u \in \mathcal{T}, u \leq t\}$$

relative to a specified set $\mathcal{T}$ of observation times.

2.4.2 Forward

A forward contract promises to exchange at a future time T an amount K for a number n of shares. Amount K , number of shares n and maturity T are determined as part of the contract. Thus the payoff is

$$nS_T - K$$

The forward price $F(T)$ is defined as the amount defined, such that the fair value of a contract exchanging a single share for $F(T)$ at time T is zero. Thus

$$F(T) = \mathbb{E} [S_T]$$

If discounting factors P , as well as dividend amounts d_i , ex dates t_i and payment dates t'_i , are known the forward price is given as

$$F(T) = P(T)^{-1} \left(S_0 - \sum_{t_i < T} P(t'_i) d_i \right) \quad (2.1)$$

This follows from replicating the contract: at time $t = 0$ the cost of the fixed amount is $P(T)F(T)$. Delivery of S_T can be secured by investing S_0 to buy one share today. However, this will also produce dividends in the time interval $(t, T]$. The value of those dividends today can be deducted from S_0 . Since the cash amount is secured by the share the discount factors should be repo rates. Formula (2.1) ignores the risk of one of the parties defaulting.

2.4.3 Future

A futures contract is similar to a forward contract. However the contract requires both parties to post collateral at regular time intervals in order to reduce default risk. If default rates are ignored the futures price can be assumed to be equal to the forward price.

2.4.4 Equity swap

An equity swap contract swaps equity returns against payments, which can be derived from interest rates. For example, with payment dates $t_1 < t_2 < \dots < t_n$ such a product may exchange payments

$$S_{t_i}/S_{t_{i-1}} - 1$$

against Libor payments $L_{t_{i-1}}^i$ plus spread s

$$\alpha_i \left(L_{t_{i-1}}^i + s \right)$$

at time t_i , where α_i is the time increment for the relevant Libor rate. Libor and time increments are discussed in more detail in Chapter 10.

2.5 Vanilla options

Owning shares represents considerable risk since share prices fluctuate and can also drop quickly in case of default risk of the issuing companies. To mitigate this risk options have been introduced. A European put option with maturity time T allows the holder to sell a share at a strike price K which is agreed as part of the contract. Thus, if the investor owns the option and a share, the downside risk is limited: should the share price drop below K at time T , the option can be exercised with a profit of $K - S_T$ such that the value of the portfolio is guaranteed to be greater than K . The allowed exercise time T is also specified in the contract. Variants of this option include European call options which give the right to buy rather than sell a share at the strike price. American options give the right to exercise the option at any time until maturity. From a pricing point of view European options depend on the share price at T only, while American option prices depend on assumptions on the entire path of the share price. American and European options are also denoted Plain Vanilla options, in contrast to Exotic options which include any other conceivable payoff contingent on the path of one or several equity processes. Vanilla options are listed on exchanges as well as traded over the counter. Options are typically cash settled. A company may increase their equity capital by issuing options. Those options are called warrants.

Options are often combined in order to satisfy an investor's view, such as in the downside protection example above. Common combinations include:

- call spread and put spread: difference between two options of the same kind but with different strikes
- straddle: sum of call and put with all other contractual features being the same
- butterfly: difference between two call or put spreads with strikes K_1, K_2 and K_2, K_3 respectively
- calendar spread: difference between two options of the same kind but with different maturities

2.5.1 Forward and discount factor

The derivatives market on stocks and stock indices is dominated by European (and American) Plain Vanilla options. For a given underlying a liquid options market may only exist for a few strikes and maturities. Prices for other options can be obtained by fitting, interpolating or extrapolating those prices. There are fundamental properties which need to be satisfied by a set of European option prices. Conversely, specifying European option prices implies important properties of the underlying equity dynamics, including discount factor, forward and marginal distribution of the equity process. In case of European options one can use call-put parity and box trades to infer discount factor and forward price:

- Call-put parity: the relation

$$(S_t - K)^+ - (K - S_t)^+ = S_t - K$$

implies that the difference between call and put with the same underlying, strike and maturity is equal to the discounted difference between forward and strike price.

- Box trade: the relation

$$(S_t - K_1)^+ - (K_1 - S_t)^+ - (S_t - K_2)^+ + (K_2 - S_t)^+ = K_2 - K_1$$

implies that the value of the box trade (difference between two call-put parity trades with different strikes) is equal to the discounted difference between the strikes. This allows one to infer the discount factor from Vanilla options.

Combining box trade and call-put parity allows one to independently derive forward prices and discount factors from European Vanilla option prices.

2.5.2 No-arbitrage conditions

If you assume European option call and put prices $Call(K, t)$ and $Put(K, t)$ are given for all maturities and strikes, then the following holds for Vanilla options:

- The function $Call(K, t)$ ($Put(K, t)$) decreases (increases) in strike K . This can be seen by considering the payoff of a call (put) spread, namely the difference between two call (put) options with the same maturity and different strikes.
- The functions $Call(K, t)$ and $Put(K, t)$ are convex in strike K . This can be seen by considering the payoff of a butterfly spread, namely the difference of two call (put) spreads with the same maturity.

Option prices typically increase with t . More specifically, if S is a martingale and discount factors are equal to 1 (zero interest rates) then

$$\begin{aligned} Call(K, t) &= \mathbb{E} \left[(\mathbb{E}_t [S_T] - K)^+ \right] \\ &\leq \mathbb{E} \left[\mathbb{E}_t \left[(S_T - K)^+ \right] \right] = Call(K, T) \end{aligned}$$

since $(x - K)^+$ is convex in x . This can be generalized to the case of affine models as discussed in Chapter 5. If

$$S_t = (F(t) - C(t))M_t + C(t)$$

where M is a martingale and F and C are deterministic functions then

$$Call(K, t) \leq aCall(bK + c, T)$$

with

$$\begin{aligned} a &= P(t, T) \frac{F(T) - C(T)}{F(t) - C(t)} \\ b &= \frac{F(t) - C(t)}{F(T) - C(T)} \\ c &= \frac{C(t)F(T) - C(T)F(t)}{F(T) - C(T)} \end{aligned}$$

2.5.3 Black-Scholes formula

The Black-Scholes model assumes a log-normally distributed asset S_t at t , namely

$$S_t = F(t)e^{\sigma\sqrt{t}X - \frac{1}{2}\sigma^2 t}$$

where $F(t) = \mathbb{E}[S_t]$ is the forward price and $\sigma^2 t$ is the variance of $\log S_t$. The parameter σ is (somewhat sloppily) referred to as volatility. Given forward $F(t)$ and volatility σ the undiscounted call option price on such an asset can be computed as

$$\begin{aligned} \mathbb{E}[(S_t - K)^+] &= \mathbb{E}[S_t \mathbf{1}_{\{S_t > K\}}] - K \mathbb{P}[S_t > K] \\ &= F(t) \mathbb{E}\left[e^{\sigma\sqrt{t}X - \frac{1}{2}\sigma^2 t} \mathbf{1}_{\{-X < d_-\}}\right] - K \mathbb{P}[-X < d_-] \\ &= BSCall(F(t), K, \sigma, t) \end{aligned}$$

The formula

$$BSCall(F(t), K, \sigma, t) = F(t)\Phi(d_+) - K\Phi(d_-) \quad (2.2)$$

with

$$d_{\pm} = \frac{\log \frac{F(t)}{K} \pm \frac{\sigma^2 t}{2}}{\sigma\sqrt{t}}$$

is referred to as the Black-Scholes formula. The corresponding formula for the put payoff $(K - S_t)^+$ is

$$BSPut(F(t), K, \sigma, t) = K\Phi(-d_-) - F(t)\Phi(-d_+) \quad (2.3)$$

The call-put parity reads

$$BSCall(F(t), K, \sigma, t) - BSPut(F(t), K, \sigma, t) = F(t) - K$$

as well as the geometric parity

$$BSPut(F(t), K, \sigma, t) = -BSCall(F(t), K, -\sigma, t)$$

2.5.4 Implied volatility and market prices

Let $(Call(K_i, t_i), i = 1, 2, \dots, m)$ and $(Put(K_j, t_j), j = 1, 2, \dots, n)$ denote available market prices of European options on a given underlying for some strike-maturity combinations. Assuming forward and discount curves $F(t), P(t)$ those prices can be described by the implied volatility surface $\hat{\sigma}$ where $\hat{\sigma}(K, t)$ is defined through

$$P(t)BSCall(F(t), K, \hat{\sigma}(K, t), t) = Call(K, t) \quad (2.4)$$

Thus the market yields $\hat{\sigma}$ on a finite set of points on the (K, t) surface. Volatilities will not only differ for different strike-maturity combinations but also for bid, offer, call and put prices referring to the same strike and maturity.

Values $\hat{\sigma} = 0$ and $\hat{\sigma} = \infty$ yield the no-arbitrage bounds

$$\begin{aligned} P(t)(F(t) - K)^+ &\leq Call(K, t) \leq P(t)F(t) \\ P(t)(K - F(t))^+ &\leq Put(K, t) \leq P(t)K \end{aligned}$$

2.5.5 Implied distribution

Fix an option expiry $t > 0$. The market implied stock distribution F^{S_t} can be obtained from (not discounted) market prices of Vanilla options as

$$F^{S_t}(K) = \mathbb{IP}[S_t < K] = 1 + \frac{\partial}{\partial K} (P(t)^{-1} Call(K, t)) = \frac{\partial}{\partial K} (P(t)^{-1} Put(K, t))$$

where $\mathbb{IP}$ denotes risk neutral probability. This has already been observed by Bronzin (1908) and Breeden and Litzenberger (1978). More specifically, in terms of the implied volatility and its first derivative, one has

$$F^{S_t}(K) = \Phi(-d_-) + \sqrt{t} K \phi(d_-) \frac{\partial \hat{\sigma}}{\partial K} \quad (2.5)$$

with Φ and ϕ denoting standard normal distribution and density. Given $\hat{\sigma}(K)$ condition $F^{S_t}(K) \in [0, 1]$ imposes bounds on the slope of $\hat{\sigma}$ at K . The density f^{S_t} is given as

$$f^{S_t}(K) = \frac{\phi(d_-)}{K \hat{\sigma} \sqrt{t}} \left(1 + \hat{\sigma} \sqrt{t} K^2 \frac{\partial^2 \hat{\sigma}}{\partial K^2} + d_+ d_- K^2 \left(\frac{\partial \hat{\sigma}}{\partial K} \right)^2 + 2d_+ K \frac{\partial \hat{\sigma}}{\partial K} \right) \quad (2.6)$$

Given $\hat{\sigma}(K)$ and $\frac{\partial \hat{\sigma}}{\partial K}(K)$ condition $f^{S_t}(K) \geq 0$ entails a lower bound on the convexity of $\hat{\sigma}$ at K .

In practice market prices are only given at discrete strikes. In order to define a distribution those prices need to be interpolated and extrapolated. The most basic method is linear interpolation combined with flat extrapolation of the smile $\hat{\sigma}$ at maturity t . From (2.5) one can see that a jump in the slope of $\hat{\sigma}$ at K leads to either a point mass at K , if it is an upward jump, or otherwise an arbitrageable distribution due to negative mass at K .

2.5.6 American options

American options are similar to European options in that they are specified by underlying, strike and maturity. However, they can be exercised any time up to the specified maturity. Generally this results in a higher price than the corresponding European option.

There is one exception, namely call options on non-dividend paying stock with non-negative interest rates. One has, with B as in (10.1),

$$\begin{aligned} \mathbb{E}_\tau \left[B_T^{-1} (S_T - K)^+ \right] &\geq \mathbb{E}_\tau \left[\left(B_T^{-1} S_T - B_T^{-1} K \right)^+ \right] \\ &\geq \left(B_\tau^{-1} S_\tau - B_\tau^{-1} K \right)^+ = B_\tau^{-1} (S_\tau - K)^+ \end{aligned}$$

where the first inequality holds as long as interest rates are positive. The second inequality is Jensen's inequality applied to the convex function

$$f(x) = (x - B_\tau^{-1} K)^+$$

where the argument of f is the martingale

$$M_t = B_t^{-1} S_t$$

(since there are no dividends) and measure $\mathbb{P}_\tau$. Thus for any stopping time $\tau \leq T$ the holding value of the call option is greater than the intrinsic value.

American option prices always increase with maturity, a property not shared by European options. This property may be violated by a pricing model if model dynamics is chosen depending on the product, as will often be the case in practice. For example, one could compute constant dividend yield from a forward with discrete dividends. The resulting American call option price with maturity after a dividend may be cheaper than the same option with shorter maturity. Similarly, choosing constant volatility from the corresponding European implied volatility surface, a negative term structure of volatilities may lead to non-increasing American option prices.

Single-stock Vanilla markets may be given in terms of American option prices. As American options are path-dependent this leads to the nontrivial problem of finding European implied volatilities, such that market prices of American options are recovered by a model calibrated to these European implied volatilities. While it is possible to calibrate to American option prices using bootstrapping forward in time, it is often more efficient to compute the early exercise premium using the desired model with stale market parameters and subtract this stale premium from actual American market prices, resulting in an approximation of corresponding actual European option prices. One should note, however, that the rest of the market may not follow this approach and will thus trade with a different implied volatility surface.

3 [Exotic Equity Derivatives

3.1 Barriers

Introducing a knock-out barrier reduces the price of a Vanilla option. An investor may be willing to take the risk of the underlying trading at the barrier level against this sure profit. This will be the case especially if gains on the remaining portfolio offset any losses due a barrier event.

3.1.1 European barriers

Hedging derivative products with payoff discontinuities introduces gap risk. In the case of the sale of a European digital call with payoff

$$\text{European digital} = \mathbf{1}_{\{S_t > K\}}$$

this risk is passed on to the buyer by charging the higher price of a call spread with payoff

$$\frac{1}{\varepsilon} \left((S_t - K + \varepsilon)^+ - (S_t - K)^+ \right)$$

The size of ε chosen by the seller depends on their risk appetite, the competitiveness of the market, and delta and notional constraints imposed by the risk department. A short seller will pay a price obtained with $\varepsilon < 0$. The theoretical prices of European digital call and put are obtained in the limit $\varepsilon \rightarrow 0$ as

$$P(T) \left(1 - F^{S_t}(K) \right) \quad \text{and} \quad P(T) F^{S_t}(K)$$

if the payout is paid at $T \geq t$. The distribution F^{S_t} of S_t can be expressed in terms of forward $F(t)$ and implied volatilities $\hat{\sigma}$ at maturity t , see (2.5). The asset-or-nothing call with payoff

$$\text{Asset-or-nothing} = S_t \mathbf{1}_{\{S_t > K\}}$$

is obtained by adding a standard call option to the digital call. Bid and offer prices for European double digital options with payout

$$\mathbf{1}_{\{K_1 < S_T < K_2\}}$$

are also obtained from the corresponding static hedge consisting of European options.

3.1.2 Continuous barriers

Continuous (or American) barriers within a derivative payoff need to be qualified further as up or down, as well as in or out barriers. Barriers can be live on a subset of the lifetime of the derivative leading to partial or discrete barriers. They may be time dependent. Finally, there may be more than one of them.

Assuming the Black-Scholes model for S formulae for standard barrier options such as

$$\text{Up-and-out call} = (S_T - K)^+ \mathbf{1}_{\{S_T^* < B\}}$$

where S^* denotes the running maximum of S , are derived and summarized in Section 18.3.7. In order to take into account European option prices (i.e. the implied volatility smile) an improved price is obtained from a two-asset approach where the price is computed from an expectation of type

$$\mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_T^{2*} < B\}} \right]$$

The implied volatility σ_1 at option strike K and maturity T is applied to the log-normal asset S^1 while barrier level B and maturity T can be used to obtain the implied volatility σ_2 for S^2 . The price is then computed from outside barrier formulae in Section 18.4.2 with correlation $\rho = 1$. This approach does not take into account the implied volatility slope at the barrier. Single-barrier options are often valued assuming local volatility, as described in Section 6.1, using finite difference methods. This can be justified as long as the difference between local and stochastic volatility prices remains negligible. For double-barrier products this is typically not the case and a stochastic volatility model (see Section 6.2) is preferable. Implementations for the stochastic volatility models from Section 6.2 may be based on Monte Carlo methods as described in Sections 17.4 and 17.4.6. The foreign exchange market is deeper in the sense that market prices for barrier products with tight bid-offer spreads are available in that market. This imposes additional modelling flexibility. Local stochastic volatility models (Section 6.3) support asset dynamics ranging from local volatility to stochastic volatility within a single model, thus taking into account the additional market information. Due to the speed of

foreign exchange markets analytical pricing may be preferred, as long as it takes into account the market smile. An approach developed for that purpose is Vanna-Volga, see Section 15.3.

Some contracts require daily instead of continuous barrier monitoring. A pricing adjustment has been derived in Section 18.1.5, namely

$$B^{\pm} = B e^{\pm \beta \sigma \sqrt{T/m}}$$

where

$$\beta = \frac{\zeta(1/2)}{\sqrt{2\pi}} \approx 0.5826$$

For example, barrier B^+ is to be used for an up-and-out barrier B of length T with m equally spaced, monitoring dates and process volatility σ . The adjustment formula can also be used in the context of partial, time-dependent and double barriers. It may be used within Monte Carlo simulations to approximate barrier prices with specified monitoring. Simulations using a discretization scheme with n monitoring times, or indeed Brownian bridge extremum simulation yielding prices with continuous barrier monitoring of, can thus be adjusted to obtain an approximation of the price with m monitoring times. In that context the relevant volatility σ may depend on path and time step.

To account for hedging costs or gap risk, barriers are shifted such that the price of the sold option is increased. The intuition is that the maximum of the absolute value of delta experienced within the live region of the product (e.g. $(S_t, t) \in [0, B) \times [0, T)$ for an up-and-out option with barrier B and maturity T) remains limited for any possible future state (S_t, σ_t, t) . For an out barrier this means that by the time the spot gets near the shifted barrier, the contractual barrier has already been breached. By that time the cost of the contractual product has been covered through dynamic hedging of the shifted product since the shifted payoff dominates the contractual payoff. For narrow bid-ask spreads the shift should be as small as possible. This can be achieved through time-dependent barriers. For example, instead of a shifted barrier $B + c$ live on the interval $[0, T]$ one may use

$$B(t) = \begin{cases} B & \text{if } t \leq t_0 \\ B e^{\lambda(t-t_0)} & \text{if } t > t_0 \end{cases}$$

with λ such that $B(T) = B + c$. For the Black-Scholes model closed form formulae for standard barrier options with time-dependent barrier can be derived.

Note that once a barrier product is traded valuation requires the barrier event be monitored. Current market data, including past closing prices of the stock, may need to be enhanced by a product specific flag for every barrier and all traded

products. Changing such a flag may trigger valuation as a different product thus avoiding manual rebooking following barrier events.

3.2 Cliquets

Trading in Vanilla options relies on a view of the performance of an asset over a specified term. Cliquets allow for locking in the performance over a specified term without fixing a single start and end date.

3.2.1 Standard cliquets

A forward-starting call option with start or reference date t , maturity T or term $T - t$, as well as strike K , has payoff

$$\text{Forward call} = \left(\frac{S_T}{S_t} - K \right)^+$$

This product reverts to (the multiple of) a standard call option at time t when the reference stock level S_t is known. Standard cliquet products are sums of forward-starting options based on a regular schedule

$$0 = t_0 < t_1 < t_2 < \dots < t_n = T$$

with monthly, quarterly, semi-annual or annual frequency. A standard or Vanilla cliquet based on forward call spreads on a stock (or stock index) S pays

$$\text{Vanilla cliquet} = \sum_{i=1}^n \left(\frac{S_{t_i}}{S_{t_{i-1}}} - K_1 \right)^+ - \left(\frac{S_{t_i}}{S_{t_{i-1}}} - K_2 \right)^+ \quad (3.1)$$

at maturity T , with $K_1 \leq 1 \leq K_2$. Alternatives include standard cliquets consisting of forward-starting calls, puts or put spreads. Quantities $K_1 - 1 \leq 0$ and $K_2 - 1 \geq 0$ are also referred to as floor and cap on the return

$$R_i = \frac{S_{t_i}}{S_{t_{i-1}}} - 1$$

The price of these products depends strongly on modelling assumptions, as discussed in Section 7.2.

3.2.2 Exotic cliquets

Other than (local) cap and floor, cliquets can also have a global strike K leading to payoffs such as

$$\text{Call on cliquet} = \left(\left(\sum_{i=1}^n (R_i - \text{floor})^+ - (R_i - \text{cap})^+ \right) - K \right)^+ \quad (3.2)$$

Alternatives include a global put, call spread or put spread on a Vanilla cliquet. Those cliquets are referred to as exotic cliquets since they require modelling of the joint distribution of returns $(R_i, i = 1, 2, \dots, n)$ in contrast to Vanilla cliquets requiring only the distribution of each of the returns R_i separately. Popular exotic cliquets include reverse cliquets defined as

$$\text{Reverse cliquet} = \left(K + \sum_{i=1}^n R_i^- \right)^+$$

and Napoleon options

$$\text{Napoleon} = (K + \min\{R_1, R_2, \dots, R_n\})^+$$

While standard (arithmetic) cliquets are based on a sum of forward-starting options, a geometric cliquet pays at T the product of capped and floored returns, or some option on it, for example

$$\text{Geometric cliquet} = \left(\left(\prod_{i=1}^n (1 + (R_i \wedge \text{cap}) \vee \text{floor}) \right) - K \right)^+ \quad (3.3)$$

For $\text{floor} \ll 0$ and $\text{cap} \gg 0$ a geometric cliquet effectively becomes a standard Vanilla option.

Discrete-time models as described in Section 7.4 have been designed specifically for the risk management of exotic cliquets.

3.2.3 Market data

Market data for cliquets can be purchased from providers such as MarkIt. Those data are computed as consensus prices for specified products obtained from participating investment banks and do not have to be based on actual transactions. It turns out that consensus prices of Vanilla cliquets indicate the use of models where the distribution of forward-starting returns R_i for $i > 1$ is similar to the distribution of R_1 which can be inferred from Vanilla options, see

Section 2.5.5. Exotic cliquet prices point towards models with stochastic volatility (see Section 6.2) calibrated to the Vanilla market. While risk departments or regulators require models capturing both Vanilla and exotic cliquet data in a single model the data may actually be incompatible. It should be noted however that actual transactions are sparse. Also, model development for the purpose of providing data is driven by the requirement to achieve market consensus. This may perpetuate standard modelling approaches.

3.3 Asians

Asian options give the right to buy or sell an average of stock fixings at specified dates instead of the stock price at maturity. The purpose is usually to spread the fixing risk over a limited period of time. An additional desired effect is that an Asian option is cheaper than the corresponding European option. An Asian feature may determine the strike of a forward-starting option. In that case the option strike is set as a fixed percentage of an average of stock fixings in the near future. Averaging around maturity is also referred to as Asian-out, while averaging for the determination of the strike is called Asian-in. Thus one has call payoff variants

$$Asian = \left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K \right)^+ = (A - K)^+ \quad (3.4)$$

$$Asian-in = \left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K \frac{1}{m} \sum_{j=1}^m S_{s_j} \right) = (A - KA_{in})^+ \quad (3.5)$$

$$Relative\ Asian = \left(\frac{\frac{1}{n} \sum_{i=1}^n S_{t_i}}{\frac{1}{m} \sum_{j=1}^m S_{s_j}} - K \right) = \left(\frac{A}{A_{in}} - K \right)^+ \quad (3.6)$$

with $s_1 < s_2 < \dots < s_m < t_1 < t_2 < \dots < t_n$. Instead of arithmetic average A one may choose geometric mean G defined as

$$G = \left(\prod_{i=1}^n S_{t_i} \right)^{1/n}$$

Averaging often applies to the entire term of the contract for the purpose of hedging downside risk for investors exposed to regular payments. For that reason commodities contracts such as oil futures and options typically have an Asianing feature.

Pricing of these contracts often relies on Monte Carlo simulation with appropriate smile models, such as local volatility for Asian options with fixed strike (3.4)

or stochastic volatility for options with Asian-in features, such as (3.5) or (3.6). In the Black-Scholes case there are approximations for cases with discrete fixings or continuous averaging. Standard methods include log-normal approximation with matching of the first two moments, compare equation (8.1). For the second moment one has the choice of matching the moments of the average or its logarithm. Alternative distributions, such as Johnson's distribution, applied by Milevski and Posner (1998) to basket options allow for matching higher moments.

The two moment matching method has been refined by Curran (1994) using geometric conditioning:

$$\mathbb{E}[(A - K)^+] = \mathbb{E}[\mathbb{E}[(A - K)^+ | G] \mathbf{1}_{\{G < K\}}] + \mathbb{E}[\mathbb{E}[A - K | G] \mathbf{1}_{\{G > K\}}]$$

The second term can be computed in closed form. For the first term, Curran (1994) suggests a discrete approximation

$$\begin{aligned} & \mathbb{E}[\mathbb{E}[(A - K)^+ | G] \mathbf{1}_{\{G < K\}}] \\ & \approx \sum_{i=1}^n \mathbb{E}\left[\mathbb{E}\left[(A - G - ih)^+ \middle| G = \frac{K_{i-1} + K_i}{2}\right] \mathbb{P}[G \in [K_{i-1}, K_i]]\right] \end{aligned}$$

with $K_i = K - ih$, together with a log-normal approximation

$$A - G | G \approx \exp(\mu(G) + \sigma(G)X)$$

with standard Gaussian X and functions μ, σ which are also given.

3.4 Compound

A compound option is an option to buy or sell a specified option at a predetermined price. The payoff of a call on a call can be written as

$$\text{Compound} = (\mathbb{E}_t[(S_T - K)^+] - k)^+$$

This product can be used to defer the decision to buy an option by a short period of time and limit the downside risk of price movements during that time period.

Closed form valuation is due to Geske (1979) and relevant for structural models which assume stock price is an option on a company's assets, see Section 12.3.1. Naive Monte Carlo methods cannot be employed due to the expectation operator within the payoff. Instead of computationally intensive nested Monte Carlo

methods additional assumptions on the joint distribution of implied volatility and stock $(S_t, \hat{\sigma}_t(K, T))$ may allow for efficient pricing of this product as

$$\text{Compound} = (P_t(T) \text{BSCall}(F_t(T), K, \hat{\sigma}_t(K, T), T - t) - k)^+$$

3.5 Lookback

Lookback products are contingent on continuously- or discretely-monitored extrema of one or more assets. Let S_t^* and $S_{*,t}$ denote the maximum and minimum of an underlying value monitored on (a subset of) interval $[0, t]$. Single-underlying calls on the minimum include

$$\begin{aligned} \text{Fixed-strike lookback} &= (S_{*,t} - K)^+ \\ \text{Floating-strike lookback} &= (S_t - kS_{*,t})^+ \\ \text{Variable-notional lookback} &= \left(\frac{S_t}{S_{*,t}} - k \right)^+ \end{aligned}$$

Corresponding products exist on the maximum and of put type. Generally, Monte Carlo with extremum simulation, as described in Section 17.4.6, applied to smile consistent models featuring local or stochastic volatility (see Chapter 6) is the appropriate pricing method for this product family.

For products traded in the past one has to face the additional complexity of monitoring the relevant extremum attained up to today. This may be more involved than in the case of barrier products since the additional required information is a number rather than a boolean.

3.6 Autocallable

Autocallable or Express products are often sold as certificates in the German retail market. They are popular in stock markets moving sideways. The key feature of autocallable products is early redemption at hit of a discretely monitored upper barrier typically set equal to initial spot S_0 . Common monitoring frequency is annual or semi-annual. If the stock trades below the barrier at all monitoring dates t_i , $i = 1, 2, \dots, I$ the final payoff is often an increasing function $f(S_{[0, t_I]})$. The payoff is thus

$$\text{Autocallable} = \sum_{i=1}^I \mathbf{1}_{\{\tau = t_i\}} (1 + c_i) + \mathbf{1}_{\{\tau > t_I\}} f(S_{[0, t_I]})$$

where τ is the discretely monitored hitting time of the upper barrier at S_0 . Typical examples for f include a down-and-out barrier with $B \ll S_0$, such as

$$f(S_{[0,t]}) = \left(\frac{S_{t_I}}{S_0} \mathbf{1}_{\{S_{t_I} < S_0\}} + \mathbf{1}_{\{S_{t_I} \geq S_0\}} \right) \mathbf{1}_{\{S_{s,t_I} > B\}}$$

Since an autocalled product returns a coupon c_i at some t_i , which is typically set higher than the risk free interest rate achievable for the same period, the investor will most likely buy another product of the same kind. The probability of early redemption is hence often structured to be close to 100%. Due to the small probability of capital loss this product used to have additional tax advantages over capital guaranteed products.

Since autocallable products are barrier products with discrete upper (and sometimes continuous lower) barriers they can be valued using finite difference schemes.

Giese (2006) shows through a hedging argument that the price of an autocallable increases with correlation of stock and interest rate processes. If stock increases early redemption becomes more likely and the seller needs to hedge their position by purchasing shorter bonds and selling longer maturity bonds. Assuming positive (negative) correlation the interest rates will on average increase (decrease). The value of the longer maturity bonds sold decreases (increases) more than the bought short maturity bonds, resulting in a net loss (gain) on the bond hedge. To reflect this extra hedging cost (gain) the fair price in a hybrid model (see Chapter 11) should be higher (lower).

3.7 Volatility products

Options are volatility products in the sense that their price depends on the market view of volatility of the underlying. Investors seeking exposure to volatility without choosing a particular option strike can invest in products contingent on realized volatility. More recently, indices aiming at representing instantaneous volatility have gained attention as indicators of market sentiment.

3.7.1 Variance swap

A variance swap pays realized (as opposed to implied) variance at time T for a previously agreed fixed amount K , namely

$$\text{Variance swap} = \frac{1}{T-s} V(s, T) - K$$

Here $V(s, T)$ refers to realized cumulative log variance in period $[s, T]$ of an asset S as observable at time T . To capture more general cases, define the expectation at t of cumulative log variance in period $[s, T]$ of asset S with respect to a time discretization $\mathcal{T}$ as

$$V_t(s, T, S, \mathcal{T}) = \mathbb{E}_t \left[\sum_{t_i \in (s, T] \cap \mathcal{T}} R_i^2 \right], \quad R_i = \log \frac{S_{t_i}}{S_{t_{i-1}}} \quad (3.7)$$

In practice $\mathcal{T}$ refers to an enumeration of closing prices on business dates of the relevant stock exchange of S . For theoretical purposes $\mathcal{T}$ may be an infinitely fine discretization of real line. Arguments S and $\mathcal{T}$ are omitted if there is no ambiguity. Also, arguments s and t default to $s = 0$ and $t = T$ (unlike for other processes where t defaults to $t = 0$) and are omitted where justified. Hence

$$V(T) = V_T(0, T, S, \mathcal{T})$$

The variance swap contract has zero cost at inception 0 if

$$K = \frac{V_0(t, T)}{T - t}$$

The time increment in years $T - t$ is, in practice, replaced by a quantity of the same order computed using trade specific day count conventions. There are variants of this variance swap definition ranging from linear returns $R_i = S_{t_i}/S_{t_{i-1}} - 1$, to subtracting the sample mean of returns thus adjusting for the bias of the estimator, capping daily returns R_i or removing ex dividend dates from the variance calculation.

Variance is additive in the sense that

$$V(T) = V(t) + V(t, T)$$

For a wide range of models, namely models of type

$$\frac{dS_t}{S_t} = (r_t - d_t)dt + \sigma_t dW_t$$

one has

$$R_i^2 \approx \sigma_{t_{i-1}}^2 (t_i - t_{i-1})$$

and hence

$$V(t, T, \mathcal{T}) \approx \int_t^T \sigma_s^2 ds$$

if $\mathcal{T}$ is sufficiently fine. Processes σ, r, d can be stochastic and do not have to be continuous. The approximation fails for models with jumps, including Lévy or Merton-type processes or processes with jumps due to discrete dividends. It also deteriorates with fewer sample dates. The effect of discrete dividends on realized variance is discussed in Section 5.8.

3.7.2 Semi-static replication of variance swaps

From Itô's lemma one has

$$\frac{dS_t}{S_t} - d\log S_t = \frac{1}{2}\sigma_t^2 dt$$

which yields upon integration a decomposition of variance into a static part $\log S_T/S_0$ and a dynamic part dS_t/S_t consisting of a position in S readjusted dynamically to be worth 1. The static position in the log contract is usually replicated using out-of-the-money options since

$$\int_0^Z \frac{(K - S_T)^+}{K^2} dK + \int_Z^\infty \frac{(S_T - K)^+}{K^2} dK = -\log\left(\frac{S_T}{Z}\right) + \frac{S_T}{Z} - 1$$

where Z is some reference level. This yields the decomposition

$$\begin{aligned} \frac{1}{2}V(T) &= \int_0^T \frac{dS_t}{S_t} - \log \frac{S_T}{S_0} \\ &= \int_0^T \frac{dS_t}{S_t} - \log \frac{Z}{S_0} + 1 - \frac{S_T}{Z} + \int_0^Z \frac{(K - S_T)^+}{K^2} dK + \int_Z^\infty \frac{(S_T - K)^+}{K^2} dK \end{aligned}$$

Thus one sets up a portfolio at inception $t = 0$ consisting of

- a long zero coupon bond position with notional $2\left(1 - \log \frac{Z}{S_0}\right)$
- a short forward position with notional $/Z$
- a strip of out-of-the-money (with respect to Z) calls and puts with notional $2K^{-2}dK$ spanning a wide range $[K_-, K_+]$

In addition, a dynamic hedge is performed such that a portfolio in stock worth $\frac{2}{T}$ at all times is held. The expected value of the dynamic position multiplied by $\frac{T}{2}$ is

$$\begin{aligned} \mathbb{E}\left[\int_0^T \frac{dS_t}{S_t}\right] &= \mathbb{E}\left[\int_0^T (r_t - d_t)dt\right] \\ &= \log \frac{F(T)}{S_0} \end{aligned} \tag{3.8}$$

where the first equality assumes continuous dividend yield. The second equality additionally requires deterministic interest rates and dividend yield. With these assumptions the price of a variance swap is the price of the (idealized) hedge, namely

$$\frac{1}{2} V(T) = \int_0^{F(T)} \frac{Put(K, T)}{K^2} dK + \int_{F(T)}^{\infty} \frac{Call(K, T)}{K^2} dK$$

This price depends to a large extent on the prices quoted for illiquid out-of-the-money puts. One can also express the price of variance as a weighted sum of implied volatilities $\hat{\sigma}(K, T)$, see Gatheral (2006). Setting $k = \log K / F(T)$ one has

$$\frac{1}{T} V(T) = \int_0^{\infty} p(k) \hat{\sigma}(k, T)^2 dk$$

with

$$p(k) = \phi(d_-(k)) \frac{\partial d_-(k)}{\partial k}$$

$$d_-(k) = \frac{-\ln k}{\hat{\sigma}(k, T)\sqrt{T}} - \frac{\hat{\sigma}(k, T)\sqrt{T}}{2}$$

Assuming stochastic volatility models (see Section 6.2), namely a dynamic equation for the process σ , closed form expressions can be derived in some cases since

$$V(T) = \int_0^T \mathbb{E}[\sigma_t^2] dt$$

An example is Heston's model, see equation (6.11).

The above assumes deterministic interest rates and only pricing based on the expected value of the dynamic position is discussed. Variance swap pricing and hedging in the context of deterministic and stochastic interest rates is the topic of Section 11.5.

3.7.3 Volatility swap

A volatility swap pays

$$Volatility\ swap = \sqrt{\frac{V(T)}{T}} - K$$

at maturity T with K typically chosen such that no premium has to be paid at inception $t = 0$. Expanding the square root around a level close to the expected

variance swap price $V_0(T)$ one obtains from

$$\sqrt{x} \approx \sqrt{x_0} + \frac{x - x_0}{2\sqrt{x_0}} - \frac{(x - x_0)^2}{8x_0^{3/2}}$$

with $x = V(T)$ and $x_0 = V_0(T)$ the first-order approximation

$$\mathbb{E} \left[\sqrt{V(T)} \right] \approx \sqrt{V_0(T)} \left(1 - \frac{1}{8} \frac{\text{Var} [V(T)]}{V_0(T)^2} \right) \quad (3.9)$$

Thus there is a convexity adjustment to the volatility swap price, see Brockhaus and Long (2000). Alternatively, one may assume that $V(T)$ is approximately log-normally distributed:

$$V(T) \stackrel{(d)}{\approx} V_0(T) e^{\alpha X - \alpha^2/2}$$

with some standard Gaussian X and volatility α . In that case

$$e^{\alpha^2} = \frac{\mathbb{E} [V(T)^2]}{V_0(T)^2} = 1 + \frac{\text{Var} [V(T)]}{V_0(T)^2}$$

and

$$\mathbb{E} \left[\sqrt{V(T)} \right] = \sqrt{V_0(T)} e^{-\alpha^2/8} = \sqrt{V_0(T)} \left(1 + \frac{\text{Var} [V(T)]}{V_0(T)^2} \right)^{-1/8} \quad (3.10)$$

Assuming stochastic volatility models (see Section 6.2) one can compute closed form expressions for the variance of $V(T)$ in some cases. For Heston's model one has formula (6.12) to compute the convexity adjustment from model parameters. Inversely, if both variance and volatility swap prices are given one can derive variance of variance.

3.7.4 Implied volatility index (VIX)

The volatility index (VIX) is an approximation of the value of a one-month variance swap on stock index S&P 500. Technically, a grid of strikes K_i and (bid and offer) Vanilla prices on these strikes and maturity one month are required. From these prices one can derive the forward $F(T)$ and (undiscounted) mid prices Q_i for out-of-the-money options. VIX is defined as

$$\text{VIX}^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q_i(K_i) - \frac{1}{T} \left(\frac{F(T)}{K_0} - 1 \right)^2$$

see Chicago Board Options Exchange (2014). Here K_0 is the highest strike with $K_0 \leq F(T)$. Note that, following Gatheral (2006),

$$\begin{aligned}
 \frac{V_0(T)}{2} &= \int_0^{F(T)} \frac{Put(K, T)}{K^2} dK + \int_{F(T)}^{\infty} \frac{Call(K, T)}{K^2} dK \\
 &= \int_0^{K_0} \frac{Put(K, T)}{K^2} dK + \int_{K_0}^{\infty} \frac{Call(K, T)}{K^2} dK \\
 &\quad + \int_{K_0}^{F(T)} \frac{Put(K, T) - Call(K, T)}{K^2} dK \\
 &= \int_0^{K_0} \frac{Put(K, T)}{K^2} dK + \int_{K_0}^{\infty} \frac{Call(K, T)}{K^2} dK + \int_{K_0}^{F(T)} \frac{K - F(T)}{K^2} dK \\
 &\approx \int_0^{K_0} \frac{Put(K, T)}{K^2} dK + \int_{K_0}^{\infty} \frac{Call(K, T)}{K^2} dK - \frac{1}{K_0^2} \frac{(K_0 - F(T))^2}{2}
 \end{aligned}$$

Hence

$$VIX_t \approx \sqrt{12 V_t \left(t, t + \frac{1}{12} \right)}$$

The CBOE has listed futures on this index thus creating a liquid market for variance swaps. There is also a liquid market in options on VIX. Valuation of a VIX future at T is model dependent due to the square root in the definition of VIX. Dupire (2005) suggests using the formula

$$IE[VIX_T]^2 = IE[VIX_T^2] - \text{Var}[VIX_T]$$

The first term is a difference of two variance swaps and the second term may be estimated from historical data. Alternatively one may use the convexity adjustment between variance and volatility swaps discussed in Section 3.7.3.

3.7.5 Variance options

The variance swap price is sensitive to jumps of the underlying. Therefore contracts often include a cap on total variance. This cap effectively adds a short call option on variance:

$$V(T) \wedge \text{cap} = V(T) - (V(T) - \text{cap})^+$$

Other than this implicit (typically out-of-the-money) option there is limited liquidity in outright options on variance. While stochastic volatility models are

primarily developed to capture the implied volatility smile of the Vanilla market the prices of variance options specify the volatility process further. Log-normal models such as SABR have no skew by definition. The distribution of variance in Heston's model has moderately positive skew. Bayer, Gatheral and Karlsmark (2013) argue that log-normal models have a better agreement with the market.

3.7.6 Volatility products and Laplace transform

If the Laplace transform $f(\lambda) = \mathbb{E}[e^{-\lambda V(T)}]$ is known then the relation

$$v^p = \frac{p}{\Gamma(1-p)} \int_0^\infty \frac{1 - e^{-\lambda v}}{\lambda^{1+p}} d\lambda$$

applied with $p = 1/2$ yields

$$\begin{aligned} V_0(T) &= -\frac{\partial}{\partial \lambda} f(0) \\ \mathbb{E}[\sqrt{V(T)}] &= \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{1 - f(\lambda)}{\lambda^{3/2}} d\lambda \\ \mathbb{E}[(V(T) - K)^+] &= \frac{1}{\pi} \operatorname{Re} \int_0^\infty \frac{1 - f(-i\lambda) e^{-i\lambda K}}{\lambda^2} d\lambda + \frac{1}{2} (V_0(T) - K) \end{aligned}$$

For Heston's model the Laplace transform is related to CIR bond prices and is given in Section 10.5, formula (10.21).

3.7.7 Corridor variance swap

Carr and Lewis (2004) suggest bounding the process S to obtain

$$V(T, m, M) = \sum_{i=1}^N \left(\log \frac{(S_{t_i} \vee m) \wedge M}{(S_{t_{i-1}} \vee m) \wedge M} \right)^2 \approx \int_0^T \sigma_t^2 \mathbf{1}_{\{S_t \in [m, M]\}} dt$$

Itô's lemma for the process $S_t(m, M) = (S_t \vee m) \wedge M$ yields

$$d \log S_t(m, M) = \mathbf{1}_{\{S_t \in [m, M]\}} \left(\left(r_t - d_t - \frac{\sigma_t^2}{2} \right) dt + \sigma_t dW_t \right)$$

where the drift can be removed by considering S/F instead of S . Summation and taking expectations yields

$$\begin{aligned} V(T, m, M) &\approx 2\mathbb{E} \left[\log \left(\frac{F(T)}{S_T} \mathbf{1}_{\{S_T \in [m, M]\}} \right) \right] \\ &= 2 \left(\log F(T) \mathbb{P} [S_T \in [m, M]] - \mathbb{E} [\log S_T \mathbf{1}_{\{S_T \in [m, M]\}}] \right) \end{aligned}$$

It turns out that only options with strikes within $[m, M]$ are required.

3.8 Multi-asset products

Portfolio diversification can be achieved by trading in stock indices rather than single stocks. Component weighting within an index is typically rule based with the aim of representing a part of the market, such as a sector or region. Derivatives on indices can protect investments in those assets.

While index volatility is related to stock volatility through correlation, multi-asset products can also be designed specifically to gain exposure to correlation.

Mountain range products are a family of complex products that mimic trading strategies such as dynamically removing outliers.

3.8.1 Baskets and rainbows

A stock index (or basket) I of stocks $S^k, k = 1, 2, \dots, l$ is defined as

$$I_t = \sum_{k=1}^l w_k \frac{S_t^k}{S_0^k}$$

where the weights w_k are constants summing up to 1. An equally weighted basket thus has weights $w_k = 1/l$. For stock indices it is common to update weights at specified times to reflect market capitalization. Stock returns S_t^k/S_0^k may be capped at constant $1 + \text{cap}$ leading to an individually capped basket.

If weights apply to the ordered list of stock returns, namely

$$\{S_t^1, S_t^2, \dots, S_t^l\} = \{S_t^{(1)}, S_t^{(2)}, \dots, S_t^{(l)}\}$$

as well as

$$\frac{S_t^{(1)}}{S_0^{(1)}} \geq \frac{S_t^{(2)}}{S_0^{(2)}} \geq \dots \geq \frac{S_t^{(l)}}{S_0^{(l)}}$$

then the corresponding weighted sum

$$I_t = \sum_{k=1}^l w_k \frac{S_t^{(k)}}{S_0^{(k)}}$$

is referred to as a rainbow. Examples include the best (worst) performing stock with weights $w_k = \mathbf{1}_{\{k=1\}}$ ($w_k = \mathbf{1}_{\{k=l\}}$) applied to the ordered list. Allowing for negative weights one can also represent outperformance of the first stock over the second with weights $w_1 = 1, w_2 = -1$. With weights $w_k = \mathbf{1}_{\{k=1\}} - \mathbf{1}_{\{k=l\}}$ applied to the ordered list one obtains the dispersion of the stock returns measured by the difference between highest and lowest returns. More exotic variants include baskets where, at regular time intervals, the best and worst performing stocks are removed from the basket, resulting in a process with lower volatility than the individual stocks.

By assuming log-normal assets, analytical prices for basket, or indeed basket Asian options can be computed using a moment matching approach as described for Asian options in Section 3.3. Note that, often, forwards within a basket are less homogenous than within Asian products. For that reason analytical approximation is more challenging. Appropriate models for path-independent products include copula models, as described in Chapter 9. For exotic path-dependent products stochastic correlation combined with local or stochastic stock dynamics may be required, see Chapter 8.

3.8.2 Correlation products

Exposure to correlation can be captured through a combination of basket and single-stock options. A common example is a dispersion trade consisting of a long basket option where single-stock volatility exposure is offset by short single-stock option positions. Similarly, one may use straddles instead of options. Weighted average covariance over a time interval $[s, T]$ can also be obtained as spread of variance swaps

Average covariance

$$= \frac{1}{(T-s) \sum_{k \neq l} w_k w_l} \left(V \left(s, T, \sum_k w_k \frac{S^k}{S_0^k} \right) - \sum_k w_k^2 V \left(s, T, S^k \right) \right)$$

with $V(s, T, S)$ denoting realized log variance of a process S within time interval $[s, T]$ observed at time T as introduced in (3.7). Therefore average covariance can

be replicated with weak model assumptions. Using the approximation

$$\log \frac{\sum_k w_k S_{t_i}^k}{\sum_k w_k S_{t_{i-1}}^k} \approx \sum_k w_k \left(\log \frac{S_{t_i}^k}{S_{t_{i-1}}^k} \right) = \sum_k w_k R_i^k$$

one obtains the covariance swap

$$\text{Covariance swap} = \frac{1}{(T-s) \sum_{k \neq l} w_k w_l} \sum_{k \neq l} \left(w_k w_l \sum_{t_i \in (s, T]} R_i^k R_i^l \right)$$

The corresponding correlation swap is

$$\text{Correlation swap} = \frac{1}{\sum_{k \neq l} w_k w_l} \sum_{k \neq l} \left(w_k w_l \frac{\sum_{t_i \in (s, T]} R_i^k R_i^l}{\sqrt{\sum_{t_h \in (s, T]} (R_h^k)^2 \sum_{t_j \in (s, T]} (R_j^l)^2}} \right)$$

This correlation estimator is biased due to the correlation of covariance and the variance estimator.

Valuation of correlation products often relies on Monte Carlo simulation with deterministic correlation. While such simplistic correlation dynamics do not seem to be justified, the main challenge is often finding correlation levels in line with market data. Due to decorrelation effects in this context model correlation parameters need to be different within different models. This point is discussed in Section 8.3. For exotic correlation products stochastic correlation dynamics (see Section 8.5) is more convincing than local correlation (Section 8.4).

3.8.3 Mountain range

High-dimensional multi-asset path-dependent products named after Himalayan mountain were introduced by Société Générale in 1998, see Quessette (2002). Following this development institutions competed by offering new products with ever increasing numbers of assets. This required them to invest in computing power. The first competitor able to offer a new product typically earned the highest margin. Often the structure aimed to achieve a high price computed with a simplified model, whereas the more realistic price was much lower. Simplified models include models without volatility or correlation skew, local rather than stochastic volatility or correlation (see Section 8.5) or models with equal correlation levels for all asset pairs, i.e. absence of correlation dispersion. In many cases the margin charged to customers may have been high enough to compensate for model deficiencies. However the cost of managing these products in terms of

development, IT infrastructure and risk management (including reputational risk) may not always have been sufficiently accounted for by the trading and sales desks pushing them.

Altiplano

The payoff is given as

$$Altiplano = c \mathbf{1}_{\left\{ \min_k \frac{S_{s,t}^k}{S_0^k} > B \right\}} + \mathbf{1}_{\left\{ \min_k \frac{S_{s,t}^k}{S_0^k} \right\}} \left(\sum_k w_k \frac{S_t^k}{S_0^k} - K \right)^+$$

where $c \gg 1$ is a high coupon whereas $B \ll 1$ is a low barrier. Parameters are often set such that the initial offering price is 100%. If correlation is high then the probability of receiving the coupon is low. In that case, if the coupon is not received, high correlation will typically result in a low basket (or sometimes basket Asian) option payoff. If correlation is low then it is more likely that the coupon will be received. The value of the product typically increases with correlation dispersion.

Annapurna

This product is based on the same event as the Altiplano product, namely no asset hitting a lower barrier. Given the event the product pays a Vanilla or basket option:

$$Annapurna = \mathbf{1}_{\left\{ \min_k \frac{S_{s,t}^k}{S_0^k} > B \right\}} \left(\sum_k w_k \frac{S_t^k}{S_0^k} - K \right)^+$$

Atlas

The Atlas product is a basket option. At specified dates t_i , $i = 1, 2, \dots, m$ both best and worst performers are removed from the basket. Denoting by R^i the index set of remaining stocks one has

$$I_t^i = \frac{1}{\#R^i} \sum_{k \in R^i} \frac{S_t^k}{S_0^k}$$

The final payoff is

$$Atlas = (I_t^m - K)^+$$

where $t \geq t_m$. The index sets are defined recursively as

$$R^0 = \{1, 2, \dots, m\}$$

as well as

$$R^i = R^{i-1} \setminus \left(\left\{ k \in R^{i-1} : \forall l \in R^{i-1} : \frac{S_{t_i}^k}{S_0^k} \geq \frac{S_{t_i}^l}{S_0^l} \right\} \cup \left\{ k \in R^{i-1} : \forall l \in R^{i-1} : \frac{S_{t_i}^k}{S_0^k} \leq \frac{S_{t_i}^l}{S_0^l} \right\} \right)$$

Note that removed assets are not considered again, regardless of their performance.

Everest

This product name refers to a worst-of option with long maturity and many assets.

Himalayan

The payoff is an option on a discrete-time average based on a different stock for each averaging date t_i , $i = 1, 2, \dots, m$, namely

$$Himalaya = \left(\frac{1}{m} \sum_{i=1}^m \frac{S_{t_i}^{k_i}}{S_0^{k_i}} - K \right)^+$$

The sequence $k_1, k_2, \dots, k_m$ is related with index sets R_i defined recursively through

$$R^0 = \{1, 2, \dots, m\}$$

$$R^i = R^{i-1} \setminus \left\{ k \in R^{i-1} : \forall l \in R^{i-1} : \frac{S_{t_i}^k}{S_0^k} \geq \frac{S_{t_i}^l}{S_0^l} \right\}$$

via

$$R^i = R^{i-1} \cup \{k_i\}$$

Thus at fixing date t_i the best performance of all stocks within the index set R^{i-1} is added to the sum. Then the index of this best performing stock is removed, defining R^i .

3.9 Dynamic strategies

Interest by retail investors in complex payoffs, such as mountain range (Section 3.8.3), with opaque risks has all but vanished. Institutions have reacted by offering simple

payoffs on a complex underlying, namely the value of a synthetic dynamic strategy. The payoff may be the value of the strategy floored by a bond for capital protection.

Dynamic strategies offer investors access to a broad range of markets. They compete with hedge funds or active fund managers in offering algorithmic trading strategies. Strategies are rule-based, thus providing transparency as no fees are paid to asset managers for actively managing a portfolio. They can be tailored to provide downside protection (CPPI), mitigate volatility risk (vol target, timer) or take advantage of market moves (autopilot).

3.9.1 Constant proportion portfolio insurance (CPPI)

The basic idea is to manage a portfolio such that the risk is related to the distance from the present value of the insured amount F_t . If the portfolio is performing well a higher percentage of the portfolio holdings is in risky assets. The portfolio I is described by

$$I_t = \alpha_t S_t + \beta_t B_t$$

with S being a risky asset and B a less risky bond. Here S and B might also be funds. The concept can be extended to cover several assets with decreasing risk. For all $t \in (0, T]$ with T typically between five and 30 years, the guaranteed amount F_t is related to the riskless rate, fees and coupons paid out of the portfolio to investors or asset managers and additional guarantee (lock-in) features.

In the classical CPPI a fixed proportion m of excess wealth is held in the asset, namely

$$\alpha_t S_t = m_t (I_t - F_t).$$

The dynamics of the portfolio is

$$\begin{aligned} dI_t &= \alpha_t dS_t + r_t (I_t - \alpha_t S_t) dt \\ &= m_t (I_t - F_t) \frac{dS_t}{S_t} + r_t ((1 - m_t) I_t + m_t F_t) dt \\ &= m_t (I_t - F_t) \frac{dS_t}{S_t} + r_t (1 - m_t) (I_t - F_t) dt + r_t F_t dt \end{aligned}$$

An alternative interpretation can be given in terms of crash risk protection. Assuming that the aim is to survive, at all times $t \in (0, T]$, a proportional crash of the asset to $S_t = (1 - c_t) S_t^-$. Then the strategy is given by

$$\alpha_t S_t (1 - c_t) + \beta_t B_t = F_t$$

or

$$c_t = \frac{1}{m_t}$$

Typically, the protected crash size is between 10% and 20%.

In the special case of a riskless floor $\frac{dF_t}{F_t} = r_t dt$ the equation simplifies to

$$\frac{d(I_t - F_t)}{I_t - F_t} = m_t \frac{dS_t}{S_t} + r_t(1 - m_t)dt$$

Thus, assuming an asset with volatility σ and drift $r - d$ the volatility of the excess wealth $I - S$ is $m\sigma$ and the drift is $(2 - m)r - d$. The Black-Scholes solution reads

$$I_t = F_t + (I_0 - F_0) \exp \left(\int_0^t \left((2 - m_u)r_u - d_u - \frac{1}{2}m_u^2\sigma_u^2 \right) du - \int_0^t m_u\sigma_u dW_u \right)$$

In order to minimize transaction costs the portfolio is typically only rebalanced if the actual proportion deviates from the target by more than a given number, namely

$$\frac{\alpha_t S_t}{I_t - F_t} \notin [m_t - \varepsilon_{lower}, m_t + \varepsilon_{upper}]$$

Other common features include lock-in percentages p_i at given dates $t_i, i = 1, \dots, n$ increasing the guaranteed amount to $p_i I_{t_i}$ if $p_i I_{t_i} > F_T$. It turns out that even without lock-ins holdings in the risky asset often vanish quickly leaving the investor with a low yielding asset for the remaining term. To counter this α_t may be floored thus increasing the shortfall risk, that is $I_T < F_T$. An analysis of shortfall risk due to discrete rebalancing and minimum risky asset holding in the context of log-normal assets can be found in Brandl (2009). CPPI on assets with jumps are discussed by Cont and Tankov (2009).

3.9.2 Volatility target

Investors are often interested in exposure to stock fluctuations. However they may not want Vega exposure. One can rebalance between a risky asset S and a riskless bond B such that the portfolio I realizes a given target volatility Σ . Define I as

$$\frac{dI_t}{I_t} = \beta_t \frac{dS_t}{S_t} + (1 - \beta_t) \frac{dB_t}{B_t}$$

With $\beta = \frac{\Sigma}{\sigma_t}$ one has

$$\frac{dI_t}{I_t} = r_t dt + \Sigma dW_t$$

Note that σ_t has to be estimated from realized data. Volatility target options have reduced sensitivity to volatility risk.

3.9.3 Timer option

Vega risk can also be reduced by allowing for random option maturity. To this end Société Générale introduced timer options in 2007. Given a variance budget V the timer call option expires as soon as realized variance reaches V . Thus the payoff is

$$(S_\tau - K)^+$$

at time τ where

$$\tau = \inf\{t : \sum_{i:t_i \leq t} R_i^2 \geq V\}, \quad R_i = \log \frac{S_{t_i}}{S_{t_{i-1}}}$$

Due to the Dambis-Dubins-Schwarz theorem the fair option price is given by the Black-Scholes formula with variance V if appropriate assumptions hold. Those assumptions include a fine time grid (t_i) for computation of realized variance, no jumps, as well as no drift (e.g. dividends, repo rate) or interest rates. While none of these assumptions are satisfied in typical contracts, matters are complicated further since, in practice, τ is capped.

3.9.4 Autopilot

An autopilot is a dynamic strategy for investing in a diversified portfolio depending on a trend indicator. Assume a fixed number n of liquid baskets. To allow for diversification the baskets may be given as a developed economy index, an emerging market index, a commodity index or a bond index.

The trend indicator is computed for every basket by comparing the annual performance with a one-year monthly average performance. If the indicator shows positive trend for basket i then the strategy invests $1/n$ of the current strategy level in basket i . Otherwise the amount is held in cash. Positive monthly performance is capped at a fixed percentage. Due to the cap this product has pronounced forward skew exposure.

3.10 Dividend products

Dividends can be traded through calendar spreads of forwards, see (2.1). While the forward price reflects implied dividends, a stock or index position yields realized dividends. Early dividend derivatives include dividend swaps and options on dividends. The variable leg of a dividend swap consists of all gross dividends paid

within a specified period against an agreed strike. Thus a dividend swap has payoff

$$\text{Dividend swap} = \sum_{t_i \in [t, T]} D_{t_i}^i - K$$

where $D_{t_i}^i$ denotes the i -th dividend going ex at time t_i , compare Chapter 5. An option will pay this amount at a specified time, greater or equal to T if positive, and expires worthless otherwise. Bühler et al. (2010) also mention OTC products, including down-and-in dividend swap with the barrier on the corresponding spot being alive during the same period $[t, T]$. Dividend yield swaps normalize the sum of dividends by the corresponding spot price.

Dividend derivatives also trade on the listed market. Futures on the dividend stream of Stoxx 50 launched on Eurex in 2008 followed by single-stock dividend futures, with options on those index futures introduced in 2010, see Eurex (2014). In the same year CBOE launched options on the S&P annual dividend index (DIVD) and the S&P dividend index (DVS).

Modelling choices for dividends are developed in Chapter 5. As discussed therein they have an impact on all equity derivatives, in particular European and American Vanilla options, as well as path-dependent products such as cliquets and volatility products such as variance swaps. Specific models for dividend derivatives require additional model parameters such as volatility and correlation of dividend yield.

4 Implied Volatility

In equities markets one has a set of bid and offer market prices for out-of-the-money call and put options on a given underlying. These options may be European or American.

In the case of European options one can visualize bid and offer implied volatilities computed from mid discount factor and forward, see (2.4). Mid discount factor and forward are derived from fixed income, spot and dividend market data and are adjusted as required by option market prices and arbitrage considerations, see Section 2.5.1. Implied volatility functions are then parametrized in real time with the aim of always staying between bid and offer. Challenges include data quality, liquidity, frequency and tightness.

Some markets only quote American option prices. Since market participants often use single-volatility models to price American options it may be sufficient to use those to define implied volatilities. More advanced calibration methods are discussed in Section 2.5.6.

This section discusses skew parametrizations commonly used by practitioners in view of strike and calendar arbitrage. It includes Lee's moment formula and the interpolation and extrapolation methods required to describe the Vanilla option market.

4.1 Skew parametrization

Implied volatility for a given maturity T can be represented as a smooth function with a limited set of parameters, e.g. at-the-money volatility, skew and convexity, and tail parameters. Even if more parameters are used it is desirable to capture volatility dynamics by modifying only few of them. Figure 4.1 compares the skew parametrization examples described below. Parameters are chosen such that all curves exhibit atm-volatility level 20%, skew -0.2 and convexity 0.2 .

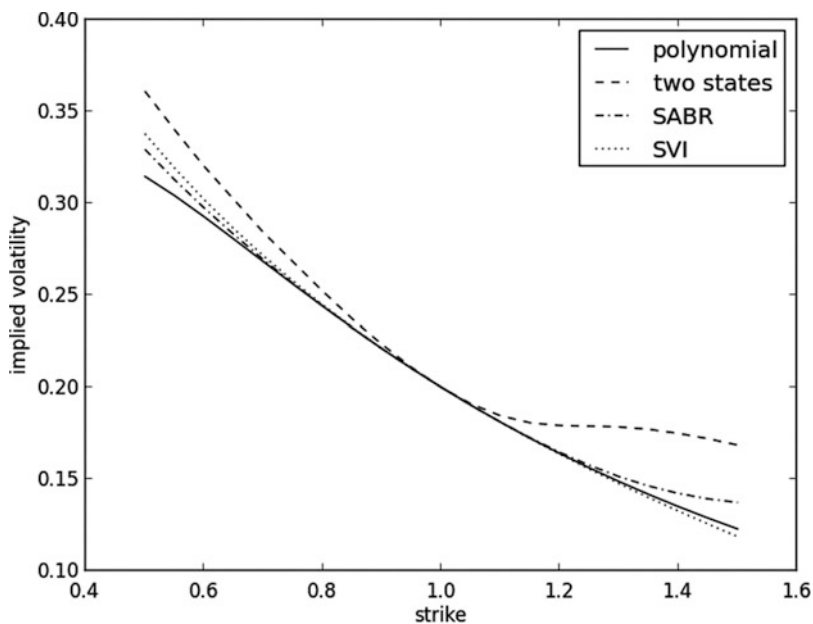


Figure 4.1 Volatility skew parametrizations

4.1.1 Exponential polynomial

It is common to express the parametrization as a function of normalized strike k with

$$k(K, F(T)) = \log \frac{K}{F(T)} \quad (4.1)$$

A simple parametrization based on log strike is given by

$$\hat{\sigma}(K) = \sigma \exp(\beta k) \quad (4.2)$$

Parameters σ and β are related to at-the-money volatility and skew respectively, namely

$$\begin{aligned} \hat{\sigma}(F(T)) &= \sigma \\ \frac{\partial \hat{\sigma}}{\partial K}(F(T)) &= \frac{\sigma}{F(T)} \beta \end{aligned}$$

Adding a quadratic term gives parametrization

$$\hat{\sigma}(K) = \sigma \exp\left(\beta k + \frac{\gamma k^2}{2}\right) \quad (4.3)$$

The extra parameter γ is related to convexity:

$$\frac{\partial^2 \hat{\sigma}}{\partial K^2}(F(T)) = \frac{\sigma}{F(T)} (\gamma + \beta (1 - F(T)^{-1}))$$

4.1.2 Two-state model

An implied volatility smile $\hat{\sigma}(K)$ can be defined implicitly through

$$\begin{aligned} BSCall(F(T), K, \hat{\sigma}(K), T) &= \frac{1}{2} BSCall(F(T), K, \hat{\sigma}_+(K), T) \\ &\quad + \frac{1}{2} BSCall(F(T), K, \hat{\sigma}_-(K), T) \\ \hat{\sigma}_{\pm}(K) &= \sigma (1 \pm w + mk) \end{aligned} \quad (4.4)$$

where σ, w, m denote parameters related with at-the-money volatility, skew and convexity. Inversion of the Black-Scholes formula is not required for pricing and risk management of European options.

This model is also known as the wow-alpha model, where wow (or vol of vol) refers to parameter w and $\alpha = \log F/K = -k$. It was introduced, to the best of our knowledge, by Susquehanna in the 1990s or earlier.

4.1.3 Hagan formula

The SABR model described in Section 6.3.1 has dynamics

$$\begin{aligned} \frac{dS_t}{S_t} &= \sigma_t S_t^{\beta-1} dW_t^S \\ d\sigma_t &= \alpha \sigma_t dW_t^\sigma \end{aligned}$$

Thus the model has parameters $S_0, \sigma_0, \alpha, \beta$. A further parameter is the correlation ρ of W^S and W^σ . Drift parameters r and d can be introduced by replacing S by $\tilde{S}$ with

$$\tilde{S}_t = S_t \exp\left(\int_0^t (r_s - d_s) ds\right) \quad (4.5)$$

Hagan et al. (2002) derive a surprisingly accurate approximation formula for implied volatility in terms of the SABR parameters, which in the special case $\beta = 1$ without local volatility reads

$$\hat{\sigma}(K, T) \approx \sigma_0 \frac{z}{\chi(z)} \left(1 + \left(\frac{\rho \alpha \sigma_0}{4} + \frac{2 - 3\rho^2}{24} \alpha \right) T \right)$$

where

$$\chi(z) = \log \frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho}$$

$$z = \frac{\alpha}{\sigma_0} \log \frac{F(T)}{K}$$

This may be considered the most important aspect of the SABR model since the formula has become a standard for implied volatility parametrization. It is easy to see that one obtains the same range of volatility smiles using the simpler formula

$$\hat{\sigma}(K) \approx \sigma_0 \frac{z}{\chi(z)} \quad (4.6)$$

since one only has to make the substitution

$$\tilde{\sigma}_0 = \sigma_0 \left(1 + \left(\frac{\rho \alpha \sigma_0}{4} + \frac{2 - 3\rho^2}{24} \alpha \right) T \right)$$

$$\tilde{\alpha} = \alpha \left(1 + \left(\frac{\rho \alpha \sigma_0}{4} + \frac{2 - 3\rho^2}{24} \alpha \right) T \right)$$

Thus implied volatilities for a given maturity T may be specified by fixing $F(T)$ and solving for parameters σ_0, α, ρ such that formula (4.6) yields a good agreement with implied volatility data.

4.1.4 SVI

Gatheral (2006) introduced the stochastic volatility inspired (SVI) parametrization

$$\hat{\sigma}(K)^2 = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right) \quad (4.7)$$

where parameters a, b, ρ, σ, m impact the skew as follows:

- a overall level of variance
- b angle between left and right asymptotes
- σ smoothness of vertex
- ρ orientation of the graph
- m translation of the graph

4.2 Tail behaviour

The parametrization examples given in Section 4.1 lead to valid implied volatilities around the forward $F(T)$. However, they typically fail to generate an arbitrage-free distribution of S_T on $[0, \infty)$. It has been shown by Lee (2004) that the volatility function $\hat{\sigma}$ implied by a distribution function F^{ST} with first moment $F(T)$ satisfies growth conditions for high and low strikes, namely

$$\begin{aligned} \limsup_{k \nearrow \infty} \frac{\hat{\sigma}^2(K)T}{k} &= \beta^* \in [0, 2] \\ \liminf_{k \searrow -\infty} \frac{\hat{\sigma}^2(K)T}{-k} &= \beta_* \in [0, 2] \end{aligned} \quad (4.8)$$

with log strike k defined in (4.1). As a consequence valid volatility parametrizations cannot have extreme growth in either wing. Moreover, the tail behaviour of volatility is related to the existence of moments of the distribution of S_T via Lee's moment formula

$$\begin{aligned} \sup\{p : \mathbb{E}[S_T^{1+p}] < \infty\} &= \frac{1}{2} \left(\frac{1}{\sqrt{\beta^*}} - \frac{\sqrt{\beta^*}}{2} \right)^2 \\ \sup\{p : \mathbb{E}[S_T^{-p}] < \infty\} &= \frac{1}{2} \left(\frac{1}{\sqrt{\beta_*}} - \frac{\sqrt{\beta_*}}{2} \right)^2 \end{aligned} \quad (4.9)$$

4.2.1 Exponential polynomial

The exponential formula (4.2) with negative skew $\beta < 0$ leads to parameter $\beta^* = 0$. The low strike limit is $\beta_* = \infty$. Hence the formula is not arbitrage-free and consequently for a functional form with slower growth has to be chosen.

If additionally convexity is allowed (4.3) then one has

$$\beta^* = \beta_* = \infty$$

and the function has to be modified for high strikes as well.

4.2.2 Two-state model

Formula (4.4) has excessive growth in both wings and is only arbitrage-free within an interval around $F(T)$.

4.2.3 Hagan formula

For strikes far from at-the-money the quality of the approximation (4.6) deteriorates. This may not be an issue when exact correspondence between model and implied volatility parameters is not required, as long as the formula represents a valid distribution. However the implied stock distribution is not arbitrage-free since

$$\frac{\hat{\sigma}^2(K)T}{k} = k \frac{\alpha^2}{\sigma_0^2} \chi^{-2} \left(-\frac{\alpha}{\sigma_0} k \right) \sim \frac{k}{\log(|k|)^2}$$

for $|k| \rightarrow \infty$. There is continued research into better implied volatility approximations or indeed exact formulae, see for example Johnson and Nonas (2009), Paulot (2009) and Islah (2009) and references therein.

4.2.4 SVI

The restriction in (4.8) is satisfied if

$$b(1 + |\rho|) \leq \frac{2}{T}$$

holds, compare Gatheral and Jacquier (2014).

4.2.5 Gaussian mixture

Brunner and Hafner (2003) suggest following a given parametrization within strike interval $[K_l, K_r]$ and using a smooth extrapolation for $K \notin [K_l, K_r]$. A candidate for the left tail should complete the distribution, such that put option prices $Put(K, T)$, as well as infinitesimal put spreads $F^{ST}(K)$ and butterflies $f_{ST}(K)$, are continuous at $K = K_l$. A similar requirement can be formulated for distributions for the right tail. A log-normal distribution cannot be used as only two parameters are available for three conditions. An alternative is a mixture of two log-normal distributions, namely

$$S_T \stackrel{(d)}{=} F(T) \left(\mathbf{1}_{\{U > \lambda\}} e^{\mu_1 + \sigma_1 X} + \mathbf{1}_{\{U \leq \lambda\}} e^{\mu_2 + \sigma_2 X} \right)$$

The disadvantage is that one has superfluous parameters. Also, patching together three different regimes seems unintuitive.

4.2.6 Delta space

Foreign exchange markets quote volatility in delta space. Since delta of a Vanilla option takes values in a bounded interval, a parametric function of delta will lead to finite limits of volatility for both low and high strikes. In particular the expectation of S_t^p will be finite for all p due to (4.9).

A simplified Black-Scholes put delta formula is

$$\Delta(K, F(T), T, \sigma) = \Phi \left(\frac{\log \frac{K}{F(T)}}{\sigma \sqrt{T}} \right)$$

In order to convert a given parametrization $\hat{\sigma}(k)$ in log strike (4.1) to a parametrization in delta space one may use the linear transformation

$$k(K, F(T), T, \sigma) = \sigma \sqrt{2\pi T} \left(\Phi \left(\frac{\log \frac{K}{F(T)}}{\sigma \sqrt{T}} \right) - \frac{1}{2} \right) \quad (4.10)$$

The transformation has been chosen such that

$$k(K, F(T), T, \sigma) \xrightarrow{\sigma \nearrow \infty} k(K, F(T))$$

holds. Lower values of σ flatten the tail. Figure 4.2 exhibits the impact of different values of the tail parameter $(\sigma \sqrt{T})^{-1}$ on the exponential polynomial smile function.

One can introduce two values σ_+, σ_- applicable for $K > F(T)$ and $K < F(T)$ respectively. This allows one to flatten upper and lower tails independently. Note that with tail parameters $\sigma_+ \neq \sigma_-$ the smile convexity is continuous but not smooth at $F(T)$. This approach can be generalized to suitable functions other than Φ . It is global in the sense that tail parameters impact the entire smile.

4.3 Time dependence

In order to describe the implied volatility surface the skew parametrizations for different maturities have to be interpolated.

Interpolation of volatilities between $(k(t_1), t_1)$ and $(k(t_2), t_2)$ where k is a suitable normalization has the disadvantage that the interpolated smile cannot be represented using the chosen parametric form at the corner dates t_1, t_2 . Arbitrage checks may be performed on a given parametric form but will be harder to verify for interpolated volatilities.

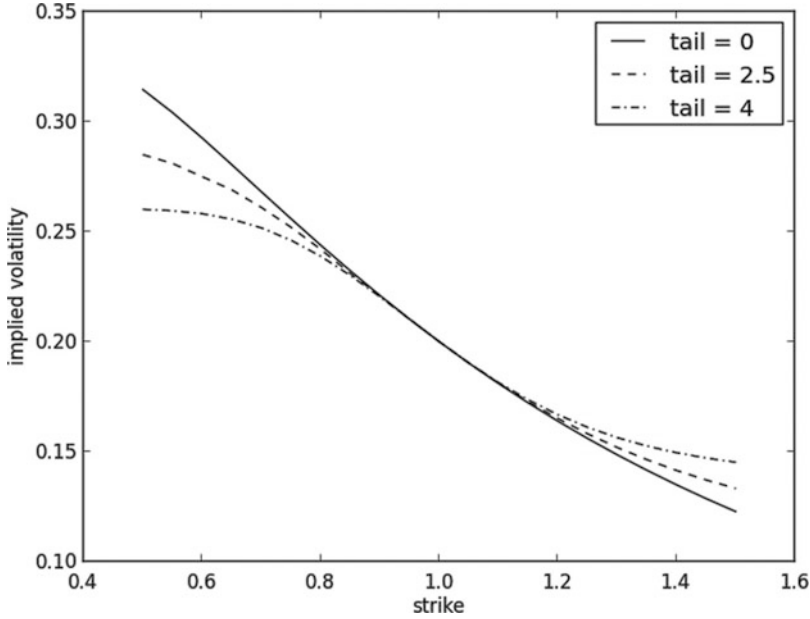


Figure 4.2 Volatility skew with tail parameter

It is useful to choose skew parametrizations which do not lead to a strong time dependence of parameters. The surface can then be represented by the appropriate interpolation of parameters. Formula (4.1) incorporates space normalization. Time normalization may be incorporated through time scaling t^p as

$$k(K, F(t), p) = \frac{\log \frac{K}{F(t)}}{t^p} \quad (4.11)$$

Time power $p = \frac{1}{2}$ assumes linear growth of log variance.

4.3.1 Two-state model

In some cases implied volatility parametrizations allow for the computation of local volatility, as introduced in Section 6.1 in closed form. The parametric smile defined through (4.4) with time-dependent parameters reads

$$\begin{aligned} BSCall(F(T), K, \hat{\sigma}(K, T), T) &= \frac{1}{2} BSCall(F(T), K, \hat{\sigma}_+(K, T), T) \\ &\quad + \frac{1}{2} BSCall(F(T), K, \hat{\sigma}_-(K, T), T) \\ \hat{\sigma}_{\pm}(K, T) &= \sigma(T) (1 \pm w(T) + m(T)k(K, F(T))) \end{aligned}$$

where $k = \log K/F(T)$ as in (4.1) and σ, w, m are time-dependent parameters.

Local volatility can be obtained from formula (6.5) in connection with derivatives of the Black-Scholes formula with respect to K (see (2.5) and (2.6)) and $\hat{\sigma}$ (or vega as given in (18.10)) as

$$\sigma_t(K) = \sqrt{\frac{a_+ + a_-}{b_+ + b_-}}$$

$$a_{\pm} = 2\phi(d_{\pm}(\hat{\sigma}_{\pm})) \left(\frac{\sigma'}{\sigma} \hat{\sigma}_{\pm} \sqrt{t} + \frac{1}{2} \frac{\hat{\sigma}_{\pm}}{\sqrt{t}} + \sigma (\pm w' - m'k) \right)$$

$$b_{\pm} = \frac{\phi(d_{\pm}(\hat{\sigma}_{\pm}))}{\hat{\sigma}_{\pm} \sqrt{t}} \left(1 + \frac{2k}{\hat{\sigma}_{\pm}} \sigma m + \left(\frac{k^2}{\hat{\sigma}_{\pm}} - \frac{1}{4} \hat{\sigma}_{\pm}^4 t^3 \right) \sigma^2 m^2 \right)$$

4.3.2 Hagan formula

Calibrated term parameters from the Hagan formula (4.6) exhibit reasonably stable values for correlation ρ while volatility of volatility α decreases with maturity. This reflects the missing mean reversion of volatility in the SABR model, see Section 6.3.1.

One may use the market forward curve $F(t)$ and appropriate interpolation and extrapolation for at-the-money volatilities $\hat{\sigma}(F(t), t)$, a fixed value for ρ and a parametric formula for α^2 . The latter may be derived from the variance of mean reverting processes, such as OUP or CIR. Since $\alpha_t^2 t$ is a measure for the total variance of volatility at time t one would typically expect this measure to increase with t .

4.3.3 SVI

Gatheral and Jacquier (2014) present the reparametrization of (4.7)

$$\sigma_0^2 = a + b \left(-\rho m + \sqrt{m^2 + \sigma^2} \right) \quad \sigma_-^2 = a + b\sigma \sqrt{1 - \rho^2}$$

$$s_0 = \frac{b}{2\sigma_0} \left(-\frac{m}{\sqrt{m^2 + \sigma^2}} + \rho \right) \quad s_{\pm} = \pm \frac{b}{\sigma_0} (1 \pm \rho)$$

The new parameters are intuitive as they represent

$$\begin{array}{ll} \sigma_0, \sigma_- & \text{at-the-money volatility, minimum volatility} \\ s_-, s_0, s_+ & \text{left skew, at-the-money skew, right skew} \end{array}$$

A variant of this reparametrization has presumably been used at Goldman Sachs and is attributed to Tim Klassen in Gatheral and Jacquier (2014).

A key result in Gatheral and Jacquier (2014) are conditions for a smooth surface with SVI time slices ensuring absence of both calendar and strike arbitrage.

5 Dividends

One of the attractive features of stocks is that the holder is entitled to dividends. Dividend levels, payment and ex-dates are typically stable and known in advance but are sometimes adjusted by the board of management.

There are various methods to incorporate dividends into equity processes. Successful models combine tractability, namely the availability of closed form formulae for European options, with realistic features such as deterministic near dividends and stochastic far dividends.

This section discusses widely used approaches, including classic deterministic and proportional dividends. A compromise between deterministic and proportional dividends is given by affine models, assuming that dividends can be written as the sum of a constant and a proportional part, where proportional refers to the stock price at the ex-dividend date.

Dividend discount models attempt to model stock prices as the sum of expected future dividend streams. It will be seen that many stock models, including affine models, can be interpreted as dividend discount models. Beginning with a more realistic assumption on dividend dynamics than in the affine model allows one to build a homogenous stock price model, which will be developed at the end of this section.

5.1 Forward

Delivery of stock at time t can be replicated by purchasing the stock today. Dividends going ex in period $(0, t]$ reduces the replication cost by the present value of those dividends. This leads to the forward price

$$F(t) = P(t)^{-1} \left(S_0 - \sum_{t_i \leq t} D_0^i \right) \quad (5.1)$$

where D_t^i is the expected value at time t of the dividend going ex at time t_i . The forward price is the risk neutral expectation of the stock price at time t .

5.2 Proportional dividends

Writing the stock price S as

$$S_t = F(t)M_t \quad (5.2)$$

where M is a continuous martingale with $M_0 = 1$, one obtains proportional dividends:

$$\begin{aligned} D_{t_i}^i &= S_{t_i-} - S_{t_i} = (F(t_i-) - F(t_i)) M_{t_i-} \\ &= \frac{P(t_i)^{-1} D_0^i}{F(t_i-)} S_{t_i-} = \frac{D_0^i}{S_0 - \sum_{j < i} D_0^j} S_{t_i-} \end{aligned} \quad (5.3)$$

5.2.1 Closed form formula

The calibration problem of European options can be translated into a calibration problem for options on the martingale. Due to (5.2) one has

$$(S_t - K)^+ = F(t) \left(M_t - \frac{K}{F(t)} \right)^+$$

In other words an option on S expiring at t with strike K can be interpreted as $F(t)$ options on M with strike $K/F(t)$. The implied volatility $\hat{\sigma}_M(k, t)$ for an option on M with strike k has to satisfy

$$\frac{Call(kF(t), t)}{P(t)F(t)} = BSCall(1, k, \hat{\sigma}_M(k, t), t)$$

Note that smooth $\hat{\sigma}_M$ imposes a discontinuity of standard implied volatility $\hat{\sigma}$ of options written on S at ex-dividend dates.

5.2.2 Binomial tree

A naive binomial tree can be implemented by setting state S_n^k after k up moves and $n - k$ down moves to

$$S_n^k = S_0 e^{(2k-n)\sigma\sqrt{\Delta}}$$

The corresponding risk neutral node probability for a down move at time $n\Delta$ is given as

$$p_n^k = \frac{S_{n+1}^{k+1} - F_n^k}{S_{n+1}^{k+1} - S_{n+1}^k}$$

where F_n^k is the one step forward price with maturity $n+1$ at space-time node (k, n) . Due to (5.1) one has

$$F_n^k = P(\Delta)^{-1} (S_n^k - D_n)$$

where D_n is the sum of the value of all dividends going ex in the time interval $(t_n, t_{n+1}]$ at t_n . In the proportional dividend case one has $D_n = c_n S_n^k$ for a constant c_n , see (5.3). Hence

$$\begin{aligned} p_n^k &= \frac{S_{n+1}^{k+1} - P(\Delta)^{-1} (S_n^k - c_n S_n^k)}{S_{n+1}^{k+1} - S_{n+1}^k} \\ &= \frac{e^{\sigma\sqrt{\Delta}} - P(\Delta)^{-1} (1 - c_n)}{e^{\sigma\sqrt{\Delta}} - e^{-\sigma\sqrt{\Delta}}} \end{aligned} \quad (5.4)$$

Thus p_n^d does not depend on the state d , which makes this method very efficient. For small Δ probabilities will not remain within $[0, 1]$ hence a better approach is to construct a tree approximating a log-normal process M .

5.3 Deterministic dividends

If deterministic dividends are assumed then the natural dynamics of an equity model with volatility σ is

$$dS_t = r_t S_t dt + \sigma_t S_t dW_t - \sum_i D_t^i \delta_{t-t_i} \quad (5.5)$$

In order to ensure positive stock prices one can replace D_t^i by $D_t^i \wedge S_{t-}$. Numerical schemes such as Finite Difference and Monte Carlo can be extended to this case, see Wilmott (1998). However calibration becomes more involved since even in the constant volatility constant case $\sigma_t = \sigma$ European options cannot be priced in closed form. Approaches such as the Dupire formula (1994) for local volatility and the semi-analytical formula due to Heston (1993) (see Chapter 6 for both volatility models) cannot be applied.

5.3.1 Binomial tree

The risk neutral probability of the binomial tree example discussed in Section 5.2, compare (5.4), is now given as

$$\begin{aligned} p_n^k &= \frac{S_{n+1}^{k+1} - P(\Delta)^{-1}(S_n^k - D_n)}{S_{n+1}^{k+1} - S_{n+1}^k} \\ &= \frac{e^{\sigma\sqrt{\Delta}} - P(\Delta)^{-1}(1 - D_n/S_n^k)}{e^{\sigma\sqrt{\Delta}} - e^{-\sigma\sqrt{\Delta}}} \end{aligned}$$

Thus deterministic dividends lead to spatially inhomogenous probabilities in contrast to the proportional case. There are alternative efficient tree methods exploiting the continuity condition

$$\Pi_{t-}(S_{t-}) = \Pi_t(S_t)$$

for all t where $\Pi_t(S)$ is the price of a derivative at time t if the stock price is S , see Vellekoop and Nieuwenhuis (2006).

5.3.2 Closed form approximation (Bos-Vandermark)

In a famous article Bos and Vandermark (2002) presented a variant of the Black-Scholes formula, namely

$$Call^{BV}(K, t) = P(t)BSCall(P(t)^{-1}S_0 - C^n(t), K + C^f(t), \sigma, t) \quad (5.6)$$

with

$$C^n(t) = P(t)^{-1} \sum_{t_i \leq t} \frac{t - t_i}{t} D_0^i$$

and

$$C^f(t) = P(t)^{-1} \sum_{t_i \leq t} \frac{t_i}{t} D_0^i$$

This formula is a good approximation for a European option price with model (5.5). Furthermore the formula is continuous, both as the valuation date moves across a dividend and maturity moves across a dividend.

5.4 Affine models

It is now standard to model dividends as affine functions of the spot at the ex-date

$$D_{t_i}^i = \alpha_i + \beta_i S_{t_i-} \quad (5.7)$$

with deterministic α_i, β_i , for example $\beta_i = 0$ for $t_i \leq 1y$ and $\alpha_i = 0$ for $t_i \geq 5y$. This is equivalent to the stock process being an affine function in the sense that

$$S_t = (F(t) - C(t))M_t + C(t) \quad (5.8)$$

holds for a martingale M with $M_0 = 1$ and a deterministic function C . One has

$$(S_t - K)^+ = (F(t) - C(t)) \left(M_t - \frac{K - C(t)}{F(t) - C(t)} \right)^+$$

In other words an option on S expiring at t with strike K can be interpreted as $F(t) - C(t)$ options on M with a modified strike. Thus the option market may be represented using an implied volatility surface $\hat{\sigma}_M$. Stock models may be defined by calibrating M to $\hat{\sigma}_M$ and applying (5.8). This approach dates back to Brockhaus et al. (2000) p7f. Further references include Overhaus et al. (2002) p116ff and Overhaus et al. (2007) p6ff. One has the following result (compare Overhaus et al. (2007) p7):

Assume a stock process S with jumps due to dividends only. The following two statements are equivalent:

1. The stock process S is positive and pays positive affine dividends (5.7) satisfying

$$\begin{aligned} \alpha_i &= 0 & \text{for all } i \geq n \text{ for some } n \\ \beta_i &\in [0, 1) & \text{for all } i \\ P(t_i)\alpha_i &\geq -\beta_i \sum_{j \geq i} P(t_j)\alpha_j \prod_{k=i}^j (1 - \beta_k)^{-1} & \text{for all } i \end{aligned}$$

2. The stock process S is an affine function (5.8) of a positive continuous martingale M with $M_0 = 1$ where C satisfies

$$\begin{aligned} C(t) &= 0 & \text{for all } t \geq T \text{ for some } T \\ \Delta F(t) &\leq \Delta C(t) \leq 0 & \text{for all } t \\ dC(t) &= rC(t) & \text{for all } t \notin \{t_1, t_2, \dots\} \end{aligned}$$

with notation $\Delta f(t) = f(t) - f(t-)$ for a given càdlàg function f .

In order to show (2) $\Rightarrow$ (1) substitute (5.8) into the dividend drop $S_{t_i-} - S_{t_i}$. This yields

$$\begin{aligned} S_{t_i-} - S_{t_i} &= (-\Delta F(t_i) + \Delta C(t_i))M_{t_i-} - \Delta C(t_i) \\ &= (-\Delta F(t_i) + \Delta C(t_i)) \frac{S_{t_i-} - C(t_i-)}{F(t_i-) - C(t_i-)} - \Delta C(t_i) \geq 0 \end{aligned}$$

and hence

$$\alpha_i = \frac{C(t_i-) \Delta F(t_i) - F(t_i-) \Delta C(t_i)}{F(t_i-) - C(t_i-)} \quad \beta_i = -\frac{\Delta(F(t_i) - C(t_i))}{F(t_i-) - C(t_i-)} \quad (5.9)$$

One computes

$$\alpha_i + \beta_i C(t_i-) = -\Delta C(t_i) = C(t_i-) - \frac{P(t_{i+1})}{P(t_i)} C(t_{i+1}-)$$

and hence

$$P(t_i)C(t_i-) = \frac{P(t_i)\alpha_i}{1-\beta_i} + \frac{P(t_{i+1})C(t_{i+1}-)}{1-\beta_i} = \dots = \sum_{j \geq i} P(t_j)\alpha_j \prod_{k=i}^j (1-\beta_k)^{-1}$$

The properties of C imply the required properties of α_i and β_i for all i as well as positivity of S .

Implication (1) $\Rightarrow$ (2) relies on the construction of C and M via induction through $n, n-1, \dots, 1$ such that (5.8) holds.

- For $t \geq t_n$ define $C(t) = 0$ as well as M_t through $S_t = F(t)M_t$.
- If C is defined for $t \geq t_i$ set

$$-\Delta C(t_i) = \alpha_i + \beta_i C(t_i-) = \frac{\alpha_i + \beta_i C(t_i)}{1-\beta_i} \quad (5.10)$$

This defines C and hence M through (5.8) on $t \geq t_{i-1}$, and one has

$$\begin{aligned} \Delta S_{t_i} &= (\Delta F(t_i) - \Delta C(t_i)) \frac{S_{t_i-} - C(t_i-)}{F(t_i-) - C(t_i-)} + \Delta C(t_i) \\ &= \dots = -\beta_i S_{t_i-} - \alpha_i \end{aligned}$$

due to

$$-\Delta F(t_i) = \alpha_i + \beta_i F(t_i-) = \frac{\alpha_i + \beta_i F(t_i)}{1-\beta_i}$$

Note that as seen above (5.10) yields via induction

$$P(t_i)C(t_i-) = \sum_{j \geq i} P(t_j)\alpha_j \prod_{k=i}^j (1-\beta_k)^{-1} \quad (5.11)$$

Hence C has the required properties. Continuity of M at t_i can also be established.

5.4.1 Proportional dividends

The case

$$C(t) = 0$$

leads to $\alpha_i = 0$ and hence proportional dividends as introduced in Section 5.2. Note that equation (5.2) is a special case of (5.8).

5.4.2 Martingale plus future dividends

With

$$C(t) = P(t)^{-1} \sum_{t_i > t} D_0^i \quad (5.12)$$

equation (5.8) implies

$$\alpha_i = P(t_i)^{-1} D_0^i \quad \text{and} \quad \beta_i = 0$$

since

$$\Delta F = \Delta C = -P(t_i)^{-1} D_0^i$$

Thus dividends are deterministic. The disadvantage of this approach is that the stock price at time t depends on dividends payable later than t . This seems reasonable since the stock price should never fall below the value of guaranteed future dividend payments. On the other hand, with this assumption a derivative maturing at t is sensitive to the yield curve beyond maturity as well as to dividends going ex after t . An attempt to remedy this limitation is to only consider dividends out to a terminal date T , namely

$$C(t) = P(t)^{-1} \sum_{T > t_i > t} D_0^i \quad (5.13)$$

In that case dividends with ex-date $t_i \geq T$ will be proportional. In the special case of T being the maturity of a derivative this has the advantage that M is log-normal if S is, with the same volatility. One may be tempted to use that volatility to price American options using a binomial tree for M . However the disadvantage of setting T to the maturity of the product is that the stock dynamics will depend on the product. It is common practice to choose volatilities depending on the product in the context of Vanilla options. However, having the dividend dynamics, and thus the forward dynamics, depend on the product leads to additional inconsistency.

5.4.3 Martingale minus past dividends

Setting

$$C(t) = -P(t)^{-1} \sum_{t_i \leq t} D_0^i$$

one also obtains $\alpha_i = P(t)^{-1} D_0^i$ and $\beta_i = 0$. This approach has the advantage that only dividends up to t are required for S_t . However negative C leads to the problem of potentially negative stock prices.

5.4.4 Taper curve

The case of cash dividends up to a fixed time T obtained with $C(t)$ as in equation (5.13) and proportional dividends thereafter can be refined. Defining an increasing function τ with $\tau_0 = 0$ and $\tau_t = 1$ for $t \geq T$ one can set

$$C(t) = P(t)^{-1} \sum_{t_i > t} (1 - \tau_{t_i}) D_0^i \quad (5.14)$$

This leads via (5.9) to affine dividends with

$$\alpha_i = P(t_i)^{-1} D_0^i \left(1 - \tau_{t_i} \frac{F(t_i-) }{F(t_i-) - C(t_i-)} \right)$$

$$\beta_i = \tau_{t_i} \frac{P(t_i)^{-1} D_0^i}{F(t_i-) - C(t_i-)}$$

Beyond T dividends will be proportional. Note that stock dynamics up to time T does not depend on dividends and interest rates beyond T due to (5.8). One may alternatively set α_i, β_i as

$$\alpha_i = (1 - \tau_{t_i}) P(t_i)^{-1} D_0^i$$

$$\beta_i = \tau_{t_i} \frac{P(t_i)^{-1} D_0^i}{F(t_i-)}$$

and derive C from (5.11). The taper curve has an impact on the difference between European and American prices. It can thus be derived to some extent from market data.

5.4.5 Bos-Vandermark revisited

The Bos-Vandermark formula (5.6) discussed in Section 5.3.2 has been developed with a view to approximating a process with deterministic dividends and log-normal dynamics between ex-dividend dates, such as (5.5). The model as defined in the Bos-Vandermark paper depends on the maturity T of the European option. To obtain a consistent model one may fix T , namely

$$C^n(t, T) = P(t)^{-1} \sum_{t_i \leq t} \frac{(T - t_i)^+}{T} D_0^i \quad C^f(t, T) = P(t)^{-1} \sum_{t_i \leq t} \frac{t_i \wedge T}{T} D_0^i$$

Inserting $C^n(t, T), C^f(t, T)$ rather than $C^n(t), C^f(t)$ into formula (5.6) implies a stock process given as

$$S_t = (P(t)^{-1} S_0 - C^n(t, T)) M_t - C^f(t, T) \quad (5.15)$$

with a log-normal process M . This gives rise to linear dividends, namely

$$S_{t_i-} - S_{t_i} = P(t_i)^{-1} D_0^i \left(\frac{(T - t_i)^+}{T} M_{t_i} + \frac{t_i \wedge T}{T} \right)$$

Substituting M via equation (5.15) one obtains

$$\begin{aligned} \alpha_i &= D_0^i \left(\frac{(T - t_i)^+}{T} \frac{C^f(t_i-, T)}{S_0 - P(t_i) C^n(t_i-, T)} + P(t_i)^{-1} \frac{t_i \wedge T}{T} \right) \\ \beta_i &= D_0^i \frac{(T - t_i)^+}{T} \frac{1}{S_0 - P(t_i) C^n(t_i-, T)} \end{aligned}$$

Note that $\alpha_i \searrow 0$ for $t_i \searrow 0$ and $\beta_i \searrow 0$ for $t_i \nearrow T$ which is the contrary of the taper curve approach introduced in Section 5.4.4.

5.5 Dividend discount models

Dividend discount models assume that the stock price can be viewed as the present value of all future dividends

$$S_t = \sum_{t_i > t} D_t^i \quad (5.16)$$

where D_t^i is the value at t of the dividend cashflow going ex at t_i . Given a stock model $(S_t, t \geq 0)$ with discrete dividends one has

$$S_{t_i-} - S_{t_i} = D_{t_i}^i \quad (5.17)$$

One may attempt to specify processes D_t^i for $t < t_i$ such that (5.16) holds. It will be seen in Section 5.5.3 that this is indeed possible in the case of affine dividends and that the resulting dividend dynamics is inhomogenous. The focus of Section 5.5.4 is a homogenous variant of this dynamics.

Rather than taking the stock process as a starting point Bernhart and Mai (2012) use ansatz (5.16) to construct stock processes from dividend processes. Similar to

van Binsbergen et al. (2010) dividends are separated into near and far dividends leading to

$$S_t = \sum_{T \geq t_i > t} D_t^i + \sum_{t_i > T} D_t^i = \sum_{T \geq t_i > t} D_t^i + P(t)^{-1} \left(\sum_{t_i > T} D_0^i \right) M_t$$

Near dividends and the martingale M are then modelled independently, leading to a framework for joint modelling of dividends and stock.

5.5.1 Calibration

The dividend discount approach (5.16) implies the calibration condition

$$P(t)F(t) = S_0 - \sum_{t_i \leq t} D_0^i = \sum_{t_i > t} D_0^i \quad (5.18)$$

If a stock does not pay dividends then one may assume a single dividend payment at infinity with value $D_t = S_t$ at time t . Otherwise one can assume D_0^i to be known for all i up to an index k . With exponential growth thereafter, namely

$$D_0^i = D_0^k e^{-q\Delta(i-k)} \quad \text{and} \quad t_i = t_k + \Delta(i-k)$$

for all $i \geq k$ and a time increment Δ one computes

$$q\Delta = \log \frac{S_0 - \sum_{i < k} D_0^i}{S_0 - \sum_{i \leq k} D_0^i} = \log \frac{P(t_{k-1})F(t_{k-1})}{P(t_k)F(t_k)}$$

from (5.16) or (5.18) with $t = 0$. Growth parameter q can be interpreted as a long term dividend yield.

5.5.2 Deterministic dividends

The deterministic dividend model (5.5) implies dividend dynamics

$$dD_t^i = r_t D_t^i dt$$

and

$$S_t = P(t)^{-1} \sum_{t_i > t} D_0^i$$

The dividend discount approach is thus not compatible with deterministic dividends as it implies deterministic stock dynamics.

5.5.3 Affine dividends

One obtains for the affine model (5.8) from (5.17)

$$D_{t_i}^i = \left(P(t_i)^{-1} D_0^i - \Delta C(t_i) \right) M_{t_i} + \Delta C(t_i)$$

For $t \leq t_i$ one may replace $D_{t_i}^i, P(t_i), M_{t_i}$ by $D_t^i, P(t), M_t$, namely

$$P(t) D_t^i = \left(D_0^i - P(t_i) \Delta C(t_i) \right) M_t + P(t_i) \Delta C(t_i)$$

Summation over all i with $t_i > t$ yields

$$\begin{aligned} P(t) S_t &= \left(P(t) F(t) - \sum_{t_i > t} P(t_i) \Delta C(t_i) \right) M_t + \sum_{t_i > t} P(t_i) \Delta C(t_i) \\ &= P(t) (F(t) - C(t)) M_t + P(t) C(t) \end{aligned}$$

Dividend dynamics

$$\begin{aligned} dD_t^i &= r_t D_t^i dt + P(t)^{-1} \left(D_0^i - P(t_i) \Delta C(t_i) \right) dM_t \\ &= r_t D_t^i dt + P(t)^{-1} D_0^i \tau_i dM_t \end{aligned} \tag{5.19}$$

together with (5.16) thus implies the affine stock model (5.8). Note that all dividends are driven by the same factor M . Also, dividends have a volatility level determined by both M and index i .

5.5.4 Homogenous dividend dynamics

Taper curve models have been developed to capture a bond-like feature of dividends, namely low volatility for near dividends and high volatility for far dividends. However, unlike in bond dynamics, the dividend volatility is held constant so that a far dividend will have high volatility until it goes ex.

Homogenous dividend dynamics assumes a taper curve τ as in Section 5.4.4, namely an increasing function with values in $[0, 1]$ and $\tau_0 = 0$ as well as

$$dD_t^i = r_t D_t^i dt + P(t)^{-1} D_0^i \tau_{t_i-t} dM_t \tag{5.20}$$

The difference with (5.19) is that the impact dM_t is scaled down as the ex-dividend date approaches. The solution is given as

$$P(t) D_t^i = D_0^i \left(1 + \int_0^t \tau_{t_i-s} dM_s \right)$$

In case of continuous τ integration by parts yields

$$P(t)D_t^i = D_0^i \left(1 + \tau_{t_i-t} M_t - \tau_{t_i} M_0 - \int_0^t M_s d\tau_{t_i-s} \right)$$

Note that $M \geq 0$ and $M_0 = 1$ guarantee positive dividends. Summation over all i with $t_i > t$ leads to

$$S_t = F(t) + P(t)^{-1} \left(\mathcal{T}_t^t M_t - \mathcal{T}_0^t M_0 - \int_0^t M_s d\mathcal{T}_s^t \right)$$

with $\mathcal{T}_s^t$ for $0 \leq s \leq t$ defined as

$$\mathcal{T}_s^t = \sum_{t_i > t} D_0^i \tau_{t_i-s}$$

Thus the stock process is the sum of an affine function and an integral of the martingale M . This approach is exemplified with different specifications of the taper curve τ :

- Dividend announcement at a fixed time interval a before the ex-date is represented by the indicator function

$$\tau_t = \mathbf{1}_{\{t > a\}}$$

One has

$$\begin{aligned} P(t)D_t^i &= D_0^i \left(1 + \int_0^t \mathbf{1}_{\{t_i-s > a\}} dM_s \right) \\ &= D_0^i M_{t \wedge (t_i-a)^+} \end{aligned}$$

and

$$\begin{aligned} P(t)S_t &= P(t) \sum_{t_i > t} D_t^i \\ &= \sum_{t_i-a > t} D_0^i M_t + \sum_{t_i > t \geq t_i-a} D_0^i M_{(t_i-a)^+} \end{aligned} \quad (5.21)$$

The situation discussed by Korn and Rogers (2005) is obtained by further specializing

$$t_i = ih, \quad a = (1 - \varepsilon)h \geq 0, \quad M_t = e^{-\mu t} X_t / X_0, \quad P(t) = e^{-rt}$$

where X is an exponential Lévy process with mean

$$\mathbb{E}[X_t] = X_0 e^{\mu t}$$

as well as

$$D_0^i = \lambda X_0 P(t_i - a) e^{\mu(t_i - a)}$$

Formula (5.21) yields

$$S_t = \lambda \left(\sum_{(i-1+\varepsilon)h > t} e^{(\mu-r)((i-1+\varepsilon)h-t)} X_t + \sum_{ih > t \geq (i-1+\varepsilon)h} e^{r(t-(i-1+\varepsilon)h)} X_{(i-1+\varepsilon)h} \right)$$

The second sum is either empty if the next dividend after t has not been announced yet or has exactly one term. Note that formula (5.21) also covers the case of longer announcement periods, namely $a > h$ or $\varepsilon < 0$. In that case the second sum consisting of all dividends announced at t may have more than one term.

- An alternative to the indicator function is the piecewise linear function

$$\tau_t = \frac{t}{a} \wedge 1$$

where a is the time distance beyond which a given dividend has maximal volatility. One obtains for $t \geq a$

$$\begin{aligned} P(t)S_t &= \left(P(a+t)F(a+t) + \sum_{a+t \geq t_i > t} D_0^i \frac{t_i - t}{a} \right) M_t \\ &\quad + a^{-1} \sum_{a+t \geq t_i > t} D_0^i \int_{(t_i-a)^+}^t M_s ds \end{aligned}$$

For $t < a$ one has to add the term

$$\sum_{a \geq t_i > t} D_0^i \left(1 - \frac{t_i}{a} \right)$$

representing the announced part of near dividends.

- With an exponential function

$$\tau_t = 1 - e^{-\lambda t}$$

one obtains

$$\begin{aligned}\mathcal{T}_s^t &= P(t) (F(t) - e^{\lambda s} C(t)) \\ d\mathcal{T}_s^t &= \lambda P(t) C(t) e^{\lambda s} ds\end{aligned}$$

where $C(t)$ is defined through

$$P(t)C(t) = \sum_{t_i > t} D_0^i e^{-\lambda t_i}$$

Thus

$$S_t = C(t) + (F(t) - e^{\lambda t} C(t)) M_t + \lambda C(t) \int_0^t e^{\lambda s} M_s ds \quad (5.22)$$

Setting $\lambda = 0$ (except for the definition of $C(t)$) this formula reverts to (5.8). Figure 5.1 exhibits implied volatilities for forward-starting options, with maturities ranging from two years to ten years and starting one year before maturity. Process volatility is set to 20% flat. There are 12 annual discrete dividends of expected value 10%, as well as a constant interest rate 5%, implying exponential decay of dividend levels beyond year 12 due to the lower interest rate. The taper curve is specified with $\lambda = 50\%$. The homogenous model leads to persistent (positive) forward skew, while the inhomogenous model exhibits almost log-normal return distribution for far one year returns. Volatility levels decrease in the homogenous case due to an exponential decay of dividends. They increase in the inhomogenous model since far dividends are more volatile than near dividends.

Standard (spot-starting) implied volatilities are almost flat across all maturities with increasing term structure converging to a level below 20% in the homogenous case, see Figure 5.2.

Bühler (2012) independently arrives at process (5.22) assuming that dividend payments $D_{t_i}^i$ can be written as

$$\begin{aligned}P(t_i)D_{t_i}^i &= D_0^i Y_{t_i} \\ dY_t &= \lambda(M_t - Y_t)dt\end{aligned}$$

with $Y_0 = M_0 = 1$ and hence

$$Y_T = e^{-\lambda(T-t)} Y_t + \lambda \int_t^T e^{-\lambda(T-s)} M_s ds$$

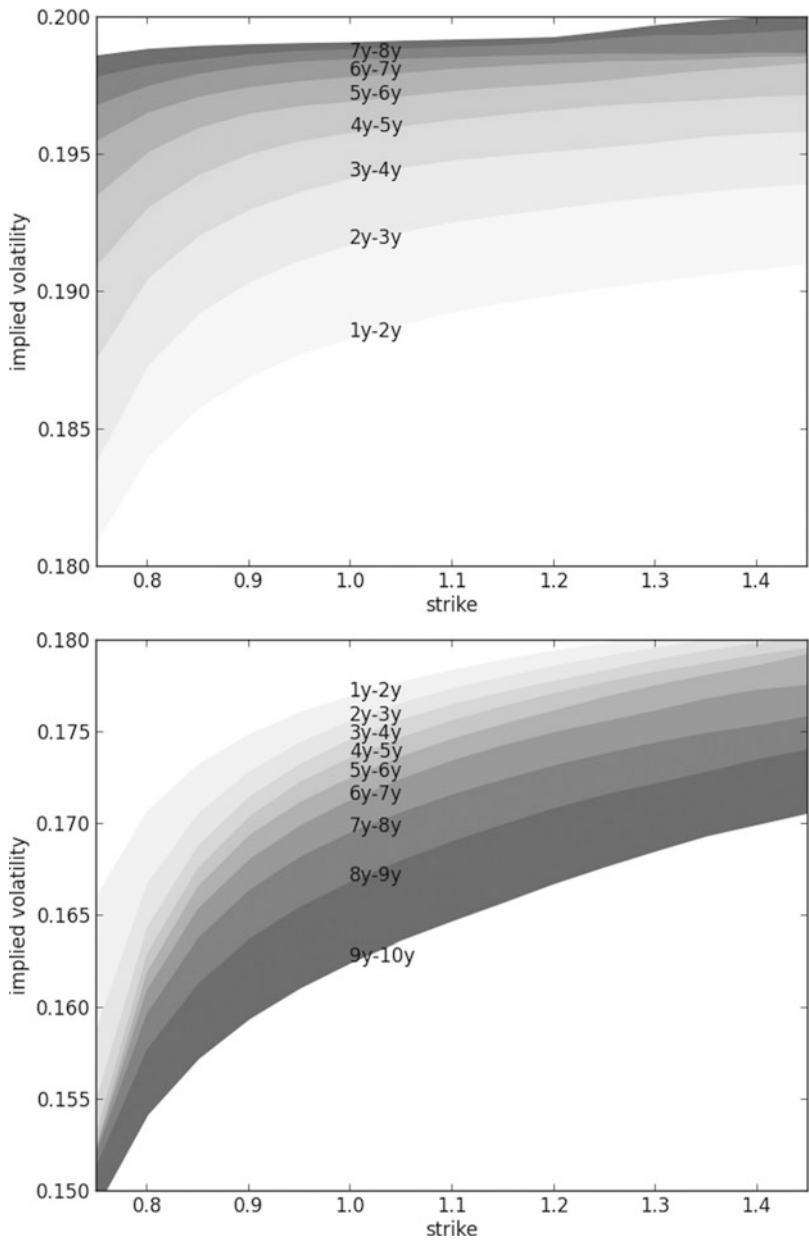


Figure 5.1 Forward volatility: inhomogenous vs. homogenous dividends

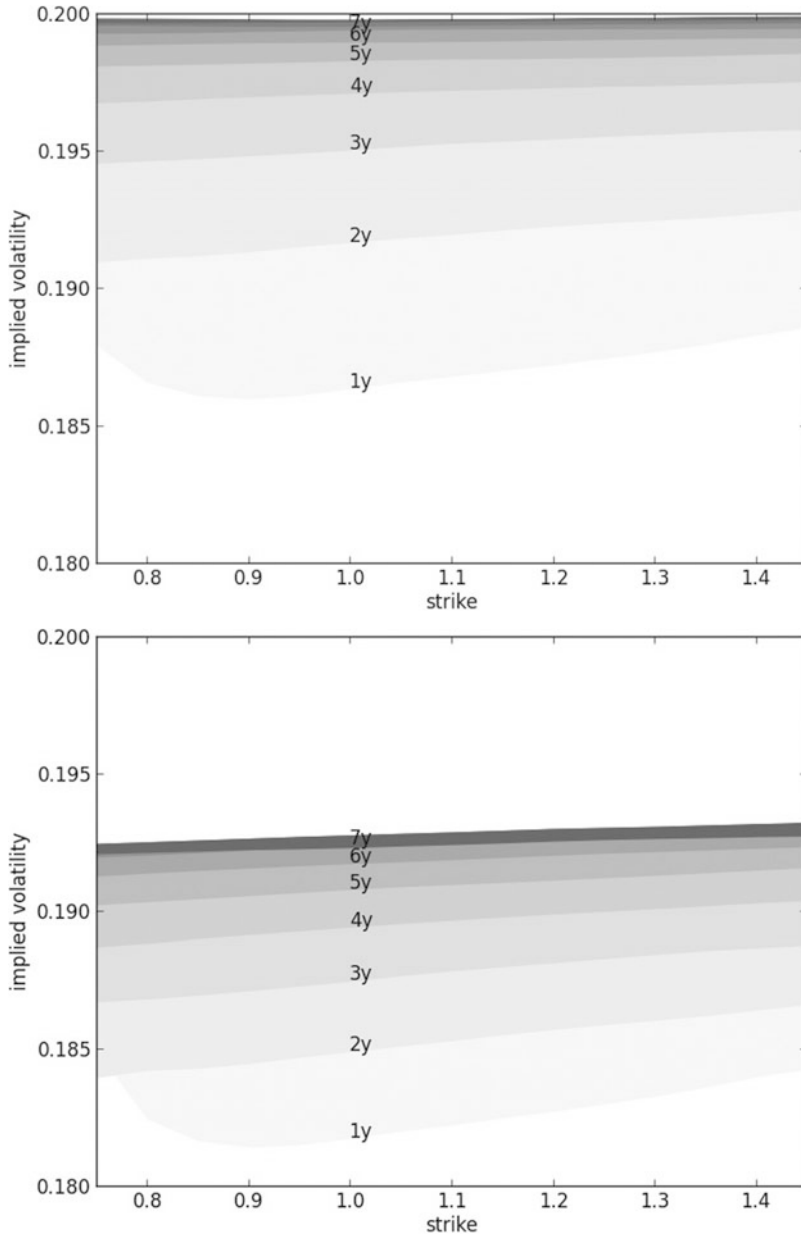


Figure 5.2 Spot volatility: inhomogenous vs. homogenous dividends

His setup allows for a stock process with inner value $\tilde{S}_0 M_t$ in the sense that

$$P(t)S_t = \tilde{S}_0 M_t + \sum_{t_i > t} P(t_i) \mathbb{E}_t \left[D_{t_i}^i \right]$$

Dividends and inner value can be decorrelated by introducing a Brownian W^Y and generalizing Y to

$$dY_t = \lambda(M_t - Y_t)dt + \alpha Y_t^p dW_t^Y$$

5.6 Stochastic dividend yield

A closed form formula for Vanilla options within a Black-Scholes model, enhanced with an Ornstein-Uhlenbeck process d for the dividend yield driven by the same Brownian motion, has been given by Lioui (2006). This result can be generalized to time-dependent parameters, allowing for correlation as well as enforcing calibration to the forward curve. Thus one has the dynamics

$$\begin{aligned} \frac{dS_t}{S_t} &= (r_t - d_t)dt + \sigma_t^S dW_t^S \\ dd_t &= \kappa_t(\theta_t - d_t)dt + \sigma_t^d dW_t^d \end{aligned}$$

where W^S, W^d are correlated and θ is determined through the forward condition $\mathbb{E}[S_t] = F(t)$. Solving this system one obtains a log-normal process S with

$$\begin{aligned} S_t = F(t) \exp & \left(\int_0^t \left(\sigma_s^S dW_s^S - \sigma_s^d(t) dW_s^d \right) \right. \\ & \left. - \frac{1}{2} \int_0^t \left((\sigma_s^S)^2 + (\sigma_s^d(t))^2 - 2\rho_s \sigma_s^S \sigma_s^d(t) \right) ds \right) \end{aligned}$$

where σ^d is given as

$$\sigma_t^d(T) = \sigma_t^d \int_t^T e^{-\int_t^s \kappa_u du} ds$$

as shown in Section 10.4. Thus the Vanilla price is given by the standard Black-Scholes formula (18.8) with variance

$$\sigma^2 t = \int_0^t \left((\sigma_s^S)^2 + (\sigma_s^d(t))^2 - 2\rho_s \sigma_s^S \sigma_s^d(t) \right) ds$$

Bühler et al. (2010) also use a correlated Ornstein-Uhlenbeck process as a driver for dividend yield. However, to model discrete dividends at times t_i they suggest the log-normal process

$$S_t = F(t) \exp \left(\int_0^t \left(\sigma_s^S dW_s^S - \frac{1}{2} (\sigma_s^S)^2 ds \right) - \sum_{t_i \leq t} (C_i + E_i y_{t_i}) \right)$$

$$E_i \in \left\{ 1, \log \frac{F(t_i-)}{F(t_i)} \right\}$$

with constants C_i chosen such that $\mathbb{E}[S_{t_i}] = F(t_i)$ holds. The two suggested values for E_i are intended to represent normal and log-normal volatility of dividend yield.

5.7 Stochastic hazard and interest rates

The affine dividend model considered in Section 5.4 can be extended to stochastic hazard and interest rates. Interest rates may be specified in terms of a numéraire N , such as a cash bond B with stochastic short rates, see (10.1) and Section 10.2. The affine stock model can be postulated as

$$S_t = \frac{P(t)N_t}{N_0} \mathbf{1}_{\{\tau > t\}} e^{H_t} ((F(t) - C(t))M_t + C(t))$$

with $H_t = \int_0^t h_s ds$ and default time τ with intensity h , see Section 12.2. Martingale M , numéraire N and integrated hazard rate H can be correlated. Note that S_t/N_t is a martingale on intervals $[t_{i-1}, t_i]$ since

$$\mathbf{1}_{\{\tau > t\}} e^{H_t} = \mathbf{1}_{\{U_t < e^{-H_t}\}} e^{H_t}$$

remains a martingale under a new measure defined through density M where U_t are independent uniform variates for all t . Also $P(t)F(t)$ and $P(t)C(t)$ are piecewise constant. Assuming linear dividends

$$S_{t_i-} - S_{t_i} = \tilde{\alpha}_i + \tilde{\beta}_i S_{t_i-}$$

one computes

$$\tilde{\alpha}_i = \frac{P(t_i)N_{t_i}}{N_0} \mathbf{1}_{\{\tau > t_i\}} e^{H_{t_i}} \alpha_i$$

$$\tilde{\beta}_i = \beta_i$$

with α_i, β_i given in terms of $F(i), C(i)$ in equation (5.9). Cash offset C can be determined from $\alpha_i = \mathbb{E}[\tilde{\alpha}_i]$ and β_i via (5.11). The cash proportion $\tilde{\alpha}_i$ is no longer deterministic but is proportional to N_{t_i} and also depends on the hazard rate. For log-normal M, N and truncated Gaussian H closed form formulae for Vanilla options can be obtained, compare Section 13.1.2. The case of deterministic H and N and a general martingale M has been discussed in Bühler (2008) and Overhaus et al. (2007).

5.8 Variance swap

Replication and pricing of variance swaps on continuous assets was presented in Section 3.7.1. In the presence of dividends this analysis needs to be refined, see Bühler (2010) and White (2013). For that purpose variance is split into variance between dividends and variance due to dividend drops. Given a discretization $t_0 = 0 < t_1 < \dots < t_n = T$ including all ex-dividend dates in $(0, T]$ one may write

$$V(T) = \sum_{i=1}^n \log^2 \frac{S_{t_i-}}{S_{t_{i-1}}} + \sum_{\tau_i \in (0, T]} \log^2 \frac{S_{t_i}}{S_{t_i-}} \quad (5.23)$$

where S_{t-} denotes the pre-dividend value if a dividend goes ex at t . Given stock dynamics

$$\frac{dS_t}{S_{t-}} = r_t dt + \sigma_t dW_t + \sum_i \left(1 - \frac{D_{t_i}^i}{S_{t_i-}}\right) \delta_{t_i}(dt)$$

where $D_{t_i}^i = S_{t_i-} - S_{t_i}$ is the dividend drop, one obtains

$$d \log S_t = \left(r_t - \frac{1}{2} \sigma_t^2\right) dt + \sigma_t dW_t + \sum_i \log \left(1 - \frac{D_{t_i}^i}{S_{t_i-}}\right) \delta_{t_i}(dt)$$

which upon integration yields

$$\log \frac{S_T}{S_0} = \int_0^T \left(r_t - \frac{1}{2} \sigma_t^2\right) dt + \int_0^T \sigma_t dW_t + \sum_{\tau_i \in (0, T]} \log \left(1 - \frac{D_{t_i}^i}{S_{t_i-}}\right)$$

The continuous part of the variance swap is well approximated by $\int_0^T \sigma_t^2 dt$. Hence

$$\begin{aligned} V(T) &= \int_0^T \sigma_t^2 dt + \sum_{\tau_i \in (0, T]}^n \log^2 \frac{S_{t_i}}{S_{t_i-}} \\ &= 2 \left(-\log \frac{S_T}{S_0} + \int_0^T r_t dt + \int_0^T \sigma_t dW_t + \sum_{\tau_i \in (0, T]} \log \left(1 - \frac{D_{t_i}^i}{S_{t_i-}} \right) \right) \\ &\quad + \sum_{\tau_i \in (0, T]}^n \log^2 \left(1 - \frac{D_{t_i}^i}{S_{t_i-}} \right) \end{aligned}$$

which is formula (46) in White (2013). Assuming deterministic interest rates and affine dividends

$$D_{t_i}^i = \alpha_i + \beta_i S_{t_i-}$$

one obtains

$$\begin{aligned} \mathbb{E}[V(T)] &= -2\mathbb{E} \left[\log \frac{S_T}{S_0} \right] - 2\log P(t) \\ &\quad + \sum_{\tau_i \in (0, T]} \mathbb{E} \left[2\log \left(1 - \beta_i - \frac{\alpha_i}{S_{t_i-}} \right) + \log^2 \left(1 - \beta_i - \frac{\alpha_i}{S_{t_i-}} \right) \right] \end{aligned}$$

If the variance contract is specified in terms of

$$R_i = \frac{S_{t_i}}{S_{t_{i-1}}} - 1 \quad (5.24)$$

rather than log returns $R_i = \log(S_{t_i}/S_{t_{i-1}})$ then the continuous part of the variance swap does not change due to

$$\log x \approx x - 1 + \mathcal{O}((x - 1)^2)$$

However the quadratic dividend adjustment changes and the resulting expectation reads

$$\begin{aligned} \mathbb{E}[V(T)] &= -2\mathbb{E} \left[\log \frac{S_T}{S_0} \right] - 2\log P(t) \\ &\quad + \sum_{\tau_i \in (0, T]} \mathbb{E} \left[2\log \left(1 - \beta_i - \frac{\alpha_i}{S_{t_i-}} \right) + \left(\beta_i + \frac{\alpha_i}{S_{t_i-}} \right)^2 \right] \end{aligned}$$

Remarks:

- If the variance contract stipulates that dividend drops are removed from variance computation then only the term which is quadratic in the dividend drop is removed as pointed out by both Bühler (2008) and White (2013). The term $\log\left(1 - \beta_i - \frac{\alpha_i}{s_{t_i-}}\right)$ remains.
- Static replication involves options maturing at T as well as t_i- (or alternatively t_i) for all ex-dividend dates $t_i \in (0, T]$ where $\alpha_i \neq 0$.
- In the proportional dividend case with $\alpha_i = 0$ for all i one obtains

$$\mathbb{E}[V(T)] = -2\mathbb{E}\left[\log\frac{S_T}{F(T)}\right] + \sum_{\tau_i \in (0, T]} \log^2(1 - \beta_i)$$

where the two terms represent continuous variance and the contribution of dividend drops respectively. This is formula (43) in White (2013). In the case of linear returns (5.24) one has the alternative expression

$$\mathbb{E}[V(T)] = -2\mathbb{E}\left[\log\frac{S_T}{F(T)}\right] + \sum_{\tau_i \in (0, T]} \beta_i^2$$

6 [Short Volatility Models

Persistent volatility can be generated through volatility functionally, depending on the stock level, through a stochastic process correlated with equity returns or through a combination of both. Additionally, the volatility process and the return distributions may exhibit jumps. Stochastic volatility models are incomplete and not Markovian in the filtration generated by the stock process. Calibration of stochastic volatility (or indeed any) model to the Vanilla market has an impact on hedging.

6.1 Local volatility

The derivatives market on stocks and stock indices is dominated by European (and American) Plain Vanilla options. This explains the importance of models which can be fitted to the market prices of those options. This section focuses on complete models for the stock S which are fitted to an arbitrage-free surface $Call(K, t)$ of call prices.

6.1.1 General nonparametric formula

Below a general formula is derived which not only leads to Dupire's formula (1994) but also allows generalizations to local stochastic volatility, as well as equity rates and equity credit hybrids. The fitting condition can be stated as

$$Call(K, t) = \mathbb{E} \left[B_t^{-1} (S_t - K)^+ \right]$$

The stock dynamics is given as

$$\frac{dS_t}{S_t} = (r_t - d_t)dt + \sigma_t dW_t, \quad \frac{dB_t}{B_t} = r_t dt \quad (6.1)$$

with deterministic yield q . Both r and σ may be stochastic. The Itô-Tanaka formula applied to the function

$$c_t = B_t^{-1} (S_t - K)^+$$

reads

$$dc_t = -r_t c_t + B_t^{-1} \left(1_{\{S_t > K\}} dS_t + \frac{1}{2} \delta(S_t - K) d\langle S \rangle_t \right)$$

Integrating over time and taking expectations one obtains

$$\begin{aligned} \mathbb{E}[c_T] &= c_0 - \int_0^T \mathbb{E} \left[r_t B_t^{-1} (S_t - K)^+ \right] dt + \int_0^T \mathbb{E} \left[(r_t - d_t) B_t^{-1} S_t 1_{\{S_t > K\}} \right] dt \\ &\quad + \frac{1}{2} \int_0^T \mathbb{E} \left[B_t^{-1} \delta(S_t - K) \sigma_t^2 \right] K^2 dt \\ &= c_0 + K \int_0^T \mathbb{E} \left[r_t B_t^{-1} 1_{\{S_t > K\}} \right] dt - \int_0^T d_t \mathbb{E} \left[B_t^{-1} S_t 1_{\{S_t > K\}} \right] dt \\ &\quad + \frac{1}{2} \int_0^T \mathbb{E} \left[B_t^{-1} \delta(S_t - K) \sigma_t^2 \right] K^2 dt \end{aligned}$$

Note that

$$Call(K, T) = \mathbb{E}[c_T] = \mathbb{E} \left[B_T^{-1} S_T 1_{\{S_T > K\}} \right] - K \mathbb{E} \left[B_T^{-1} 1_{\{S_T > K\}} \right]$$

Hence, deriving the previous equation with respect to T and omitting the arguments (K, T) of $Call$ one obtains

$$\begin{aligned} \frac{\partial Call}{\partial T} &= K \mathbb{E} \left[r_T B_T^{-1} 1_{\{S_T > K\}} \right] - d_T \mathbb{E} \left[B_T^{-1} S_T 1_{\{S_T > K\}} \right] \\ &\quad + \frac{1}{2} \mathbb{E} \left[B_T^{-1} \delta(S_T - K) \sigma_T^2 \right] K^2 \\ &= K \mathbb{E} \left[r_T B_T^{-1} 1_{\{S_T > K\}} \right] - d_T \left(Call - K \frac{\partial Call}{\partial K} \right) \\ &\quad + \frac{1}{2} \frac{\partial^2 Call}{\partial K^2} \mathbb{E} \left[\sigma_T^2 | S_T = K \right] K^2 \end{aligned}$$

This leads to a general formula which is valid for models with stochastic volatility and interest rates, namely

$$\begin{aligned} &\mathbb{E} \left[\sigma_T^2 | S_T = K \right] \\ &= 2 \frac{\frac{\partial Call}{\partial T} - K \mathbb{E} \left[r_T B_T^{-1} 1_{\{S_T > K\}} \right] + d_T \left(Call - K \frac{\partial Call}{\partial K} \right)}{\frac{\partial^2 Call}{\partial K^2} K^2} \end{aligned} \quad (6.2)$$

6.1.2 Dupire

Assuming deterministic interest rates r and a local volatility function $\sigma_t = \sigma_t(S_t)$ one obtains the famous Dupire's formula (1994)

$$(\sigma_T(K))^2 = 2 \frac{\frac{\partial \text{Call}}{\partial T} + (r_T - d_T)K \frac{\partial \text{Call}}{\partial K} + d_T \text{Call}}{\frac{\partial^2 \text{Call}}{\partial K^2} K^2} \quad (6.3)$$

In practice one often uses variants of this formula:

- If call prices can be expressed as options on a continuous positive martingale M , as in Section 5.4, then the above local volatility assumption leads to a local volatility function $\sigma_t = \sigma_t(M_t)$ satisfying (6.3) with $r = d = 0$.
- There is a corresponding formula in terms of implied volatility $\hat{\sigma}(K, T)$ and its derivatives, see Gatheral (2006):

$$(\sigma_T(K))^2 = \frac{\frac{\partial w}{\partial T}}{1 - \frac{k}{w} \frac{\partial w}{\partial k} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) \left(\frac{\partial w}{\partial k} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial k^2}} \quad (6.4)$$

where

$$w(k, T) = \left(\hat{\sigma}(F(T)e^k, T) \right)^2 T \quad \text{and} \quad F(T) = S_0 e^{\int_0^T (r_t - d_t) dt}$$

- An alternative formula in terms of $v(k, T) = \sqrt{w(k, T)}$ reads

$$(\sigma_T(K))^2 = \frac{2v \frac{\partial v}{\partial T}}{1 + (d_+ + d_-) \frac{\partial v}{\partial k} + d_+ d_- \left(\frac{\partial v}{\partial k} \right)^2 + v \frac{\partial^2 v}{\partial k^2}} \quad (6.5)$$

These formulae can be used if implied volatilities are given as parametric functions. An example is presented in Section 4.3.1. When using this model one should be aware of its shortcomings:

- Hedging: the local volatility model is complete. Hence delta hedging should be sufficient for any claim. However the model is used in such a way that for every pricing and sensitivity calculation a new local volatility surface is generated. Native local volatility delta, i.e. delta obtained by freezing the local volatility function, is often worse than Black-Scholes delta.
- Volatility surface: European options are only given for a limited number of strikes and maturities. Local volatility depends on interpolation and extrapolation and requires an arbitrage-free implied volatility surface. In fact the existence of a local volatility function is equivalent to the absence of arbitrage.

- Time homogeneity: the shape of the local volatility function is inherently time dependent.
- Model dynamics: the model predicts flattening of skew and convexity of returns through time. This holds for both unconditional skew as measured by forward implied volatility and the return distribution $\frac{S_T}{S_t} | \mathcal{F}_t$ conditional on a typical state.
- Calibration: the model does not allow calibration to products other than European options.

Models such as stochastic volatility, local stochastic volatility and jump models have been developed in order to address those shortcomings.

6.1.3 Shifted log-normal

An example for a local volatility function leading to a tractable model is

$$\sigma_t(S_t) = \sigma_t \left(1 + \beta \left(\frac{F(t)}{S_t} - 1 \right) \right) \quad (6.6)$$

with $\beta \in [0, 1)$ and a deterministic function σ_t . With

$$\frac{dM_t}{M_t} = (1 - \beta)\sigma_t dW_t \quad M_0 = 1$$

the shifted log-normal process

$$S_t = F(t) \left(\frac{M_t}{1 - \beta} - \frac{\beta}{1 - \beta} \right)$$

satisfies

$$\begin{aligned} dS_t &= S_t(r_t - d_t)dt + F(t)M_t\sigma_t dW_t \\ &= S_t(r_t - d_t)dt + ((1 - \beta)S_t + \beta F(t))\sigma_t dW_t \end{aligned}$$

and hence has a local volatility function (6.6). The special cases $\beta = 0$ and $\beta \nearrow 1$ lead to log-normal and Gaussian models, respectively. Option pricing is performed with the generalized Black-Scholes formula

$$C(K, T) = \frac{P(T)}{1 - \beta} BSCall \left(F(T), \beta F(T) + (1 - \beta)K, (1 - \beta) \sqrt{\frac{1}{T} \int_0^T \sigma_t^2 dt}, T \right)$$

This model allows for negative stock prices.

Other tractable local volatility models can be found in Carr et al. (1999).

6.1.4 Implied volatility from local volatility

Dupire's formula requires a smooth arbitrage-free implied volatility specification. Alternative methods for local volatility calibration have been developed with a view to guaranteeing robustness.

Reghai (2006) begins with the following approximation for implied volatility from local volatility

$$\hat{\sigma}(K, T)^2 T \approx \int_0^T \mathbb{E}[\sigma_t(S_t)^2 | S_T = K] dt \quad (6.7)$$

A motivation for this formula can be found in Gatheral (2006). In the case of time-dependent volatility it follows from the distribution of Brownian bridges that S_t given $S_T = K$ is log-normally distributed with mean $m_t(K, T)$ and log variance $v_t(K, T)$ given as

$$m_t(K, T) = F(t) \left(\frac{K}{F(T)} \right)^\alpha e^{\frac{1}{2}\alpha \int_t^T \sigma_s^2 ds}$$

$$v_t(K, T) = \alpha \int_t^T \sigma_s^2 ds$$

with

$$\alpha = \frac{\int_0^t \sigma_s^2 ds}{\int_0^T \sigma_s^2 ds}$$

Given a local volatility function $\sigma_t(x)$ a sequence of volatility term structures σ^i and corresponding processes S^i is defined through

$$\sigma_t^0 = \sigma_t(F(t))$$

$$\frac{dS_t^i}{S_t^i} = (r_t - d_t)dt + \sigma_t^i dW_t$$

$$\sigma_t^{i+1} = \mathbb{E}[\sigma_t(S_t^i) | S_T^i = K]$$

This algorithm converges quickly. The limit is the most likely path given the terminal value of the local volatility process and can be used to compute the right hand side of (6.7) and thus implied volatility from local volatility. A better

approximation, taking into account local volatility convexity, is given by formula

$$\begin{aligned}\hat{\sigma}(K, T)^2 T &\approx \int_0^T \mathbb{E}[\sigma_t(S_t)^2 | S_T = K] dt \\ &+ \int_0^T \text{Var}[\sigma_t(S_t)^2 | S_T = K] \frac{\partial \sigma_t}{\partial S}(\mathbb{E}[\sigma_t(S_t)^2 | S_T = K]) dt\end{aligned}$$

see Reghai et al. (2012). Gatheral and Wang (2012) expresses implied volatility using a most likely path given as a solution to a variational problem.

Let an algorithm mapping a local volatility surface to an implied volatility surface be summarized as a functional Φ . In order to invert this functional a fixed point iterative algorithm can be used, namely

$$\begin{aligned}\sigma^0(K, T) &= \hat{\sigma}(K, T) \\ \sigma^{i+1}(K, T) &= \sigma^i(K, T) \frac{\hat{\sigma}(K, T)}{\Phi(\sigma^i)(K, T)}\end{aligned}$$

where each iteration has to be performed with a suitable choice of (K, T) grid and corresponding interpolation and extrapolation. Reghai (2006) presents this algorithm as a robust and efficient alternative to Dupire's formula. In addition it has a natural generalization to the case of stochastic interest rates, see Section 11.3.1.

6.1.5 Constant elasticity of variance

An important example for parametric local volatility is the CEV model with stock S satisfying

$$\frac{dS_t}{S_t} = (r - q)dt + \sigma e^{\alpha t} S_t^{\rho-1} dW_t \quad (6.8)$$

where $S_t = 0$ is assumed to be an absorbing state. Only $\rho \in (0, 1)$ is covered here. The case $\rho = 1$ is the Black-Scholes model, while $\rho = 0$ is related to Bachelier's Gaussian model. For $\rho > 1$ the process (with $r = q = 0$) is a local martingale but not a martingale. This case is discussed in Lewis (2000). Note that it is sufficient to consider the special case

$$d\tilde{S}_t = \tilde{S}_t^\rho d\tilde{W}_t \quad (6.9)$$

since S can be obtained from $\tilde{S}$ through the space-time change

$$\begin{aligned}S_t &= e^{(r-q)t} \tilde{S}_{\tau_t} \\ \tau_t &= \int_0^t \sigma^2 e^{2((r-q)(\rho-1)+\alpha)s} ds\end{aligned}$$

with Brownian motion $\tilde{W}$ given as

$$\tilde{W}_{\tau_t} = \int_0^t \sigma e^{((r-q)(\rho-1)+\alpha)s} dW_s$$

since

$$\begin{aligned} dS_t &= (r-q)S_t dt + e^{(r-q)t} d(\tilde{S}_t)_t \\ &= (r-q)S_t dt + e^{(r-q)t} (d\tilde{S})_{\tau_t} \tau'_t \\ &= (r-q)S_t dt + e^{(r-q)t} \tilde{S}_{\tau_t}^\rho (d\tilde{W})_{\tau_t} \tau'_t \\ &= (r-q)S_t dt + e^{(r-q)t} \tilde{S}_{\tau_t}^\rho d(\tilde{W}_t)_t \\ &= (r-q)S_t dt + e^{(r-q)t} \tilde{S}_{\tau_t}^\rho \sigma e^{((r-q)(\rho-1)+\alpha)t} dW_t \\ &= (r-q)S_t dt + \sigma e^{\alpha t} S_t^\rho dW_t \end{aligned}$$

Let X denote a process with dimension δ stopped at zero, namely

$$dX^{(\delta)} = 2\sqrt{X_t^{(\delta)}} dW_t + \delta \mathbf{1}_{\{X_t^{(\delta)} > 0\}} dt$$

For $\delta \geq 0$ this process is known as squared Bessel process stopped at zero. An application of Itô's lemma shows that $\tilde{S} = (aX)^b$ with $a > 0$ and $b \neq 0$ solves (6.9) if

$$\delta + 2(b-1) = 0, \quad 2|b|\sqrt{a} = 1, \quad \rho = 1 - \frac{1}{2b}$$

holds. For negative b the Brownian W in (6.9) has to be replaced by $-W$. The distribution of a squared Bessel process at $t \geq 0$ with $\delta \geq 0$ is noncentral chi-squared, namely

$$X_t^{(\delta)} \stackrel{(d)}{=} t \cdot V^{(\delta, X_0^{(\delta)}/t)}$$

where $V^{(\delta, \kappa)}$ denotes a noncentral chi-squared random variable with δ degrees of freedom and noncentrality parameter κ , see Delbaen (2002). The corresponding distribution function is denoted $Q(\cdot; \delta, \kappa)$. The above allows one to derive (part of) the closed form formula for an option on $\tilde{S}$ satisfying (6.9), due to Cox (1996), for parameters $0 < \rho < 1$, namely

$$\tilde{C}(K, T) = S_0 Q\left(2w; 2 + \frac{1}{1-\rho}, 2z\right) - K \left(1 - Q\left(2w; 2 - \frac{1}{1-\rho}, 2z\right)\right)$$

with

$$z = \frac{S_0^{2(1-\rho)}}{2(1-\rho)^2 T} \quad w = \frac{K^{2(1-\rho)}}{2(1-\rho)^2 T}$$

The corresponding formula for options on S satisfying (6.8) is obtained as

$$C(K, T) = e^{-qt} \tilde{C} \left(\frac{K}{e^{(r-q)t}}, \sigma^2 \frac{e^{2((r-q)(\rho-1)+\alpha)T} - 1}{2((r-q)(\rho-1)+\alpha)T} \right) \quad (6.10)$$

6.2 Stochastic volatility

The local volatility model has often been criticized for its dynamic properties. Stochastic volatility models assume an autonomous process for instantaneous volatility. This process will typically be correlated with stock returns since increasing stock prices lead to lower volatility. The resulting stock model will be incomplete in the sense that volatility risk cannot be diversified through positions in stock.

6.2.1 Heston

Heston's model (1993b) is defined by

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - q)S_t dt + \sigma_t dW_t^S \\ d\sigma_t^2 &= \kappa(\theta - \sigma_t^2)dt + \alpha\sigma_t dW_t^\sigma \end{aligned}$$

with two correlated Brownian motions W^S and W^σ . This model is famous since it admits efficient Vanilla option pricing and thus calibration. The two-dimensional process $(\log S_t/F(t), \sigma_t^2)$ with $F(t) = S_0 e^{(r-d)t}$ is affine in the sense of (16.1) and (16.2). The coefficients are $r = \lambda = 0$ as well as

$$\begin{aligned} \mu(x) &= \begin{pmatrix} -\frac{1}{2}x_2 \\ \kappa(\theta - x_2) \end{pmatrix} \\ \sigma(x)\sigma^T(x) &= \begin{pmatrix} 1 & \alpha\rho \\ \alpha\rho & \alpha^2 \end{pmatrix} x_2 \end{aligned}$$

Equations (16.3) have been solved explicitly by Heston (1993a). The solution is

$$\begin{aligned} B_1(u, t) &= iu \\ B_2(u, t) &= \frac{\beta(u) + D(u)}{\alpha^2} \left(\frac{e^{D(u)t} - 1}{c(u)e^{D(u)t} - 1} \right) \end{aligned}$$

$$A(u, t) = \frac{\kappa\theta}{\alpha^2} \left((\beta(u) + D(u))t - 2 \log \left(\frac{c(u)e^{D(u)t} - 1}{c(u) - 1} \right) \right)$$

where

$$\begin{aligned}\beta(u) &= \kappa - \rho\alpha u i \\ c(u) &= \frac{\beta(u) + D(u)}{\beta(u) - D(u)} \\ D(u)^2 &= \beta(u)^2 + u(i + u)\alpha^2\end{aligned}$$

This model is arguably the most important stochastic volatility model, not least due to its numerical tractability. It also allows one to incorporate some empirically observable properties of stock dynamics, namely

- Short volatility is mean reverting.
- Stock returns and volatility increments are negatively correlated.
- Volatility of variance increases with the level of volatility. The model assumes proportionality to σ_t . Research by Gatheral (2006) on the dynamics of volatility products indicates, however, that the truth is closer to proportionality to σ_t^2 .

Empirical data show that volatility does not fall below a fairly stable bound since one rarely observes equity volatilities below 10%. Heston's model does not capture this property.

Five parameters $\sigma_0, \sigma_\infty, \kappa, \rho, \alpha$ will often not be sufficient to fit an implied volatility surface with the required accuracy. There are two common extensions to Heston's model:

- Jumps in stock and jumps in volatility introduced by Bates (1996) lead to improved calibration performance.
- Term structure of parameters, as discussed by Mikhailov and Nögel (2003), add flexibility at the expense of the model's explanatory power.

The first two moments of integrated variance

$$V(T) = \frac{1}{T} \int_0^T \sigma_t^2 dt$$

can be computed. Since $\bar{v}_t = \mathbb{E}[\sigma_t^2]$ satisfies

$$d\bar{v}_t = \kappa (\theta^2 - \bar{v}_t) dt$$

with solution

$$\bar{v}_t = (\sigma_0^2 - \theta^2) e^{-\kappa t} + \theta^2$$

one has

$$\mathbb{E}[V(T)] = \frac{1 - e^{-\kappa T}}{\kappa T} (\sigma_0^2 - \theta^2) + \theta^2 \quad (6.11)$$

Note that this expression does not depend on the volatility of volatility α and the correlation with equity returns ρ . The second moment has been given in Brockhaus and Long (2000) as

$$\begin{aligned} \text{Var}[V(T)] = \frac{\alpha^2 e^{-2\kappa T}}{2\kappa^3 T^2} & \left[\left(2 \left(-1 + e^{2\kappa T} - 2e^{\kappa T} \kappa T \right) (\sigma_0^2 - \theta^2) \right) \right. \\ & \left. + \left(-1 + 4e^{\kappa T} - 3e^{2\kappa T} + 2e^{2\kappa T} \kappa T \right) \theta^2 \right] \quad (6.12) \end{aligned}$$

For higher moments see Dufresne (2001). This formula can be extended to time-dependent parameters if $\sigma_0 = \theta$, see Piterbarg (2005). The Laplace transform of $V(T)$ has first been derived in the context of short rate models, see Section 10.5, equation (10.21).

6.2.2 Schöbel-Zhu

Stein and Stein (1991) and Schöbel and Zhu (1999) consider an Ornstein-Uhlenbeck process for the volatility process, namely

$$d\sigma_t = \kappa(\theta - \sigma_t)dt + \alpha dW_t^\sigma$$

Applying Itô's lemma shows that the variance process σ_t^2 satisfies

$$d\sigma_t^2 = 2\kappa \left(\frac{\alpha^2}{2\kappa} + \theta\sigma_t - \sigma_t^2 \right) dt + 2\alpha\sigma_t dW_t^\sigma$$

Hence the model is affine in $(X_t = \log(S_t/F(t)), \sigma_t, v_t = \sigma_t^2)$ with coefficients $r = \lambda = 0$ as well as

$$\begin{aligned} \mu(x) &= \begin{pmatrix} -\frac{1}{2}x_3 \\ \kappa(\theta - x_2) \\ \alpha^2 + 2\kappa(\theta x_2 - x_3) \end{pmatrix} \\ \sigma(x)\sigma^T(x) &= \begin{pmatrix} x_3 & \alpha\rho x_2 & 2\alpha\rho x_3 \\ \alpha\rho x_2 & \alpha^2 & 2\alpha^2 x_2 \\ 2\alpha\rho x_3 & 2\alpha^2 x_2 & 4\alpha^4 x_3 \end{pmatrix} \end{aligned}$$

If $\theta = 0$ then the variance process is the square-root process of Heston's model. Nevertheless, the model cannot be considered a generalization of Heston since two parameters κ, α determine the three parameters, mean reversion, volatility of volatility and long term variance of the square-root process σ_t^2 , in the corresponding Heston model.

Equations (16.3) can be solved in closed form, see Kahl (2004). Denoting by $\psi_{\kappa, \theta, \alpha, \rho}^H(u, \log S_0, \sigma_0^2, t)$ the Fourier transform of the Heston model, the Fourier transform $\psi_{\kappa, \theta, \alpha, \rho}^{SZ}(u, \log S_0, \sigma_0, \sigma_0^2, t)$ of the Schöbel-Zhu model can be computed as

$$\psi_{\kappa, \theta, \alpha, \rho}^{SZ}(u, \log S_0, \sigma_0, \sigma_0^2, t) = \psi_{2\kappa, \alpha^2/(2\kappa), 2\alpha, \rho}^H(u, \log S_0, \sigma_0^2, t) e^{A_z + B_z \sigma_0}$$

with

$$\begin{aligned} B_z(u, t) &= \frac{\kappa \theta}{\alpha^2} \frac{\beta(u) - D(u)}{D(u)} \left(\frac{1 - e^{-\frac{1}{2} D(u) t}}{1 - c(u)^{-1} e^{-D(u) t}} \right) \\ A_z(u, t) &= \frac{\kappa^2 \theta^2}{2\alpha^2} \frac{\beta(u) - D(u)}{D^3(u)} \left(\beta(u)(D(u)t - 4) + D(u)(D(u)t - 2) + \hat{A}(t) \right) \\ \hat{A}(t) &= 4e^{-\frac{1}{2} D(u) t} \frac{\frac{D(u)^2 - 2\beta(u)^2}{\beta(u) + D(u)} e^{-\frac{1}{2} D(u) t} + 2\beta(u)}{1 - c(u)^{-1} e^{-D(u) t}} \end{aligned}$$

Volatility is Gaussian and therefore attains negative values. This does not pose numerical problems as in the Heston model. Crossing the zero volatility line merely changes the sign of correlation. The model admits a semi-analytical formula for Vanilla options and is thus as tractable as Heston's model.

6.2.3 Wishart

Bru (1991) introduced Wishart processes Σ satisfying

$$d\Sigma_t = \left(\Omega \Omega^T + M \Sigma_t + \Sigma_t M^T \right) dt + \sqrt{\Sigma_t} dW_t^\sigma Q + Q^T (dW_t^\sigma)^T \sqrt{\Sigma_t}$$

Here Ω, M, Q are matrices. Additionally, Ω is assumed invertible and related to Q through

$$\Omega \Omega^T = \beta Q^T Q$$

with a constant $\beta \geq n - 1$, see Bru (1991), p747. Fonseca, Grasselli and Tebaldi (2006) suggest a single-asset process

$$\frac{dS_t}{S_t} = r dt + \text{trace} \sqrt{\Sigma_t} dW_t^S$$

where both W^S and W^σ are standard matrix valued Brownian motions. The correlation between $W^{S,k,i}$ and $W^{\sigma,k,j}$ is given by R_{ij} for all k . Such a correlation structure can be constructed through

$$W_t^\sigma = W_t^S R^T + \bar{W}_t \sqrt{1 - RR^T}$$

The authors claim that this model allows for additional flexibility over Heston's model when calibrating both short and long implied volatility skew.

The special case of diagonal M, Q, R leads to the Wishart process being diagonal as well, thus leading to a multi-dimensional Heston process. This model improves the standard Heston model only marginally. A two-dimensional Heston process has independently been considered by Kainth (2007).

Benabid et al. (2008) show that for the case of integer β Wishart processes can be constructed from β independent multi-dimensional OUP as

$$\begin{aligned} \Sigma_t &= \sum_{i=1}^{\beta} X_t^k X_t^{k,T} \\ dX_t^k &= M^T X_t^k dt + Q^T dW_t^k \end{aligned}$$

General β processes can be obtained through a change of measure. This approach is relevant for Monte Carlo implementation. Benabid focuses on two-dimensional OUP and diagonal M, Q and derives implied volatility skew expansions.

6.2.4 Scott-Chesney

Scott-Chesney (1989) suggest a log-normal mean reverting (or exponential Ornstein-Uhlenbeck) volatility process, namely

$$d \log \sigma_t = \kappa (\log \sigma_\infty - \log \sigma_t) dt + \alpha dW_t^{\log \sigma}$$

An alternative model with similar dynamics is described by

$$\frac{d\sigma_t}{\sigma_t} = \kappa (\sigma_\infty - \sigma_t) dt + \alpha dW_t^\sigma$$

For these models closed form approximations, or semi-analytical formulae for Vanilla options, are not known. However they benefit from more realistic volatility distribution than Heston's model. Calibration to typical equity markets often requires correlation close to -100% .

6.2.5 Double Heston

The mean reversion level may be modelled as a (slowly) mean reverting process, namely

$$\begin{aligned}\frac{dS_t}{S_t} &= (r_t - d_t)dt + \sqrt{v_t^1} dW^S \\ dv^1 &= \kappa^1 (v_t^2 - v_t^1) dt + \alpha_1 \sqrt{v_t^1} dW^1 \\ dv^2 &= \kappa^2 (\theta - v_t^2) dt + \alpha_2 \sqrt{v_t^2} dW^2\end{aligned}$$

where $\kappa^1 \gg \kappa^2$. This model is also discussed in the context of variance curve models, see Section 7.3.2. Two mean reversion speeds allow for a better joint calibration of short and long implied volatility skew.

6.3 Local stochastic volatility

Assume deterministic interest rates r , as well as volatility given as product of two functions, namely

$$\frac{dS_t}{S_t} = (r_t - d_t)dt + g_t(S_t)\sigma_t dW_t \quad (6.13)$$

where g is a deterministic function of t and S_t and σ is a stochastic process. Considering σ as given the aim is to determine g such that the model is calibrated to the market of European options.

Local stochastic volatility models allow for the production of a range of prices for exotic options while being calibrated to Vanilla options. This can be achieved by first calibrating stochastic volatility as best as possible, applying a damping factor to volatility of volatility and then computing the local volatility correction. Depending on the damping factor, the stochastic local volatility model will be closer to either local volatility or to stochastic volatility. Such an approach is common for modelling of foreign exchange processes since first-generation exotic options such as single- or double-barrier are liquid enough to prohibit the use of local volatility and stochastic volatility models only.

Important parametric volatility specifications and nonparametric approaches are discussed below. A survey with useful further references has been compiled by Homescu (2014).

6.3.1 SABR

The acronym SABR refers to the parameters α, β, ρ of the model, which is defined as

$$\begin{aligned} dS_t &= \sigma_t S_t^\beta dW_t^S \\ d\sigma_t &= \alpha \sigma_t dW_t^\sigma \end{aligned}$$

where ρ is the correlation of W^S and W^σ . Special cases include CEV (constant elasticity of variance, $\alpha = 0$) and log-normal volatility ($\alpha = \beta = 1$). The model is thus an example for a local-stochastic volatility model where the implied volatility smile is generated by the product $g(S_t)\sigma_t$ of a function of stock $g(S_t) = S_t^{\beta-1}$ (local volatility) and an autonomous log-normal process σ_t (stochastic volatility). The parameter choice allows for dynamics ranging from local volatility to stochastic volatility while still calibrating to a given at-the-money implied volatility level and skew. The model does not feature mean reversion. Consequently the volatility distribution does not converge to a stationary distribution for large t .

6.3.2 Linear local volatility

Fix constants $\bar{S}, \bar{\beta}$ and a Heston process without drift $\tilde{S}$ with stochastic volatility $\tilde{\sigma}$. The affine transformation

$$S = \frac{1}{1 - \bar{\beta}} \left(\tilde{S} - \bar{\beta} \bar{S} \right) \quad \sigma = \frac{\tilde{\sigma}}{1 - \bar{\beta}}$$

leads to

$$dS_t = \left((1 - \bar{\beta}) S_t + \bar{\beta} \bar{S} \right) \sigma dW_t^S$$

compare Section 6.1.3. Thus one can extend Heston's approach to Vanilla pricing to dynamics (6.13) with

$$g(S_t) = (1 - \bar{\beta}) + \bar{\beta} \frac{\bar{S}}{S_t}$$

at the expense of allowing for negative prices. Assuming $\sigma_0 = \sigma_\infty = \bar{\lambda}$ one has

$$\begin{aligned} d\sigma_t^2 &= \kappa(1 - \sigma_t^2)dt + \bar{\alpha}\sigma dW_t^\sigma \\ g(S_t) &= \bar{\lambda} \left((1 - \bar{\beta}) + \bar{\beta} \frac{\bar{S}}{S_t} \right) \end{aligned}$$

with $\sigma_0 = 1$. Volatility level is manipulated via $\bar{\lambda}$, skew is generated through correlation ρ and $\bar{\beta}$, and convexity is related to $\bar{\alpha}$. Mean reversion κ dampens convexity (and skew if $\rho \neq 0$). By fixing $\kappa > 0$ and $\rho = 0$ one may infer $\bar{\lambda}, \bar{\beta}, \bar{\alpha}$ for a given maturity, such that level, skew and convexity are matched.

In order to have a consistent model for all maturities Piterbarg (2005) suggests time-dependent parameters λ, β, α and shows how to compute effective parameters $\bar{\lambda}, \bar{\beta}, \bar{\alpha}$ such that the implied volatility smile for a given term is approximately equal.

6.3.3 Quadratic local volatility

Analytical approximation of Vanilla option prices in the case of quadratic g , namely

$$g(S_t, t)S_t = \lambda_t \left((1 - \beta_t)S_t + \beta_t \bar{S} + \frac{1}{2}\gamma_t(S_t - \bar{S})^2 \right)$$

was attempted by Andersen and Hutchings (2009). Note that S is, in most cases, a strict local martingale but not a martingale.

For constant parameters and the linear case $\gamma = 0, \rho \neq 0$ revert to the previous Section 6.3.2. In case of zero correlation $\gamma \neq 0, \rho = 0$ one can use

$$\mathbb{E}[(S_T - K)^+] = \mathbb{E}[\mathbb{E}[(S_T - K)^+ | V(T)]]$$

where the inner integral is the quadratic local volatility price derived by Andersen (2008b). The distribution of integrated variance $V(T)$ is also known, see Dufresne (2001). Combining these results one may approximate

$$Call(\rho, \gamma) \approx Call(\rho, 0) + Call(0, \gamma) - Call(0, 0)$$

where $Call(\rho, \gamma)$ refers to the call price with constant parameters $\alpha, \kappa, \rho, \lambda, \beta, \gamma$. Time dependent $\alpha, \lambda, \beta, \gamma$ can be calibrated using effective parameters following Piterbarg (2005).

6.3.4 Nonparametric local volatility

Equations (6.2) and (6.3) imply

$$\sigma_t^d(K)^2 = g_t(K)^2 \mathbb{E}[\sigma_t^2 | S_t = K] \quad (6.14)$$

where σ^d refers to Dupire's local volatility function. Let S^σ now denote the process with volatility σ , i.e. $g = 1$. This process is generally not calibrated to the market. Instead it will lead to option prices $Call^\sigma$ different from the market prices $Call$. Formula (6.3) with $Call^\sigma$ instead of $Call$ leads to a local volatility function $\sigma^{\sigma, d} = \sigma_t^{\sigma, d}(S_t^\sigma)$. This function satisfies

$$\sigma_t^{\sigma, d}(K)^2 = \mathbb{E}[\sigma_t^2 | S_t^\sigma = K] \quad (6.15)$$

due to (6.2). Now assume that the processes S and S^σ have the same time copula. In other words, there exists a deterministic function $D^\sigma = D_t(S)$ which is monotonic in S for all t such that

$$S_t^\sigma = D_t(S_t)$$

Equations (6.14) and (6.15) imply

$$g_t(K) = \frac{\sigma_t^d(K)}{\sqrt{\mathbb{E}[(\sigma_t^\sigma)^2 | S_t = K]}} = \frac{\sigma_t^d(K)}{\sqrt{\mathbb{E}[(\sigma_t^\sigma)^2 | S_t^\sigma = D_t(K)]}} = \frac{\sigma_t^d(K)}{\sigma_t^{\sigma,d}(D_t(K))}$$

This shows that the unknown local volatility correction g is given as a ratio of two local volatility functions, namely the market local volatility function σ^d and the local volatility function $\sigma^{\sigma,d}$ obtained from stochastic volatility prices, evaluated at a modified strike $D_t(K)$. The function D , assuming it exists, is given as

$$D_t(K) = \left(F^{S_t^\sigma}\right)^{-1}\left(F^{S_t}(K)\right) \quad (6.16)$$

where F^X denotes the distribution function of the random variable X .

There is no reason for S and S^σ to have the same time copula. However, the above can be interpreted as the first iteration as follows; let $g^0 = 1$. Compute g^{i+1} from g^i via

$$g_t^{i+1}(K) = g_t^i(K) \frac{\sigma_t^d(K)}{\sigma_t^{g^i\sigma,d}(D_t^{g^i\sigma}(K))}$$

Here $\sigma_t^{g\sigma,d}$ is the local volatility function corresponding to local stochastic volatility prices $Call^{g\sigma}$ and $D^{g\sigma}$ is the quantile map between processes $S^{g\sigma}$ and S . This algorithm typically converges in a few iterations. The method is a variant of the method described in Section 6.1.4 with Φ being the functional defined as

$$\Phi(g)(K, t) = \sigma_t^{g\sigma,d}(D_t^{g\sigma}(K))$$

Remarks:

- Local stochastic volatility can also be derived on the basis of Gyöngy's theorem, see Piterbarg (2006).
- The quantile map D in equation (6.16) can also be expressed as the solution of an ODE. Forde (2006) shows that the solution is given as an harmonic average

$$D_t(x) = \int_{x_0}^x \frac{dy}{\tilde{\sigma}_t(y)}$$

where x_0 is a suitable initial value. More details on the implementation of this approach can be found in Henry-Labordère (2009).

- The iterative scheme can be implemented using Monte Carlo methods for the computation of $Call^{g\sigma}$.
- A direct implementation can be based on a forward PDE solver with computation of local volatility at every time step, see Ren et al. (2007).

6.4 Jump diffusion

6.4.1 Independent returns

Exponential Lévy models assume that the increments of $\log S_t$ are independent. In that case the distribution of log increments is known from European options since

$$\mathbb{E} \left[\exp \left(u \log \frac{S_T}{S_t} \right) \right] = \frac{\mathbb{E} \left[\exp (u \log S_T) \right]}{\mathbb{E} \left[\exp (u \log S_t) \right]}$$

Inversion of the Laplace transform in order to construct the distribution of increments is possible in principle. More practical approaches rely on specific Lévy processes, such as Brownian motion with drift and jumps

$$d \log S_t = \mu_t dt + \sigma_t dW_t + j_t dN_t$$

where μ, σ are deterministic, N is a counting process with deterministic intensity h and jump sizes j_s, j_t are independent for all $s \neq t$ and drawn from a family of jump distributions. Drift μ has to satisfy

$$F(T) = S_0 \exp \left(\int_0^T \left(\mu_t + \frac{1}{2} \sigma_t^2 + h_t \left(\mathbb{E} \left[e^{j_t} \right] - 1 \right) \right) dt \right)$$

The classical Merton model assumes homogenous Gaussian jumps j_t with mean μ_j and volatility σ_j for all t . European options can be priced in closed form in that case since

$$\log S_T \stackrel{(d)}{=} \log F(T) - \frac{1}{2} \int_0^T \sigma_t^2 dt - \left(e^{\mu_j + \sigma_j^2/2} - 1 \right) H_T + \int_0^T \sigma_t dW_t + \sum_{i=1}^{N_T} j_i$$

where j_i are independent Gaussians with parameters μ_j, σ_j and $H_T = \int_0^T h_t dt$. Note that $\log S_T$ conditional on N_T is Gaussian with mean and volatility given as

$$\begin{aligned}\mu(N_T) &= \log F(T) - \frac{1}{2} \int_0^T \sigma_t^2 dt - \left(e^{\mu_j + \sigma_j^2/2} - 1 \right) H_T + \mu_j N_T \\ \sigma^2(N_T) &= \int_0^T \sigma_t^2 dt + \sigma_j^2 N_T\end{aligned}$$

One obtains

$$\begin{aligned}\mathbb{E}[(K - S_T)^+] &= \sum_{n \geq 0} \mathbb{P}[N_T = n] \text{BSPut} \left\{ \exp \left(\mu(n) + \frac{1}{2} \sigma^2(n) \right), K, \sigma(n), T \right\} \\ &= \sum_{n \geq 0} \frac{e^{-H_T} H_T^n}{n!} \text{BSPut} \left\{ F(T) \exp \left(- \left(e^{\mu_j + \sigma_j^2/2} - 1 \right) H_T + \left(\mu_j + \frac{\sigma_j^2}{2} \right) n \right), \right. \\ &\quad \left. K, \sigma(n), T \right\}\end{aligned}$$

The case of jump to default is obtained with $\mu_j = -\infty$ and leads to

$$\begin{aligned}\mathbb{E}[(K - S_T)^+] &= e^{-H_T} \text{BSPut} \left(F(T) e^{H_T}, K, \sqrt{\int_0^T \sigma_t^2 dt}, T \right) \\ &\quad + (1 - e^{-H_T}) K\end{aligned}\tag{6.17}$$

where the first term relates to $n = 0$ and the second is the sum over $n > 0$. Note that e^{-H_T} is the survival probability and K is the payout in case of default, hence $S_T = 0$.

An alternative to Gaussian jumps are double exponential jumps with jump density

$$f_j(x) = p\eta_+ e^{-\eta_+ x} \mathbf{1}_{\{x > 0\}} + (1 - p)\eta_- e^{\eta_- x} \mathbf{1}_{\{x < 0\}}$$

where $\eta_+ > 1$ and $\eta_- > 0$. Kou and Wang derive analytical formulae for the price of European and American, lookback, barrier and perpetual American options, see Kou and Wang (2004) and references therein.

Another direction of research focuses on time-homogenous pure jump Lévy processes L with infinitely many jumps (infinite activity). Examples include Barndorff-Nielsen's normal inverse Gaussian (NIG) model, the Variance Gamma process and the CGMY process family named after and introduced by Carr, Geman,

Madan and Yor (2002). Stock processes can be constructed from such processes via exponentiation as

$$S_t = F(t) \frac{\exp L_t}{\mathbb{E}[\exp L_t]}$$

Negative jump mean results in negative skew allow for the calibration of implied volatilities for short maturities. With increasing maturity the skew decay is often faster than that observed in equities markets.

Since returns cannot be assumed to be independent due to volatility clustering, as pointed out in Section 1.3, these processes are not recommended for path-dependent products.

6.4.2 Stochastic clock

Volatility clustering can be incorporated within Lévy processes L assuming an increasing process C and setting

$$S_t = F(t) \frac{\exp L_{C_t}}{\mathbb{E}[\exp L_{C_t}]}$$

The Variance-Gamma process where L is Brownian motion with drift and C is a Gamma process has independent increments and thus does not exhibit persistent volatility. In order to model persistent volatility Carr et al. (2003) consider CGMY processes L where the clock C is the time integral of a mean reverting process, such as the CIR process introduced in Section 10.5.

6.4.3 Volatility with jumps

Bates (1996) extends Heston's model allowing for return jumps. This results in an affine model as discussed in Chapter 16. Return jumps allow for short skew calibration, which tends to be too flat in continuous stochastic volatility models.

Jumps in volatility can be incorporated in order to capture the variance swap distribution observed in volatility markets. In that case both return and volatility are assumed to jump simultaneously.

6.5 Non-Markovian models

In a one-factor setting the only complete continuous Markovian model which can be calibrated to Vanilla options, is Dupire's local volatility model.

A larger family of processes which can be calibrated to Vanilla options, and to some extent more exotic products, are stochastic volatility models. This family is

larger since model completeness is no longer assumed. Stochastic volatility models are not Markovian in the sense that the distribution of S_T given $\mathcal{F}_t$, namely all information up to time t , is different from the distribution of S_T given S_t . In order to get equality the state σ_t has also to be given. In short, the process (S_t, σ_t) is Markovian while (S_t) is not. Note that this even holds if the correlation between dS_t and $d\sigma_t$ is ± 1 .

The family of calibrated processes can alternatively be enlarged by giving up the Markov property. Research in this direction is sparse.

6.5.1 Hobson-Rogers

A complete non-Markovian model has been discussed by Hobson and Rogers (1998). Their proposed dynamics is

$$\begin{aligned}\frac{dS_t}{S_t} &= (r_t - d_t)dt + \sigma_t(Z_t)dW_t \\ Z_t &= \lambda \int_0^\infty e^{-\lambda s} \log\left(\frac{S_t}{S_{t-s}}\right) ds\end{aligned}$$

The process (S_t, Z_t) is again Markovian. For a given parameter λ it should be possible to find $\sigma_t(z)$ such that Vanilla option prices are matched. Forward-starting option prices will differ for calibrated models with different λ due to different autocorrelation. Hobson and Rogers also specify a volatility function as

$$\sigma^2(z) = a(1 + bZ_t + cZ_t^2) \vee d$$

where a impacts volatility level, b governs skew and λ is a memory parameter. Note that the model requires past stock values S_t , $t < 0$.

6.6 Calibration and hedging stochastic volatility

Introduction of incomplete models is often motivated by the need to price exotic options. These models have to be calibrated to the Vanilla market. The main practical problem is their use within risk management:

- Changing the price of a Vanilla option (vega bucket) does not have a meaningful impact on the calibration result.
- The model sensitivities cannot easily be expressed in terms of the risks of a portfolio of Vanilla options.

A link between changes in the Vanilla market and model sensitivities are established below, following a presentation by Knudsen (2002). Typically there are more market

(or Vanilla) instruments with prices P_k^{mkt} , $k = 1, 2, \dots, N$ than model parameters σ_l , $l = 1, 2, \dots, n$:

$$N > n$$

Hedging a position with value P_0 within the model leads to the problem of finding n market instruments such that

$$\Pi^{mdl} = P_0^{mdl} - \sum_{l=1}^n P_l^{mdl}$$

has no sensitivity to σ . This is also referred to as model parameter hedging.

If market prices change one needs $\frac{\partial P_l^{mdl}}{\partial P_k^{mkt}}$, $l = 0, 1, 2, \dots, n$. One has

$$\frac{\partial P_l^{mdl}}{\partial P_k^{mkt}} = \sum_{l=1}^n \frac{\partial P_l^{mdl}}{\partial \sigma_l} \frac{\partial \sigma_l}{\partial P_k^{mkt}}$$

However these sensitivities depend on the calibration procedure. If a particular instrument has not been included in the calibration then the sensitivity to this instrument will be zero.

Let calibration consist in solving a least square problem with all market instruments included:

$$\sum_{k=1}^N w_k \left(P_k^{mdl} - P_k^{mkt} \right)^2 = \min!$$

Then, for $l = 1, 2, \dots, n$,

$$h_l \left(\sigma, P^{mkt} \right) = \sum_{k=1}^N w_k \left(P_k^{mdl} - P_k^{mkt} \right) \frac{\partial P_k^{mdl}}{\partial \sigma_l} = 0$$

Due to implicit function theorem σ is a function of P^{mkt} and

$$\frac{\partial \sigma}{\partial P^{mkt}} = - \left(\frac{\partial h}{\partial \sigma} \right)^{-1} \frac{\partial h}{\partial P^{mkt}}$$

provided that the inverse exists. Also

$$\begin{aligned} \frac{\partial h_l}{\partial P_k^{mkt}} &= -w_k \frac{\partial P_k^{mdl}}{\partial \sigma_l} \\ \frac{\partial h_l}{\partial \sigma_m} &= \sum_{k=1}^N w_k \left(\frac{\partial P_k^{mdl}}{\partial \sigma_l} \frac{\partial P_k^{mdl}}{\partial \sigma_m} + \left(P_k^{mdl} - P_k^{mkt} \right) \frac{\partial^2 P_k^{mdl}}{\partial \sigma_l \partial \sigma_m} \right) \end{aligned}$$

These formulae express the model parameter sensitivity to market prices in terms of model price sensitivity to model parameters. If the model is well calibrated the second derivatives can be ignored.

A simple example with $N = 2$ and $n = 1$ illustrates the role of both model and calibration. Assume two Vanilla calls with strikes 100% and 105% and maturity one year. Let the model be Black-Scholes. A calibration routine computes the single model parameter volatility $\hat{\sigma}$ by minimizing

$$w_1 \left(P_1^{mdl}(\hat{\sigma}) - P_1^{mkt} \right)^2 + w_2 \left(P_2^{mdl}(\hat{\sigma}) - P_2^{mkt} \right)^2$$

If $w_2 = 0$ then exposure to $\hat{\sigma}$ (105%, one year) is not hedged against while at-the-money volatility exposure is captured. If both $w_1 > 0$ and $w_2 > 0$ exposure to both prices is captured to some extent. This exhibits the role of calibration. However a market move with P_1^{mkt} and P_2^{mkt} changing in opposite directions may not be captured. This is due to the limitation of the Black-Scholes model which does not allow for volatility skew. In summary, a market move orthogonal to the manifold defined as solutions of the least square problem is not hedged against.

Another example is a market consisting of Vanilla options and a portfolio including path-dependent products, such as a forward-starting option. The chosen model is local volatility. A vega hedge against changes of all Vanilla options does not necessarily immunize against changes of the path-dependent option price. Hybrid risks, such as sensitivity to interest rate or hazard rate volatility and correlation, are also not captured.

7] Implied Volatility Dynamics

7.1 Implied volatility delta

There is a natural order of market data speed, with spot levels changing faster than at-the-money volatility, at-the-money volatility changing more rapidly than volatility skew and volatilities being more volatile than dividend forecasts. Hedging performance can be improved by assuming a link between different market parameters, see Andreasen and Høge (2014). For example, when calculating a price with a new spot, or computing the sensitivity to a stock move (delta) using a spot shift, one may assume that this move is accompanied by a volatility move in the opposite direction or a change in expected dividends in the same direction. Thus, a delta hedge also hedges part of vega if stock and volatility are correlated. If delta and vega are hedged separately one has to be careful not to double count vega exposure. This section discusses pricing approaches assuming a spot move but no new volatility information.

7.1.1 Deterministic smile dynamics

Deterministic smile dynamics assumes that the implied volatility surface depends on spot only. Thus there is a function $\hat{\sigma}_{S_t,t}(K, T)$ which denotes the implied volatility surface observed at time t if the spot level is S_t . There are two important special cases:

- **Sticky strike** refers to

$$\hat{\sigma}_{S_t,t}(K, T) = \hat{\sigma}(K, T)$$

The dynamics of a Vanilla option is thus described by the Black-Scholes model, which is the only complete model with sticky strike dynamics. Otherwise an arbitrage can be constructed by fixing two call options $Call_1, Call_2$ with implied volatilities $\sigma_1 > \sigma_2$. The P&L of a delta hedged position over the time period dt

is given by

$$dP\&L(Call_i)_t = \frac{1}{2} ((\delta S_t)^2 - \sigma_i^2 S_t^2 dt) \Gamma_t^i$$

An arbitrage can be constructed by trading $\Gamma^2 Call_1 - \Gamma^1 Call_2$ since the corresponding $P\&L$ has no Γ, Θ and is positive:

$$dP\&L(\Gamma^2 Call_1 - \Gamma^1 Call_2)_t = \frac{\Gamma_t^1 \Gamma_t^2}{2} S_t^2 (\sigma_1^2 - \sigma_2^2) dt > 0$$

- **Sticky delta** refers to a dynamics of Vanilla options depending on moneyness and term, namely

$$\hat{\sigma}_{S_t, t}(K, T) = \hat{\sigma}(K/S_t, T - t)$$

The only complete models with sticky delta dynamics are models which assume independent returns. This follows with proportional forward dynamics, i.e. $F_t(T) = S_t c$ for a constant $c = c(t, T)$ from

$$\begin{aligned} \mathbb{E}_t \left[\left(\frac{S_T}{S_t} - K \right)^+ \right] &= \frac{1}{S_t} BSCall(F_t(T), KS_t, \hat{\sigma}_{S_t, t}(KS_t, T), T - t) \\ &= BSCall(c, K, \hat{\sigma}(K, T - t), T - t) \end{aligned}$$

Notwithstanding these theoretical findings it is useful to make a deliberate choice of implied volatility dynamics for the purpose of sensitivity computation. This choice should reflect the average implied volatility dynamics observed in the market. Alternatively, it could be derived from a theoretical model considered to be the most realistic.

In the case of implied volatility parametrization in terms of $k = k(K, F(t))$ as in (4.1), one has (at least) two choices for the computation of price and delta. The first approach consists of freezing the forward at a reference level $F(t)^*$ and computing

$$C(K, t, S_0) = P(t) BSCall(F(t, S_0), K, \hat{\sigma}(k(K, F(t)^*), t), t)$$

Alternatively one can update $F(t) = F(t, S_0)$ within the volatility parametrization

$$C(K, t, S_0) = P(t) BSCall(F(t, S_0), K, \hat{\sigma}(k(K, F(t, S_0)), t), t)$$

The former approach leads to sticky strike or Black-Scholes smile dynamics, while the latter is sticky delta. Note that when fitting volatility the reference forward parameter should always be reset to the latest available forward.

7.1.2 Sticky delta and stochastic volatility

Stochastic volatility models (Section 6.2), possibly with proportional jumps but without a local volatility component, are sometimes referred to as sticky delta since

$$\hat{\sigma}_{S_t, \sigma_t, t}(K, T) = \hat{\sigma}_{\sigma_t}(K/S_t, T - t) \quad (7.1)$$

However if dS and $d\sigma$ are correlated then integration yields

$$\hat{\sigma}_{S_t, t}(K, T) = \hat{\sigma}_{\mathbb{E}[\sigma_t | S_t]}(K/S_t, T - t) \neq \hat{\sigma}(K/S_t, T - t)$$

Although spot dependence of implied volatility is in moneyness terms only there is a local effect, in that, if spot is high at time t then it is likely that volatility will be low at the same time. This is assuming negative correlation between log returns and volatility increments.

7.1.3 Real implied volatility dynamics

Volatility reacts slowly on spot moves. In quiet markets volatility is quoted by strike and is updated much less frequently than spot. This indicates sticky strike dynamics. When markets are volatile then implied volatility will be updated more frequently and dynamics may resemble sticky delta.

The implied volatility surface can be described by principal components level, term slope, skew and skew decay. When analyzing the time series of those parameters one observes a relatively stable correlation of these components with spot level and with mean reversion.

Realistic models should exhibit stochastic implied volatility dynamics, in the sense that the smile dynamics may allow for both sticky strike and sticky delta dynamics, as well as random changes between the two. To some extent local stochastic volatility models capture this behaviour, see Section 6.3. A model specifically designed for stochastic skew is due to Jäckel and Kahl (2007).

7.2 Forward volatility

Determining the price $Call(t, T, K)$ for the payoff

$$\left(\frac{S_T}{S_t} - K \right)^+$$

is at the core of equity derivatives modelling since it depends on volatility dynamics and thus modelling assumptions. It may seem natural to use a standard implied volatility surface in order to specify those prices as

$$\hat{\sigma}(t, T, K) = \hat{\sigma}(KF(t), T - t)$$

where the right hand side can be implied from Vanilla option prices. However this approach ignores the effect of persistent volatility as observed in stock time series (see Section 1.3), which is the topic of this section.

7.2.1 Variance additivity

If a log-normal process is assumed then forward volatility levels are obtained from the additivity of variance; one has

$$\hat{\sigma}(t, T)^2(T - t) = \hat{\sigma}(T)^2 T - \hat{\sigma}(t)^2 t$$

and no strike dependency is required. In the case of independent returns additivity holds for the logarithm of the Laplace transform of $\log S$:

$$\log \mathbb{E} \left[\exp \left(u \log \frac{S_T}{S_t} \right) \right] = \log \mathbb{E} [\exp (u \log S_T)] - \log \mathbb{E} [\exp (u \log S_t)]$$

In the general case one has additivity of realized variance. Given a partition $0 = t_0 < t_1 < \dots < t_n = T$ of the interval $[0, T]$ and realized variances

$$V(t, T) = \sum_{t_i \in (t, T]} \left(\log \left(\frac{S_{t_i}}{S_{t_{i-1}}} \right) \right)^2$$

then one has additivity of variance in the sense that

$$V(t, T) = V(T) - V(t)$$

holds where the argument 0 is omitted for spot-starting variance.

7.2.2 Level and skew

Assuming that the volatility surface is of the form

$$\hat{\sigma}(T, K) = \sigma(T) + s(T) \frac{K - S_0}{S_0}$$

one can show (see Demeterfi et al. (1999)) that up to first order the variance swap price is given by

$$V(T) \approx \sigma(T)^2 T (1 + 3s(T)^2 T)$$

Assuming the same holds true for forward-starting variance swaps one obtains a relationship between forward volatility and forward skew, namely

$$V(t, T) \approx \sigma(t, T)^2 (T - t) (1 + 3s(t, T)^2 (T - t))$$

where the left hand side is given due to additivity of variance. This relationship ignores convexity effects and can thus only hold for strikes not too far from the spot level.

7.2.3 Convexity

It is well known that the implied volatility smile for maturity $T - t$ is not a good indicator for the forward volatility smile of the return S_T/S_t . In fact, the forward volatility smile is more convex. In order to understand this effect assume a time-homogenous stochastic volatility model where volatility is stochastic but not a local function of spot S_t .

- Fix t, T, K and denote by σ_t the stochastic volatility process with mean $\bar{\sigma}_t$.
- Let $\hat{\sigma}_s = \hat{\sigma}_s(t, T, K)$ denote the Black-Scholes implied volatility for the forward-starting option observed at time $s \in [0, t]$.
- The aim is to determine $\hat{\sigma}_0$ as a function of mean and variance of σ_t .

Note that $\hat{\sigma}_t$ is a function of σ_t , compare equation (7.1). Thus, conditioning on σ_t one has

$$Call = \mathbb{E} [BSCall(\hat{\sigma}_t(\sigma_t))]$$

Here $Call$ denotes the model price of the forward-starting option without discounting. Deterministic arguments $F_t(T)/S_t, K$ and $T - t$ of $BSCall$ are omitted. Expanding σ_t around $\bar{\sigma}_t$ one obtains (the first-order term vanishes)

$$Call \approx BSCall(\hat{\sigma}_t(\bar{\sigma}_t)) + \frac{1}{2} \frac{\partial^2 BSCall(\hat{\sigma}_t(\sigma))}{\partial \sigma^2} \Big|_{\sigma=\bar{\sigma}_t} \text{Var} [\sigma_t] \quad (7.2)$$

The second derivative is given by

$$\frac{\partial^2 BSCall(\hat{\sigma}_t)}{\partial \sigma^2} = \Upsilon(\hat{\sigma}) \left(\frac{d_+ d_-}{\hat{\sigma}} \left(\frac{\partial \hat{\sigma}_t}{\partial \sigma} \right)^2 + \frac{\partial^2 \hat{\sigma}_t}{\partial \sigma^2} \right)$$

since

$$\frac{\partial^2 BSCall(\hat{\sigma})}{\partial \hat{\sigma}^2} = \frac{\partial BSCall(\hat{\sigma})}{\partial \hat{\sigma}} \frac{d_+ d_-}{\hat{\sigma}} \quad (7.3)$$

Alternatively, expanding the Black-Scholes formula viewed at $t = 0$ around

$$\hat{\sigma}_t = \hat{\sigma}_t(\bar{\sigma}_t)$$

up to the first order one has, due to (7.3)

$$\begin{aligned} Call &= BSCall(\hat{\sigma}_0) \\ &\approx BSCall(\hat{\sigma}_t) + \Upsilon(\hat{\sigma}_t)(\hat{\sigma}_0 - \hat{\sigma}_t) \end{aligned} \quad (7.4)$$

Equating (7.2) and (7.4) leads to

$$\hat{\sigma}_0 \approx \hat{\sigma}_t + \frac{\text{Var}[\sigma_t]}{2} \left(\frac{d_+(\hat{\sigma}_t)d_-(\hat{\sigma}_t)}{\hat{\sigma}_t} \left(\frac{\partial \hat{\sigma}_t}{\partial \sigma} \right)^2 + \frac{\partial^2 \hat{\sigma}_t}{\partial \sigma^2} \right)$$

where $\hat{\sigma}_t = \hat{\sigma}_t(\bar{\sigma}_t)$. This can be simplified using the approximation

$$\frac{\partial \hat{\sigma}_t}{\partial \sigma} \approx 1 \quad \text{and} \quad \frac{\partial^2 \hat{\sigma}_t}{\partial \sigma^2} \approx 0$$

Assuming a time-homogenous stochastic volatility model one has

$$\hat{\sigma}_t \approx \hat{\sigma}(S_0 K, T - t)$$

where the right hand side is spot implied volatility. Thus the implied volatility $\hat{\sigma}(K, t, T)$ of a forward-starting option can be approximated as

$$\hat{\sigma}(K, t, T) \approx \hat{\sigma}(S_0 K, T - t) + \frac{\text{Var}[\sigma_t]}{2\hat{\sigma}(S_0 K, T - t)^3} \left(\left(\log \frac{F(T)}{F(t)K} \right)^2 - \frac{1}{4} \hat{\sigma}(S_0 K, T - t)^4 \right)$$

At-the-money forward volatility in particular can be slightly lower than the corresponding spot volatility. In-the-money and out-of-the-money forward volatility will be higher due to the convexity increase of forward-starting smile being a multiple of $(\log K)^2$.

7.3 Modelling implied volatility

A first attempt at modelling the dynamics of implied volatility is due to Schönbucher (1998), who models the joint dynamics of spot and single implied volatility. This approach leads to a constraint on the short time to maturity implied volatility limit. Schweizer and Wissel (2008) provide more recent research in this direction.

Dupire (1996) observes that instantaneous variance is tradeable and thus a martingale with respect to a martingale measure. Bühler (2006) expands this idea by applying HJM theory with the appropriate (no) drift condition. Applications focus on modelling options on variance swaps. Important examples derived from Bühler's variance curve functionals can be expressed as multi-factor stochastic volatility models, such as double Heston or double log-normal, where double refers to the stochastic mean reversion level. Bergomi (2005) proposes a two-factor exponential Ornstein-Uhlenbeck process as a variance curve model.

In discrete time one may assume independent returns and specify forward volatility directly. Since with that approach (spot) implied volatilities are not part of the calibration procedure some dependency between returns is required to calibrate to standard Vanilla options. A complete non-Markovian discrete-time market model has been presented by the author, see Brockhaus (2004). Bergomi (2005) describes a discrete-time version of his model also leading to a discrete-time market model. Variants of these approaches have presumably also been used by other practitioners, see Behvand (2010).

7.3.1 Schönbucher

Schönbucher (1998) studies the dynamics of implied volatility $\hat{\sigma} = \hat{\sigma}(K, T)$. The starting point is a stock process

$$\frac{dS_t}{S_t} = r_t dt + \sigma_t dW_t^S$$

with stochastic volatility σ_t and an implied volatility process with dynamics

$$d\hat{\sigma} = \mu_t^{\hat{\sigma}} dt + \alpha_t dW_t^{\hat{\sigma}}$$

where W^S and $W^{\hat{\sigma}}$ may be correlated. Itô's formula for the call price *Call* reads

$$\begin{aligned} d\text{Call} &= \text{Call}_t dt + \text{Call}_S dS + \text{Call}_{\hat{\sigma}} d\hat{\sigma} + \frac{1}{2} \text{Call}_{SS} d\langle S \rangle + \frac{1}{2} \text{Call}_{\hat{\sigma}\hat{\sigma}} d\langle \hat{\sigma} \rangle + \text{Call}_{S\hat{\sigma}} d\langle \hat{\sigma}, S \rangle \\ &= \left(\text{Call}_t + rS\text{Call}_S + \mu^{\hat{\sigma}} \text{Call}_{\hat{\sigma}} + \frac{1}{2} \sigma^2 S^2 \text{Call}_{SS} + \frac{1}{2} \alpha^2 \text{Call}_{\hat{\sigma}\hat{\sigma}} + \rho \sigma \alpha \text{Call}_{S\hat{\sigma}} \right) dt \\ &\quad + \sigma S \text{Call}_S dW^S + \alpha \text{Call}_{\hat{\sigma}} dW^{\hat{\sigma}} \end{aligned}$$

The call price also satisfies the Black-Scholes equation

$$dCall = \left(Call_t + rSCall_S + \frac{1}{2} \hat{\sigma}^2 S^2 Call_{SS} \right) dt + \hat{\sigma} SCall_S dW$$

Equating the expectation

$$\mathbb{E} [dCall] = rCall dt$$

of both equations yields the equation for the implied volatility drift $\mu^{\hat{\sigma}}$

$$2Call_{\hat{\sigma}} \mu^{\hat{\sigma}} = (\hat{\sigma}^2 - \sigma^2) S^2 Call_{SS} - \alpha^2 Call_{\hat{\sigma}\hat{\sigma}} - 2\rho\sigma\alpha Call_{S\hat{\sigma}}$$

This can be expanded for the case of the European call option to

$$2\hat{\sigma} \mu^{\hat{\sigma}} = \frac{1}{\tau} (\hat{\sigma}^2 - \sigma^2) - \alpha^2 d_+ d_- - 2\rho\sigma\alpha \frac{d_-}{\sqrt{\tau}}$$

If the limit $\lim_{t \rightarrow T} \mu_t^{\hat{\sigma}}$ remains bounded, a solution to this equation exists. This no-bubble restriction can be written as quadratic polynomial in σ (or quartic polynomial in $\hat{\sigma}$)

$$\hat{\sigma}^2 \sigma^2 - 2\rho\sigma\alpha\hat{\sigma}f - \hat{\sigma}^4 + f^2\alpha^2 = 0$$

where $f = \log \frac{S}{K}$ with solution

$$\sigma = \frac{\rho\alpha}{\hat{\sigma}} + \sqrt{\hat{\sigma}^2 - \frac{f^2}{\hat{\sigma}^2} \alpha^2 (1 - \rho^2)}$$

Babbar (2001) constructs a nontrivial example for this single option case which satisfies the above conditions.

Schönbucher (1998) also considers the case of several options with implied volatilities $\hat{\sigma}(K_i, t_i)$ assuming there is only one option per maturity, as well as a continuum of implied volatilities $\hat{\sigma}(K(t), t)$. Ledoit et al. (1998) study the dynamics of implied volatility $\hat{\sigma}_t(f, \tau)$ expressed in terms of moneyness f and time to maturity $\tau = T - t$ leading to alternative drift conditions. Schweizer and Wissel (2008) expand the analysis to convex payoffs instead of the specific call option payoff.

Cont et al. (2002) define dynamics of implied volatility in terms of the dynamics of principal components of the surface, as well as stock dynamics without consideration of arbitrage at the short time to maturity limit.

7.3.2 Variance curve

Let instantaneous variance $v_t(T)$ be defined as

$$v_t(T) = \frac{\partial}{\partial T} \mathbb{E}_t [V(t, T)]$$

with $V(t, T)$ being log variance of a stock S as defined in Section 3.7.1, see (3.7). Bühler (2006) studies $v_t(T)$ in analogy to forward rate $f_t(T)$. Since variance $V(T)$, and hence forward variance $V(t, T)$ and $\frac{\partial}{\partial T} V(t, T)$, can (in theory) be replicated through a hedging portfolio one has the HJM (no) drift condition

$$dv_t(T) = \sigma_t^v(T) dW_t \quad (7.5)$$

with respect to a martingale measure, where $\sigma^v(T)$ is a volatility process and W a Brownian motion, as already observed by Dupire (1996). Using Musiela parametrization (see Section 10.3.1) this gives

$$d\tilde{v}_t(\tau) = \frac{\partial}{\partial \tau} \tilde{v}_t(\tau) dt + \tilde{\sigma}_t^v(\tau) dW \quad (7.6)$$

For practical purposes finite-dimensional representation in terms of a time-homogenous Markov process Z is desirable, namely

$$\begin{aligned} \tilde{v}_t(\tau) &= G(Z_t, \tau) \\ dZ_t &= \mu^Z(Z_t) dt + \sigma^Z(Z_t) dW_t \end{aligned}$$

Due to Itô's lemma (7.6) requires

$$\frac{\partial}{\partial \tau} G = \mu^Z \frac{\partial}{\partial z} G + \frac{1}{2} \frac{\partial^2}{\partial z^2} G$$

An example is

$$G(z, x) = z_2 + (z_1 - z_2) e^{-\kappa x}$$

which requires drift

$$\mu^Z(z) = (-\kappa(z_1 - z_2), 0)^T$$

Volatility can be freely specified. Bühler (2006) suggests

$$\sigma^Z(z) = \begin{pmatrix} \nu z_1^\alpha & 0 \\ \eta \rho z_2 & \eta \bar{\rho} z_2 \end{pmatrix}$$

with $\bar{\rho}^2 = 1 - \rho^2$. In other words

$$\begin{aligned} dZ_t^1 &= \kappa(Z_t^2 - Z_t^1)dt + \alpha(Z_t^1)^\beta dW_t^1 \\ \frac{dZ_t^2}{Z_t^2} &= \alpha_2 dW_t^2 \end{aligned}$$

where W^1 and W^2 have correlation ρ . Parameters $\beta = 1/2, \alpha_2 = 0$ lead to Heston's model

$$\frac{dS_t}{S_t} = (r_t - d_t)dt + \alpha \sqrt{Z_t^1} dW_t^3$$

where W^3 may be correlated with W^1 . This approach can be extended further such that Z_2 mean reverts to a stochastic process Z_3 , namely

$$\begin{aligned} dZ_t^1 &= \kappa_1(Z_t^2 - Z_t^1)dt + \alpha_1(Z_t^1)^{\beta_1} dW_t^1 \\ dZ_t^2 &= \kappa_2(Z_t^3 - Z_t^2)dt + \alpha_2(Z_t^2)^{\beta_2} dW_t^2 \\ \frac{dZ_t^3}{Z_t^3} &= \alpha_3 dW_t^3 \end{aligned}$$

Volatility typically mean reverts quickly towards a slowly mean reverting process, i.e. $\kappa_1 \gg \kappa_2$. Special case $\beta_1 = \beta_2 = 1/2, \alpha_3 = 0$ is also known as double Heston, and $\beta_1 = \beta_2 = 1, \alpha_3 = 0$ as double log-normal, see Gatheral (2008). The double Heston model is also discussed in Section 6.2.5.

7.3.3 Bergomi

Bergomi (2009) proposes variance dynamics with log-normal mean reverting variance defined through a volatility function

$$\sigma_t^v(T) = \alpha v_t(T) e^{-\kappa(T-t)}$$

A solution can be given in terms of a standard Ornstein-Uhlenbeck process X as

$$\begin{aligned} v_t(T) &= v_0(T) \exp \left(\alpha e^{-\kappa(T-t)} X_t - \frac{\alpha^2}{2} e^{-2\kappa(T-t)} \mathbb{E}[X_t^2] \right) \\ dX_t &= -\kappa X_t dt + dW_t \end{aligned} \tag{7.7}$$

with $X_0 = 0$. The natural extension to two factors has dynamics

$$\frac{dv_t(T)}{v_t(T)} = \alpha_1 e^{-\kappa_1(T-t)} dW_t^1 + \alpha_2 e^{-\kappa_2(T-t)} dW_t^2$$

where W^1 and W^2 can be correlated. The solution is now expressed in terms of two standard Ornstein-Uhlenbeck processes X^1, X^2 driven by correlated Brownians W^1, W^2 as

$$v_t(T) = v_0(T) \exp \left(\sum_{i \in \{1,2\}} \alpha_i e^{-\kappa_i(T-t)} X_t^i - \frac{1}{2} \sum_{i,j \in \{1,2\}} \alpha_i \alpha_j e^{-(\kappa_i + \kappa_j)(T-t)} \mathbb{E} \left[X_t^i X_t^j \right] \right) \quad (7.8)$$

The corresponding spot process has volatility $\sigma_t^S = \sqrt{v_t(t)}$.

7.4 Discrete-time models

7.4.1 Forward volatility specification

In the absence of a smile model one may specify forward volatility for the purpose of pricing and risk management of Vanilla cliquets. This can be achieved by specifying forward volatility level and skew separately. Such an approach is motivated by a principal component analysis of implied volatility dynamics.

The forward volatility level can be specified by choosing a backbone implied volatility term structure $(\hat{\sigma}(t), t \in [0, T])$. This term structure may be derived from

- the most liquid options struck at spot S_0 , namely $\hat{\sigma}(t) = \hat{\sigma}(t, S_0)$,
- the volatilities $\hat{\sigma}(t) = \hat{\sigma}(t, F(t))$ interpolated at the mean of the distribution $F(t)$, namely the forward, or from
- variance swaps: $\hat{\sigma}(t) = \sqrt{\mathbb{E} \left[\frac{V(t)}{t} \right]}$.

The reference forward volatility $\hat{\sigma}(t, T)$ is obtained via variance additivity, namely

$$\hat{\sigma}(t, T)^2(T-t) = \hat{\sigma}(T)^2 T - \hat{\sigma}(t)^2 t$$

This volatility is interpreted as volatility at the reference strike $K(t, T)$, which may be set as $K(t, T) = 1$ or forward $K(t, T) = F(T)/F(t)$ in the sense that

$$\hat{\sigma}(t, T, K(t, T)) = \hat{\sigma}(t, T)$$

In order to obtain forward volatilities $\hat{\sigma}(t, T, K)$ for strikes different from the reference strike $K(t, T)$ one may define a skew function such that

$$\hat{\sigma}(t, T, K) = \hat{\sigma}(t, T) + s(t, T, K)$$

Assuming skew remains unchanged set

$$s(t, t + \tau, K) = \hat{\sigma}(\tau, K(\tau)K) - \hat{\sigma}(\tau, K(\tau))$$

where $K(t)$ is the interpolation level for the backbone, namely $K(t) = S_0$ or $K(t) = F(t)$. A multiplicative scheme can be defined as

$$\hat{\sigma}(t, T, K) = \hat{\sigma}(t, T)s(t, T, K)$$

where skew

$$s(t, t + \tau, K) = \frac{\hat{\sigma}(\tau, K(\tau)K)}{\hat{\sigma}(\tau, K(\tau))}$$

is given as the volatility ratio observed today for a given moneyness. If a skew increase or decay through time is predicted this approach can be modified.

Specification of forward volatilities allows one to price forward-starting options, and thus Vanilla cliquets, using the Black-Scholes formula. Note that assuming forward skew is equal to spot skew ignores the effect of stochastic volatility discussed in Section 7.2.3 and may therefore lead to mispricing. Thus the above forward volatility specification is not recommended. Instead, one may add convexity using the adjustment derived therein to account for the stochastic volatility effect. Alternatively, one has semi-analytic formulae for forward-starting options for affine models, see Section 16.2.2.

7.4.2 Independent returns

Assuming that forward volatilities are given one can derive the unconditional implied distributions

$$F_i(K) = \mathbb{P} \left[\frac{S_{t_i}}{S_{t_{i-1}}} \leq K \right]$$

for the relevant dates $0 = t_0 < t_1 < t_2 < \dots < t_n$ of a trade. This enables the sampling of a process consistent with those volatilities. The algorithm proceeds as follows:

1. Sample independent uniform variables $(U_i, i = 1, 2, \dots, n)$.
2. For all i , apply the deterministic mapping D_i to U_i , where

$$D_i(x) = F_i^{-1}(x)$$

and set $S_{t_i} = S_0 \prod_{k \leq i} D_k(U_k)$.

In the absence of volatility skew this algorithm is equivalent to discrete-time Black-Scholes sampling. It yields a process with independent returns. Due to the construction of the process there will not be consistency with other market information, such as spot implied volatility for maturities $t_i, i > 1$ or forward-starting options with different return length.

In order to calibrate the discrete model not only to forward-starting but also to spot-starting option prices extra degrees of freedom are required, namely some dependence between different returns. A general approach following Brockhaus (2004) is given below.

7.4.3 Discrete complete model

Fix a regular schedule $0 = t_0 < t_1 < \dots < t_n = T$ with corresponding returns

$$R_i = \frac{S_{t_i}}{S_{t_{i-1}}}$$

Let $\hat{\sigma}_t(K, t, T)$ denote the implied volatility for the Vanilla option with strike $S_t K$ and maturity T observed at time t .

First step: Today's market implied distribution F_{R_1} of R_1 can be derived from market prices of Vanilla options, namely

$$F_{R_1}(K) = \frac{\partial}{\partial K} Put(S_0 K, t_1)$$

where $Put(K, t)$ is the (not discounted) price of a put option with strike K and maturity t . These prices can be represented using today's implied volatilities. In order to sample from this distribution one may use a standard uniform variate U_1 , namely

$$\frac{S_{t_1}}{S_0} = D(U_1) = F_{R_1}^{-1}(U_1) \quad (7.9)$$

Second step: Assuming deterministic smile dynamics as described in Section 7.1.1 one has, due to equation (7.9),

$$\hat{\sigma}_{t_1}(t_1, t_2, K) = \hat{\sigma}(S_{t_1}, K) = \hat{\sigma}(U_1, K) \quad (7.10)$$

in the sense that there exists a function (again denoted by $\hat{\sigma}$) with arguments (U_1, K) representing the left hand side. Generating the distribution of R_2 requires the quantile mapping $D(u_1, \cdot)$ implied by the volatility function $\hat{\sigma}(u_1, \cdot)$ in (7.10) for all u_1 in $[0, 1]$. One may thus write

$$\frac{S_{t_2}}{S_{t_1}} = D(U_1, U_2)$$

where U_1, U_2 are independent uniform variates. The two volatility functions $\hat{\sigma}(K)$ and $\hat{\sigma}_u(K)$ characterize the set of arbitrage-free distributions of (S_{t_1}, S_{t_2}) with deterministic smile dynamics.

General step: Generalizing this argument one sets

$$\frac{S_{t_i}}{S_{t_{i-1}}} = D(U_1, \dots, U_{i-1}, U_i)$$

where the quantile mapping D is derived from a volatility function

$$\hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) = \hat{\sigma}(R_1, \dots, R_{i-1}, K) = \hat{\sigma}(U_1, \dots, U_{i-1}, K)$$

via

$$\begin{aligned} & D^{-1}(U_1, \dots, U_{i-1}, K) \\ &= \frac{\partial}{\partial K} BS\left(\frac{F(t_i)}{F(t_{i-1})}, K, \hat{\sigma}(U_1, \dots, U_{i-1}, K), (t_i - t_{i-1})\right) \end{aligned} \quad (7.11)$$

where -1 refers to the inverse with respect to the last variable. In order to match the Vanilla market one has the constraints

$$\mathbb{E}\left[(S_{t_i} - K)^+\right] = \mathbb{E}\left[\left(S_0 \prod_{j=1}^i D(U_1, \dots, U_{j-1}, U_j) - K\right)^+\right] = \text{Call}(K, t_i)$$

for all i and K where $\text{Call}(K, t)$ denotes the market price of a (not discounted) call with strike K and maturity t . Unconditional forward-starting option prices are given as

$$\mathbb{E}\left[(R_i - K)^+\right] = \mathbb{E}\left[BS\left(\frac{F(t_i)}{F(t_{i-1})}, K, \hat{\sigma}(U_1, \dots, U_{i-1}, K), t_i - t_{i-1}\right)\right]$$

Note that if forward volatility is deterministic

$$\hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) = \hat{\sigma}(U_1, \dots, U_{i-1}, K) = \hat{\sigma}_i(K)$$

then the asset returns $R_i, i = 1, 2, \dots, n$ are independent.

The key assumption made here is that the discrete process $(S_{t_i}, i = 0, 1, \dots, n)$ is adapted to its own filtration. This condition is satisfied by local volatility but not by stochastic volatility models. The model is not assumed to be Markovian. One cannot expect to find a specification such that the Vanilla market is matched exactly. Instead, a parametrized specification of forward volatility has to be calibrated to market prices. Some well known models can be represented as discrete-time models within this framework. They can be characterized in terms of their smile (dependence on K) and their memory or autocorrelation properties (dependence of next return on past returns):

- **Black-Scholes:** The Black-Scholes model does not exhibit smile. Also, it does not feature autocorrelation. Hence

$$\hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) = \sigma_i$$

- **Local volatility:** Dupire's local volatility model generates smile through spot dependence only, namely

$$\begin{aligned} \hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) &= \hat{\sigma}_{t_{i-1}}(S_{t_{i-1}}) \\ &= \sigma_{t_{i-1}} \left(S_0 \prod_{j=1}^{i-1} D(U_1, \dots, U_{j-1}, U_j) \right) \end{aligned}$$

There is no dependency on K . Thus, conditional on the start value $S_{t_{i-1}}$, the return R_i is log-normally distributed.

- **Independent returns:** Processes with independent returns are also known as exponential Lévy processes. Those models have no memory in the sense that smile is only generated through dependency on K :

$$\hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) = \sigma_i(K)$$

- **Hobson-Rogers:** The Hobson-Rogers model described in Section 6.5.1 takes into account the spot history with exponential time-decay of impact, namely

$$\begin{aligned}
 \hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) &= \hat{\sigma}_{t_{i-1}} \left(\sum_{j=-\infty}^{i-2} e^{-\lambda(t_{i-1}-t_j)} \log \left(\frac{S_{t_{i-1}}}{S_{t_j}} \right) \right) \\
 &= \hat{\sigma}_{t_{i-1}} \left(\sum_{j=-\infty}^{i-2} e^{-\lambda(t_{i-1}-t_j)} \sum_{k=j+1}^{i-1} \log R_k \right) \\
 &= \hat{\sigma}_{t_{i-1}} \left(\sum_{k=-\infty}^{i-1} \log R_k \sum_{j=-\infty}^{k-1} e^{-\lambda(t_{i-1}-t_j)} \right) \\
 &= \hat{\sigma}_{t_{i-1}} \left(\frac{1}{1 - e^{-\lambda\Delta}} \sum_{k=-\infty}^{i-1} \log R_k e^{-\lambda(t_{i-1}-t_{k-1})} \right) \\
 &= \hat{\sigma}_{t_{i-1}} \left(\frac{1}{1 - e^{-\lambda\Delta}} \sum_{k=1}^{i-1} \log D(U_1, \dots, U_{k-1}, U_k) e^{-\lambda(t_{i-1}-t_{k-1})} \right)
 \end{aligned}$$

if $t_i = \Delta i$ for all i . Thus the impact of returns on volatility decays exponentially with time. Return $R_k = D(U_1, \dots, U_{k-1}, U_k)$ depends mainly on U_k . The impact of the other arguments is indirect through volatility dynamics and will be less pronounced. It follows that the impact of the U_k also decays with time. As in the local volatility model there is no dependency on strike K .

7.4.4 Volatility factorization

The complete market model described in Brockhaus (2004) assumes the factorization

$$\hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) = \hat{\sigma}(U_1, \dots, U_{i-1}, K) = f_i(U_1, \dots, U_{i-1}) g_i(K) \quad (7.12)$$

Strike dependence: The smile function g_i is determined by at-the-money volatility, skew and further parameters. Those further parameters may determine convexity, reference forward and tail behaviour. Candidates are given in Section 4.1. Alternatively, one may derive forward volatility g_i though linear displacement from a given spot parametrization, as described in Section 7.4.1.

Noise dependence: The number of arguments of the volatility function f_i increases with i . In order to keep the number of arguments constant one can define Gaussian

factors $(Z_1, Z_2, \dots, Z_n)$ obtained as linear combinations from the U_i as

$$Z_i = \sum_{j=1}^i w_{ij} \Phi^{-1}(U_i)$$

normalised to unit variance through

$$\sum_{j=1}^i w_{ij}^2 = 1$$

and set

$$f_i(Z_i) = \exp\left(\alpha_i Z_i - \frac{\alpha_i^2}{2}\right) \quad (7.13)$$

with α_i specifying volatility of volatility. Weights can be derived as autocorrelation functions

$$w_{ij} = \rho(t_i, t_j)$$

from processes such as Ornstein-Uhlenbeck

$$dZ_t = -\kappa Z_t dt + dW_t$$

with autocorrelation between Z_s and Z_t given as

$$\rho(s, t) = \frac{e^{-\kappa(t-s)} - e^{-\kappa(t+s)}}{\sqrt{1 - e^{-2\kappa t}} \sqrt{1 - e^{-2\kappa s}}}$$

or fractional Brownian motion, formally

$$dZ_t = (dW_t)^\kappa$$

with autocorrelation

$$\rho(s, t) = \frac{s^\kappa + t^\kappa - (t-s)^\kappa}{2(st)^{\kappa/2}}$$

Calibration: The model is specified through functions $f_i(Z_i)$ and $g_i(K)$. The first function is parametrized through volatility of volatility α_i and mean reversion

κ . The second is parametrized through expected at-the-money volatility σ_i and skew s_i .

The skews s_i for $g_i(K)$ may be derived from spot skews. The sign of α_i will be equal to the sign of s_i and thus be negative for equities markets. Given κ the volatility of volatility levels α_i allow for the matching of at-the-money spot skews $\frac{\partial}{\partial K} \hat{\sigma}(K(t_i), t_i)$. Volatilities σ_i allow matching with at-the-money spot volatilities $\hat{\sigma}(K(t_i), t_i)$. The level of κ will determine spot and forward implied volatility convexity.

Incomplete extension: Volatility of volatility α_i can also be derived from forward volatility convexity, if known, since forward-starting options are given integrals one-dimensional Gaussian integral. In that case additional stochastic volatility factors V_i allow for modification of spot skew via

$$R_i = D(Z_{i-1}) \left(\rho_i \Phi^{-1}(U_i) + \sqrt{1 - \rho_i^2} \Phi^{-1}(V_i) \right) \quad (7.14)$$

Correlation ρ_i impacts spot skew while leaving forward skew unchanged. Note that this model no longer complete.

7.4.5 Bergomi

Bergomi (2005) proposes a discrete version of the two-factor variance curve model presented in Section 7.3.3. The stock process is assumed to satisfy

$$\frac{dS_t}{S_t} = (r_t - d_t)dt + \sigma_i \left(\frac{S_t}{S_{t_{i-1}}} \right)^{1-\beta_i} dW_t$$

within period $[t_{i-1}, t_i]$ where σ_i is chosen, such that the stock process is consistent with the variance process, namely

$$\begin{aligned} v_{t_{i-1}}(t_i) &= \mathbb{E}_{t_{i-1}} \left[\frac{1}{t_i - t_{i-1}} \int_{t_{i-1}}^{t_i} \sigma_i^2 \left(\frac{S_s}{S_{t_{i-1}}} \right)^{2(1-\beta_i)} ds \right] \\ &= \frac{\sigma_i^2}{t_i - t_{i-1}} \mathbb{E} \left[\int_0^{t_i - t_{i-1}} \tilde{S}_s^{2(1-\beta_i)} ds \right] \end{aligned}$$

where $\tilde{S}_s = S_{t_{i-1}+s}/S_{t_{i-1}}$. Thus σ_i^2 is a deterministic linear function of the random variable $v_{t_{i-1}}(t_i)$. The skew parameter β_i can be chosen such that forward skew

$$\frac{\partial}{\partial K} \hat{\sigma}_{t_{i-1}}(KS_{t_{i-1}}, t_i)$$

is also a deterministic function of $v_{t_{i-1}}(t_i)$.¹ This parametrization is a special case of the factorization described in Section 7.4.4, see equation (7.12) where $g_i(K)$ is the implied volatility smile at time t_i of a CEV process starting in 1 at time t_{i-1} with instantaneous volatility 1. The discrete factor Z is derived from the Ornstein-Uhlenbeck processes in equations (7.7) or (7.8). Bergomi (2005) allows for decorrelation between the Brownian increment determining return R_i and the Brownian motions driving the Ornstein-Uhlenbeck processes up to t_i , as in equation (7.14). The variance level for period $[t_{i-1}, t_i]$ is log-normal and specified through an exponential function $f_i(Z_i)$ as (7.13).

7.4.6 Behvand

Behvand (2010) discusses a model which is referenced, somewhat cheekily, as a private communication from within Merrill Lynch Quantitative Risk Management. At-the-money forward volatility f_i for period $[t_{i-1}, t_i]$ is determined recursively from f_{i-1} through

$$\frac{f_i}{v_i} = \frac{f_{i-1}}{v_{i-1}} \exp \left(\alpha \sqrt{\Delta_i} \Phi^{-1}(U_{i-1}) - \frac{1}{2} \alpha^2 \Delta_i + \gamma R_{i-1} - \kappa \Delta_i \right) + (1 - e^{-\kappa \Delta_i})$$

with $\Delta_i = t_i - t_{i-1}$ with deterministic starting value σ_1 . Here, deviating from Behvand's notation α denotes volatility of volatility and v is mean reversion level. Thus

$$\hat{\sigma}_{t_{i-1}}(t_{i-1}, t_i, K) = \hat{\sigma}(U_1, \dots, U_{i-1}, K) = f_i(U_1, \dots, U_{i-1}) g_i(K)$$

The multiplicative skew function $g_i(K)$ is derived from spot implied volatilities as described in Section 7.4.1. Thus the model is a complete model with multiplicative volatility structure, as the example discussed in Section 7.4.4. Autocorrelation is derived recursively instead of through a mean reverting factor process.

¹ This should lead to dependency between the distribution of R_i and $v_{t_{i-1}}(t_i)$ and thus between R_i and $S_{t_{i-1}}$ in contrast to the claimed decoupling of short forward skew and spot/vol correlation.

8 Correlation

There exists a liquid options market for stock indices and, since the components are often stocks of important companies, there are also options on those stocks. In many financial institutions the responsibility for market data (forward, volatility) is separated between stock and index trading desks. Index exposure will be hedged with indices rather than component stocks. If only Vanilla options are traded this is a valid approach and there is no correlation exposure. Forward discrepancies between index and sum of stocks will typically be small and can be attributed to market inefficiencies.

If multi-stock products, such as basket, rainbow and outperformance products, are considered then one has to make assumptions on stock correlation. In order to avoid internal arbitrage, index options modelled as a basket of component stocks have to match index option prices.

Similarly, there is a liquid market in correlation products on the most important indices, such as the world basket, consisting of S&P 500, Stoxx 50 and Nikkei 225, representing US, European and Asian markets. In addition to equally weighted baskets there is a market for rainbow options including best- and worst-of options, on those three indices. It will not be sufficient to assume a single correlation matrix to match observed prices for those products.

These problems can be addressed at various levels. For Vanilla options on stocks and indices the concept of implied volatility can be enhanced through the introduction of implied correlation. The local volatility model can be extended so that the smile of a given index is matched. Stochastic volatility models can be extended such that correlation is also stochastic. There are also natural extensions of copula and cliquet models to the multi-asset situation.

Let a basket $I_t = \sum_{k=1}^n w_k S_t^k$ be given, consisting of underlyings S^k with constant volatilities σ^k , and a fixed correlation matrix ρ^{kl} . Approximating the basket at time t with a log-normal process with the same first and second moment leads to the

basket forward F^I and volatility σ^I satisfying

$$F^I(t) = \sum_{k=1}^n w^k F^k(t)$$

$$F^I(t)^2 e^{\sigma^I(t)^2 t} = \sum_{k,l=1}^n w^k w^l F^k(t) F^l(t) e^{\rho^{kl} \sigma^k \sigma^l t} \quad (8.1)$$

A first-order approximation yields the approximate formula

$$F^I(t)^2 \sigma^I(t)^2 \approx \sum_{k,l=1}^n w^k w^l F^k(t) F^l(t) \rho^{kl} \sigma^k \sigma^l \quad (8.2)$$

Alternative approximations for the sum of log-normal assets include higher moments, see Ju (2002) and references therein, geometric conditioning as suggested by Curran (1994) and numerical methods. These methods work better in the context of Asian options, since the expectations of the components can be very different in the basket case.

When pricing single-stock options the volatilities have to be chosen as implied volatility $\hat{\sigma}^k(K, t)$ for the relevant strike and maturity of the product. Assuming a smile model for the single stocks with constant correlation matrix it turns out that a good approximation for the resulting basket implied volatility is obtained by choosing an effective strike level for single-stock volatilities. This strike level can be derived from the moneyness k of the basket option with strike K , With

$$K = k F^I(t)$$

one defines the basket volatility smile $\hat{\sigma}^I$ from equation (8.1) through line break:

$$\begin{aligned} & \hat{\sigma}^I(K, t)^2 t \\ &= \hat{\sigma}^I(k F^I(t), t)^2 t \\ &= \log \left(\sum_{k,l=1}^n w^k w^l F^k(t) F^l(t) \exp \left(\rho^{kl} \hat{\sigma}^k(k F^k(t), t) \hat{\sigma}^l(k F^l(t), t) \right) \right) - 2 \log F^I(t) \\ &\approx \frac{1}{F^I(t)^2} \sum_{k,l=1}^n w^k w^l F^k(t) F^l(t) \rho^{kl} \hat{\sigma}^k(k F^k(t), t) \hat{\sigma}^l(k F^l(t), t) \end{aligned}$$

A more rigorous treatment can be found in Jäckel (2012) and Avellaneda et al. (2002).

8.1 Implied correlation

The positive relationship between Black-Scholes volatility and option price allows one to derive implied volatility from market prices of options, see Section 2.5.4. Similarly, given a single-stock smile-consistent pricing formula for basket options with a fixed correlation matrix, one can derive an implied correlation since basket volatility increases with increasing correlation. Examples for single-stock smile-consistent pricing formulae include Black-Scholes prices with basket volatility, as in equations (8.1) or (8.2), as well as prices obtained with multi-asset local volatility with fixed correlation for the driving Brownian motions or multi-asset Gauss copula models. Implied correlation computed with $\rho^{kl} = \rho$ for $k \neq l$ often exhibits negative skew.

In order to imply a correlation matrix with $n(n-1)/2$ different entries from a single basket price one can use convex combinations of correlation matrices. One approach combines a reference correlation matrix ρ^{kl} with either the independence matrix ($\rho^{kl} = 0$ for $k \neq l$) or the dependence matrix ($\rho^{kl} = 1$ for $k \neq l$). If basket prices are known, as in the case of stock indices or world index, one can derive a weight $w \in [0, 1]$ depending on basket strike and maturity. The correlation surface is then given as

$$\rho_t^{kl}(K) = w_t(K)^+ + (1 - |w_t(K)|) \rho^{kl} \quad (8.3)$$

with $w_t(K) \in [-1, 1]$. This approach is applicable to reference matrices with predominantly positive entries, as is the case for stocks and indices. A matrix with constant off diagonal entries ρ is positive definite for

$$\rho \in \left[-\frac{1}{n-1}, 1 \right]$$

where n is the dimension of the matrix. This allows one to slightly extend the range of matrices, see Reghai (2010).

A disadvantage of formula (8.3) is that the correlation smile is not smooth in $w = 0$. An alternative approach is based on eigenvalue decomposition. Let

$$\rho = U^T D U$$

where U is a matrix of eigenvectors and D is a diagonal matrix with positive eigenvalues, such that $d_{11} \geq d_{22} \geq \dots \geq d_{nn} > 0$. One can define covariance matrices c_λ through

$$c_\lambda = U^T D^\lambda U$$

where D^λ is the diagonal matrix with entries

$$D_{kk}^\lambda = d_{kk}^\lambda$$

Denoting the corresponding correlation matrix by ρ_λ one has

$$\begin{aligned}\rho_1^{kl} &= \rho^{kl} \\ \rho_0^{kl} &= 0 \\ \rho_\infty^{kl} &= 1\end{aligned}$$

for $k \neq l$, assuming $d_{11} > d_{22}$ for the last statement. This approach assumes correlation smile can be described by changing eigenvalues, while eigenvectors remain constant. While the resulting correlation curves are smooth in λ they are not necessarily monotonic. Figure 8.1 exhibits the correlation smile obtained with matrix

$$\rho = \begin{pmatrix} 1 & 0.1 & 0.5 \\ 0.1 & 1 & 0.9 \\ 0.5 & 0.9 & 1 \end{pmatrix}$$

which has approximate eigenvalues 2.074, 0.915 and 0.011 and orthogonal matrix

$$U = \begin{pmatrix} 0.379 & -0.875 & -0.301 \\ 0.615 & 0.481 & -0.625 \\ 0.692 & 0.052 & 0.720 \end{pmatrix}$$

The x-axis is rescaled somewhat arbitrarily to the weight range $[-1, 1]$ via

$$\lambda(w) = \exp\left(3\Phi^{-1}\left(\frac{w+1}{2}\right)\right)$$

Note that eigenvalue decomposition is neither uniquely defined nor continuously dependent on the correlation matrix, which may lead to instabilities in a changing correlation environment.

8.2 Correlation term structure

As a first step to correlation modelling note that, in a Gaussian framework, it is possible to incorporate a term structure of correlation. An implied correlation term structure derived as described above is related to covariance of log returns via

$$\text{Cov}\left[\log S_t^k, \log S_t^l\right] = \hat{\rho}^{kl}(t) \hat{\sigma}^k(t) \hat{\sigma}^l(t) t$$

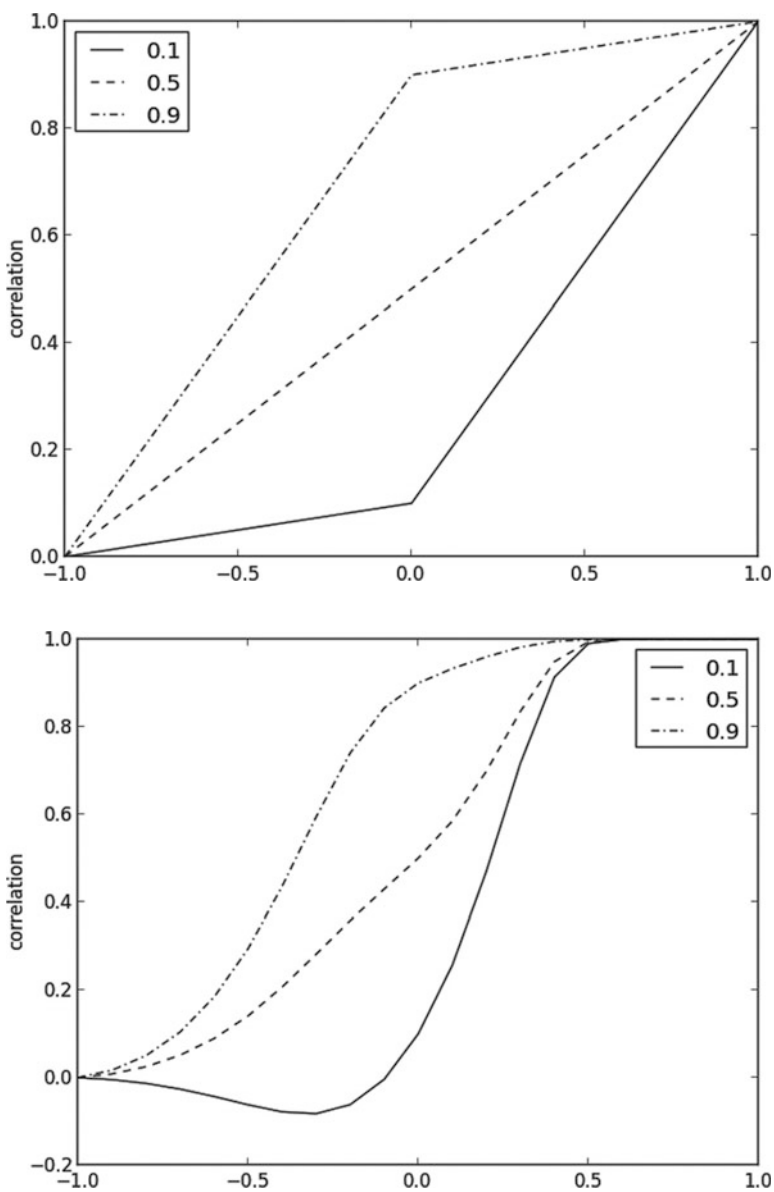


Figure 8.1 Correlation smile: linear versus smooth

Covariance is additive in the sense that

$$\begin{aligned} & \text{Cov} \left[\log \left(\frac{S_T^k}{S_t^k} \right), \log \left(\frac{S_T^l}{S_t^l} \right) \right] \\ &= \text{Cov} \left[\log S_T^k, \log S_T^l \right] - \text{Cov} \left[\log S_t^k, \log S_t^l \right] \end{aligned}$$

Defining forward correlation as

$$\text{Cov} \left[\log \left(\frac{S_T^k}{S_t^k} \right), \log \left(\frac{S_T^l}{S_t^l} \right) \right] = \hat{\rho}^{kl}(t, T) \hat{\sigma}^k(t, T) \hat{\sigma}^l(t, T) (T - t)$$

one obtains the formula

$$\hat{\rho}^{kl}(t, T) = \frac{\hat{\rho}^{kl}(T) \hat{\sigma}^k(T) \hat{\sigma}^l(T) T - \hat{\rho}^{kl}(t) \hat{\sigma}^k(t) \hat{\sigma}^l(t) t}{\hat{\sigma}^k(t, T) \hat{\sigma}^l(t, T) (T - t)}$$

for forward correlation. This is the natural extension of the concept of forward volatility in a log-normal context to higher dimensions. Note that forward correlation depends on volatility. The approximate formula

$$\hat{\rho}^{kl}(t, T) \approx \frac{\hat{\rho}^{kl}(T) T - \hat{\rho}^{kl}(t) t}{T - t}$$

only holds if time dependence of volatility can be neglected. If a Black-Scholes model is used with volatility interpolation dependent on the product then correlation term structure can be combined with the convex combination approach for implied correlation:

1. Define strike levels K_t^k for component stocks S^k and derive implied volatility term structures $\hat{\sigma}^k(t_i)$ at relevant product dates t_i .
2. Derive a term structure of correlation from basket options struck at a reference strike K_t , such as the basket forward.
3. Define a surface for the weights $w_t(K)$ in equation (8.3) from basket options.
4. For the basket process at strike level K , derive the term structure of correlation for dates t_i .
5. Deriving forward correlation between dates t_{i-1} and t_i yields a log-normal process with constant correlation, drift and volatility between benchmark dates.

If a Gauss copula model driven by stock processes X^k with $dX_t^k = \sigma_t^k dW^k$, as in Section 9.9, is used, then the above approach can also be applied. Note that in the first step no component strike levels are required and implied correlations for steps 2 and 3 are derived using a Gauss copula model for basket options. The only remaining product-dependent parameter is the basket strike level K determining correlation term structure.

8.3 Decorrelation

Volatility term structure leads to decorrelation as exhibited in the example below. Consider two assets S^1, S^2 at dates $t = 1$ and $T = 2$. The first asset has flat volatility $\sigma_1^1 = \sigma_2^1 = 25\%$ and the second asset has volatility $\sigma_1^2 = 40\%, \sigma_2^2 = 30\%$ and hence forward volatility

$$\sigma_{12}^2 = \sqrt{(\sigma_2^2)^2 2 - (\sigma_1^2)^2} = 14.14\%$$

The correlation is set to a flat value $\rho = \rho_1 = \rho_2$. It turns out that the forward correlation is

$$\rho_{12} = \frac{\rho \sigma_2^1 \sigma_2^2 2 - \rho \sigma_1^1 \sigma_1^2}{\sigma_{12}^1 \sigma_{12}^2} = \rho \frac{\sigma_2^2 2 - \sigma_1^2}{\sigma_{12}^2} = \rho \sqrt{2}$$

Thus the highest possible correlation value for $|\rho|$ attainable is $\frac{1}{\sqrt{2}} = 70.71\%$ which requires maximal correlation values

$$\rho_1 = \rho_{12} = 1$$

This shows that term structure of volatility leads to decorrelation.

A similar effect can be observed in local and stochastic volatility models. Thus, to achieve consistent pricing of calibration instruments, such as index options, the correlation of driving Brownian motions has to be adjusted if different models are used for correlation calibration and pricing.

In the case of multi-asset models with stochastic volatility one has to define three correlation matrices, $\rho_{ij}^S, \rho_{ij}^{\sigma}$ and $\rho_{ij}^{S\sigma}$, referring respectively to the correlation of stock returns, volatility increments respectively the combination of stock return of asset i and volatility increment of asset j . Calibration requirements may be given in terms of a stock return correlation matrix ρ_{ij}^S , as well as stock-volatility correlation vector $\rho_{ii}^{S\sigma}$ for the same stock. As pointed out by Jäckel and Kahl (2009), the three matrices can be filled taking intuition from two independent Gaussian vectors X and Y , where X has correlation ρ^S and

$$Z_i = \rho_{ii}^{S\sigma} X_i + \sqrt{1 - (\rho_{ii}^{S\sigma})^2} Y_i$$

With Z driving volatility one obtains

$$\begin{aligned} \rho_{ij}^{S\sigma} &= \rho_{ij}^S \rho_{jj}^{S\sigma} \\ \rho_{ij}^{\sigma} &= \rho_{ii}^{S\sigma} \rho_{jj}^{S\sigma} \rho_{ij}^S + \sqrt{1 - (\rho_{ii}^{S\sigma})^2} \sqrt{1 - (\rho_{jj}^{S\sigma})^2} \rho_{ij}^Y \end{aligned}$$

Jäckel and Kahl (2009) suggest specifying correlation matrix ρ^Y as a convex combination of independence and dependence matrices, namely

$$\rho_{ij}^Y = \beta + (1 - \beta)\delta_{ij} \quad (\beta \in [-1/(n-1), 1])$$

Another natural choice, based on the intuition that correlation dispersion of stock returns and volatility increments should be similar, is

$$\rho_{ij}^Y = \beta \mathbf{1}_{\{\beta > 0\}} + \delta_{ij} \mathbf{1}_{\{\beta < 0\}} + (1 - |\beta|)\rho_{ij}^S \quad (\beta \in [-1, 1])$$

compare (8.3). In both cases choice of β achieves the desired basket volatilities.

8.4 Langnau's local correlation

A natural extension of the local volatility model to a multi-asset situation leads to local correlation. In a first attempt, Ilinski (2001) derived correlation from pair basket prices with different weights. Those baskets are typically not traded and a price is therefore not available. To remedy this problem Austing (2011, 2012) proposed the specification of those basket prices in order to derive local correlation.

An alternative approach due to Langnau (2009) implies local correlation from basket local volatility through a local weight function w . Assume an n -asset local volatility setting, namely

$$\frac{dS_t^k}{S_t^k} = \mu_t^k dt + \sigma_t^k(S_t^k) dW_t^k \quad (i = 1, \dots, n) \quad (8.4)$$

A weighted basket of assets is defined through

$$I = \sum_{k=1}^n w_k S^k \quad (8.5)$$

Assume furthermore that the processes (S, I) are adapted to the filtration of W . With respect to I , the local volatility process (if it exists) should be of the form

$$\frac{dI_t}{I_t} = \mu_t^I dt + \sigma_t^I(I_t) dW_t^I \quad (8.6)$$

with some adapted Brownian motion W^I . These three equations impose restrictions on drift μ^I and volatility σ^I .

With regards to the drift, the dividend yields (aka repo rates) q^k of the S^k have to be equal. If cash dividends are modelled, then they have to be consistent:

$$\begin{aligned}\mu_t^k S_t^k dt &= (r_t - d_t) S_t^k dt - \sum_i d_i^k \delta_{t_i^k}(dt) \\ \mu_t^l I_t dt &= (r_t - d_t) I_t dt - \sum_{k,i} w_k d_i^k \delta_{t_i^k}(dt)\end{aligned}$$

where d_i^k is the cash dividend of stock k payable at time t_i . The quadratic variation of I has to satisfy

$$\begin{aligned}d\langle I \rangle_t &= \sigma_t(I_t)^2 I_t^2 dt \\ &\stackrel{!}{=} \sum_{k,l} w_k w_l \sigma_t^k(S_t^k) \sigma_t^l(S_t^l) S_t^k S_t^l \rho_t^{k,l} dt\end{aligned}$$

where ρ is an adapted process of correlation matrices. There may be no solution (in case of arbitrage) or many solutions. An approach leading to a solution if there is no arbitrage and all correlations are positive is discussed below.

Fix a deterministic process of correlations ρ . Define instantaneous variance v_I from (8.6) and stock variances from (8.4) as a function of the instantaneous correlation matrix of dW_t . If the W^k are perfectly correlated, this leads to v_I , in the independent case one has v_0 , and with ρ one obtains v_ρ . In other words, omitting subscripts t and differentials dt ,

$$\begin{aligned}v_I &= \sigma(I)^2 I^2 \\ v_\rho &= \sum_{k,l} w_k w_l \sigma^k(S^k) \sigma^l(S^l) S^k S^l \rho^{k,l} \\ v_1 &= \sum_{i,j} w_k w_l \sigma^k(S^k) \sigma^l(S^l) S^k S^l \\ v_0 &= \sum_i w_k^2 (\sigma^k(S^k) S^k)^2\end{aligned}$$

Define a set

$$A = \{v_I > v_\rho\}$$

The solution is given as a convex combination of ρ and two extreme matrices, namely perfect correlation and independence with the mixture depending on S as

follows. On A set

$$\begin{aligned}\rho(S) &= p\rho + (1-p) \\ p &= \frac{v_I - v_1}{v_\rho - v_1}\end{aligned}$$

On the complement A^c set (for all off diagonal elements)

$$\begin{aligned}\rho(S) &= p\rho \\ p &= \frac{v_I - v_0}{v_\rho - v_0}\end{aligned}$$

This defines a local correlation process $\rho(S)$ if p takes values in $[0, 1]$. In that case it also defines a solution. On A one has

$$v_{\rho(S)} = pv_\rho + (1-p)v_1 = v_I$$

Similarly for A^c :

$$v_{\rho(S)} = pv_\rho + (1-p)v_0 = v_I$$

This approach is practical for products on single stocks which are components of one index. It has to be extended in order to be applicable to different, possibly overlapping, baskets. Reghai (2010) extends the approach by allowing for different local correlation processes depending on the nature of the product. Note that the assumption that correlation between two stocks depends on all stocks within an index is somewhat unsatisfactory.

8.5 Stochastic correlation

Correlation can also be modelled as a stochastic process in its own right. Figure 8.2 displays the correlation of Bayer and Deutsche Telekom returns based on one year windows (dashed line, inverted scale) as well as DAX closing (solid line). Correlation typically increases with decreasing index level which is in line with observed negative (implied) correlation skew, discussed in Section 8.1 above.

Sections 8.5.1 and 8.5.2 follow Brockhaus and Giese (2010).

8.5.1 Two stocks

In the case of two stocks it is natural to require the single correlation process ρ to be mean reverting and bounded. A candidate for such a process is the Jacobi process

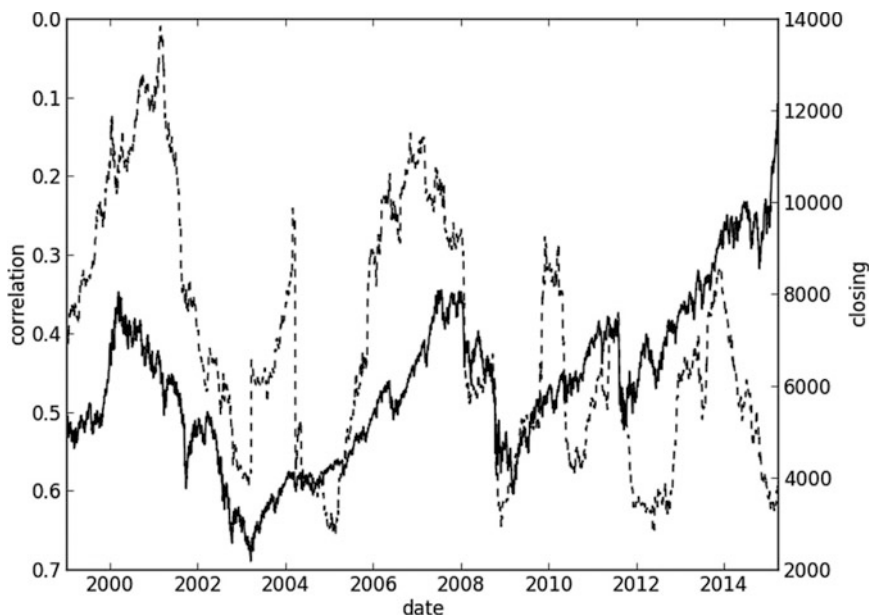


Figure 8.2 Stock return correlation versus index level

defined as

$$dX_t = \kappa(m - X_t)dt + \eta\sqrt{(X_t - a)(b - X_t)}dW_t$$

with $a < X_0 < b$. The boundaries are unattainable if, and only if,

$$\kappa \geq \frac{\eta^2(b-a)}{2(m-a)} \text{ and } \kappa \geq \frac{\eta^2(b-a)}{2(b-m)}$$

With $a = -1$ and $b = 1$ one obtains the stochastic correlation process ρ

$$d\rho_t = \kappa(\rho_\infty - \rho_t)dt + \eta\sqrt{1 - \rho_t^2}dW_t$$

with parameter constraints

$$\kappa \geq \max\left(\frac{\eta^2}{\rho_\infty + 1}, \frac{\eta^2}{\rho_\infty - 1}\right),$$

and properties

$$\lim_{t \rightarrow \infty} \mathbb{E}[\rho_t] = \rho_\infty, \quad \lim_{t \rightarrow \infty} \text{Var}[\rho_t] = \frac{1 - \rho_\infty^2}{\frac{2\kappa}{\eta^2} + 1}$$

Mean reversion κ and volatility of correlation η can be calibrated to time series data using maximum likelihood estimators or generalized method-of-moment estimators. They can also be obtained by calibration to index options.

Following van Emmerich (2006) the two-asset model is given as

$$\begin{aligned}\frac{dS_t^k}{S_t^k} &= (r_t - d_t^k)dt + \sigma_t^k dW_t^k, & k \in \{1, 2\} \\ d\rho_t &= \kappa(\rho_\infty - \rho_t)dt + \eta\sqrt{1 - (\rho_t - a)(b - \rho_t)}dW_t^3\end{aligned}$$

with $-1 \leq a < \rho_0 < b \leq 1$. The three Brownian motions can be correlated with correlation matrix

$$\begin{pmatrix} 1 & \rho_t & \rho_1 \\ \rho_t & 1 & \rho_2 \\ \rho_1 & \rho_2 & 1 \end{pmatrix}$$

where ρ_1, ρ_2 are deterministic. Note that the correlation matrix must be positive semi-definite which is equivalent to

$$\rho_1\rho_2 - \sqrt{(1 - \rho_1^2)(1 - \rho_2^2)} \leq \rho_t \leq \rho_1\rho_2 + \sqrt{(1 - \rho_1^2)(1 - \rho_2^2)}$$

Therefore one obtains a well-defined two-asset model with stochastic correlation via

$$\begin{aligned}a &= \rho_1\rho_2 - \sqrt{(1 - \rho_1^2)(1 - \rho_2^2)} \\ b &= \rho_1\rho_2 + \sqrt{(1 - \rho_1^2)(1 - \rho_2^2)}\end{aligned}$$

8.5.2 Boortz's common factor model for many stocks

Generalization of the above approach to n dimensions with $n(n-1)/2$ stochastic stock correlations is not straightforward. The resulting symmetric matrix with $n(n-1)/2 + n = (n+1)n/2$ dimensions needs to be transformed into a positive semi-definite matrix for each path and time t , resulting in a considerable loss of intuition and high computational cost. Boortz (2008) introduces stochastic correlation by assuming

$$dW_t^{S^k} = \sqrt{1 - (\beta_t^k)^2} dW_t^k + \beta_t^k dW_t^m \quad (8.7)$$

where the $W^k, k = 1, \dots, n$ and W^m are independent Brownian motions and $\beta_t^k, k = 1, \dots, n$ are stochastic processes on $[-1, 1]$. With this setup the instantaneous pairwise correlation between the returns of S^k and S^l at time t is

$$\rho_t^{kl} = \beta_t^k \beta_t^l$$

and the matrix (ρ_t^{kl}) is a correlation matrix. Equation (8.7) implies that the k -th stock process S^k is driven by an idiosyncratic factor W^k and the common (or market) factor W^m , which also impacts all other stock processes. The stochastic factors $\beta_t^k, k = 1, \dots, n$ can be modelled as independent Jacobi processes

$$d\beta_t^k = \kappa^k(\beta_\infty^k - \beta_t^k)dt + \eta^k \sqrt{1 - (\beta_t^k)^2} d\hat{W}_t^k, \quad \beta_0^k \in (-1, 1)$$

where $\hat{W}^1, \dots, \hat{W}^n$ are Brownian motions. Furthermore one sets

$$d\langle \hat{W}^k, W^m \rangle_t = \eta^k dt$$

where $\eta^k \in (-1, 1)$ are constants for $k = 1, \dots, n$. Consequently the correlations are also correlated with the stock price processes via the common factor W^m . This allows one to model observed negative correlation between stock returns and correlations as exhibited in Figure 8.2.

For more than two assets a given correlation matrix cannot be represented as one- (or common-) factor correlation matrix. An efficient algorithm that finds an approximating one-factor matrix has been given by Andersen et al. (2003).

8.5.3 A finite state correlation model for many stocks

The following variant of Langnau (2009) and Boortz (2008) addresses some of the limitations of their approaches. Define two correlation states ρ^+, ρ^- , as in Langnau (2009), such that

$$\rho = \frac{1}{2}(\rho^+ + \rho^-)$$

Also assume a common factor M , as in Boortz (2008). Choose M to be an Ornstein-Uhlenbeck process, namely

$$dM_t = -\kappa M_t dt + dW_t^M$$

M represents the state of the economy. Hence, in order to have negative correlation skew, as is common in equities markets, the aim is to have a higher probability of low correlation state ρ^- with increasing M . In a down movement of the market represented by lower M the more likely correlation state is ρ^+ . For the correct

expected correlation the expected probability for both states has to be $1/2$. This can be achieved by defining probabilities as mapped process M , namely

$$p_t^+ = 1 - F^{M_t}(M_t), \quad p_t^- = F^{M_t}(M_t)$$

In the case of $\rho_{kl}^\pm = \beta_k^\pm \beta_l^\pm$, as in Boortz (2008), one can set

$$dW_t^{S^k} = \beta_{k,t} dW_t^M + \sqrt{1 - \beta_{k,t}^2} dW_t^k$$

where, for all t , one draws vectors $\beta_t \in \{\beta^+, \beta^-\}$ with probabilities p_t^+, p_t^- , respectively. In the general case one can allow for several factors indexed by m such that

$$\rho_{kl}^\pm \approx \sum_m \beta_{k,m}^\pm \beta_{l,m}^\pm,$$

(see Andersen et al. (2003)) and simulate

$$dW_t^{S^k} = \sum_m \beta_{k,m,t} dW_t^{M,m} + \sqrt{1 - \beta_{k,m,t}^2} dW_t^k$$

with $W^{M,1} = W^M$. Note that spot correlation skew can be increased by moving ρ^+, ρ^- further apart. Parameter κ allows the specification of a forward correlation skew. Correlation convexity can be modelled by introducing a third correlation state. Term structure of correlation level and skew can also be incorporated.

9 [Copulas

Copulas have many applications within financial modelling, including the representation of the joint distribution of

- several default times, as discussed in Section 12.3,
- various assets at a given future time horizon,
- one asset at several future time horizons (time copula), see Section 9.9.

This section provides a basic mathematical background as well as a discussion of relevant examples. A good survey has been provided by Frees and Valdez (1997). A collection of articles on copulas in the context of finance can be found in Rank (2006). Textbooks include Joe (1997) and Nelsen (1998).

9.1 Definition

Let a vector of random variables $X = (X^i, i = 1, 2, \dots, N)$ be given, with (invertible) distribution functions

$$F^i(x) = \mathbb{P} \left[X^i \leq x \right].$$

Denoting the N -variate distribution function by F , i.e.

$$F(x_1, x_2, \dots, x_N) = \mathbb{P} \left[X^1 \leq x_1, X^2 \leq x_2, \dots, X^N \leq x_N \right]$$

the copula C^F of (the distribution of) X is defined as

$$C^F(x_1, \dots, x_N) = F \left((F^1)^{-1}(x_1), \dots, (F^N)^{-1}(x_N) \right)$$

In other words, C^F is the distribution of $(F^i(X^i), i = 1, 2, \dots, N)$. In order to gain some intuition it may be instructive to compute the copulae of the following pairs (X^1, X^2) :

1. $X^1 = X^2$
2. $X^1 = -X^2$
3. $X^2 = aX^1 + b$
4. $X^1 = f^1(Y^1)$ and $X^2 = f^2(Y^2)$, where (Y^1, Y^2) have copula C and f^1, f^2 are two continuous, strictly increasing functions
5. (X^1, X^2) bivariate Gaussian with correlation ρ

9.2 Dependence measures

9.2.1 Linear correlation

Linear correlation, or Pearson correlation, or simply correlation, is defined as:

$$\rho(X^1, X^2) = \frac{\text{IE}[(X^1 - \bar{X}^1)(X^2 - \bar{X}^2)]}{\sqrt{\text{IE}[(X^1 - \bar{X}^1)^2] \text{IE}[(X^2 - \bar{X}^2)^2]}}$$

where $\bar{X} = \text{IE}[X]$. The correlation of (X^1, X^2) is invariant under positive linear transformations, namely

$$\rho(X^1, X^2) = \rho(\alpha^1 X^1 + \beta^1, \alpha^2 X^2 + \beta^2)$$

if $\alpha^1 > 0, \alpha^2 > 0$. Generally, if f^1 and f^2 are increasing and continuous then

$$\rho(X^1, X^2) \neq \rho(f^1(X^1), f^2(X^2))$$

9.2.2 Kendall's Tau

Kendall's Tau is defined as:

$$\rho_K(X^1, X^2) = \text{IP}[(X^1 - \tilde{X}^1)(X^2 - \tilde{X}^2) > 0] - \text{IP}[(X^1 - \tilde{X}^2)(X^2 - \tilde{X}^1) < 0]$$

Here $(\tilde{X}^1, \tilde{X}^2)$ is an independent copy of (X^1, X^2) . The intuition is that first a reference point is sampled from the joint distribution. Then an independent second sample is counted as positive if it is in the right upper or left lower quadrant, or negative otherwise.

If f^1, f^2 are increasing and continuous, then $\rho_K(X^1, X^2) = \rho_K(f^1(X^1), f^2(X^2))$. Hence Kendall's Tau is a function of the copula. Indeed

$$\rho_K(X^1, X^2) = 4 \int \int C(u, v) dC(u, v) - 1$$

Let $(X_i^1, X_i^2), i = 1, 2, \dots, N$ be a sequence of independent bivariate random variables with distribution (X^1, X^2) . Kendall's Tau can be estimated as

$$\hat{\rho}_K(X^1, X^2) = \binom{N}{2}^{-1} \sum_{i < j} \text{sign}((X_i^1 - X_j^1)(X_i^2 - X_j^2))$$

The relationship between Kendall's Tau ρ_K and linear correlation for a Gaussian pair (X^1, X^2) is

$$\rho(X^1, X^2) = \sin\left(\frac{\pi}{2} \rho_K(X^1, X^2)\right)$$

9.2.3 Spearman's Rho

Spearman's Rho ρ_S is defined as

$$\rho_S(X^1, X^2) = \rho(F^1(X^1), F^1(X^2)) = 12 \mathbb{E} \left[\left(F^1(X^1) - \frac{1}{2} \right) \left(F^2(X^2) - \frac{1}{2} \right) \right]$$

Spearman's Rho is a function of the copula:

$$\rho_S(X^1, X^2) = 12 \int \int (C(u, v) - uv) dudv$$

9.2.4 Tail dependence

One motivation for the introduction of copulas is the observation that, for a given dependence level (in terms of one of the above dependence measures), default times τ^k modelled with multivariate Gaussian distribution do not assign enough probability to the event of several defaults occurring together, i.e.

$$\mathbb{IP}[\tau^1 < t, \tau^2 < t]$$

In order to measure the mass of tail events two coefficients λ_u and λ_l for upper and lower tail dependence have been introduced, with

$$\lambda_u = \lim_{q \nearrow 1} \mathbb{IP}[F_2(X_2) > q | F_1(X_1) > q]$$

$$\lambda_l = \lim_{q \searrow 0} \mathbb{IP}[F_2(X_2) < q | F_1(X_1) < q]$$

9.3 Archimedean copulas

Let $X = (X^1, X^2, \dots, X^N)$ be conditionally independent, given Y with $Y \geq 0$, and let

$$\mathbb{P}\left[X^i \leq x^i \mid Y\right] = \left(h^i(x^i)\right)^Y$$

Then the distribution of X is

$$F(x^1, \dots, x^N) = \mathcal{L}^Y \left(\sum_{i=1}^N g^i(x^i) \right)$$

where $\mathcal{L}^Y$ is the Laplace transform of Y and $h^i = \exp -g^i$.

The vector X can be simulated as

$$X^i = \left(g^i\right)^{-1} \left(-\frac{1}{Y} \log U^i \right)$$

where $Y = F_Y^{-1}(U^0)$ and $U^0, U^1, \dots, U^N$ are independent, uniform random variates. This can be seen by noting the conditional independence of the X^i , given Y and

$$\mathbb{P}\left[X^i \leq x^i \mid Y\right] = \exp\left(-g^i(x^i)Y\right) = \left(h^i(x^i)\right)^Y$$

Therefore the unconditional distribution is

$$\mathbb{P}\left[X^1 \leq x^1, \dots, X^N \leq x^N\right] = \mathbb{E}\left[\exp\left(-Y \sum_{i=1}^N g^i(x^i)\right)\right]$$

In order to have the X^i uniformly distributed one needs

$$g^1 = \dots = g^N = \left(\mathcal{L}^Y\right)^{-1}$$

which is the special case of Archimedean copulas. Important examples include:

- Independence: $g(x) = -\log(x)$, with bivariate distribution

$$C(u, v) = uv$$

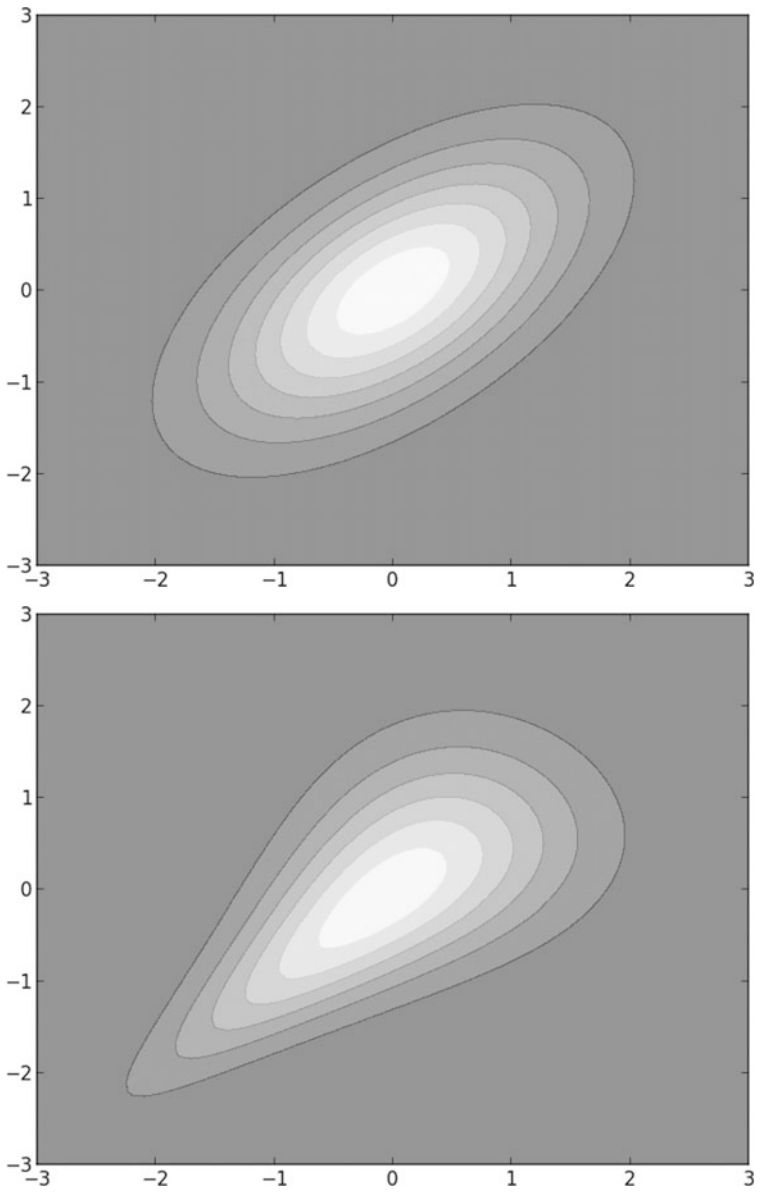
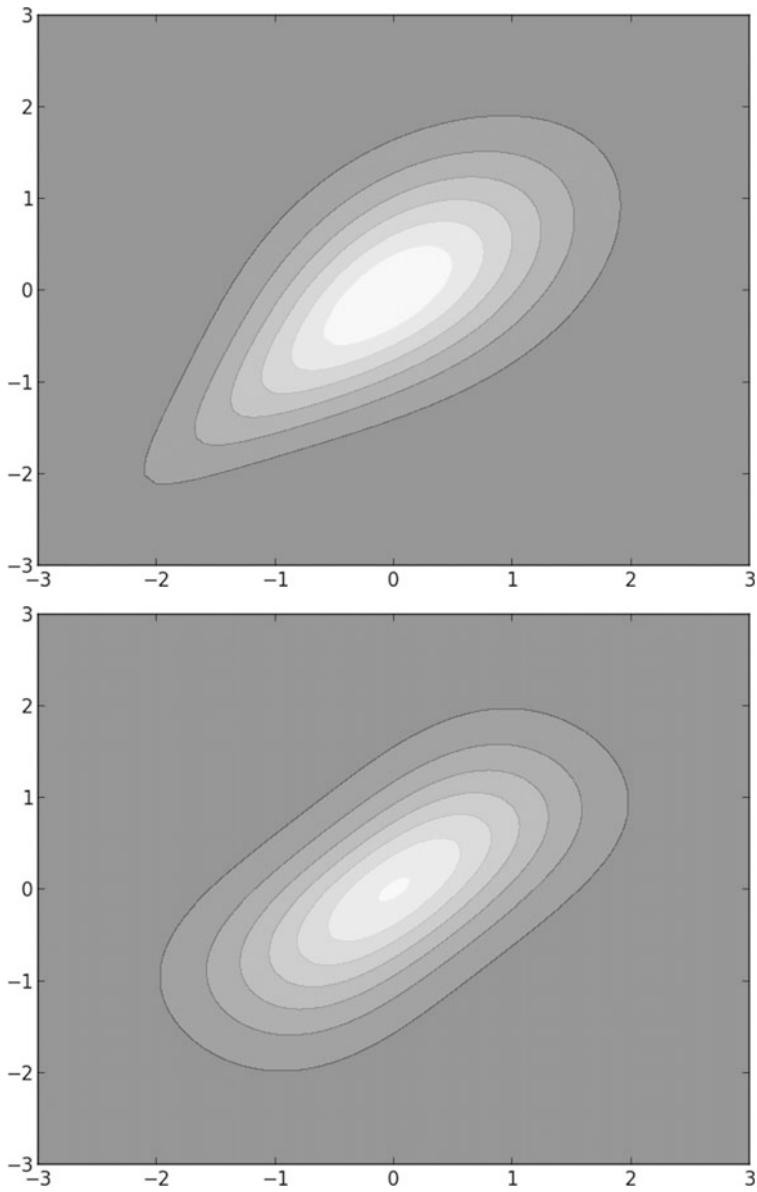


Figure 9.1 Copula densities: Gauss, Clayton, Gumbel, Frank

**Figure 9.1** (*Continued*)

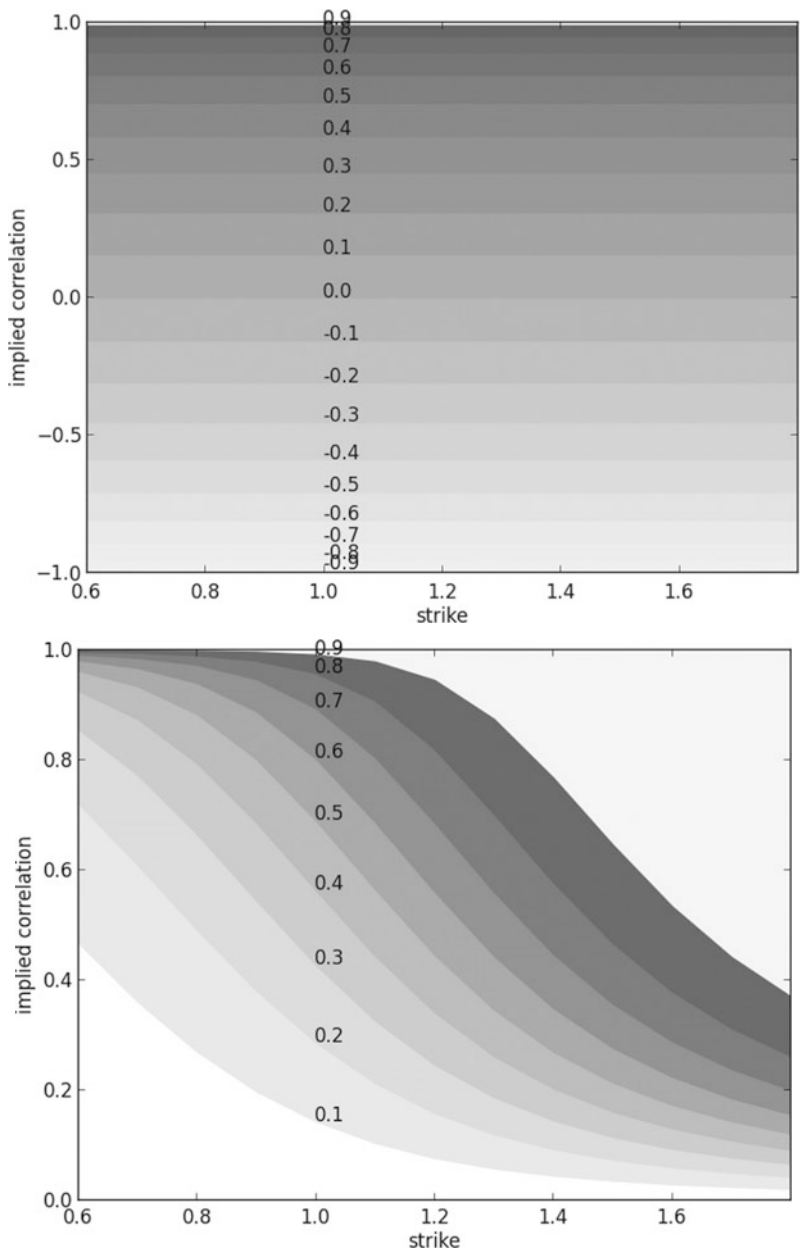


Figure 9.2 Implied correlation: Gauss, Clayton, Gumbel, Frank

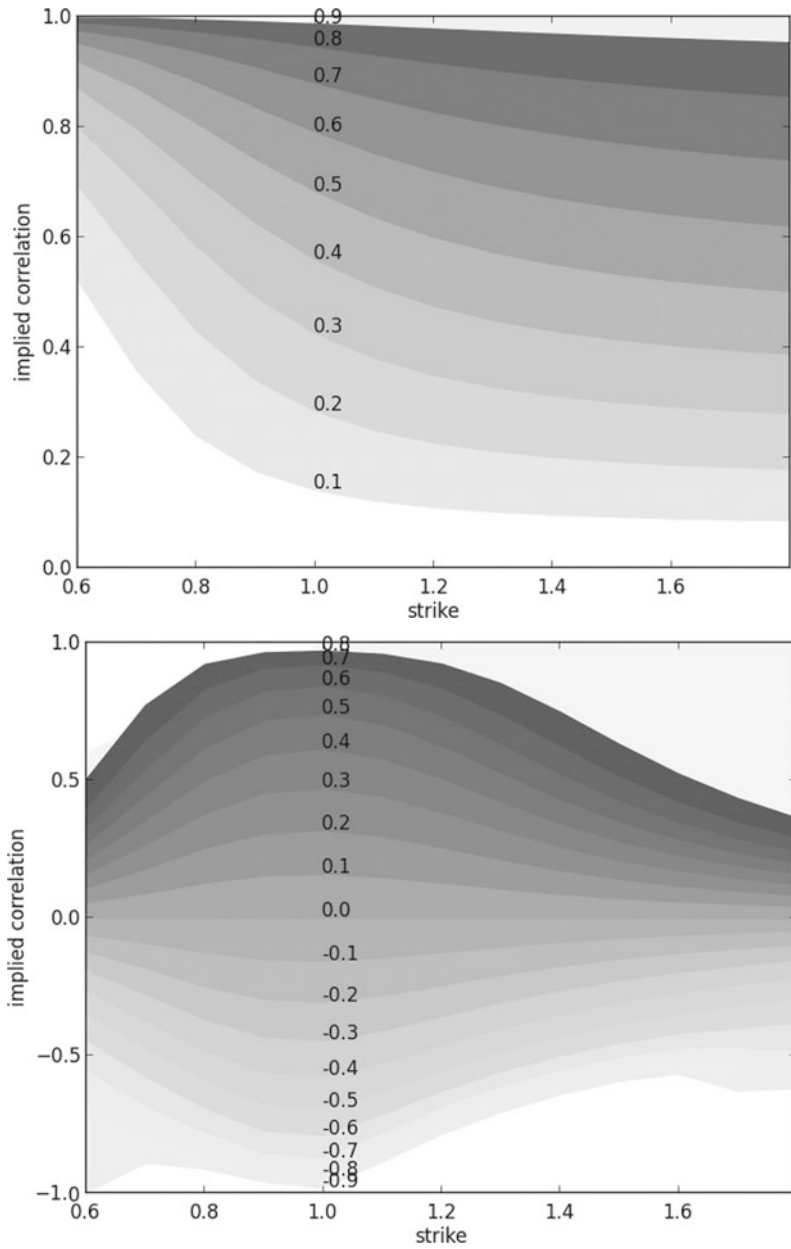


Figure 9.2 (Continued)

- Clayton: $g(x) = x^{-\theta} - 1$, $\theta > 1$, with Kendall's Tau $\rho_K = \frac{\theta}{\theta+2}$, bivariate distribution

$$C(u, v) = (u^{-\theta} + v^{-\theta})^{-1/\theta}$$

and lower tail dependence $\lambda_l = 2^{-1/\theta}$.

- Gumbel: $g(x) = (-\log(x))^\theta$, $\theta \geq 1$, with Kendall's Tau $\rho_K = 1 - \theta^{-1}$, bivariate distribution

$$C(u, v) = \exp\left(-\left[(-\log u)^\theta + (-\log v)^\theta\right]^{1/\theta}\right)$$

and lower tail dependence $\lambda_l = 2 - 2^{1/\theta}$.

- Frank: $g(x) = \log \frac{e^{\theta x} - 1}{\theta}$, $\theta \in \mathbb{R}$, with Kendall's Tau

$$\rho_K = 1 - \frac{4}{\theta} \left(D_1(|\theta|) + \frac{\theta^+}{2} - 1 \right) \quad \text{with} \quad D_1(x) = x^{-1} \int_0^x \frac{u}{e^u - 1} du$$

and bivariate distribution

$$C(u, v) = \frac{1}{\theta} \log \left(1 + \frac{(e^{\theta u} - 1)(e^{\theta v} - 1)}{e^\theta - 1} \right)$$

Note that for $N > 2$ one cannot specify dependence for each pair of variables X^k, X^l , $k \neq l$ separately. A generalization adding more parameters has been given by Rogge and Schönbucher (2003).

9.3.1 Implied correlation

In order to compare Archimedean copulas with the elliptical Gaussian copula it is instructive to consider a pair of standard Gaussians X^1, X^2 with given copula C , rather than uniforms $(U^1, U^2) = (N(X^1), N(X^2))$ with distribution C . Densities with Gaussian margins and Kendall's $\rho_K = 0.4$ are given in Figure 9.1.

It can be seen that (inverted) Gumbel and Clayton tail off slower than Gauss and Frank towards $(-3, -3)$. Therefore out-of-the-money basket and worst-of-two put option prices will be higher with Gumbel and Clayton. This can also be evidenced by considering implied correlation as introduced in Section 8.1. One year basket options on two log-normal assets with forwards 1, weights 50%, volatilities 20% and strikes ranging from 60% to 180% are considered. Each line refers to a copula with specified Kendall's tau ρ_K . Note that Clayton and Gumbel lead to correlation skew. Only positive dependence is supported. Frank allows for positive and negative ρ_K and leads to correlation smile. Gauss produces no skew but gives a correlation level.

9.3.2 Estimation

Let (X^1, X^2) have Archimedean copula with generator g . The random variable $Z = F^X(X^1, X^2)$ is related with g via

$$F^Z(x) = x - \frac{g(x)}{g'(x)}$$

The distribution $\hat{F}^Z$ of Z can be estimated using the random variates

$$Z_i = \frac{1}{N-1} \#\{(X_j^1, X_j^2) : X_j^1 < X_i^1 \text{ and } X_j^2 < X_i^2\}$$

Finally one chooses $\hat{g}$ which gives the best fit.

9.4 Marshall-Olkin copula

This section is based on Embrechts et al. (2003).

The bivariate Marshall-Olkin copula is derived from independent shocks with Poisson arrival times of intensities $\lambda_1, \lambda_2, \lambda_{12}$ affecting the first event, the second event or both. The copula is given as

$$C(u_1, u_2) = u_1 u_2 \left(u_1^{-\alpha_1} \wedge u_2^{-\alpha_2} \right)$$

where $\alpha_i = \lambda_{12}/(\lambda_i + \lambda_{12})$. This copula has mass on the curve $u_1^{\alpha_1} = u_2^{\alpha_2}$. Spearman's rho ρ_S and Kendall's tau ρ_K are given as

$$\rho_S = \frac{3\alpha_1\alpha_2}{2\alpha_1 + 2\alpha_2 - \alpha_1\alpha_2}$$

$$\rho_K = \frac{\alpha_1\alpha_2}{\alpha_1 + \alpha_2 - \alpha_1\alpha_2}$$

Upper tail dependence is given as

$$\lambda_u = \alpha_1 \wedge \alpha_2$$

In the multivariate case one has Poisson arrival times Y_s with intensity λ_s for every $s \in \{1, 2, \dots, N\}$. Those arrival times can be simulated as $Y_s = -\log \frac{U_s}{\lambda_s}$ with independent uniform variates U_s . The arrival time of a shock to index k is $X_k = \min_{s:k \in s} Y_s$. The corresponding uniform variate is $U_k = e^{-\lambda_k X_k}$ with $\lambda_k = \sum_{s:k \in s} \lambda_s$.

An advantage of this copula is that it allows for synchronous defaults if the margins are the same.

9.5 t -copula

The t -copula with parameters (ν, ρ) is the copula of

$$\sqrt{\frac{\nu}{Y}}X$$

where X is a multivariate Gaussian vector with zero mean, unit variance and correlation ρ and Y is independent from X with chi-squared distribution and ν degrees of freedom. In order to simulate from the copula note that a uniform is obtained using univariate cumulative t -distribution function t_ν as

$$U_k = t_\nu \left(\sqrt{\frac{\nu}{Y}} X_k \right)$$

The copula belongs to the class of symmetric elliptical copulas, namely the class of copulas derived from distributions with elliptical level sets. For large ν this copula converges to the Gaussian copula. One may use parametrization (μ, ρ) with $\mu = \nu^{-1}$ in order to get the Gaussian copula with parameters $(0, \rho)$. Kendall's tau ρ_τ does not depend on ν , hence

$$\rho_K = \sin \left(\frac{\pi}{2} \rho \right)$$

The t -copula has upper and lower tail dependence given as

$$\lambda_u = \lambda_l = 2t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right)$$

It produces a symmetric correlation smile. An advantage of this copula is that for more than two random variables it allows one to specify correlation levels for all pairs. It thus extends the Gaussian copula. Most other copulas only allow for a homogenous correlation structure, in the sense that there is only one dependence parameter describing the dependence of all pairs X_k, X_l for $k \neq l$.

9.6 Factor copula

9.6.1 Gaussian factor copula

In the Gaussian case a one-factor copula is defined through

$$X^k = \beta_k M + \sqrt{1 - \beta_k^2} Y_k$$

with $\beta_k^2 \leq 1$ and independent Gaussians $M, Y_1, Y, Y_2, \dots, Y_N$. The corresponding correlation matrix is

$$\rho^{kl} = \beta_k \beta_l$$

This family can be enriched by allowing the β_k to depend on M , i.e. $\beta_k = \beta_k(M)$. Note that the X^k are no longer Gaussian, which leads to less tractable models.

9.6.2 Non-Gaussian factor copula

An alternative method to generate correlation skew is a two-step method. Let (X_1^1, X_1^2) denote standard Gaussians with correlation ρ_1 . Now define a correlation function $\rho_2(X_1^1, X_1^2)$ for the second period. Using two independent Gaussians Z^1, Z^2 define

$$X_2^1 = Z^1, \quad X_2^2 = \rho_2(X_1^1, X_1^2)Z^1 + \sqrt{1 - (\rho_2(X_1^1, X_1^2))^2}Z^2$$

One obtains two Gaussians $X^1 = X_1^1 + X_2^1$ and $X^2 = X_1^2 + X_2^2$ with variance 2 :

$$\mathbb{E} \left[e^{\lambda X^k} \right] = \mathbb{E} \left[e^{\lambda X_1^k} \mathbb{E} \left[e^{\lambda X_2^k} \mid X_1^1, X_1^2 \right] \right] = \mathbb{E} \left[e^{\lambda X_1^k} e^{\lambda^2/2} \right] = e^{\lambda^2}$$

However, the copula of (X^1, X^2) is non-Gaussian. One has for all λ, μ

$$\mathbb{E} \left[e^{(\lambda X^1 + \mu X^2)} \right] = \mathbb{E} \left[e^{(\lambda X_1^1 + \mu X_2^1)} e^{\frac{1}{2}(\lambda^2 + \mu^2 + 2\rho_2(X_1^1, X_2^1)\lambda\mu)} \right] \neq e^{(\lambda^2 + \mu^2 + 2\rho\lambda\mu)}$$

In applications to equities the function ρ_2 may reflect the observation that correlation increases if both assets decrease. Typical functions are

$$\rho_2(x, y) = f(x + y)$$

$$\rho_2(x, y) = f(x \vee y)$$

$$\rho_2(x, y) = f(x \wedge y)$$

where f is a decreasing function with values in $[-1, 1]$.

9.6.3 Dynamic factor copula

The factor copula approach has a natural extension to more steps and more than two assets

$$X_i^k, k = 1, 2, \dots, N, i = 1, 2, \dots, I$$

such that

- for all k the random vector $X^k = (X_i^k, i = 1, 2, \dots, I)$ consists of independent standard Gaussians,
- the random vector $X_1 = (X_1^k, k = 1, 2, \dots, N)$ is multivariate Gaussian with a correlation ρ_1 , and
- for $i > 1$ the random vector $X_i = (X_i^k, k = 1, 2, \dots, N)$ conditional on the ‘past’ $(X_j, j = 1, 2, \dots, i - 1)$ is Gaussian with a correlation matrix $\rho_i = \rho_i(X_j, j = 1, 2, \dots, i - 1)$.

Applications include a multi-asset extension of the Black-Scholes copula model from Section 9.9, or the complete market model described in Section 7.4.3.

9.7 Convex combination

A convex combination of copulas is again a copula. This allows one to combine given copulas to obtain new ones. Practically, this amounts to convex combinations of prices; given N market prices $(\pi^i, i = 1, 2, \dots, N)$ for different (copula-dependent) products and M copulas, one obtains a matrix of model prices π_j^i for product i priced with copula j . Employing standard methods one can derive a convex combination for the copula, minimizing the discrepancy between model and market prices.

9.8 Model-independent arbitrage bounds

The computation of model-independent arbitrage bounds for derivatives prices, contingent on one or more assets observed at one or more future times, is a topic of active research. In that context it is typically assumed that the marginal distributions are known from market data, such as prices of Vanilla options. For basket options $(S_t^1 + S_t^2 - K)^+$ analytical expressions for optimal copulas giving sharp bounds have been obtained by Laurence and Wang (2005). For forward-starting options $(S_T - S_t)^+$ Hobson and Neuberger (2012) find an upper bound as well as a corresponding model by means of Skorokhod embedding. A survey on Skorokhod embedding has been compiled by Hobson (2010). Beiglböck et al. (2013) present an alternative approach based on a duality theory of optimal transport.

9.9 Gauss copula model

A Gauss copula approach for path-dependent derivatives on several assets has presumably been in use by practitioners since well before 2000. References include

presentations by the author (2002, 2006) and Jäckel's presentation (2004). The mathematical problem has been stated (and solved) by Carr et al. (1999).

9.9.1 Copula of Black-Scholes model

Consider the Hull-White extension of the Black-Scholes model with time-dependent parameters, namely

$$\frac{dS_t}{S_t} = r_t dt + \sigma_t dW_t \quad (9.1)$$

The solution of this equation is given by

$$S_t = F(t) \exp \left(X_t - \frac{1}{2} (\hat{\sigma}(t))^2 t \right)$$

with

$$X_t = \int_0^t \sigma_u dW_u \quad (9.2)$$

and forward $F(t) = S_0 \exp \left(\int_0^t r_u du \right)$. The link between instantaneous volatility σ and implied volatility $\hat{\sigma}$ in this model is given by

$$(\hat{\sigma}(t))^2 t = \int_0^t \sigma_u^2 du$$

The time copula of S is a Gaussian copula with correlation

$$\rho_{tT} = \frac{\hat{\sigma}(t)}{\hat{\sigma}(T)} \sqrt{\frac{t}{T}}$$

9.9.2 Black-Scholes copula model

It seems natural to construct a stochastic process S by combining market implied distributions

$$F^{S_t}(K) = \mathbb{IP} [S_t < K]$$

given through put spreads with the Black-Scholes copula. A stock process S can be defined via

$$\begin{aligned} S_t &= D_t(X_t) \\ D_t(x) &= \left(F^{S_t} \right)^{-1} \left(F^{X_t}(x) \right) \end{aligned} \quad (9.3)$$

where X_t is the process specified in (9.2). This process has two desirable properties, namely

- The process is calibrated to the European options market.
- The process has the same copula as the Black-Scholes model.

In order to see this, note that the processes S_t and X_t have the same time copula since, for all t , $D_t(x)$ is increasing in x . Also one has

$$\mathbb{P}[S_t < K] = \mathbb{P}\left[F^{X_t}(X_t) < F^{S_t}(K)\right] = F^{S_t}(K)$$

since $F^{X_t}(X_t)$ is uniformly distributed. If there is no volatility dependence on K in the market implied volatilities, namely $\hat{\sigma}(K, t) = \hat{\sigma}(t)$, then the two processes (9.1) and (9.3) coincide.

The mathematical structure of the generated diffusion can be derived from (9.2) and (9.3). Itô's lemma yields

$$\begin{aligned} dS_t &= \frac{\partial D_t}{\partial x}(X_t) dX_t + \frac{1}{2} \frac{\partial^2 D_t}{\partial x^2}(X_t) d\langle X \rangle_t + \frac{\partial D_t}{\partial t}(X_t) dt \\ &= \left(\sigma_t \frac{\partial D_t}{\partial x}(X_t) \right) dW_t + \left(\frac{\sigma_t^2}{2} \frac{\partial^2 D_t}{\partial x^2}(X_t) + \frac{\partial D_t}{\partial t}(X_t) \right) dt. \end{aligned}$$

On the other hand argument X_t of D_t can be replaced by $D_t^{-1}(S_t)$ where D_t^{-1} denotes the inverse of $D_t(x)$ with respect to x . Therefore, reinserting all arguments, the equation is of the form

$$\frac{dS_t}{S_t} = \mu_t(S_t) dt + \sigma_t(S_t) dW_t.$$

Hence, the diffusion has the same structure as the local volatility model, except that the drift is not the riskless rate r but has a local component:

$$\mu \neq r \tag{9.4}$$

Note that this inequality violates the no-arbitrage condition.

The local volatility in this model is given as

$$\begin{aligned} \sigma_t(K) &= \frac{\sigma_t \frac{\partial D_t}{\partial x}(D^{-1}(K))}{K} = \frac{\sigma_t f^{X_t}(D_t^{-1}(K))}{K f^{S_t}(K)} \\ &= \frac{\sigma_t f^{X_t}\left((F^{X_t})^{-1}(F^{S_t}(K))\right)}{K f^{S_t}(K)} \end{aligned}$$

with F^Z and f^Z denoting the distribution and density of Z for a random variable Z , respectively. Using the fact that X_t is Gaussian with variance $(\hat{\sigma}(t))^2 t$ this expression simplifies to

$$\sigma_t(K) = \frac{\sigma_t}{\hat{\sigma}(t)} \frac{\phi\left(\Phi^{-1}\left(\frac{\partial \text{Put}(K,t)}{\partial K}\right)\right)}{\sqrt{tK} \frac{\partial^2 \text{Put}_t(K)}{\partial K^2}}$$

The dependency of σ on the process $(X_t, t \geq 0)$ is reflected in the factor

$$\sigma_t/\sigma(t) = \sigma_t / \sqrt{1/t \int_0^t \sigma_u^2 du}.$$

9.9.3 Black-Scholes copula versus local volatility model

In the last section it was shown that combining market implied marginal distribution and Gaussian copula à la Black-Scholes leads to a model with arbitrage. One can invert this problem, see Carr et al. (1999) to the problem of finding mapping functions $D_t(x)$ (or, equivalently, local volatility surfaces), such that the Black-Scholes copula model coincides with Dupire, namely $\mu = r$. The solution is a necessary condition in terms of a partial differential equation (PDE) on local volatility

$$\frac{\sigma^2}{2} \frac{\partial^2 \sigma}{\partial S^2} + rS \frac{\partial \sigma}{\partial S} + \frac{\partial \sigma}{\partial t} = r\sigma$$

By fixing the distribution at time T and the instantaneous volatility σ defining the copula, it follows from this necessary condition that the model is fully specified. Thus, it is only possible to calibrate to options expiring at T and, additionally, a term structure of option prices, such that matching the distribution at a second date $t < T$ will typically violate the condition. An inspection of the special cases proposed in Carr et al. (1999) shows that good calibration of the skew for one maturity T can be achieved. However, there are no parameters left to calibrate a second maturity t .

Given an implied volatility surface $\hat{\sigma}(K, t)$ there is no reason why the Black-Scholes copula model and Dupire should coincide since the Dupire process generally does not have a Gaussian time copula.

10 Fixed Income

Fixed income is at the core of financial modelling. Before considering assets such as stocks, currencies, commodities or credit events it is necessary to value future cash flows.

Fixed income models can be divided into three broad categories, namely

- Short rate models are based on forward rate or zero coupon bond dynamics. They also include Heath-Jarrow-Morton models. Short rate models often allow for closed form solutions for liquid products. They are typically not rich enough to calibrate to all required market data.
- Libor market models are based on a readily tradeable asset. They typically require high computation effort and are based on Monte Carlo simulation.
- Markov Functional models attempt to remedy some shortcomings of short rate models without giving up their computational efficiency. They are typically employed for products with early exercise.

10.1 Market

10.1.1 Short rate, forward rate and zero coupon bond

The price at t of receiving one unit of currency at time T is denoted $P_t(T)$. Such a product is also known as a zero coupon bond. The process $P_t(T)$ typically decreases in T with values in $(0, 1]$ and $P_T(T) = 1$. It is also assumed smooth in T . The forward rate is defined as

$$f_t(T) = -\frac{\partial \log P_t(T)}{\partial T}$$

In other words

$$P_t(T) = \exp\left(-\int_t^T f_t(s) ds\right)$$

For $T = t$ one obtains the short rate

$$r_t = f_t(t)$$

10.1.2 Cash account

Cash account B is defined as

$$B_t = \exp\left(\int_0^t r_s ds\right) \quad (10.1)$$

This represents the strategy of investing a unit for a short time $[t, t + dt]$ into a zero coupon bond with value $P_t(t + dt)$ and continuously rolling into new bonds.

10.1.3 Libor

Fix an interval $[S, T]$. The simple compounded forward rate $L^{[S, T]}$ for the interval is defined by

$$1 + L_t^{[S, T]}(T - S) = \frac{P_t(S)}{P_t(T)} \quad (10.2)$$

Simple compounded forward rates are also denoted as Libor (short for London interbank offer rate). This definition is motivated by the trading strategy:

- At t , sell a bond with notional 1 maturing at S
- At t , buy a bond with notional $1 + L_t^{[S, T]}(T - S)$ maturing at T

where $L_t^{[S, T]}$ is fixed at t such that no cash is required at t .

Given a time discretization $0 \leq T_0 < T_1 < \dots < T_n$ one may write L^i for $L^{[T_i, T_{i+1}]}$ and α_i for the time increment $T_{i+1} - T_i$. Time increments α_i for the purpose of interest rate specification are determined using an agreed day count convention. Equation (10.2) can now be rewritten (with argument $t = 0$ omitted) as

$$P(T_{i+1}) = \frac{P(T_i)}{1 + \alpha_i L^i} \quad (10.3)$$

This allows for bootstrapping a discount curve from Libor rates.

10.1.4 Swap

A standard interest rate swap allows one to swap a fixed swap rate s against a floating Libor rate. Thus the value of a receiver (= receive fixed) swap at $t \leq T_0$ is

$$\sum_{i=0}^{n-1} P_t(T_{i+1}) \alpha_i (s - L_t^i) = sA_t - P_t(T_0) + P_t(T_n)$$

where $A_t = A_t(T_n)$ is the value of an annuity given as

$$A_t = \sum_{i=0}^{n-1} \alpha_i P_t(T_{i+1})$$

The par swap rate is the value $s_t = s_t(T_n)$, such that the swap has value zero at time t , namely

$$s_t = \frac{P_t(T_0) - P_t(T_n)}{A_t} \quad (10.4)$$

With $t = T_0 = 0$ and $n = i$ one has

$$P(T_i) = 1 - s(T_i)A(T_i) = 1 - s(T_i)A(T_{i-1}) - s(T_i)\alpha_i P(T_i)$$

and hence

$$P(T_i) = \frac{1 - s(T_i)A(T_{i-1})}{1 + s(T_i)\alpha_i}$$

allowing one to bootstrap a discount curve from swap rates.

10.1.5 Overnight index swap

Before the Lehman default in 2008 market participants bootstrapped discount curves using forward rate contracts for maturities up to around two years and swap rates for longer terms. Awareness of credit risk resulted in a widening spread between overnight and Libor rates. As a result a plethora of discount curves appeared, dependent on the swap frequency assumed within the bootstrapping algorithm.

Overnight index swaps (OIS) swap fixed payments periodically (annually, semi-annually or once, if shorter than six months) against the geometric average of daily fixings of an overnight rate (EONIA, Fed Funds). Maturities vary from one week to 30 years. For OIS swap rates bootstrapping, as described in Section 10.1.4,

is valid and results in a discount curve P^{OIS} . Since cash collateral accrues at the overnight rate it is common practice to discount collateralized cash flows using this curve. For example, if Libor payments within a standard swap are collateralized then Libor rates can be derived from basis swaps swapping standard Libor L against OIS Libor L^{OIS} : The basis spread $s = s(T_n)$ satisfies

$$\begin{aligned}\sum_{i=0}^{n-1} \alpha_i L^i P^{OIS}(T_{i+1}) &= \sum_{i=0}^{n-1} \alpha_i (L^{OIS,i} + s) P^{OIS}(T_{i+1}) \\ &= 1 - P^{OIS}(T_n) + sA^{OIS}(T_n)\end{aligned}$$

From the L^i one can compute a (forwarding) curve P through (10.3). Equation (10.4) does not hold if forwarding and discounting curves are different.

10.1.6 Tenor basis swap

A tenor basis swap has two floating legs with different payment frequencies. The (shorter) tenor has a (positive) tenor spread added to the relevant rate. Tenor basis spread s is defined through

$$\sum_{T_i \in \tau} \alpha_i L^{\tau,i} P(T_{i+1}) = \sum_{T'_j \in \tau'} \alpha_j (L^{\tau',j} + s) P(T'_{j+1})$$

where τ, τ' denote two time discretizations of $[0, T]$. Given a forwarding curve for Libor L^τ one can derive a corresponding curve for $L^{\tau'}$ from basis spread term structure data.

Basis swap data may only exist in reference currency USD. In that case cross currency swap data may also be required in order to derive forwarding for specific tenors, see Section 15.1. It is now standard to maintain a multi-curve framework for the purpose of representing swap market information, see Kenyon and Stamm (2012) for details. For equity derivatives the relevant discounting curve depends on funding requirements, compare Section 14.6.

10.1.7 Cap and floor

A caplet is a call option on a Libor rate with payment at the end of the period. A floorlet is the corresponding put. Typically one trades strings of caplets (floorlets) which are called cap (floor). In connection with a swap a cap can be used to cap the (Libor) rate. The value of the i -th caplet at fixing time T_i is

$$P_{T_i}(T_{i+1}) \alpha_i \left(L_{T_i}^i - K \right)^+ = (1 - \alpha_i - P_{T_i}(T_{i+1}))^+$$

Thus a call on Libor (=caplet) is equivalent to a put on a bond.

Caps and floors are quoted using a sum of Black-Scholes values with notional α_i , forwards

$$L_0^i = \frac{1}{\alpha_i} \left(\frac{P_0(T_i)}{P_0(T_{i+1})} - 1 \right)$$

and discount factors $P_0(T_{i+1})$. Variances are $\sigma^2 \alpha_i$ with the same value σ used for every caplet within a given cap. Thus, for a given currency one has a surface of volatilities specified by two parameters, cap term T_n and strike K . In order to compute the implied volatility for the i -th caplet with strike K one has to compute the price difference between two caps with strike K maturing at T_{i+1} and T_i using corresponding implied volatilities $\sigma(K, T_{i+1})$ and $\sigma(K, T_i)$. This may require volatility interpolation. The caplet volatility is the implied volatility of the resulting caplet price.

10.1.8 Swaption

A swaption allows the holder to enter a swap at a fixed swap rate specified in the swaption contract. The intrinsic value of a receiver swaption at maturity $T = T_0$ with fixed rate s is

$$\left(\sum_{i=0}^{n-1} P_T(T_{i+1}) \alpha_i (s - L_T^i) \right)^+ = A_T (s - S_T)^+$$

where S_T denotes the par swap rate at time T . A swaption may be considered an option on a coupon bond with strike 1. The coupons payable at T_i are $s\alpha_{i-1}$ for $i = 1, 2, \dots, n-1$ and a final coupon $s\alpha_{n-1} + 1$ at T_n .

Swaptions are quoted using the Black-Scholes formula with forward A_0 and discount factor 1. Thus for a given currency one has a volatility cube with three parameters, maturity T_0 , swap length $T_n - T_0$ and strike s . This last is typically quoted relative to today's par swap rate s_0 .

10.2 Short rate

10.2.1 Parametrizations

There are different parametrizations, depending on whether one models $r, f(T)$ or $P(T)$. Given that all dynamics is driven by the same d -dimensional vector W of

Brownian motions one can define

$$\begin{aligned} dr_t &= \mu_t^r dt + \sigma_t^r dW_t \\ \frac{dP_t(T)}{P_t(T)} &= \mu_t^P(T) dt + \sigma_t^P(T) dW_t \\ df_t(T) &= \mu_t^f(T) dt + \sigma_t^f(T) dW_t \end{aligned}$$

The connection between those parametrizations is

$$\mu_t^f(T) = \frac{\partial \sigma_t^P(T)}{\partial T} \sigma_t^P(T) - \frac{\partial \mu_t^P(T)}{\partial T} \quad \text{and} \quad \sigma_t^f(T) = -\frac{\partial \sigma_t^P(T)}{\partial T} \quad (10.5)$$

$$\mu_t^r(T) = \frac{\partial f_t(T)}{\partial T} + \mu_t^f(T) \quad \text{and} \quad \sigma_t^r = \sigma_t^f(T) \quad (10.6)$$

$$\mu_t^P(T) = r_t - \int_t^T \mu_t^f(u) du + \frac{1}{2} \sigma_t^P(T)^2 \quad \text{and} \quad \sigma_t^P(T) = -\int_t^T \sigma_t^f(u) du \quad (10.7)$$

At this point it is not assumed that the underlying measure is a martingale measure or that the model is free of arbitrage. The remainder of this section is devoted to the derivation of equations (10.5), (10.6) and (10.7).

- In order to show (10.5) note that Itô's lemma allows one to write $\log P_t(T)$ as stochastic integral and differentiate with respect to T .
- Equation (10.6) can be derived as follows; one has

$$r_t = f_t(t) = f_0(t) + \int_0^t \mu_s^f(t) ds + \int_0^t \sigma_s^f(t) dW_s$$

as well as

$$\mu_s^f(t) = \mu_s^f(s) + \int_s^t \frac{\partial \mu_s^f(u)}{\partial T} du$$

and similarly for σ . Inserting yields

$$\begin{aligned} r_t &= f_0(t) + \int_0^t \mu_s^f(s) ds + \int_0^t \int_s^t \frac{\partial \mu_s^f(u)}{\partial T} du ds \\ &\quad + \int_0^t \sigma_s^f(s) dW_s + \int_0^t \sigma_s^f(t) \int_s^t \frac{\partial \sigma_s^f(u)}{\partial T} du dW_s \end{aligned}$$

Changing the order of integration leads to (10.6).

- For equation (10.7) let $X_t = -\int_t^T f_t(u) du$. Then

$$\begin{aligned}
 X_t &= \int_t^T (-f_u(u) + f_u(u) - f_t(u)) du \\
 &= -\int_t^T f_u(u) du + \int_t^T \int_t^u \mu_v^f(u) dv du + \int_t^T \int_t^u \sigma_v^f(u) dW_v du \\
 &= -\int_t^T f_u(u) du + \int_t^T \int_v^T \mu_v^f(u) du dv + \int_t^T \int_v^T \sigma_v^f(u) du dW_v \\
 &= X_0 + \int_0^t f_u(u) du - \int_0^t \int_v^T \mu_v^f(u) du dv - \int_0^t \int_v^T \sigma_v^f(u) du dW_v
 \end{aligned}$$

and hence

$$dX_t = \left(f_t(t) - \int_t^T \mu_t^f(u) du \right) dt - \left(\int_t^T \sigma_t^f(u) du \right) dW_t$$

Itô's lemma implies

$$\begin{aligned}
 \frac{dP_t(T)}{P_t(T)} &= dX_t + \frac{1}{2} d\langle X \rangle_t \\
 &= \left(r_t - \int_t^T \mu_t^f(u) du + \frac{1}{2} \left(\int_t^T \sigma_t^f(u) du \right)^2 \right) dt - \left(\int_t^T \sigma_t^f(u) du \right) dW_t
 \end{aligned}$$

as claimed.

10.2.2 Affine term structure

If the short rate process r is an affine process, in the sense of Chapter 16, namely

$$\mu_t^r = m_t^0 + m_t^1 r_t, \quad (\sigma_t^r)^2 = s_t^0 + s_t^1 r_t \quad (10.8)$$

then due to (16.3) one has

$$P_t(T) = \exp(A_t(T) + r_t B_t(T))$$

with

$$\begin{aligned}
 \frac{\partial B_t(T)}{\partial t} &= 1 - m_t^1 B_t(T) - \frac{1}{2} s_t^1 B_t(T)^2 \\
 \frac{\partial A_t(T)}{\partial t} &= -m_t^0 B_t(T) - \frac{1}{2} s_t^0 B_t(T)^2
 \end{aligned} \quad (10.9)$$

The equation for $B(T)$ is a Riccati equation. From the solution to this equation $A(T)$ can be determined via integration.

To illustrate the approach, consider the Gaussian model

$$dr_t = \theta dt + \sigma^r dW_t$$

Equations (10.9) have the solution

$$B_t(T) = t - T$$

$$A_t(T) = \theta(T - t) + \frac{1}{6}(\sigma^r)^2(T - t)^3$$

and hence

$$P_t(T) = \exp\left(\theta(T - t) + \frac{1}{6}(\sigma^r)^2(T - t)^3 - r_t(T - t)\right)$$

If θ, σ are time dependent then $A_t(T)$ is

$$A_t(T) = \int_t^T \theta_s(s - T)ds + \frac{1}{2} \int_t^T (\sigma_s^r)^2(s - T)^2 ds$$

This model is known as the Ho-Lee model.

10.3 Heath-Jarrow-Morton

Heath et al. (1992) model the forward rate

$$df_t(T) = \mu_t^f(T)dt + \sigma_t^f(T)dW_t$$

$$f_0(T) = -\frac{\partial \log P_0(T)}{\partial T}$$

where P_0 denotes the initial yield curve. In contrast to short rate models the initial yield curve is thus matched by construction. Note that due to

$$P_t(T) = \exp\left(-\int_t^T f_t(u)du\right)$$

$f(T)$ determines the entire yield curve dynamics. The key result of Heath-Jarrow-Morton is that no arbitrage implies the existence of a d -dimensional process λ such that

$$\mu_t^f(T) = \sigma_t^f(T) \int_t^T \sigma_t^f(u)du - \sigma_t^f(T)\lambda_t \quad (10.10)$$

In order to show this revisit equation (10.7), namely

$$\begin{aligned}\frac{dP_t(T)}{P_t(T)} &= \mu_t^P(T)dt + \sigma_t^P(T)dW_t \\ \mu_t^P(T) &= r_t - \int_t^T \mu_t^f(u)du + \frac{1}{2}\sigma_t^P(T)^2 \\ \sigma_t^P(T) &= - \int_t^T \sigma_t^f(u)du.\end{aligned}$$

In other words the risk premium is given as

$$- \int_t^T \mu_t^f(u)du + \frac{1}{2}\sigma_t^P(T)^2.$$

The fundamental theorem of asset pricing states that an equivalent martingale measure exists. It follows from a representation theorem that there is λ , such that

$$- \int_t^T \mu_t^f(u)du + \frac{1}{2}\sigma_t^P(T)^2 = \sigma_t^P(T)\lambda_t$$

Derivation with respect to T yields equation (10.10).

The importance of this result is that, with respect to the martingale measure, the forward rate is specified only through initial yield curve and volatility $\sigma_t^f(T)$. Drift is given as

$$\mu_t^f(T) = \sigma_t^f(T) \int_t^T \sigma_t^f(u)du$$

Thus $f_t(T)$ satisfies

$$\begin{aligned}df_t(T) &= \sigma_t^f(T) \left(\int_t^T \sigma_t^f(u)du \right) dt + \sigma_t^f(T)dW_t \\ f_0(T) &= - \frac{\partial \log P_0(T)}{\partial T}\end{aligned}\tag{10.11}$$

and bond dynamics is

$$\frac{dP_t(T)}{P_t(T)} = r_t dt + \sigma_t^P(T)dW_t\tag{10.12}$$

integrating one obtains

$$P_t(T) = P_0(T) \exp \left(\int_0^t \left(r_u - \frac{1}{2} \sigma_u^P(T)^2 \right) du - \int_0^t \sigma_u^P(T) dW_u \right) \quad (10.13)$$

Setting $T = t$ yields the cash bond distribution

$$B_t = P_0(t)^{-1} \exp \left(\int_0^t \sigma_u^P(t) dW_u^P + \frac{1}{2} \int_0^t \sigma_u^P(t)^2 du \right) \quad (10.14)$$

Dividing equation (10.13) by $P_t(t) = 1$, expressed by using (10.13) again with $T = t$, yields the bond distribution

$$P_t(T) = \frac{P_0(T)}{P_0(t)} \exp \left(-\frac{1}{2} \int_0^t \left(\sigma_s^P(T)^2 - \sigma_s^P(t)^2 \right) ds - \int_0^t \left(\sigma_s^P(T) - \sigma_s^P(t) \right) dW_s \right) \quad (10.15)$$

Note that the short rate r does not appear in equations (10.14) and (10.15).

10.3.1 Musiela parametrization

The forward rate $f_t(T)$ and its volatility $\sigma_t^f(T)$ may also be specified by t and term $\tau = T - t \geq 0$, namely

$$\begin{aligned} \tilde{f}_t(\tau) &= f_t(t + \tau) \\ \tilde{\sigma}_t^f(\tau) &= \sigma_t^f(t + \tau) \end{aligned}$$

One has

$$\begin{aligned} d\tilde{f}_t(\tau) &= \frac{\partial}{\partial t} df_t(t + \tau) \\ &= \left(\frac{\partial \tilde{f}_t(\tau)}{\partial \tau} + \tilde{\sigma}_t^f(\tau) \left(\int_0^\tau \tilde{\sigma}_t^f(u) du \right) \right) dt + \tilde{\sigma}_t^f(\tau) dW_t \end{aligned}$$

10.3.2 Gaussian short rates

Consider constant volatility $\sigma_t^f(T) = \sigma^f$. Then

$$\mu_t^f(T) = (\sigma^f)^2(T - t)$$

and hence

$$\begin{aligned} f_t(T) &= f_0(T) + \int_0^t (\sigma^f)^2 (T-s) ds + \int_0^t \sigma^f dW_s \\ &= f_0(T) + (\sigma^f)^2 t \left(T - \frac{t}{2} \right) + \sigma^f W_t \end{aligned}$$

This implies

$$r_t = f_t(t) = f_0(t) + \frac{1}{2}(\sigma^f)^2 t^2 + \sigma^f W_t$$

Taking the differential

$$dr_t = \left(\frac{\partial f_0(t)}{\partial T} + (\sigma^f)^2 t \right) dt + \sigma^f dW_t$$

shows that this is the Ho-Lee model introduced in Section 10.2.2.

Generalizing this approach let $\sigma_t^f(T)$ now be an arbitrary d -dimensional deterministic function. Then both forward rate $f_t(T)$ and short rate r_t are Gaussian and bonds $P_t(T)$ are log-normally distributed. For that reason these models are known as Gaussian short rate models.

10.3.3 Quasi Gaussian

In general the short rate process implied by a given volatility function $\sigma_t^f(T)$ and (10.11) is not Markovian. However this is the case if (the transpose of) $\sigma_t^f(T)$ is separable into a product of a deterministic vector function and a, possibly, stochastic matrix process in the sense that

$$\sigma_t^f(T)^T = g(T)^T H_t$$

where $g > 0$ is a vector of k deterministic functions and H is a matrix of processes of dimension $k \times k$. Models such as Hull-White and Ho-Lee satisfy this condition. Following Andreasen (2001) the model can, without loss of generality, be defined as

$$P_t(T) = \frac{P_0(T)}{P_0(t)} e^{-G(t,T)^T X_t - \frac{1}{2} G(t,T)^T Y_t G(t,T)}$$

where

$$\begin{aligned} g^i(t) &= \int_0^t e^{-\kappa_s^i} ds \\ g^i(t, T) &= \frac{g^i(T)}{g^i(t)} \\ G^i(t, T) &= \int_t^T g^i(t, s) ds \end{aligned}$$

as well as

$$\begin{aligned} dX_t &= \left(Y_t(1, 1, \dots, 1)^T - [\kappa_t]X_t \right) dt + V_t dW_t \\ dY_t &= \left(V_t V_t^T - [\kappa_t]Y_t - Y_t[\kappa_t] \right) dt \end{aligned}$$

with $X_0 = Y_0 = 0$. The volatility matrix process V is related to H via

$$V_t = [g_t]^{-1} H_t$$

and can be deterministic (leading to a Gaussian Markovian model) or stochastic. Andreasen (2001) suggests using a shifted Heston model, as described in Section 6.3.2.

10.4 Hull-White

Due to its numerical tractability and ease of calibration this model is widely used in the context of hybrid models. The Ornstein-Uhlenbeck process as a model for the short rate

$$dr_t = (\theta - \kappa r_t) dt + \sigma^r dW_t$$

has been introduced by Vasicek. This model is Gaussian and affine. The discussion in Section 10.2.2 shows that bond prices have dynamics

$$P_t(T) = \exp(A_t(T) + r_t B_t(T)) \quad (10.16)$$

with

$$\begin{aligned} B_t(T) &= -e^{\kappa t} \int_t^T e^{-\kappa s} ds = -\frac{1 - e^{-\kappa(T-t)}}{\kappa} \\ A_t(T) &= \theta \int_t^T B_s(T) ds + \frac{1}{2}(\sigma^r)^2 \int_t^T B_s(T)^2 ds \end{aligned}$$

The corresponding model with time-dependent parameters κ, θ, σ^r is known as the Hull-White model and has the solution

$$B_t(T) = -e^{\int_0^t \kappa_s ds} \int_t^T e^{-\int_0^s \kappa_u du} ds$$

$$A_t(T) = \int_t^T \theta_s B_s(T) ds + \frac{1}{2} \int_t^T (\sigma_s^r)^2 B_s(T)^2 ds$$

Parameter θ can be chosen such that the initial yield curve is matched. Alternatively one can follow Heath-Jarrow-Morton with an application of Itô's lemma to (10.16) which shows that

$$\sigma_t^P(T) = -\sigma_t^r B_t(T)$$

The distribution of cash bond B_t and bond $P_t(T)$ is known due to equations (10.14) and (10.15), respectively.

Explicit computation of θ , forward rates $f_0(t)$ and short rates r_t can often be avoided. In cases where θ is given one can consider the process $x_t = e^{\int_0^t \kappa_s ds} r_t$ satisfying

$$dx_t = e^{\int_0^t \kappa_s ds} (\theta_t dt + \sigma_t^r dW_t)$$

Hence short rates satisfy

$$r_T = e^{-\int_t^T \kappa_s ds} r_t + \int_t^T e^{-\int_s^T \kappa_u du} (\theta_s ds + \sigma_s^r dW_s) \quad (10.17)$$

10.4.1 Caplets

Given Black-Scholes volatilities σ_i for the i -th caplet one has

$$\begin{aligned} \sigma_i^2 \alpha_i &= \int_0^{T_i} \left(\sigma_t^P(T_{i+1}) - \sigma_t^P(T_i) \right)^2 dt \\ &= \left(\int_0^{T_i} (\sigma_t^r)^2 \Lambda_t^2 dt \right)^2 \left(\int_{T_i}^{T_{i+1}} \Lambda_t^{-1} dt \right)^2 \end{aligned}$$

with $\Lambda_t = \exp \int_0^t \kappa_s ds$. Assuming piecewise constant κ allows one to compute Λ_t for arbitrary t and hence

$$V(T_i) = \int_0^{T_i} (\sigma_t^r)^2 \Lambda_t^2 dt$$

for all i . Using an appropriate interpolation for V allows a simulation of the yield curve dynamics P on any given time grid such that caplet volatilities are matched.

Mean reversion parameter κ impacts the short rate (and due to (10.16) yield curve) autocorrelation

$$\text{Corr}[r_t, r_T] = \sqrt{\frac{V(t)}{V(T)}}$$

This has an impact on the exercise boundary of callable products, such as Bermudan swaptions. It can hence be calibrated if prices for such products are known.

10.4.2 Swaption

A standard approach for analytical swaption valuation due to Jamshidian (1989) requires a solution for the short rate value $r^* = r_t$ such that

$$\sum_{i=1}^n c_i P_t(T_i, r^*) = 1 \quad (10.18)$$

holds where $c_i = s\alpha_{i-1}$ for $i < n$ and $c_n = s\alpha_{n-1} + 1$, see Section 10.1.8. Since all $P_t(T_i)$ are decreasing in r_t such a solution exists, and furthermore

$$\left(\sum_{i=1}^n c_i P_t(T_i, r_t) \right)^+ = \sum_{i=1}^n c_i (P_t(T_i, r_t) - P_t(T_i, r^*))^+$$

Hence the swaption price can be obtained as a sum of bond option prices with strikes given implicitly via (10.18).

10.5 Cox-Ingersoll-Ross

The square root process

$$dr_t = (\theta - \kappa r_t)dt + \sigma \sqrt{r_t} dW_t \quad (10.19)$$

leads to equation

$$\frac{\partial B_t(T)}{\partial t} = 1 + \kappa B_t(T) - \frac{1}{2} \sigma^2 B_t(T)^2$$

The solution is due to Cox, Ingersoll and Ross and reads

$$\begin{aligned}
 -B_t(T) &= \frac{2(e^{\gamma(T-t)} - 1)}{(\gamma + \kappa)(e^{\gamma(T-t)} - 1) + 2\gamma} \\
 e^{A_t(T)} &= \left(\frac{2\gamma e^{\frac{1}{2}(\kappa + \gamma)(T-t)}}{(\gamma + \kappa)(e^{\gamma(T-t)} - 1) + 2\gamma} \right)^{2\theta/\sigma^2} \\
 \gamma &= \sqrt{\kappa^2 + 2\sigma^2}
 \end{aligned} \tag{10.20}$$

The scaling properties of equation (10.19) yield the transform

$$\mathbb{E} \left[e^{-\lambda \int_0^T r_t dt} \right] = \exp(A_0(T) + \lambda r_0 B_0(T)) \tag{10.21}$$

with generalized parameter

$$\gamma = \sqrt{\kappa^2 + 2\lambda\sigma^2}$$

In order to match the initial yield curve Brigo and Mercurio (2001) suggest adding a deterministic offset ϕ , namely

$$r_t = r_t^\alpha + \phi_t$$

where now r^α refers to the homogenous CIR process defined through equation (10.19), with parameters $\alpha = (x_0, \theta, \kappa, \sigma)$ where x_0 is the starting value, i.e. $r_0^\alpha = x_0$. For r to match the initial yield curve requires

$$P_0(t) = \mathbb{E} \left[e^{-\int_0^t r_u du} \right] = \exp \left(A_0(t) + x_0 B_0(t) - \int_0^t \phi_u du \right)$$

This implies bond distribution

$$P_t(T) = \frac{P_0(T)}{P_0(t)} \exp(A_0(T) - A_0(t) + x_0(B_0(T) - B_0(t)) + A_t(T) + r_t^\alpha B_t(T))$$

The offset can be computed as

$$\begin{aligned}
 \phi_t &= \frac{\partial}{\partial t} (-\log D(t) + A_0(t) + x_0 B_0(t)) \\
 &= f_0(t) - \frac{2\theta(e^{\gamma t} - 1)}{(\kappa + \gamma)(e^{\gamma t} - 1) + 2\gamma} - x_0 \frac{4\gamma^2 e^{\gamma t}}{((\kappa + \gamma)(e^{\gamma t} - 1) + 2\gamma)^2}
 \end{aligned}$$

where the sum of the last two terms are the negative forward rate of the CIR process r^α . CIR parameters $\alpha = (x_0, \theta, \kappa, \sigma)$ are chosen such that market prices of caps are reproduced as best as possible. Brigo and Mercurio (2001) suggest restricting x_0 to $0 < x_0 < r_0$ and hence $\phi_0 > 0$. Enforcing $\phi_t \geq 0$ for all t and hence positive rates r seems to be too restrictive.

10.6 Markov Functional

A fixed income model is called a Markov Functional model if there is a numéraire N , an equivalent martingale measure IP and a Markov process X , such that

- The process X is a Markov process with respect to IP.
- The numéraire N is a deterministic function of t and X :

$$N_t = N_t(X_t) \quad (t \geq 0)$$

Note that this fully determines the model. In particular, zero coupon bonds $P_t(T)$ maturing at time T are also deterministic functions of X_t , since

$$P_t(T) = P_t(T, X_t) = N_t(X_t) \mathbb{E} \left[\frac{1}{N_T(X_T)} \middle| X_t \right] \quad (10.22)$$

10.6.1 Calibration to caplets

Choose a time discretization $0 = T_0 < T_1 < T_2 < \dots < T_n$ as well as numéraire

$$N_t = P_t(T_n)$$

Given caplet prices (and hence Libor digital prices) the numéraire functional $N_t(X_t)$ on the time grid can be computed via induction beginning with

$$N_{T_n}(X_{T_n}) = P_{T_n}(T_n) = 1$$

If functionals $N_{T_k} = N_{T_k}(X_{T_k})$ are given for all $k > i$ one computes N_{T_i} in two steps:

- **Step 1:** Assuming $L_{T_i}^i = L_{T_i}^i(X_{T_i})$ is an increasing function of X_{T_i} one determines $L_{T_i}^i$ such that model and market price for caplets coincide as follows; let

$$K = L_{T_i}^i(x)$$

The model price satisfies

$$V_0^{i,K,mdl} = N_0 \mathbb{E} \left[\frac{1_{[K,\infty)}(L_{T_i}^i(X_{T_i}))}{N_{T_{i+1}}(X_{T_{i+1}})} \right] = N_0 \mathbb{E} \left[\frac{1_{[x,\infty)}(X_{T_i})}{N_{T_{i+1}}(X_{T_{i+1}})} \right] = V_0^{i,x,mdl}$$

where the last equation defines $V_0^{i,x,mdl}$. The left hand side $V_0^{i,K,mdl}$ requires knowledge of the function $x \mapsto L_{T_i}^i(x)$, while the right hand side $V_0^{i,x,mdl}$ does not. The link between x and K is given by

$$V_0^{i,x,mdl} \stackrel{!}{=} V_0^{i,K,mkt}$$

In practice the function $L_{T_i}^i$ can be determined as follows: given x compute $V_0^{i,x,mdl}$. The corresponding strike K is obtained by inserting $V_0^{i,x,mdl}$ into the inverse of the function $K \mapsto V_0^{i,K,mkt}$.

- **Step 2:** One concludes by computing $N_{T_i}(X_{T_i})$ from $L_{T_i}^i(X_{T_i})$ and the induction assumption. Equations (10.2) and (10.22) yield

$$1 + L_{T_i}^i(X_{T_i})(T_{i+1} - T_i) = \frac{1}{P_{T_i}^{T_{i+1}}} = \left(N_{T_i}(X_{T_i}) \mathbb{E} \left[\frac{1}{N_{T_{i+1}}(X_{T_{i+1}})} \middle| X_{T_i} \right] \right)^{-1}$$

11] Equity-interest Rate Hybrids

The problem of modelling assets with stochastic drift has been studied in order to risk manage long dated foreign exchange derivatives. In that context advanced yield curve dynamics is often combined with simplified foreign exchange smile dynamics. In the equity context the aim is to extend smile consistent models to the stochastic interest rate case. Interest rates dynamics is often limited to single-factor short rate models, such as Hull-White's extension of the Vasicek model or the Cox-Ingersoll-Ross square root process, discussed in Sections 10.4 and 10.5.

Especially for long dated derivatives part of the equity volatility stems from rates dynamics. This has an impact on long dated products with pronounced equity vega. Thus most derivatives will have some exposure to interest rate volatility and a correlation between equities and rates. The sensitivity to their correlation is known as hybrid risk and cannot easily be hedged. Instead, this market risk is often accounted for through a capital reserve related to the exposure. Hence hybrid risk needs to be monitored.

There are products with specific equity and rates optionality which require stochastic modelling of both components. Examples include swap structures where floating interest rate coupons are contingent on equity performance. Such products also require specific hybrid models.

11.1 Constant equity volatility

11.1.1 Log-normal model

Let cash bond B and stock S be jointly log-normally distributed. One has

$$\begin{aligned}\mathbb{IE}\left[B_t^{-1}\right] &= P(t) \\ \mathbb{IE}\left[B_t^{-1}S_t\right] &= P(t)F(t)\end{aligned}$$

Hence B_t and S_t can be written as

$$B_t^{-1} = P(t) \exp \left(-\sigma_B W_t^B - \frac{\sigma_B^2 t}{2} \right)$$

$$S_t = F(t) \exp \left(\rho_{B,S} \sigma_S \sigma_B t + \sigma_S W_t^S - \frac{\sigma_S^2 t}{2} \right)$$

where σ_S, σ_B and $\rho_{B,S}$ denote annualized volatilities and correlation of $\log S_t$ and $\log B_t$, and W^B, W^S are two Brownian motions with correlation $\rho_{B,S}$. Applying equation (18.12) one obtains a variant of the Black-Scholes formula, namely

$$\mathbb{E} \left[B_t^{-1} (S_t - K)^+ \right] = P(t) \left\{ F(t) \Phi \left(d_+ + \rho_{B,S} \sigma_B \sqrt{t} \right) - K \Phi \left(d_- + \rho_{B,S} \sigma_B \sqrt{t} \right) \right\} \quad (11.1)$$

with $d_{\pm}$ given as

$$d_{\pm} = \frac{1}{\sigma_S \sqrt{t}} \left(\log \frac{F(t)}{K} \pm \frac{1}{2} \sigma_S^2 t \right)$$

Note that volatility σ_S is the total volatility of $\log S$, which includes the contribution of $\log B$. The next aim is to express the formula in terms of parameters $\sigma_{S/B}$ and $\rho_{B,S/B}$, referring to volatility of $\log \left(B_t^{-1} S_t \right)$ and corresponding correlation with $\log B_t$ instead of σ_S and $\rho_{B,S}$. One can write W^S in terms of two Brownian motions driving B and S/B as

$$\sigma_S W_t^S = \sigma_B W_t^B + \sigma_{S/B} W_t^{S/B}$$

where

$$\sigma_S^2 = \sigma_B^2 + \sigma_{S/B}^2 + 2\rho_{B,S/B} \sigma_B \sigma_{S/B} \quad (11.2)$$

The correlation $\rho_{B,S}$ of $\log S$ and $\log B$ satisfies

$$\begin{aligned} \rho_{B,S} \sigma_B \sigma_S t &= \mathbb{E} \left[\sigma_B W_t^B \sigma_S W_t^S \right] = \mathbb{E} \left[\sigma_B W_t^B \left(\sigma_B W_t^B + \sigma_{S/B} W_t^{S/B} \right) \right] \\ &= \sigma_B^2 t + \rho_{B,S/B} \sigma_B \sigma_{S/B} t \end{aligned}$$

and hence

$$\rho_{B,S} = \frac{\sigma_B + \rho_{B,S/B} \sigma_{S/B}}{\sqrt{\sigma_B^2 + \sigma_{S/B}^2 + 2\rho_{B,S/B} \sigma_B \sigma_{S/B}}} \quad (11.3)$$

11.1.2 Hull-White

An application is in the case of Gaussian rates, as in Section 10.3.2, with time-dependent equity volatility σ , as well as correlation ρ and deterministic dividend yield q , namely

$$\frac{d(S_t/B_t)}{S_t/B_t} = -d_t dt + \sigma_t dW_t^S$$

In the special case of one-factor Hull-White interest rates, as in Section 10.4, one has

$$\begin{aligned} (\sigma_t^B)^2 t &= \int_0^t (v_s(T) - v_s(t))^2 ds \\ (\sigma_t^{S/B})^2 t &= \int_0^t \sigma_s^2 ds \\ \rho_t^{B,S/B} \sigma_t^B \sigma_t^{S/B} t &= \int_0^t \rho_s \sigma_s (v_s(T) - v_s(t)) ds \end{aligned}$$

with

$$v_s(T) = \sigma_t^r e^{\kappa t} \int_t^T e^{-\kappa s} ds$$

11.2 Gauss copula

The Gauss copula model for stocks S with deterministic interest rates, as discussed in Section 9.9, specifies

$$S_t = D_t(X_t)$$

where X is a log-normal process

$$X_t = F(t) e^{\int_0^t \sigma_s dW_s - \frac{1}{2} \int_0^t \sigma_s^2 ds}$$

with forward $F(t)$ and instantaneous volatility σ_t , see (9.3). Functions D_t are chosen such that

$$C(K, t) = P(T) \mathbb{E}[(S_T - K)^+]$$

holds for all K and t where $C(K, t)$ are market prices for call options.

A Hull-White hybrid process Y as defined in Section 11.1.2 may be set up such that

$$\mathbb{E} \left[B_t^{-1} (Y_t - K)^+ \right] = P(t) \mathbb{E} \left[(X_t - K)^+ \right]$$

holds for all t and all K . In fact X_t is the distribution of Y_t with respect to the t -forward measure. It follows that

$$\mathbb{E} \left[B_t^{-1} (D_t(Y_t) - K)^+ \right] = P(t) \mathbb{E} \left[(D_t(X_t) - K)^+ \right]$$

holds. Thus the Gauss copula model for stocks with deterministic interest rates can be extended to Gaussian rates. The mapping function D_t does not change if both Black-Scholes and log-normal hybrid models feature the same stock volatility.

The copula approach in the context of hybrids was also discussed in Jäckel's presentation (2004).

11.3 Local equity volatility

Assume local volatility as well as stochastic interest rates, namely

$$\begin{aligned} \frac{dS_t}{S_t} &= (r_t - d_t)dt + \sigma_t(S_t)dW_t^S \\ dr_t &= \mu_t^r(r_t)dt + \sigma_t^r(r_t)dW_t^r \end{aligned} \quad (11.4)$$

with deterministic functions $\sigma_t(x)$, $\mu_t^r(r)$, $\sigma_t^r(r)$ and two correlated Brownian motions W^S, W^r . In Section 6.1 a general relationship between European option prices and conditional variances has been given, see (6.2):

$$\mathbb{E} \left[\sigma_T^2 | S_T = K \right] = 2 \frac{\frac{\partial \text{Call}}{\partial T} - K \mathbb{E} \left[r_T B_T^{-1} \mathbf{1}_{\{S_T > K\}} \right] + d_T \left(\text{Call} - K \frac{\partial \text{Call}}{\partial K} \right)}{\frac{\partial^2 \text{Call}}{\partial K^2} K^2}$$

In the case of a local volatility function $\sigma_T = \sigma_T(K)$ one also has

$$\mathbb{E} \left[\sigma_T^2 | S_T = K \right] = \sigma_T^2(K)$$

This local volatility function will depend on the dynamics of r . Specializing further to deterministic interest rates $\bar{r}$, this becomes Dupire's local volatility function $\bar{\sigma}_T(K)$, see (6.3):

$$\bar{\sigma}_T^2(K) = 2 \frac{\frac{\partial \text{Call}}{\partial T} + (\bar{r}_T - d_T)K \frac{\partial \text{Call}}{\partial K} + d_T \text{Call}}{\frac{\partial^2 \text{Call}}{\partial K^2} K^2}$$

Comparing the equations for $\sigma_T(K)$ and $\bar{\sigma}_T(K)$ gives

$$\sigma_T^2(K) = \bar{\sigma}_T^2(K) - 2 \frac{\mathbb{E} \left[B_T^{-1} (r_T - \bar{r}_T) \mathbf{1}_{\{S_T > K\}} \right]}{\frac{\partial^2 C}{\partial K^2} K} \quad (11.5)$$

where S is the process with stochastic interest rates, defined in (11.4). Calculation of the correction term is not straightforward as it implies knowledge of function σ itself.

11.3.1 Hull-White

Balland (2005) considers the case of Heath-Jarrow-Morton interest rate dynamics with Gaussian forward rates r , namely

$$dr_t(T) = \gamma(t, T) dW_t$$

with respect to the T -forward measure and deterministic volatility function γ , compare Section 10.3. Hull-White interest rates with constant volatility σ and mean reversion κ are an important special case with

$$\begin{aligned} \gamma(t, T) &= \sigma_r e^{-\kappa(T-t)} \\ \Gamma(t, T) &= \int_t^T \gamma(t, s) ds = \frac{\sigma_r}{\kappa} \left(1 - e^{-\kappa(T-t)} \right) \end{aligned}$$

see Section 10.4. Using Malliavin calculus Blacher shows

$$\begin{aligned} \sigma_T(K)^2 &= \bar{\sigma}_T(K)^2 - 2 \int_0^t \gamma(t, T) \Gamma(t, T) dt \\ &\quad - 2 \int_0^T \rho_t^{rS} \gamma(t, T) \mathbb{E}_T \left[\sigma_t(S_t) \frac{Y_T}{Y_t} \middle| S_T = K \right] dt \end{aligned} \quad (11.6)$$

with notation $\sigma'_t(x) = \frac{\partial}{\partial x} \sigma_t(x)$ and where

$$\frac{dY_t}{Y_t} = \sigma_t(S_t) \sigma'_t(S_t) dW_t - \left(\rho_t^{rS} \Gamma(t, T) + \sigma_t(S_t) \right) \sigma_t(S_t) \sigma'_t(S_t) dt$$

with $Y_0 = 1$. In order to compute $\sigma_t(x)$ approximately equation (11.6) can be modified using the approximation

$$\mathbb{E}_T \left[\sigma_t(S_t) \frac{Y_T}{Y_t} \middle| S_T = K \right] \approx \sigma_t(K) \quad (11.7)$$

leading to the implicit equation for $\sigma(K)$

$$\sigma_T(K)^2 \approx \bar{\sigma}_T(K)^2 - 2 \int_0^T \gamma(t, T) \Gamma(t, T) dt - 2 \int_0^T \rho_t^{rS} \gamma(t, T) \sigma_t(K) dt$$

A discrete-time version of this equation based on a time grid $t_0 = 0 < t_1 < t_2 < \dots < t_n = T$ reads

$$\begin{aligned} \sigma_{t_j}(K)^2 &\approx \bar{\sigma}_{t_j}(K)^2 - 2 \sum_{i=1}^j \int_{t_{i-1}}^{t_i} \gamma(t, t_j) \Gamma(t, t_j) dt - 2 \sum_{i=1}^j \int_{t_{i-1}}^{t_i} \rho_t^{rS} \sigma_t(K)^2 \gamma(t, t_j) dt \\ &\approx \bar{\sigma}_{t_j}(K)^2 - \sum_{i=1}^j \alpha_i - \sum_{i=0}^j \sigma_{t_i}(K) \beta_i \end{aligned} \quad (11.8)$$

for all K and $j = 1, 2, \dots, n$ with constants α_i, β_i not dependent on K . These equations can be solved through fixed point iteration or bootstrapping.

Fixed point iteration begins with the initial guess $\sigma^0 = \bar{\sigma}$. Surface σ^{k+1} is computed from surface σ^k via

$$\sigma_{t_j}^{k+1}(K)^2 = \bar{\sigma}_{t_j}^k(K)^2 - \sum_{i=1}^j \alpha_i - \sum_{i=0}^j \sigma_{t_i}^k(K) \beta_i$$

for all K and $j = 1, 2, \dots, n$.

Bootstrapping is based on the quadratic equation

$$\sigma_{t_j}(K)^2 + \sigma_{t_j}(K) \beta_j = \bar{\sigma}_{t_j}(K)^2 - \sum_{i=1}^j \alpha_i - \sum_{i=0}^{j-1} \sigma_{t_i}(K) \beta_i =: A(j, K)$$

with positive solution

$$\sigma(K, t_j) = -\frac{\beta_j}{2} + \frac{1}{2} \sqrt{\beta_j^2 - A(j, K)}$$

The bootstrapping procedure proceeds forward in time as $A(j, K)$ depends on $\sigma_{t_i}(K)$, $i < j$.

An alternative to the approximation (11.7) attempts to compute

$$\mathbb{E} \left[B_T^{-1} (r_T - \bar{r}_T) \mathbf{1}_{\{S_T > K\}} \right] = P(T) \mathbb{E}_T \left[r_T \mathbf{1}_{\{S_T > K\}} \right] + \bar{r}_T \frac{\partial C(K, t)}{\partial K}$$

in (11.5) directly via copula methods. In the Hull-White case $(r_T, \log B_T)$ is Gaussian. Following the Hull-White hybrid copula model in Section 11.2 one obtains

$$\mathbb{E}_T \left[r_T \mathbf{1}_{\{S_T > K\}} \right] = \rho \sqrt{\text{Var} [r_T]} \varphi(D_T^{-1}(K))$$

Reghai (2006) suggests a fixed point algorithm in the same spirit, as described in Section 6.1.4. Observing that in the case of volatility term structure implied volatility is given as

$$\hat{\sigma}(K, T)^2 T = \int_0^T (\sigma_t^2 + \Gamma_t(T)^2 + 2\rho_t \sigma_t \Gamma_t(T)) dt$$

one has an iterative scheme for equity mean and log-normal variance at time t , given $S_T = K$. Note that one does not condition on r_T or B_T . This defines an algorithm Φ computing $\hat{\sigma}$ from equity local volatility $\sigma_t(K)$ in the hybrid case. Inverting Φ using fixed point method yields the result.

Finally Ren et al. (2007) suggest solving the corresponding multi-dimensional forward equation for density to recover local volatility σ in the same way as proposed for the stochastic local volatility model, introduced in Section (6.3).

11.4 Stochastic equity volatility

Stochastic volatility models can also be extended to the case of stochastic interest rates. This section focuses on dynamics of type

$$\begin{aligned} \frac{dS_t}{S_t} &= (r_t - d_t)dt + \sigma_t dW_t^S \\ d\sigma_t &= \mu_t^\sigma(\sigma_t)dt + \sigma_t^\sigma(\sigma_t)dW_t^\sigma \\ dr_t &= \mu_t^r(r_t)dt + \sigma_t^r(r_t)dW_t^r \end{aligned}$$

with deterministic functions $d_t, \mu_t^\sigma(s), \sigma_t^\sigma(s), \mu_t^r(r), \sigma_t^r(r)$ driven by three correlated Brownian motions, W^S, W^σ, W^r . The solution can be written as

$$S_t = P(t)F(t)B_tM_t$$

with deterministic forward $F(t)$ and discount factor $P(t)$, cash bond B and a process M , which is a martingale with respect to numéraire B , namely

$$M_T = \exp \left(\int_0^T \sigma_t dW_t^S - \frac{1}{2} \int_0^T \sigma_t^2 dt \right)$$

11.4.1 Independence

Assuming independence between W^r and (W^S, W^σ) results in independence of B and M . Thus the Laplace transform applied to the T -forward measure has the form

$$\mathbb{E}_T \left[e^{-\lambda \log S_T} \right] = e^{-\lambda \log(P(t)F(t))} \mathbb{E}_T \left[e^{-\lambda \log B_T} \right] \mathbb{E} \left[e^{-\lambda \log M_T} \right]$$

As a result, given the Laplace transform for the stochastic volatility model with deterministic interest rates, one can extend the analysis to the hybrid case if the transform for the cash bond is known.

11.4.2 Heston

Heston's model is affine in $(\log S_t, \sigma_t^2)$. Assuming, Vasicek or Cox-Ingersoll-Ross short rate process r_t , the combined system $(\log S_t, \sigma_t^2, r_t)$ is only affine if r is independent of the Brownians (W^S, W^σ) driving S and σ . To allow for correlation between rates and stock while maintaining an affine system Giese (2006) and Andreasen (2006) suggest a coupling factor σ_{Sr} , namely

$$\begin{aligned} \frac{dS_t}{S_t} &= (r_t - d_t)dt + \sigma_t dW_t^S + \sigma_{Sr} dW_t^r \\ d\sigma_t^2 &= \kappa^\sigma (\theta^\sigma - \sigma_t^2)dt + \sigma^\sigma \sigma_t dW_t^\sigma \\ dr_t &= \kappa^r (\theta^r - r_t)dt + \sigma^r r_t^p dW_t^r \end{aligned}$$

while still assuming independence between (W^S, W^σ) and W^r . Here $p = 0$ and $p = \frac{1}{2}$ lead to Vasicek and CIR rates, respectively. The dynamics of $F_t(T) = S_t/P_t(T)$ is

$$\begin{aligned} \frac{dF_t(T)}{F_t(T)} &= ((\sigma^r)^2 B^2(t, T)) dt + \sigma_t dW_t^S + (\sigma_{Sr} + \sigma^r B(t, T)) dW_t^r \\ d\sigma_t &= \kappa (\theta - \sigma_t^r) dt + \sigma^\sigma dW_t^\sigma \end{aligned}$$

With respect to the T -forward measure one obtains

$$\begin{aligned} \frac{dF_t(T)}{F_t(T)} &= \sigma_t dW_t^S + (\sigma_{Sr} + \sigma^r B(t, T)) dW_t^{T,r} \\ d\sigma_t &= \kappa (\theta - \sigma_t^r) dt + \sigma^\sigma dW_t^\sigma \end{aligned}$$

where

$$dW_t^r = dW_t^{T,r} - \sigma^r B(t, T)dt$$

Thus, with respect to the T -forward measure, $F(T)$ is the product of two independent processes, namely a standard Heston and a log-normal process with volatility term structure $\sigma^r B(t, T)$.

This model leads to stochastic instantaneous correlation

$$\rho_t^{Sr} = \frac{\sigma_{Sr} r_t^p}{\sqrt{\sigma_t^2 + \sigma_{Sr}^2 r_t^{2p}}}$$

Solving for the coupling factor yields

$$\sigma_{Sr} = \frac{\rho_t^{Sr} \sigma_t}{r_t^p \sqrt{1 - (\rho_t^{Sr})^2}}$$

To obtain an average instantaneous correlation $\bar{\rho}_{Sr}$ for a time horizon T Giese (2006) suggests the approximation

$$\sigma_{Sr} \approx \frac{\bar{\rho}^{Sr} \mathbb{E} \left[\sqrt{\frac{1}{T} \int_0^T \sigma_t^2 dt} \right]}{\mathbb{E} \left[\sqrt{\frac{1}{T} \int_0^T r_t^p dt} \right] \sqrt{1 - (\bar{\rho}^{Sr})^2}}$$

An approximation for the two expectations is discussed in Section 3.7.3. A detailed discussion of this model can be found in Grzelak and Oosterlee (2011).

11.4.3 Schöbel-Zhu

Van Haastrecht et al. (2009) extend the Schöbel-Zhu model to allow for Hull-White interest rates, as introduced in Section 10.4. This is achieved by considering the T -forward stock $F_t(T) = S_t/P_t(T)$ with dynamics (ignoring dividends)

$$\begin{aligned} \frac{dF_t(T)}{F_t(T)} &= \left((\sigma^r)^2 B^2(t, T) + \rho^{rS} \sigma_t \sigma^r B(t, T) \right) dt + \sigma_t dW_t^S + \sigma^r B(t, T) dW_t^r \\ d\sigma_t &= (\theta^\sigma - \kappa^\sigma \sigma_t^\sigma) dt + \sigma^\sigma dW_t^\sigma \end{aligned}$$

With respect to the T -forward measure one has the two-dimensional system

$$\begin{aligned} \frac{dF_t(T)}{F_t(T)} &= \sigma_t dW_t^{T,S} + \sigma^r B(t, T) dW_t^{T,r} \\ d\sigma_t &= \left(\theta^\sigma + \rho^{rS} \sigma^\sigma \sigma^r B(t, T) - \kappa^\sigma \sigma_t^\sigma \right) dt + \sigma^\sigma dW_t^{T,\sigma} \end{aligned}$$

where

$$\begin{aligned}dW_t^r &= dW_t^{T,r} - \sigma^r B(t, T)dt \\dW_t^S &= dW_t^{T,S} - \rho^{rS} \sigma^r B(t, T)dt \\dW_t^\sigma &= dW_t^{T,\sigma} - \rho^{r\sigma} \sigma^r B(t, T)dt\end{aligned}$$

This system is of the same type as the standard Schöbel-Zhu model. Hence the process $(\log F_t(T), \sigma_t, \sigma_t^2)$ is affine leading to equations of type (16.3) which can be solved, see van Haastrecht et al. (2009).

11.4.4 Multi-factor models with stochastic interest rates

Gnoatto and Grasselli (2014) extend stochastic volatility models based on Wishart processes to support stochastic interest rates as well. The short rate is assumed to satisfy

$$r_t = h + \text{trace}(H\Sigma_t)$$

where H is positive definite and symmetric. This allows one to specify a multi-currency framework with stochastic short rates and foreign exchange rates with stochastic volatility. Exchange rates X^i relative to a reference currency with interest r^0 satisfy

$$\frac{dX_t^i}{X_t^i} = (r_t^0 - r_t^i)dt - \text{trace}(A_i\sqrt{\Sigma_t}dW_t^X)$$

In the same measure the exchange rate between currency i and j $X^{ij} = X^j/X^i$ satisfies

$$\frac{dX_t^{ij}}{X_t^{ij}} = (r_t^i - r_t^j + \text{trace}((A_i - A_j)\Sigma_t A_i))dt + \text{trace}((A_i - A_j)\sqrt{\Sigma_t}dW_t^X)$$

11.5 Dynamic hedging of variance swaps

11.5.1 Deterministic interest rates

Variance swaps, including their semi-static replication as a sum of a static and a dynamic position, have been discussed in Section 3.7.1. A hedging strategy for the dynamic position $\frac{dS_t}{S_t}$ is given below, assuming deterministic interest rates and no dividends. The corresponding strategy with stochastic interest rates is discussed

subsequently. The effect of dividends on variance swaps is discussed in Section 5.8. A strategy investing in ϕ^S units of stock S and in ϕ^B units of cash bond B is self-financing if

$$dX_t = \phi_t^S dS_t + r_t \left(X_t - \phi_t^S S_t \right) dt = \phi_t^S dS_t + \left(X_t - \phi_t^S S_t \right) \frac{dB_t}{B_t}$$

holds. One has

$$d \frac{X_t}{B_t} = \frac{\phi_t^S S_t}{B_t} \sigma_t^S dW_t^S$$

Let interest rates be deterministic. Then

$$\mathbb{E} \left[\int_0^T \frac{dS_u}{S_u} \middle| \mathcal{F}_t \right] = \int_0^t \frac{dS_u}{S_u} + \int_t^T r_u du$$

and

$$d\mathbb{E} \left[\int_0^T \frac{dS_u}{S_u} \middle| \mathcal{F}_t \right] = \frac{dS_t}{S_t} - r_t dt = \sigma_t^S dW_t^S$$

Setting $\phi_t^S = \frac{B_t}{S_t}$ leads to

$$\frac{X_T}{B_T} - \frac{X_0}{B_0} = \int_0^T \frac{dS_t}{S_t} - \mathbb{E} \left[\int_0^T \frac{dS_t}{S_t} \right]$$

Dividing ϕ^S and ϕ^B by B_T and setting X_0 equal to the expectation on the right hand side, namely

$$\begin{aligned} \phi_t^S &= \frac{B_t}{B_T S_t} \\ \phi_t^B &= \frac{1}{B_T} \left(\frac{X_t}{B_t} - 1 \right) \\ X_0 &= \frac{1}{B_T} \mathbb{E} \left[\int_0^T \frac{dS_t}{S_t} \right] = \frac{1}{B_T} \int_0^T r_t dt = -P(T) \log P(T) \end{aligned}$$

satisfies the hedging requirement

$$X_T = \int_0^T \frac{dS_t}{S_t}$$

11.5.2 Stochastic interest rates

The hedging strategy for $\frac{dS_t}{S_t}$ in the case of stochastic interest rates has been discussed by Hörfelt and Torné (2010). Let the bond $P(T)$ be driven by Brownian motion W^P and have volatility $\sigma^P(T)$. A self-financing strategy investing in ϕ^S units of stock S and in ϕ^P units of bond $P(T)$ is self-financing if

$$dX_t = \phi_t^S dS_t + \phi_t^P dP_t(T) + r_t \left(X_t - \phi_t^S S_t - \phi_t^P P_t(T) \right) dt$$

holds. One has

$$d \frac{X_t}{P_t(T)} = \frac{\phi_t^S S_t}{P_t(T)} \sigma_t^S dW_t^S + \left(\phi_t^P - \frac{X_t}{P_t(T)} \right) \sigma_t^P(T) dW_t^P$$

where W^S and W^P are Brownian motions with respect to the T -forward measure. On the other hand

$$\mathbb{E}_T \left[\int_0^T \frac{dS_u}{S_u} \middle| \mathcal{F}_t \right] = \int_0^t \frac{dS_u}{S_u} + \mathbb{E}_T \left[\int_t^T \left(r_u + \rho_u \sigma_u^P(T) \sigma_u^S \right) du \middle| \mathcal{F}_t \right]$$

Due to (10.1) one has

$$\begin{aligned} d\mathbb{E}_T \left[\int_0^T \frac{dS_u}{S_u} \middle| \mathcal{F}_t \right] &= \frac{dS_t}{S_t} - d\log P_t(T) + (\dots) dt \\ &= \sigma_t^S dW_t^S - \sigma_t^P(T) dW_t^P \end{aligned}$$

since the drift term has to vanish. This leads to the hedging strategy

$$\begin{aligned} \phi_t^S &= \frac{P_t(T)}{S_t} \\ \phi_t^P &= \frac{X_t}{P_t(T)} - 1 \end{aligned}$$

Again due to (10.1) the initial cost to set up the hedge is

$$X_0 = -\log P(T) + \mathbb{E}_T \left[\int_0^T \sigma_u^P(T) \left(\rho_u \sigma_u^S - \frac{1}{2} \sigma_u^P(T) \right) du \right]$$

which is equal to (3.8).

12] Credit

Default can impact a specific trade between two counterparties in three ways, through default of either counterparty as well as default of a firm related to the underlyings. In this section the focus is on modelling default times in the context of credit derivatives. Default of the underlying in the case of equity derivatives is discussed in Chapter 13, while counterparty default is covered in Chapter 14.

12.1 Market

12.1.1 Default and recovery

Default time τ and recovery rate R are defined in terms of

- default event: bankruptcy, failure to pay, restructuring
- reference entity: firm name
- reference asset: bond, loan including seniority

For standard contracts such as credit default swaps (CDS) the determination of default time including settlement is described in the corresponding term sheet and will typically be based on standard documentation as provided by ISDA.

The actual recovery rate can only be determined after the event. Often R is set to a constant, such as 40% or 30%, derived from empirical data. If random recovery is required a beta distribution can be used.

The market consensus on survival probabilities

$$p(t) = \mathbb{P} [\tau > t]$$

can be derived from bond spreads or credit default swaps if the recovery rate is given. Where many default events are considered default correlations can be inferred to some extent from portfolio structures, such as first-to-default (FtD), n -th-to-default and collateralized debt obligation (CDO).

12.1.2 Bond spreads

Consider a risky zero coupon bond maturing at T with recovery rate R promising to pay one currency unit. In the case of default at time $\tau \leq T$ the recovered amount is R times the value of a corresponding default free bond. Thus the fair value of the risky bond is $\text{IE}[X]$ where

$$X = P_0(T) (\mathbf{1}_{\{\tau > T\}} + R\mathbf{1}_{\{\tau \leq T\}})$$

This expression covers the case of stochastic interest rates if IP is the T -forward measure. Taking expectations yields

$$\text{IE}[X] = P_0(T) (1 - (1 - R)\text{IP}[\tau < T])$$

The bond spread $s(T)$ is defined as

$$\text{IE}[X] = P_0(T)e^{-s(T)T}$$

Therefore

$$s(T) = -\log \frac{1 - (1 - R)(1 - p(T))}{T} \quad (12.1)$$

12.1.3 Credit default swap

A CDS consists of two legs, referred to as premium and default. The premium leg represents regular insurance premium payments payable until the earlier of maturity or default time. The default leg makes good losses on specified assets in case of default within the protected time period. The legs are defined by

- premium leg: payment dates and amounts, size and payment date of accrued coupon
- default leg: start and end of protection, payment delay in case of default.

A CDS can be physically settled or cash settled:

- physically settled: upon default the protection buyer receives the notional N of the reference assets and gives up those assets. The cost for the protection seller is of the order of the non recoverable fraction $N(1 - R)$. The recovery rate R is not fixed in the contract.
- cash settled: the reference asset is used to determine the recovery rate. The amount paid by the protection seller in case of default is $N(1 - R)$.

A CDS is similar to a portfolio consisting of a riskless coupon bond (short) and a risky coupon bond (long), both with notional N and trading at par. The higher coupon received on the risky bond reflects its financial risk.

The mark-to-market value V_t of an idealized CDS at t maturing at T is (assuming no default by time t) given by the formula

$$V_t = Ns \sum_{i: t_i \in (t, T]} p_t(t_i) P_t(t_i) (t_i - t_{i-1}) - N(1 - R) \int_t^T P_t(u) \left(-\frac{\partial}{\partial u} p_t(u) \right) du \quad (12.2)$$

where $P_t(u)$ and $p_t(u)$ denote a risk free discount factor and survival probability from today t to u , respectively. Independence of p and P is assumed. At inception $t = 0$ the spread $s = s(T)$ is chosen such that $V_0 = 0$.

The survival probability curve ($p(t), t \geq 0$) can be bootstrapped from a set of CDS par spreads $s(T_i)$ if assumptions determining the interpolation of curve points between time points T_i are specified. A good choice is log-linear interpolation, since extrapolation from T_i to T_{i+1} does not change the curve on $[0, T_i]$. This does not hold for spline interpolation. Alternatively, one can fit a curve from the Nelson-Siegel family.

Assuming constant interest rates r and a continuous premium payment stream of sdt instead of discrete payments of $s(t_i - t_{i-1})$ at times t_i one has

$$V_0 = \mathbb{E}[c_T]$$

with

$$\begin{aligned} c_T &= s \int_0^{\tau \wedge T} e^{-rt} dt - (1 - R)e^{-r\tau} \mathbf{1}_{\{\tau < T\}} \\ &= \frac{s}{r} \left(1 - e^{-r\tau} \mathbf{1}_{\{\tau < T\}} - e^{-rT} \mathbf{1}_{\{\tau > T\}} \right) - (1 - R)e^{-r\tau} \mathbf{1}_{\{\tau < T\}} \\ &= \frac{s}{r} - \left(\frac{s}{r} + 1 - R \right) e^{-r\tau} \mathbf{1}_{\{\tau < T\}} - \frac{s}{r} e^{-rT} \mathbf{1}_{\{\tau > T\}} \end{aligned}$$

The special case $r = 0$ yields

$$\begin{aligned} c_T &= s(\tau \wedge T) - (1 - R)\mathbf{1}_{\{\tau < T\}} \\ &= sT\mathbf{1}_{\{\tau > T\}} + (s\tau - (1 - R))\mathbf{1}_{\{\tau < T\}} \end{aligned}$$

Taking expectations one has

$$\mathbb{E}[c_T] = \frac{s}{r} - \left(\frac{s}{r} + 1 - R \right) \mathbb{E}[e^{-r\tau} \mathbf{1}_{\{\tau < T\}}] - \frac{s}{r} e^{-rT} \mathbb{P}[\tau > T]$$

and

$$\mathbb{E}[c_T] = (sT + 1 - R)\mathbb{IP}[\tau > T] + s\mathbb{E}[\tau \mathbf{1}_{\{\tau < T\}}] - 1 + R$$

for $r = 0$. The par spread $s(T)$ is defined such that $\mathbb{E}[c_T] = 0$, that is

$$s(T) = \frac{r(1 - R)\mathbb{E}[e^{-r\tau} \mathbf{1}_{\{\tau < T\}}]}{1 - \mathbb{E}[e^{-r\tau} \mathbf{1}_{\{\tau < T\}}] - e^{-rT}\mathbb{IP}[\tau > T]} \quad (12.3)$$

for $r > 0$. For $r = 0$ one obtains

$$s(T) = \frac{(1 - R)\mathbb{IP}[\tau < T]}{\mathbb{E}[\tau \mathbf{1}_{\{\tau < T\}}] + T\mathbb{IP}[\tau > T]} \quad (12.4)$$

Formulae (12.3) and (12.4) can be used if survival probabilities are given exogenously.

12.2 Reduced form models

The hazard rate $h(t)$ describes the probability of default in $[t, t + dt]$, given no default before t :

$$h(t)dt = \frac{p(t) - p(t + dt)}{p(t)}$$

In other words,

$$h(t) = -\frac{\partial \log p(t)}{\partial t} \quad \text{or} \quad p(t) = \exp\left(-\int_0^t h(u)du\right)$$

Denote the spread of a CDS maturing at T by $s(T)$. If spreads $s(T)$ are large relative to r one has qualitatively

$$\frac{\partial}{\partial T}(s(T)T) \approx (1 - R)h(T)$$

If both hazard rate h and interest rate r are constant then

$$h = \frac{s}{1 - R} \quad \text{and} \quad s = \frac{R - 1}{T} \log p(T)$$

hold for the CDS with a continuous premium stream. This can be derived from (12.3) or (12.4). Models based on a hazard rate are called reduced form models.

12.2.1 Stochastic hazard rates

In the context of credit derivatives credit spreads are often modelled as stochastic processes. This leads to a survival probability curve dynamics similar to the yield curve dynamics derived from interest rate dynamics, see Chapter 10. Given a survival probability curve $p_t(T)$ known at time t the corresponding forward hazard rates for time T as observed at t satisfy

$$h_t(T) = -\frac{\partial \log p_t(T)}{\partial T} \quad \text{and} \quad p_t(T) = \exp\left(-\int_t^T h_t(u) du\right)$$

Hazard rates may be specified as stochastic process h_t . Consistency with survival probabilities $p(t)$ requires

$$p(T) = p_0(T) = \mathbb{E}\left[e^{-\int_0^T h_t dt}\right]$$

analogous to the consistency between short rates r_t and discount factors $P_0(T)$. Common hazard rate dynamics include CIR dynamics, see Section 10.5. Gaussian hazard rates as in the Hull-White model from Section 10.4 have the disadvantage that they can become negative.

If interest rates r are stochastic but independent from h then calibration to CDS data or risky bonds follows from consistency with today's survival curve. For the case of correlated Hull-White hazard and interest rates Brigo and Alfonsi (2005) compute the moments of the Gaussian vector

$$\left(\int_0^T (r_t + h_t) dt, h_T\right)$$

allowing them to obtain risky discount factors $\mathbb{E}\left[\exp\left(-\int_0^t (r_u + h_u) du\right)\right]$ as well as CDS spreads $s(T)$ via

$$s(T) = (1 - R) \frac{\int_0^T \mathbb{E}\left[h_t \exp\left(-\int_0^t (r_u + h_u) du\right)\right] dt}{\int_0^T \mathbb{E}\left[\exp\left(-\int_0^t (r_u + h_u) du\right)\right] dt}$$

For the correlated CIR case they suggest using bivariate Hull-White with the same starting point, drift and correlation as a proxy. Volatility of r in the proxy model is chosen such that the expectation of

$$\exp\left(\int_0^T r_t dt\right)$$

is the same in both models and analogously for h instead of r .

12.3 Structural models

12.3.1 Merton's model

Merton's asset value model (1974) is a one-period model for the capital structure of a firm defined by its assets and debts. It formulates a link between default probability, recovery rate and stock price (including volatility smile) from asset volatility and debt to asset ratio and is the basis for many important models, including Li's Gauss copula model for default times, see Section 12.3.6, counterparty credit risk models with correlated default, see Chapter 14, and the CreditGrades model, see Section 12.3.3.

At time 0 a loan L is made to the firm, which in turn promises to pay back debt B at time T . Default happens if the firm's asset values A_T at time T fall below B . In that case only A_T can be recovered by the lenders. This is reflected in the loan amount L given to the firm in the first place, namely

$$L = P_0(T) \mathbb{E} [A_T \wedge B] < P_0(T) B$$

where $P_0(T)$ is the riskless discount factor from today to T . The survival probability $p(T)$ is

$$p(T) = \mathbb{IP} [A_T > B] \quad (12.5)$$

and the bond spread $s(T)$ satisfies

$$e^{-s(T)T} B = \mathbb{E} [A_T \wedge B] \quad (12.6)$$

The recovery rate R is the fraction of B which can be recovered in case of default

$$RB = \mathbb{E} [A_T | A_T < B] \quad (12.7)$$

From the lenders' point of view the expected shortfall is the value of a put option on A with strike B . One has

$$\begin{aligned} \mathbb{E} [(B - A_T)^+] &= \mathbb{IP} [A_T < B] \mathbb{E} [(B - A_T)^+ | A_T < B] \\ &= (1 - p(T)) (B - \mathbb{E} [A_T | A_T < B]) \\ &= (1 - p(T)) (1 - R) B \end{aligned}$$

Thus expected shortfall is the non recoverable fraction times default probability.

Assuming log-normal assets with mean A_0 and volatility σ_A all quantities can be computed in closed form. From (12.5) one has

$$p(T) = \Phi(d_-) \quad (12.8)$$

with

$$d_{\pm} = \frac{1}{\sigma_A \sqrt{T}} \left(\log \left(\frac{A_0}{B} \right) \pm \frac{1}{2} \sigma_A^2 T \right)$$

Also (12.6) and (12.7) imply

$$e^{-s(T)T} = \frac{A_0}{B} \Phi(-d_+) + \Phi(d_-) \quad (12.9)$$

and

$$R = \frac{A_0}{B} \frac{\Phi(-d_+)}{\Phi(-d_-)}$$

respectively. Recovery is typically close to 100%. Assuming constant interest r and (12.8) for all T then one can also compute CDS spread term structure using (12.3) or (12.4), as well as (18.30). This is somewhat inconsistent since recovery R is assumed constant in the first two formulae. Due to the continuous nature of the asset process one obtains zero credit spreads for short maturities T .

Shareholders own a call option on the assets with strike B :

$$S_t = P_t(T) \mathbb{E}_t \left[(A_T - B)^+ \right] = \text{Call}_t(A_t) \quad (12.10)$$

In the case of log-normal assets this allows one to derive a relationship between asset volatility and equity volatility:

$$dS_t = (\dots)dt + \sigma_t^S S_t dW_t^S = (\dots)dt + \frac{\partial \text{Call}_t(A_t)}{\partial A} \sigma_t^A A_t dW_t^A$$

Hence, if the option delta is approximately one and the value at time t of the loan close to B , then

$$\sigma_t^S = \sigma_t^A \frac{A_t \frac{\partial \text{Call}_t(A_t)}{\partial A}}{S_t} \approx \sigma_t^A \frac{S_t + B}{S_t} \quad (12.11)$$

This yields the well known equity volatility smile and is the basis of the CreditGrades model introduced by Finkelstein et al. Avoiding the approximation in (12.11) an equity call is, in fact, an option on an option:

$$\text{Call}_0(S_0, K, t) = P_0(t) \mathbb{E} \left[(S_t - K)^+ \right] = P_0(t) \mathbb{E} \left[(\text{Call}_t(A_t, B, T) - K)^+ \right]$$

The semi-analytical solution (with log-normal A) is due to Geske (1979) and reads

$$\begin{aligned} Call_0(S_0, K, t) = & A_0 \Phi_2 \left(d_+^*, d_+; \sqrt{\frac{t}{T}} \right) \\ & - D(T) B \Phi_2 \left(d_-^*, d_-; \sqrt{\frac{t}{T}} \right) - D(t) K \Phi(d_-^*) \end{aligned} \quad (12.12)$$

Here $d_{\pm}, d_{\pm}^*$ are related to the moneyness and term $A_0/B, T$ and $A_*/K, t$, respectively, with A_* denoting the critical value of A_t such that

$$Call_t(A_*, B, T) = K$$

The volatility surface implied by this model depends on asset volatility, loan ratio and the due date of the loan. Starting from an implied volatility surface it is in fact possible to make statements on the capital structure of the firm (asset volatility, loan ratio). This has been discussed in Hull et al. (2004). when choosing two liquid option prices, namely 25% and -25% delta put options maturing in two months, they solve equation (12.12) for loan ratio and asset volatility, given a term T . Repeating this exercise with different terms T leads to a credit spread term structure ($s(T)$) using formula (12.9). This model predicts a positive relationship between credit spreads and at-the-money volatilities and between credit spreads and volatility skew.

12.3.2 Merton's model with continuous barrier

A generalization to a continuous setting consists in defining default time τ in terms of an asset value A hitting a lower barrier, see Black and Cox (2004):

$$\tau = \inf\{t \geq 0 : A_t \leq B\}$$

For a log-normal asset one obtains from (18.22)

$$p(t) = \mathbb{P}[\tau > t] = \Phi(d_-) - \sqrt{\frac{A_0}{B}} \Phi(-d_+)$$

with

$$d_{\pm} = \frac{1}{\sigma_A \sqrt{t}} \log \frac{A_0}{B} \pm \frac{\sigma_A \sqrt{t}}{2}$$

As in the case of a European barrier only live at maturity, one observes zero credit spreads for short maturities. In order to remedy this problem various modifications have been suggested:

- A time-dependent barrier (B_t) consistent with given survival probabilities ($p(t)$) can be computed numerically, see Hull and White (2000).
- Modelling the asset process A as a jump diffusion process as proposed by Zhou (2001) allows one to realistically calibrate both credit spreads and equity options. One disadvantage of the model is its lack of tractability. This problem can be removed by assuming that any jump triggers default. This approach has been pursued in CSFB's proprietary CUSP model.
- A random barrier has been introduced by Finkelstein et al. within the CreditGrades model, as described in the next section.
- Garreau and Kercheval (2014) model assets as exponential Lévy processes. Default time is defined as the first hitting time of a constant barrier. They show that the resulting model is a reduced form model where the hazard rate is related to the Lévy measure of the asset process.

12.3.3 CreditGrades

A remedy for zero credit spread for short maturities has been proposed by Finkelstein et al. The context is again a log-normal asset process A and a continuous barrier, as in Black and Cox (2004). The basic idea is that the default barrier is assumed to be random. This reflects the intuition that an investor does not have reliable information on the capital structure of a firm and thus on factors determining the exact timing of a default event. It is natural to start with a log-normal barrier

$$B = \bar{B}e^{\lambda Z - \lambda^2/2}$$

where the standard Gaussian Z is assumed to be independent of the asset process A . This model has been copyrighted under the name CreditGrades. The default time is again defined as

$$\tau = \inf\{t \geq 0 : A_t \leq B\}$$

Mathematically the survival probability is given by a partial barrier formula due to

$$\begin{aligned} p(T) &= \mathbb{IP}[\forall t \in [0, T] : A_t > B] \\ &= \mathbb{IP}\left[\forall t \in [0, T] : A_0 e^{\sigma_A W_t - \frac{1}{2}(\sigma_A^2 t + \lambda^2)} > \bar{B}e^{-\lambda^2}\right] \\ &= \mathbb{IP}\left[\forall t \in \left[\frac{\lambda^2}{\sigma_A^2}, T + \frac{\lambda^2}{\sigma_A^2}\right] : A_0 e^{\sigma_A W_t - \frac{1}{2}\sigma_A^2 t} > \bar{B}e^{-\lambda^2}\right] \end{aligned}$$

For $\lambda \neq 0$ one has the puzzling effect of $p(0) < 1$, which could be stated as ‘waking up and being already dead’. From (18.32) with parameters

$$v = -\frac{\sigma_A}{2} \quad t = \frac{\lambda^2}{\sigma_A^2} \quad T = T + \frac{\lambda^2}{\sigma_A^2} \quad x = \frac{1}{\sigma_A} \left(\log \frac{A_0}{B} + \lambda^2 \right)$$

the hitting probability of the partial barrier is given as

$$p(T) = \Phi_2 \left(b - \frac{\lambda}{2}, b - \frac{\sigma_T}{2}; \frac{\lambda}{\sigma_T} \right) - e^{\lambda b} \Phi_2 \left(b + \frac{\lambda}{2}, -b - \frac{\sigma_T}{2}; -\frac{\lambda}{\sigma_T} \right)$$

$$b = \lambda + \frac{\log \frac{A_0}{B}}{\lambda}, \quad \sigma_T^2 = \sigma_A^2 T + \lambda^2 \quad (12.13)$$

In practice one uses as approximation the simpler full barrier formula

$$p(T) \approx \Phi \left(\frac{b\lambda}{\sigma_T} - \frac{\sigma_T}{2} \right) - e^{\lambda b} \Phi \left(-\frac{b\lambda}{\sigma_T} - \frac{\sigma_T}{2} \right) \quad (12.14)$$

obtained from (18.22) with the same parameters.

From survival probabilities given in (12.13) or (12.14) one can derive bond spreads in connection with formula (12.1). CDS spreads can also be computed using formulae (12.3) or (12.4) using (18.31). Figure 12.1 displays a CDS spread term structure for various levels of B and λ , recovery at 30% and $\sigma_A = 20\%$.

There are standard methods for obtaining debt per share D from published data. Expected recovery on debt for all the firm's assets is typically set to $\mathbb{E}[L] = 50\%$, with volatility $\lambda = 30\%$. The default barrier is given as $B = LD$. Parameters for the firm's assets (per share) A_0 and σ_A are not observable. They are fixed as

$$A_0 = S_0 + \bar{B}, \quad \sigma_A = \sigma_S^* \frac{S^*}{S^* + \bar{B}}$$

with S^* and σ_S^* denoting reference share price and volatility. The motivation is based on Merton's model, compare formulae (12.10) and (12.11). The stock level S^* is known and volatility σ_S^* can be inferred from empirical data or the Vanilla market. Recovery R depends on the seniority of the bond. Typical values are between $R = 30\%$ and $R = 50\%$. This allows one to determine credit spreads $s(T)$ for specific bonds or CDS from market data.

The model explains part of the volatility smile as due to credit risk:

$$\sigma_S(S^*) = \sigma_A \frac{S^* + \bar{B}}{S^*}$$

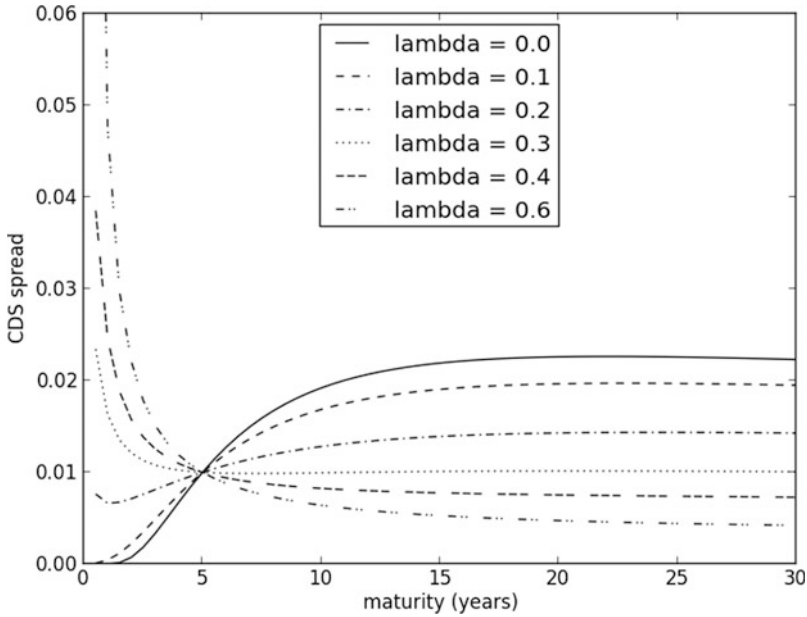


Figure 12.1 Credit spread with CreditGrades default barrier

However, the model is not rich enough to match the Vanilla market. In particular it does not reflect the term structure of volatility level and skew. Note that the process A is not equivalent to the firm's value. The formulae serve to quote spreads in the same way as the Black-Scholes formula serves to quote Vanilla prices.

12.3.4 Merton's model for many periods

Merton's one-period structural model states that default at t is equivalent to the value of a firm's assets A_t falling below a barrier level B_t . In practice, a term structure of survival probabilities ($p(t), t \geq 0$) can be derived from CDS data. Assuming log-normal A_t this allows one to compute B_t such that

$$q(t) = 1 - p(t) = \mathbb{P}[A_t \leq B_t],$$

namely

$$B_t = F_{A_t}^{-1}(q(t)) = A_0 \exp \left(\sigma_A \sqrt{t} \Phi^{-1}(q(t)) - \frac{\sigma_A^2 t}{2} \right)$$

where F_{A_t} is the distribution function of A_t . Note that this is an inconsistent default definition since the events

$$\{A_t \leq B_t\} = \{\text{default before } t\}$$

do not increase with t . This problem can be solved by setting $A_t = A$ for all t and treating the random variable A as a hidden variable. With default time τ defined as

$$\tau = q^{-1}(F_A(A)) \quad (12.15)$$

one obtains

$$\{A \leq B_t\} = \{\tau \leq t\}$$

12.3.5 Merton's model for many assets

If two assets A^1, A^2 are considered, it is natural to assume some correlation ρ between them. One assumes default of asset i before time t if $A_t^i < B_t^i$. This allows one to compute the probability of the event

$$\{\text{both assets have defaulted before } t\}$$

It turns out that the probability of that event is given by

$$\mathbb{P}[A_t^1 \leq B_t^1, A_t^2 \leq B_t^2] = \Phi_2(\Phi^{-1}(q^1(t)), \Phi^{-1}(q^2(t)); \rho)$$

where $\Phi_2(x, y; \rho)$ is the bivariate cumulative Gaussian distribution function. This defines the distribution function of the pair of default times (τ^1, τ^2) on the diagonal $t \mapsto (t, t)$ as

$$\mathbb{P}[\tau^1 \leq t, \tau^2 \leq t] = \Phi_2(\Phi^{-1}(q^1(t)), \Phi^{-1}(q^2(t)); \rho)$$

12.3.6 Li's model

An obvious extension of this definition to off-diagonal events is given by

$$\mathbb{P}[\tau^1 \leq s, \tau^2 \leq t] = \Phi_2(\Phi^{-1}(q^1(t)), \Phi^{-1}(q^2(t)); \rho)$$

While this is not very interesting for events related to the defaults for a given time horizon t , it does lead to an algorithm for the simulation of default times. Let

(X_1, X_2) denote a pair of random variables with distribution $(x, y) \mapsto \Phi_2(x, y; \rho)$. Then the pair

$$(\tau^1, \tau^2) = \left((q^1)^{-1}(\Phi(X_1)), (q^2)^{-1}(\Phi(X_2)) \right) \quad (12.16)$$

has the above distribution. This approach can be easily extended to many default times. The copula (see Chapter 9) of the joint distribution of the default times is Gaussian. Definition (12.16) is consistent with (12.15).

12.4 Portfolio credit derivatives

12.4.1 First-to-default

Fix a portfolio of K names with default times $\tau_1, \tau_2, \dots, \tau_K$ and notionals $N_1, N_2, \dots, N_K$. A first-to-default (FtD) contract is similar to a CDS:

- The protection seller receives a fixed coupon quarterly, until default happens.
- In case of default, the protection seller has to cover the losses of the protection buyer and the contract ends.

The main difference is the definition of default:

- The default time τ^{FtD} is the first default of a given set of K names:

$$\tau^{FtD} = \min_{k=1,2,\dots,K} \tau_k.$$

- If name k defaults first, i.e. $\tau^{FtD} = \tau_k$, then the losses of name k are covered by the protection seller. This amount is given by $N_k(1 - R_k)$, the not recoverable part of the notional N_k of name k in the portfolio.

Similarly, there is an n -th-to-default. Note that the sum of all n -th-to-default contracts is equivalent to the sum of all CDS contracts.

If correlation between the default times is high, then all defaults happen together. In that case the FtD premium is similar to the second-to-default (2tD) premium. If correlation is low, then the FtD premium will be much higher than the 2tD premium.

12.4.2 Binomial expansion

In the case of K independent defaults with probability $q(t) = (1 - p(t))$ by time t each one has

$$\mathbb{IP} \left[\tau^{NtD} > t \right] = \sum_{k=0}^{N-1} \binom{K}{k} p(t)^{K-k} q(t)^k =: B(N-1, K, p(t))$$

This allows one to compute the loss distribution in the case of independence. In the case of perfect dependence, all default times are the same since one effectively has a portfolio with one defaultable asset, i.e. $K = 1$ or

$$\mathbb{IP} \left[\tau^{NtD} > t \right] = p(t) = B(0, 1, p(t))$$

By rescaling one obtains an interpolation: Let $K = 100$.

- Diversity 50 means that one has 50 assets with twice the individual notional: the total notional of the portfolio is the same. In other words, every asset appears twice and different assets are independent. The loss distribution (with recovery set to zero) is

$$\mathbb{IP} [L \leq x] = B \left(\frac{x}{2}, \frac{K}{2}, p(t) \right)$$

- Diversity 1 means that one only has one asset with 100 times the individual notional. In other words, all assets are equal:

$$\mathbb{IP} [L \leq x] = B \left(\frac{x}{K}, \frac{K}{K}, p(t) \right)$$

High diversity corresponds to low correlation.

12.4.3 Factor model

In practice one often assumes independence, given a common market factor Z . Thus one computes the default probability $q^k(t, z)$ of name k to time t given the market factor Z is at z :

$$q^k(t, z) = \mathbb{IP} [\tau_k \leq t \mid Z = z]$$

In the case of Merton's asset value model with Gaussian X_k one has from (12.15)

$$\tau_k = \left(q^k \right)^{-1} (\Phi(X_k))$$

Now, assuming that asset returns are driven by a common factor Z and idiosyncratic risks Z_k , i.e.

$$X_k = \beta_k Z + \sqrt{1 - \beta_k^2} Z_k,$$

one has

$$\begin{aligned} q^k(t, z) &= \mathbb{P} \left[\left(q^k \right)^{-1} \left(\Phi(X_k) \right) \leq t \mid Z = z \right] \\ &= \mathbb{P} \left[X_k \leq \Phi^{-1}(q^k(t)) \mid Z = z \right] \\ &= \mathbb{P} \left[Z_k \leq \frac{\Phi^{-1}(q^k(t)) - \beta_k z}{\sqrt{1 - \beta_k^2}} \right] = \Phi \left(\frac{\Phi^{-1}(q^k(t)) - \beta_k z}{\sqrt{1 - \beta_k^2}} \right) \end{aligned} \quad (12.17)$$

In order to incorporate correlation skew, namely the dependence of correlation on the state of the common factor Z , one may assume $\beta_k = \beta_k(Z)$. Note that this adds numerical complexity since the X_k are no longer Gaussian. A remedy is discussed in Section 9.6.

12.4.4 Large pool model

The portfolio loss is given by

$$L_t = \sum_{k=1}^K N_k (1 - R_k) \mathbf{1}_{\{\tau_k \leq t\}} \quad (12.18)$$

In the case of a large, homogenous and independent portfolio the average of the indicators $\mathbf{1}_{\{\tau_k \leq t\}}$ converges to its mean $1 - p(t)$ due to the law of large numbers. Thus, assuming the maximum loss is 1 the conditional loss fraction is given by

$$\mathbb{P} [L_t \leq x \mid Z] \approx \mathbf{1}_{\{q(t, Z) \leq x\}}$$

with $q(t, z)$, as in (12.17), and hence

$$\mathbb{P} [L_t \leq x] = \Phi \left(\frac{1}{\beta} \left(\sqrt{1 - \beta^2} \Phi^{-1}(x) - \Phi^{-1}(q(t)) \right) \right) \quad (12.19)$$

A better approximation based on the central limit theorem has been proposed by Shelton (2004). The distribution of L_t conditional on Z is assumed Gaussian with

mean and variance

$$\mathbb{E}[L_t|Z] = \sum_{k=1}^K N_k(1 - R_k)q^k(t, Z) \quad (12.20)$$

$$\text{Var}[L_t|Z] = \sum_{k=1}^K N_k^2(1 - R_k)^2 p^k(t, Z)q^k(t, Z) \quad (12.21)$$

This approximation does not require a homogenous portfolio, it supports correlation skew and can be extended to two (or more) portfolios L^1, L^2 via

$$\text{Cov}[L_t^1|Z, L_t^2|Z] = \sum_{k=1}^K N_k^1 N_k^2 (1 - R_k)^2 p^k(t, Z)q^k(t, Z)$$

12.4.5 Collateralized debt obligations

A collateralized debt obligation (CDO) is defined by attachment and detachment points K_1 and K_2 . Instead of a single default several losses are required to fully receive the insured sum. Also, the first losses are not insured if $K_1 > 0$. The insured quantity is a call spread on the loss L_t , as defined in (12.18):

$$L_t^{[K_1, K_2]} = \frac{1}{K_2 - K_1} ((L_t - K_1)^+ - (L_t - K_2)^+)$$

The protection seller receives at t_k the premium

$$s^{[K_1, K_2]}(T) \left(1 - L_{t_k}^{[K_1, K_2]}\right) (t_k - t_{k-1})$$

The protection seller pays at t the amount

$$L_{t-dt}^{[K_1, K_2]} - L_t^{[K_1, K_2]}$$

Defining the CDO survival probabilities $p^{[K_1, K_2]}(t)$ as

$$1 - p^{[K_1, K_2]}(t) = \frac{1}{K_2 - K_1} \mathbb{E}[(L_t - K_1)^+ - (L_t - K_2)^+]$$

(or indeed the expected value of a put spread on L_t) a CDO can be priced similar to a CDS with zero recovery, see formula (12.2):

$$V_0^{[K_1, K_2]} = s^{[K_1, K_2]}(T) \int_0^T p^{[K_1, K_2]}(u) D(u) du - \int_0^T D(u) \left(-\frac{\partial}{\partial u} p^{[K_1, K_2]}(u) \right) du$$

In the case of conditional Gaussian loss distribution with mean $\mu(t, Z) = \mathbb{E}[L_t | Z]$ and volatility $\sigma^2(t, Z) = \text{Var}[L_t | Z]$, as in (12.20) and (12.21), one has

$$\begin{aligned} p^{[K_1, K_2]}(t, Z) &= \frac{1}{K_2 - K_1} \left((K_2 - \mu(t, Z)) \Phi \left(\frac{K_2 - \mu(t, Z)}{\sigma(t, Z)} \right) + \sigma(t, Z) \phi \left(\frac{\mu(t, Z) - K_2}{\sigma(t, Z)} \right) \right. \\ &\quad \left. - (K_1 - \mu(t, Z)) \Phi \left(\frac{K_1 - \mu(t, Z)}{\sigma(t, Z)} \right) - \sigma(t, Z) \phi \left(\frac{\mu(t, Z) - K_1}{\sigma(t, Z)} \right) \right) \end{aligned}$$

from the Bachelier formula (18.7). Unconditional probabilities $p^{[K_1, K_2]}(t)$ can be obtained efficiently using Gauss-Hermite quadrature with few points.

12.4.6 Copulas

The joint distribution of default times cannot be assumed to follow a Gaussian copula. However the Gaussian copula can be used to quote CDO tranches; the implied correlation for a tranche $[K_1, K_2]$ is called compound correlation. A more meaningful measure is the base correlation related to tranches $[0, K]$ since spreads $s^{[0, K]}$ increase with correlation.

In the run up to the 2007–2008 credit crisis effort was directed to constructing copulas to match a given curve of base correlations. This was often attempted by assuming independence, given Z and constructing suitable correlation factor dependence $\rho(Z) = \beta^2(Z)$, as described in 12.4.3.

A copula model for default times is not a dynamic model. For example, if default times follow a Gaussian copula at time 0 and no default occurs up to time t then the joint distribution is no longer Gaussian, as has already been pointed out by Schmidt and Ward (2002). Also, correlation has to be distinguished from clustering of defaults. If spreads, and thus survival probabilities, are different then perfect correlation implies that default of one asset implies default of the other asset, but at a different time. For this reason copula models may understate the probability of default clustering, leading to unrealistically low spreads for senior tranches. This was one of the criticisms of credit modelling, and hence the quantitative research community, following the credit crisis.

13] Defaultable Equity

Derivatives that require incorporation of default of the underlying's issuer include derivatives on single stock, in particular convertible bonds.

Survival probabilities $p(t)$ of a firm can be derived from credit default swaps (CDS) or risky bonds, following Chapter 12. When the firm issuing the stock defaults one may assume that the stock price becomes worthless. However, depending on the default definition the default event may be triggered by a restructuring of the firm. This can result in the stock price actually increasing. Excluding default triggered by a restructuring event it is assumed below that stock becomes worthless without recovery. Credit worsening of a firm will be reflected in a lower stock price. Hence modelling default probability should allow for negative correlation with the stock price.

13.1 Reduced form models

Hazard rate or reduced form models assume a process h defining the instantaneous default probability, see Chapter 12. Defaultable stock is thus modelled with dynamics

$$\frac{dS_t}{S_{t-}} = (r_t - d_t + h_t) dt + \sigma_t dW_t - dN_t \quad (13.1)$$

where N is a Cox process with intensity h_t at time t . Various modelling approaches for volatility process σ and hazard rate dynamics are discussed below.

13.1.1 Black-Scholes and independent default

Assume stock price dynamics given as

$$S_t = \tilde{S}_t \mathbf{1}_{\{\tau > t\}} \quad (13.2)$$

with a log-normal process $\tilde{S}$ with mean $\tilde{F}$ and volatility $\tilde{\sigma}$. Also, τ is independent of the process $\tilde{S}$. It follows that the expectation of $\tilde{S}_t$ satisfies

$$\tilde{F}(t) = \frac{F(t)}{p(t)} \quad (13.3)$$

where $F(t)$ is the forward price. Also

$$\begin{aligned} \mathbb{E}[(S_t - K)^+] &= p(t)BSCall\left(\frac{F(t)}{p(t)}, K, \tilde{\sigma}, t\right) \\ &= BSCall(F(t), p(t)K, \tilde{\sigma}, t) \end{aligned}$$

as well as

$$\begin{aligned} \mathbb{E}[(K - S_t)^+] &= (1 - p(t))K + p(t)BSPut\left(\frac{F(t)}{p(t)}, K, \tilde{\sigma}, t\right) \\ &= (1 - p(t))K + BSPut(F(t), p(t)K, \tilde{\sigma}, t) \end{aligned}$$

These modified Black-Scholes formulae allow one to imply volatilities $\tilde{\sigma}$ from the standard implied volatility surface. The volatility skew of $\tilde{S}$ is less pronounced than the skew of S since the jump to default introduces negative skew.

Volatility models developed in Chapter 6 and Section 7.4 can thus be applied to $\tilde{S}$. Similarly, dividend models from Chapter 5 can be extended to the defaultable case. Dividends thus become risky. Paid dividends will be higher than announced dividend amounts since the forward price includes default risk, while dividend payments are conditional on no prior default:

$$D_0(t) = \mathbb{E}\left[B_t^{-1} D_t(t) \mathbf{1}_{\{\tau > t\}}\right]$$

While independent default can be implemented easily it suffers from being unrealistic since the event cannot be anticipated. Accordingly the hedger is fully exposed to the default event.

13.1.2 Black-Scholes and Gaussian hazard rate

Given a positive stochastic hazard rate h with integral

$$H_T = \int_0^T h_t dt$$

assume the distribution of the defaultable asset S_T to be given as

$$S_T = F(T)e^{\sigma_S X - \sigma_S^2/2} \mathbf{1}_{\{\tau > T\}} e^{H_T}$$

$$\{\tau > T\} = \{U < e^{-H_T}\}$$

The expectation of S_T is the forward $F(T)$, which can be seen by evaluating the expectation conditional on (X, H_T) . This forward price includes the possibility of default.

The following computations rely on an approximation via a truncated Gaussian

$$H_T \approx (a_T + \sigma_H Y)^+ = G_T^+$$

where Y is a second Gaussian variate correlated with X with coefficient ρ . This approximation is particularly useful if the hazard rate process h is Gaussian, such as the processes discussed in Sections 10.3.2 and 10.4. In that case volatility parameters can be obtained in closed form. Given survival probability $p(T)$ and volatility σ_H one can solve for a_T since

$$\begin{aligned} p(T) &= \mathbb{E} \left[e^{-H_T} \right] = \mathbb{P} [a_T + \sigma_H Y \leq 0] + \mathbb{E} \left[e^{-a_T - \sigma_H Y} \mathbf{1}_{\{a_T + \sigma_H Y > 0\}} \right] \\ &= \Phi \left(-\frac{a_T}{\sigma_H} \right) + e^{-a_T + \sigma_H^2/2} \Phi \left(\frac{a_T}{\sigma_H} - \sigma_H \right) \end{aligned} \quad (13.4)$$

is decreasing in a_T . The call option price with defaultable equity can now be written as the sum of four terms, namely

$$\begin{aligned} \mathbb{E} \left[(S_T - K)^+ \right] &= \mathbb{E} \left[S_T \mathbf{1}_{\{G_T \leq 0, S_T > K\}} \right] + \mathbb{E} \left[S_T \mathbf{1}_{\{G_T > 0, S_T > K\}} \right] \\ &\quad - K \mathbb{P} [G_T \leq 0, S_T > K] - K \mathbb{P} [G_T > 0, S_T > K] \end{aligned}$$

The first term evaluates to

$$\begin{aligned} \mathbb{E} \left[S_T \mathbf{1}_{\{G_T \leq 0, S_T > K\}} \right] &= F(T) \mathbb{E} \left[\mathbf{1}_{\{G_T \leq 0\}} e^{\sigma_S X - \sigma_S^2/2} \mathbf{1}_{\{-X < d_-\}} \right] \\ &= F(T) \mathbb{P} [a_T + \sigma_H Y + \rho \sigma_S \sigma_H \leq 0, -X < d_+] \\ &= F(T) \Phi_2 \left(d_+, -\frac{a_T + \rho \sigma_H \sigma_S}{\sigma_H}; -\rho \right) \end{aligned}$$

where

$$d_{\pm} = \frac{1}{\sigma_S} \left(\log \frac{F(T)}{K} \pm \frac{1}{2} \sigma_S^2 \right)$$

For the second term an integration over U conditional on (X, Y) yields

$$\begin{aligned}
 & \mathbb{E} \left[S_T \mathbf{1}_{\{G_T > 0, S_T > K\}} \right] \\
 &= F(T) \mathbb{E} \left[\mathbf{1}_{\{G_T > 0\}} e^{\sigma_S X - \sigma_S^2/2} \mathbf{1}_{\{-X - G_T/\sigma_S < d_-\}} \right] \\
 &= F(T) \mathbb{P} \left[G_T + \rho \sigma_S \sigma_H > 0, -X - (G_T + \rho \sigma_H \sigma_S)/\sigma_S < d_+ \right] \\
 &= F(T) \mathbb{P} \left[-\sigma_H Y < a_T + \rho \sigma_S \sigma_H, -\sigma_S X - \sigma_H Y < \sigma_S d_+ + a_T + \rho \sigma_H \sigma_S \right] \\
 &= F(T) \Phi_2 \left(\frac{d_+ \sigma_S + a_T + \rho \sigma_H \sigma_S}{\sigma_{SH}}, \frac{a_T + \rho \sigma_H \sigma_S}{\sigma_H}, \frac{\rho \sigma_S + \sigma_H}{\sigma_{SH}} \right)
 \end{aligned}$$

with

$$\sigma_{SH}^2 = \sigma_S^2 + 2\rho \sigma_H \sigma_S + \sigma_H^2$$

denoting the variance of $\log S_T + H_T$. For the third term one has

$$\begin{aligned}
 \mathbb{P} \left[G_T \leq 0, S_T > K \right] &= \mathbb{P} \left[a_T + \sigma_H Y \leq 0, -X < d_- \right] \\
 &= \Phi_2 \left(d_-, -\frac{a_T}{\sigma_H}; -\rho \right)
 \end{aligned}$$

The last term is given as

$$\begin{aligned}
 & \mathbb{P} \left[G_T > 0, S_T > K \right] \\
 &= \mathbb{P} \left[G_T > 0, -X - \frac{G_T}{\sigma_S} < d_-, U < e^{-G_T} \right] \\
 &= \mathbb{E} \left[e^{-G_T} \mathbf{1}_{\{G_T > 0, -X - \frac{G_T}{\sigma_S} < d_-\}} \right] \\
 &= e^{-a_T + \sigma_H^2/2} \mathbb{P} \left[G_T - \sigma_H^2 > 0, -X + \rho \sigma_H - \frac{G_T - \sigma_H^2}{\sigma_S} < d_- \right] \\
 &= e^{-a_T + \sigma_H^2/2} \mathbb{P} \left[-\sigma_H Y < a_T - \sigma_H^2, -\sigma_S X - \sigma_H Y < \sigma_S d_- - \rho \sigma_H \sigma_S + a_T - \sigma_H^2 \right] \\
 &= e^{-a_T + \sigma_H^2/2} \Phi_2 \left(\frac{\sigma_S d_- - \rho \sigma_H \sigma_S + a_T - \sigma_H^2}{\sigma_{SH}}, \frac{a_T - \sigma_H^2}{\sigma_H}, \frac{\rho \sigma_S + \sigma_H}{\sigma_{SH}} \right)
 \end{aligned}$$

Adding the four terms leads to the call formula

$$\begin{aligned}
& \mathbb{E}[(S_T - K)^+] \\
& F(T)\Phi_2\left(d_+, -\frac{a_T + \rho\sigma_H\sigma_S}{\sigma_H}; -\rho\right) \\
& + F(T)\Phi_2\left(\frac{d_+\sigma_S + a_T + \rho\sigma_H\sigma_S}{\sigma_{SH}}, \frac{a_T + \rho\sigma_H\sigma_S}{\sigma_H}; \frac{\rho\sigma_S + \sigma_H}{\sigma_{SH}}\right) \\
& - K\Phi_2\left(d_-, -\frac{a_T}{\sigma_H}; -\rho\right) \\
& - Ke^{-a_T + \sigma_H^2/2}\Phi_2\left(\frac{\sigma_S d_- - \rho\sigma_H\sigma_S + a_T - \sigma_H^2}{\sigma_{SH}}, \frac{a_T - \sigma_H^2}{\sigma_H}; \frac{\rho\sigma_S + \sigma_H}{\sigma_{SH}}\right) \quad (13.5)
\end{aligned}$$

with

$$\sigma_{SH}^2 = \sigma_S^2 + 2\rho\sigma_H\sigma_S + \sigma_H^2$$

The corresponding negative put price is obtained by changing the sign of the first and last arguments (correlation) within the bivariate cumulative function and correcting for the payoff in case of default through term $(1 - p(T))K$ with $p(T)$ given in (13.4).

If $\sigma_H \ll a_T$ then one has from (13.4)

$$p(T) \approx e^{-a_T + \sigma_H^2/2}$$

as well as from (13.5)

$$\begin{aligned}
\mathbb{E}[(S_T - K)^+] & \approx F(T)\Phi\left(\frac{d_+\sigma_S + a_T + \rho\sigma_H\sigma_S}{\sigma_{SH}}\right) \\
& - Ke^{-a_T + \sigma_H^2/2}\Phi\left(\frac{\sigma_S d_- - \rho\sigma_H\sigma_S + a_T - \sigma_H^2}{\sigma_{SH}}\right) \quad (13.6)
\end{aligned}$$

The corresponding negative put price is obtained by changing the sign of the arguments within the cumulative function and correcting for the payoff in case of default through term $(1 - e^{-a_T + \sigma_H^2/2})K$.

Figure 13.1 exhibits implied volatilities of Vanilla options on defaultable stock with Gaussian hazard rate with different values of correlation (forward 100, volatility 20%, maturity two years, interest rate 0%, survival probability 95%). Hazard rate volatility is 2% (left graph) and 4% (right graph). Hazard rate mean a_T is calibrated to survival probability. With higher hazard rate volatility the truncated formula (13.5) deviates from the approximating Gauss formula (13.6). Note that the lower volatility bound given by the diffusion volatility 20% is violated by the

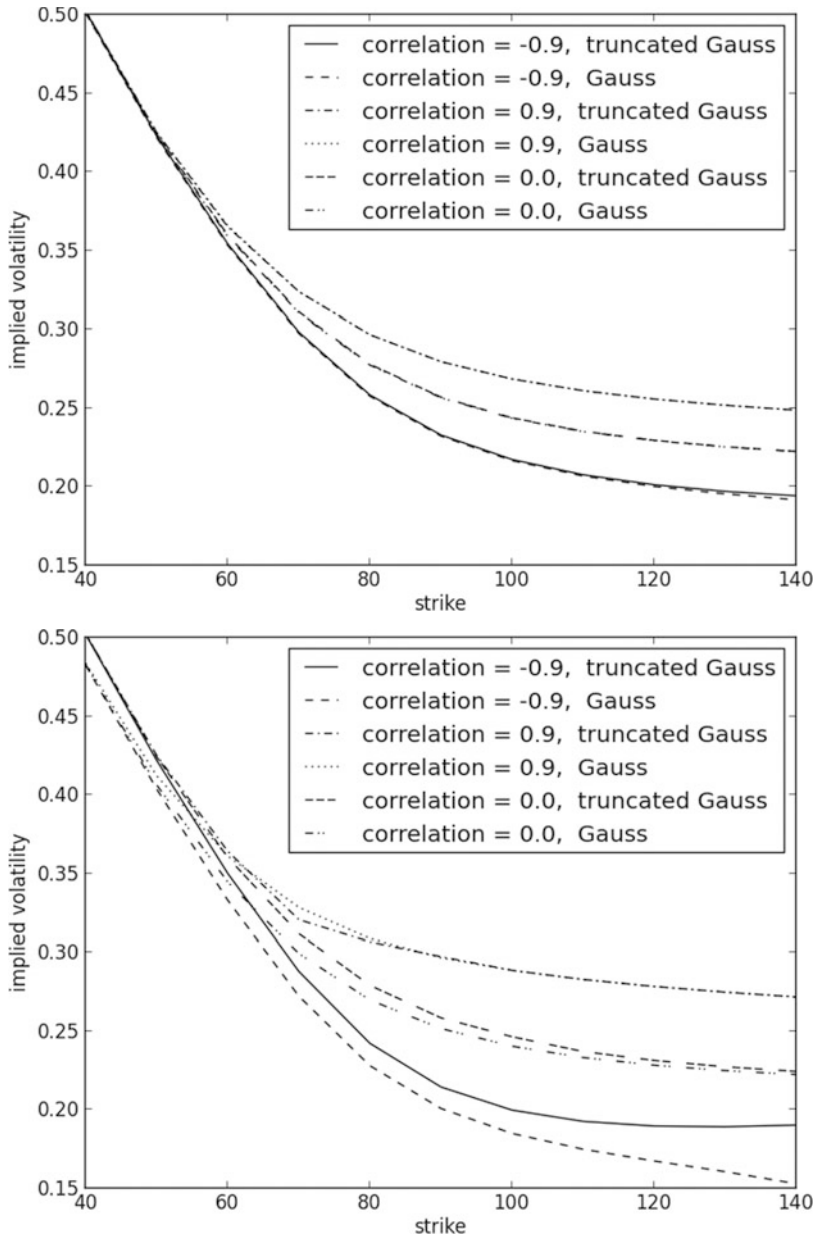


Figure 13.1 Defaultable equity with Gaussian hazard (volatility 2% and 4%)

approximating formula for correlation -90%. The approximating formula generally exhibits lower volatility levels since negative hazard cancels some of the volatility introduced by the jumps.

13.1.3 Local volatility and local hazard rate

This section generalizes Section 6.1.1 by adding a local hazard rate, as suggested by Andersen and Buffum (2004). The stock dynamics is given as

$$\frac{dS_t}{S_{t-}} = (r_t - d_t + h_t(S_{t-})) dt + \sigma_t(S_{t-}) dW_t - dN_t \quad (13.7)$$

where N is a Cox process with intensity $h_t(S_{t-})$. Thus, stock defaults at the first jump of N . The Itô-Tanaka formula applied to the function $c(t) = (S_t - K)^+$ reads

$$dc(t) = 1_{\{S_{t-} > K\}} dS_t + \frac{1}{2} \delta(S_{t-} - K) d\langle S \rangle_t - c(t-) dN_t$$

Integrating over time and taking expectations one obtains

$$\begin{aligned} & \mathbb{E}[c(T)] \\ &= c(0) + \int_0^T (r_t - d_t) \mathbb{E}[S_{t-} 1_{\{S_{t-} > K\}}] dt + \int_0^T \mathbb{E}[S_{t-} 1_{\{S_{t-} > K\}} h_t(S_{t-})] dt \\ &+ \frac{1}{2} \int_0^T \mathbb{E}[\delta(S_{t-} - K)] \sigma_T^2(K) K^2 dt - \int_0^T \mathbb{E}[c(t-) h_t(S_{t-})] dt \end{aligned}$$

Expressing quantities in terms of call prices C and their derivatives and deriving with respect to T one obtains (writing C instead of $C(K, T)$ for the call price)

$$\frac{\partial C}{\partial T} = -d_T C - (r_T - d_T) K \frac{\partial C}{\partial K} + \frac{1}{2} K^2 \sigma^2(T, K) \frac{\partial^2 C}{\partial K^2} + K \int_K^\infty \frac{\partial^2 C}{\partial K^2}(k) h_T(k) dk$$

compare (6.2). Integration by parts for the last term leads to the alternative expression

$$\begin{aligned} \frac{\partial C}{\partial T} &= - \left(d_t - K \frac{\partial h}{\partial K} \right) C - (r - q + h) K \frac{\partial C}{\partial K} + \frac{1}{2} K^2 \sigma^2(T, K) \frac{\partial^2 C}{\partial K^2} \\ &+ K \int_K^\infty C(k) \frac{\partial^2 h}{\partial K^2}(k) dk \end{aligned}$$

Andersen and Buffum suggest a finite difference implementation for the pricing of hybrid products such as convertible bonds. For CEV model with affine hazard rate

parametrization given as

$$\begin{aligned}\sigma_t(S_t) &= a_t S_t^\beta \\ h_t(S_t) &= b_t + c\sigma_t^2(S_t) = b_t + ca_t^2 S_t^{2\beta}\end{aligned}$$

a closed form representation for survival probabilities $p(t)$ and European option prices has been derived by Carr and Linetsky (2006). Time-dependent parameters allow one to calibrate initial risk free yield curve, equity forwards, survival probabilities and at-the-money implied volatilities. There are parameters left to match implied volatility skew for two maturities.

Note that the proposed dynamics introduces stock delta and vega into CDS spreads, in the sense that equity option delta can be hedged with CDS contracts and CDS delta with stocks. Also, equity and credit derivatives need to be calibrated jointly. In order to disentangle this joint calibration the hazard rate would have to be given as a function of the Brownian driver, namely as $h_t(W_t)$.

13.1.4 Stochastic volatility and stochastic hazard rate

As an alternative to a local hazard rate as in (13.7) one may correlate default probability with stock return using a stochastic hazard rate model. This can be combined with stochastic volatility. Carr and Wu (2009) assume that both variance σ^2 and hazard rate processes h are mean reverting stochastic processes, namely (13.1) with

$$\begin{aligned}d\sigma_t^2 &= (\theta_\sigma^2 - \kappa_\sigma \sigma_t^2) dt + \alpha_\sigma \sigma_t dW_t^\sigma \\ h_t &= \beta \sigma_t^2 + z_t \\ dz_t &= (\theta_z - \kappa_z z_t) dt + \alpha_z \sqrt{z_t} dW_t^z\end{aligned}$$

The Brownian motion W^z is independent of the correlated pair of Brownians (W, W^σ) . The coupling factor β (compare Section 11.4.2 for stochastic interest instead of hazard rates) allows one to incorporate positive correlation between equity volatility and credit spreads. This leads to instantaneous correlation between default intensity and equity variance

$$\rho_t^{h,\sigma} \equiv \frac{d\langle h, \sigma^2 \rangle_t}{\sqrt{d\langle h \rangle_t} \sqrt{d\langle \sigma^2 \rangle_t}} = \frac{\beta \alpha_\sigma \sigma_t}{\sqrt{\beta^2 \alpha_\sigma^2 \sigma_t^2 + \alpha_z^2 z_t}}$$

as well as instantaneous correlation between default intensity and stock price

$$\rho_t^{h,S} \equiv \frac{d\langle h, S \rangle_t}{\sqrt{d\langle h \rangle_t} \sqrt{d\langle S \rangle_t}} = \frac{\beta \rho \alpha_\sigma \sigma_t}{\sqrt{\beta^2 \alpha_\sigma^2 \sigma_t^2 + \alpha_z^2 z_t}} = \rho \rho_t^{h,\sigma}$$

Survival probabilities can be obtained as

$$\begin{aligned} p(t) &= \mathbb{E} \left[e^{-\int_0^t h_u du} \right] = \mathbb{E} \left[e^{-\beta \int_0^t \sigma_u^2 du} \right] \mathbb{E} \left[e^{-\int_0^t z_u du} \right] \\ &= \exp \left(A_0^{\beta \sigma^2}(t) + \beta \sigma_0^2 B_0^{\beta \sigma^2}(t) + A_0^z(t) + z_0 B_0^z(t) \right) \end{aligned}$$

with $(A^{\beta \sigma^2}, B^{\beta \sigma^2})$ and (A^z, B^z) as in equation (10.20), with parameters (θ, κ, σ) therein being $(\beta \theta, \kappa_\sigma, \sqrt{\beta} \alpha_\sigma)$ and $(\theta_z, \kappa_z, \alpha_z)$. This allows one to derive CDS spreads from equation (12.2), as well as bond spreads using equation (12.1) and can be used for calibration. Using notation H_T for $\int_0^T h_t dt$ and Z_T for $\int_0^T z_t dt$ the undiscounted complex payoff

$$\mathbf{1}_{\{\tau > T\}} \exp \left(iu \log \frac{S_T}{F(T)} \right)$$

has value

$$\begin{aligned} \phi(u) &= \mathbb{E} \left[\exp \left(-H_T + iu \left(H_T + \int_0^T \sigma_t dW_t - \frac{1}{2} \int_0^T \sigma_t^2 dt \right) \right) \right] \\ &= \mathbb{E}_{iu\sigma dW} \left[\exp \left(-(1-iu)H_T - \frac{iu+u^2}{2} \int_0^T \sigma_t^2 dt \right) \right] \\ &= \mathbb{E}_{iu\sigma dW} \left[\exp \left(-(1-iu)Z_T - \left((1-iu)\beta + \frac{iu+u^2}{2} \right) \int_0^T \sigma_t^2 dt \right) \right] \\ &= \mathbb{E} \left[\exp(-(1-iu)Z_T) \right] \mathbb{E}_{iu\sigma dW} \left[\exp \left(- \left((1-iu)\beta + \frac{iu+u^2}{2} \right) \int_0^T \sigma_t^2 dt \right) \right] \end{aligned}$$

where the density of $\mathbb{P}_{iu\sigma dW}$ with respect to $\mathbb{P}$ is given by the martingale M with $M_0 = 1$ and

$$\frac{dM_t}{M_t} = iu\sigma_t dW_t$$

While Z is independent of M the dynamics of σ^2 with respect to $\mathbb{P}_{iu\sigma dW}$ has modified CIR parameter

$$\tilde{\kappa}_\sigma = \kappa_\sigma - iu\rho\alpha_\sigma$$

The transform of integrated CIR processes Z_T and $\int_0^T \sigma_t^2 dt$ follows from equation (10.21).

Note that the equity variance process impacts the price of the CDS contract if $\beta > 0$. Hence the model needs to be simultaneously calibrated to CDS rates and

equity options. Carr and Wu (2009) calibrate the model for ten American stocks with different credit ratings using maximum likelihood methods. Surprisingly, their results indicate zero value for the parameter β , in which case the CDS spreads are independent of the pure stock price (excluding default) and volatility processes. Nonetheless, option implied volatility still depends on the default intensity process since intensity enters directly the price of the stock options.

13.2 Structural models

In Merton's classical model, as introduced in Section 12.3, the stock price S_t is interpreted as a call option on the assets A with strike B and maturity T , see (12.10). There is a link between credit spreads and implied volatilities, in the sense that one cannot calibrate both separately. For the case of a continuous barrier Sepp (2006) discusses time-dependent barrier B , as well as log-normal, jump diffusion and stochastic volatility dynamics for the asset process A . The stock dynamics is defined as

$$S_t = (A_t - B_t) \mathbf{1}_{\{\tau > t\}}$$

$$\tau = \inf\{t : A_t \leq B_t\}$$

If both asset and barrier have standard risk neutral equity drift, namely

$$\mathbb{E} \left[\frac{dA_t}{A_t} \right] = \frac{dB_t}{B_t} = (r_t - d_t) dt$$

then the stock process has dynamics

$$\frac{dS_t}{S_t} = (r_t - d_t) dt + \sigma_t^A \frac{S_t + B_t}{S_t} dW_t$$

on $t < \tau$. Thus S exhibits negative volatility skew. For deterministic asset volatility σ_t^A the stock process is a shifted log-normal process stopped at 0. The price of a call with strike K and maturity t is

$$\begin{aligned} P(t) \mathbb{E} [(S_t - K)^+] \\ &= P(t) \mathbb{E} [(A_t - B_t - K)^+ \mathbf{1}_{\{\tau > 0\}}] \\ &= e^{-\int_0^t d_u du} \mathbb{E} \left[\left(A_0 e^{\int_0^t \sigma_u^A dW_t - \frac{1}{2} \int_0^t (\sigma_u^A)^2 du} - B_0 - K e^{-\int_0^t (r_u - d_u) du} \right)^+ \mathbf{1}_{\{\tau > 0\}} \right] \end{aligned}$$

In the case of constant σ^A the price can be computed from formula (18.34). Sepp (2006) also obtains closed form formulae for both survival probabilities and Vanilla

options in the case of Heston stochastic volatility with zero correlation, i.e.

$$d\left(\sigma^A\right)_t^2 = \kappa \left(\left(\sigma^A\right)_\infty^2 - \left(\sigma^A\right)_t^2 \right) dt + \alpha \sigma_t^A dW_t^\sigma$$

with W and W^σ independent, as well as double exponential jump diffusion, namely

$$\frac{dA_t}{A_t} = (r_t - d_t - h\mathbb{E}\left[e^J - 1\right]) \left(\sigma_t^A\right)^2 dt + \sigma_t^A dW_t + (e^J - 1) dN_t$$

where N_t has deterministic intensity $h\left(\sigma_t^A\right)^2$ and J has double exponential distribution.

14] Counterparty Credit Risk

Since the Lehman default on September 15, 2008 the credit quality of issuers of retail products has received much attention. Arguably, the largest losses to institutions during the crisis were due to credit value adjustment (CVA) rather than to actual default.

As for reporting, the international financial reporting standard (IFRS) 13, effective since January 1, 2013, requires institutions to incorporate CVA as well as own credit risk or debt value adjustment (DVA) based on exit pricing. Exit pricing of an asset can be described as hypothetical pricing by market participants and relies on available market prices, such as credit default swaps.

Capital requirements of financial institutions are laid out in the Basel III framework by the Bank for International Settlements. This framework was finalized in December 2010 and introduced a CVA volatility charge reflecting the risk due to deterioration of credit worthiness of counterparties. The rules therein are implemented in the European Union via the capital requirements directive (CRD IV) and have had to be applied since January 1, 2014.

These changes in the regulatory and market environment have resulted in sizable investments in new CVA teams at banks, including trading, quants, risk management and IT. Software companies have incorporated a CVA reporting capability and new consulting businesses are ready to implement CVA at financial institutions.

14.1 Sources of credit risk

Single- and multi-name credit risk and contracts protecting against the default of a firm or default losses to a portfolio have been discussed in Chapter 12. The incorporation of default of the issuer of an underlying within the underlyings' dynamics and effects on derivatives pricing are the topic of Chapter 13. This section focuses on default of the counterparty. Typically, there will be two counterparties to a trade so one may interpret a given trade as a first-to-default product.

The most important counterparty credit risk (CCR) of a financial institution is typically related to fixed income and credit-linked products, such as interest rate and credit default swaps. For that reason regulators incentivize institutions to reduce exposure through margin payments specified in standardized agreements, such as ISDA's credit support annex (CSA) or trading with centralized counterparties (CCP) by imposing prohibitive capital requirements for bespoke over-the-counter (OTC) transactions. Many equity and currency OTC derivatives remain exposed to CCR. Examples of exposure to the seller's default include a bank issuing an equity-linked note, selling an equity certificate to retail clients, selling protection via a CDS contract or selling a forward contract. Default of the buyer can be relevant in cases such as institutions paying regular coupons for an equity-linked swap, buyers of products with deferred premium payment and forward contracts. There is typically no exposure to retail clients' default.

14.2 Credit valuation adjustment

Let a portfolio consisting of derivatives traded with the same counterparty be given, with value V_t at time t . This value is understood as not including counterparty default. Cashflows strictly before t are also not included. The credit value adjustment CVA is defined as the difference between risky value $\hat{V}$, including counterparty credit risk, and riskless value V :

$$\hat{V} = V + \text{CVA}$$

It will thus be negative. To allow for stochastic interest rates fix a possibly random numéraire N . Relevant examples include riskless cash account $N = B$ and riskless bond $N = P(T)$. Default of the counterparty occurs at a random time τ , with deterministic recovery rate R on the portfolio. With these assumptions (unilateral) credit valuation adjustment CVA for the period $[s, T]$ is defined as

$$\text{CVA}^{[s, T]} = -(1 - R)N_0 \mathbb{E} \left[\frac{1}{N_\tau} 1_{\{s < \tau \leq T\}} V_\tau^+ \right] \quad (14.1)$$

Superscript $[s, T]$ is omitted when considering all future cash flows of the portfolio. If independence of τ and (V, N) is assumed then one has

$$\begin{aligned} \text{CVA}^{[s, T]} &= -(1 - R)N_0 \int_s^T \mathbb{E} \left[\frac{1}{N_t} V_t^+ \right] h_0(t) p_0(t) dt \\ &\approx -(1 - R) \sum_{i=1}^{n-1} \mathbb{E} \left[\frac{1}{N_{(t_i + t_{i+1})/2}} V_{(t_i + t_{i+1})/2}^+ \right] (p(t_i) - p(t_{i+1})) \end{aligned}$$

where $s = t_1 < t_2 < \dots t_n = T$ is a discretization of $[s, T]$. The terms of the last expression are also known as loss, given default LGD for $1 - R$ and exposure at default EAD(i) for the expectation.

If own default DVA at time $\tilde{\tau}$ with recovery $\tilde{R}$ is to be included one may adjust the price V without counterparty risk to

$$\hat{V} + \text{CVA} + \text{DVA}$$

with bilateral CVA and DVA defined as

$$\text{CVA}^{[s, T]} = -(1 - R)N_0 \text{IE} \left[\frac{1}{N_\tau} 1_{\{s < \tau \leq T, \tau < \tilde{\tau}\}} V_\tau^+ \right] \quad (14.2)$$

$$\text{DVA}^{[s, T]} = -(1 - \tilde{R})N_0 \text{IE} \left[\frac{1}{N_{\tilde{\tau}}} 1_{\{s < \tilde{\tau} \leq T, \tilde{\tau} < \tau\}} V_{\tilde{\tau}}^- \right] \quad (14.3)$$

Note that this definition is symmetric, in the sense that the CVA of one counterparty is the DVA of the other counterparty. Unilateral CVA as in (14.1) is the special case of bilateral CVA as in (14.2) with $\tilde{\tau} = \infty$.

14.2.1 Standard products

- A fixed cash flow of size 1 at time T has a CVA of

$$\text{CVA} = -(1 - R)N_0 \text{IE} \left[\frac{1}{N_\tau} 1_{\{0 < \tau \leq T\}} P_\tau(T) \right]$$

In the case of independence between N and τ this simplifies to

$$\text{CVA} = -(1 - R)(1 - p_0(T))P_0(T)$$

This is the difference between riskless and risky bond, compare Section 12.1.2.

- A stock with price process S deliverable at time T has CVA

$$\text{CVA} = -(1 - R)N_0 \text{IE} \left[\frac{1}{N_\tau} 1_{\{0 < \tau \leq T\}} F_\tau(T) \right]$$

where $F_t(T)$ denotes the forward price at time t . In the case of independence between τ and (N, S) this simplifies to

$$\text{CVA} = -(1 - R)(1 - p_0(T))P_0(T)F_0(T)$$

- A forward contract with payout $S_T - K$ at time T adjusts with univariate CVA given as

$$\text{CVA} = -(1 - R)N_0 \text{IE} \left[\frac{1}{N_\tau} 1_{\{0 < \tau \leq T\}} (F_\tau(T) - K)^+ \right]$$

In the case of independence between τ and (N, S) one has

$$\begin{aligned} \text{CVA} &= -(1-R) \int_0^T \mathbb{E} \left[\frac{1}{N_t} (F_t(T) - K)^+ \middle| \tau = t \right] \mathbb{P}[\tau \in dt] \\ &= -(1-R) P_0(T) \int_0^T \text{BSCall} \left(F_0(T), K, \hat{\sigma}_0 \left(K \frac{F_0(t)}{F_0(T)}, t \right), t \right) h_0(t) p_0(t) dt \end{aligned}$$

Here proportional dividends were assumed in the sense that

$$F_t(T) = \mathbb{E}_t[S_T] = S_t \frac{F_0(T)}{F_0(t)}$$

Parameter $\hat{\sigma}_s(K, t)$ refers to the implied volatility observed at time s for an option with strike K and maturity T . Thus the CVA of the forward contract is given as the weighted sum of call options with a maturity range up to contract expiry. In the bilateral case with own default $\tilde{\tau}$ and independence between τ , $\tilde{\tau}$ and (N, S) one has

$$\begin{aligned} \text{CVA} &= -(1-R) \int_0^T \mathbb{E} \left[\frac{1}{N_t} (F_t(T) - K)^+ \middle| \tau = t, \tilde{\tau} > t \right] \mathbb{P}[\tau \in dt, \tilde{\tau} > t] \\ &= -(1-R) P_0(T) \int_0^T \text{BSCall} \left(F_0(T), K, \hat{\sigma}_0 \left(K \frac{F_0(t)}{F_0(T)}, t \right), t \right) h_0(t) \hat{\sigma}_0 p_0(t) \tilde{p}_0(t) dt \end{aligned}$$

as well as

$$\begin{aligned} \text{DVA} &= -(1-R) \int_0^T \mathbb{E} \left[\frac{1}{N_t} (F_t(T) - K)^- \middle| \tilde{\tau} = t, \tau > t \right] \mathbb{P}[\tilde{\tau} \in dt, \tau > t] \\ &= (1-\tilde{R}) P_0(T) \int_0^T \text{BSPut} \left(F_0(T), K, \hat{\sigma}_0 \left(K \frac{F_0(t)}{F_0(T)}, t \right), t \right) \\ &\quad \tilde{h}_0(t) p_0(t) \tilde{p}_0(t) dt \end{aligned}$$

- A call option with payoff $(S_T - K)^+$ at time T has

$$\text{CVA} = -(1-R) N_0 \mathbb{E} \left[\frac{1}{N_\tau} 1_{\{0 < \tau \leq T\}} \text{BSCall}(F_\tau(T), K, \hat{\sigma}_\tau(K, T), T - \tau) \right]$$

Note that this expression depends on implied volatility dynamics $\hat{\sigma}$. In the case of independence between τ and (N, S) one has

$$\text{CVA} = -(1-R) P_0(T) \int_0^T \mathbb{E} \left[\text{BSCall}(F_0(T), K, \hat{\sigma}_t(K, T), T) \right] h_0(t) p_0(t) dt$$

where the only random quantity is $\hat{\sigma}_t$. Assuming no smile dynamics $\hat{\sigma}_t = \hat{\sigma}$, in other words Black-Scholes dynamics for the option, this further simplifies to

$$\text{CVA} = -(1 - R)P_0(T)(1 - p_0(T))\text{BSCall}(F_0(T), K, \hat{\sigma}(K, T), T)$$

In other words, the price including counterparty risk is given via risky discounting.

14.3 Wrong way risk

Practitioners often start from their favourite models and inject counterparty default as an independent event. The task is then to compute the exposure at default (EAD) at discrete future times relying on methods borrowed from American Monte Carlo approaches. As simplification defaults are often only allowed at cash flow dates.

This approach ignores correlation between portfolio value and default. This is questionable due to typical negative correlation between CDS spreads and stocks and indices. Generally, any portfolio leading to counterparty exposure is correlated with the counterparties' credit to some extent: an increase in the market price V of the portfolio, and thus the exposure, leads to a higher default probability due to increased liabilities of the counterparty. Such a position is called wrong way risk.

Approaches for modelling this correlation have already been discussed in Chapter 12 for credit derivatives and Chapter 13 for defaultable equity.

- Stochastic hazard rate (reduced form): correlation is limited by hazard rate volatility, default cannot be anticipated.
- Local hazard rate (reduced form): CDS spreads carry equity delta, default cannot be anticipated.
- Copula-based models (default time and stock or stock driver): strong dependence, unrealistic dynamics.
- Asset-based models: strong dependence, default can be anticipated.

14.4 Structural models

Assume a portfolio of derivatives on the same underlying S with price process

$$V = \sum_i V^i \quad (14.4)$$

The price of every derivative V^i is defined by a maturity time (= payment time) t_i and a European payoff function $f_i(S) = f_i(S_{t_i})$. Thus path-dependent products or products with deferred payment are not covered. One has

$$\begin{aligned} \text{CVA} &= -(1-R)\mathbb{E} \left[\left(\mathbb{E}_\tau \left[\sum_{0 < \tau \leq t_i} \frac{1}{N_{t_i}} f_i(S_{t_i}) \right] \right)^+ \right] \\ &= -(1-R) \int_0^T \mathbb{E} \left[\left(\sum_{0 < t \leq t_i} \mathbb{E}_t \left[\frac{1}{N_{t_i}} f_i(S_{t_i}) \middle| \tau = t \right] \right)^+ \right] h_0(t) p_0(t) dt \end{aligned}$$

14.4.1 Time-dependent deterministic barrier

In order to assess the impact of correlation between counterparty default and stock dynamics assume an asset-based approach combining the ideas of Merton and Black-Scholes. This approach has also been discussed in Brigo and Tarenghi (2004). Counterparty default is defined as the first hitting time

$$\tau = \inf\{t : A_t < B_t\}$$

with log-normal asset

$$A_t = F_t^A \exp \left(\sigma_A W_t^A - \frac{\sigma_A^2 t}{2} \right)$$

and time-dependent barrier B , as introduced in Section 12.3. The only source of randomness of the portfolio V is a log-normal stock process

$$S_t = F_t^S \exp \left(\sigma_S W_t - \frac{\sigma_S^2 t}{2} \right)$$

driven by a Brownian motion W , as defined in Section 18.2. The Brownians W and W^A are correlated with parameter ρ . Let b_t be defined as

$$F_t^A \exp \left(\sigma_A b_t - \frac{\sigma_A^2 t}{2} \right) = B_t$$

Then one has

$$\begin{aligned}
 & \mathbb{E}_t [f(S_T) | \tau = t] \\
 &= \mathbb{E}_Z \left[f \left(F_0(T) e^{\sigma(\rho b_t + \bar{\rho} \sqrt{t} Z + W_T - W_t) - \frac{\sigma^2}{2} T} \right) \right] \\
 &= \mathbb{E}_Z \left[f \left(\left(F_0(T) e^{\sigma(\rho b_t + \bar{\rho} \sqrt{t} Z) - \frac{\sigma^2}{2} t} \right) e^{\sigma(W_T - W_t) - \frac{\sigma^2}{2} (T-t)} \right) \right] \\
 &= F_t(f, T, Z)
 \end{aligned}$$

with a function $F_t(f, T, Z)$ where Z is an independent Gaussian and $\bar{\rho}^2 = 1 - \rho^2$. Thus Black-Scholes formulae can be applied to value portfolio components as functions of a Gaussian random variable Z :

$$\text{CVA} = -(1 - R) \int_0^T P(t) \mathbb{E} \left[\left(\sum_{0 < t \leq t_i} F_t(f_i, t_i, Z) \right)^+ \right] h_0(t) p_0(t) dt$$

This can be generalized to a Gauss copula model for equity, namely

$$S_t = D_t(W_t)$$

In that case $F_t(f, T, Z)$ can be approximated using Gauss-Hermite integration with weights (w_i) and points (x_i), namely

$$F_t(f, T, Z) \approx \sum_i w_i f \left(D_t \left(\rho B_t + \bar{\rho} \sqrt{t} Z + \sqrt{T-t} x_i \right) \right)$$

Variate Z may be simulated to avoid discretization errors due to f not being smooth. The distribution of loss given default for the portfolio defined in (14.4) is given as

$$V_t^+ | \tau = t \stackrel{(d)}{=} \left(\sum_{0 < t \leq t_i} F_t(f_i, t_i, Z) \right)^+$$

where randomness is due to that part of the movement of S_t which is not fixed by the given default time and hence asset level A_t . This also allows one to simulate loss quantiles.

As an example consider a stylized equity swap. Semiannually stock is exchanged against cash, both with notional 100000. Stock volatility is 20%, hazard is 1%, interest and recovery is zero, maturity is 3y. Figure 14.1 displays loss quantiles for different values of correlation.

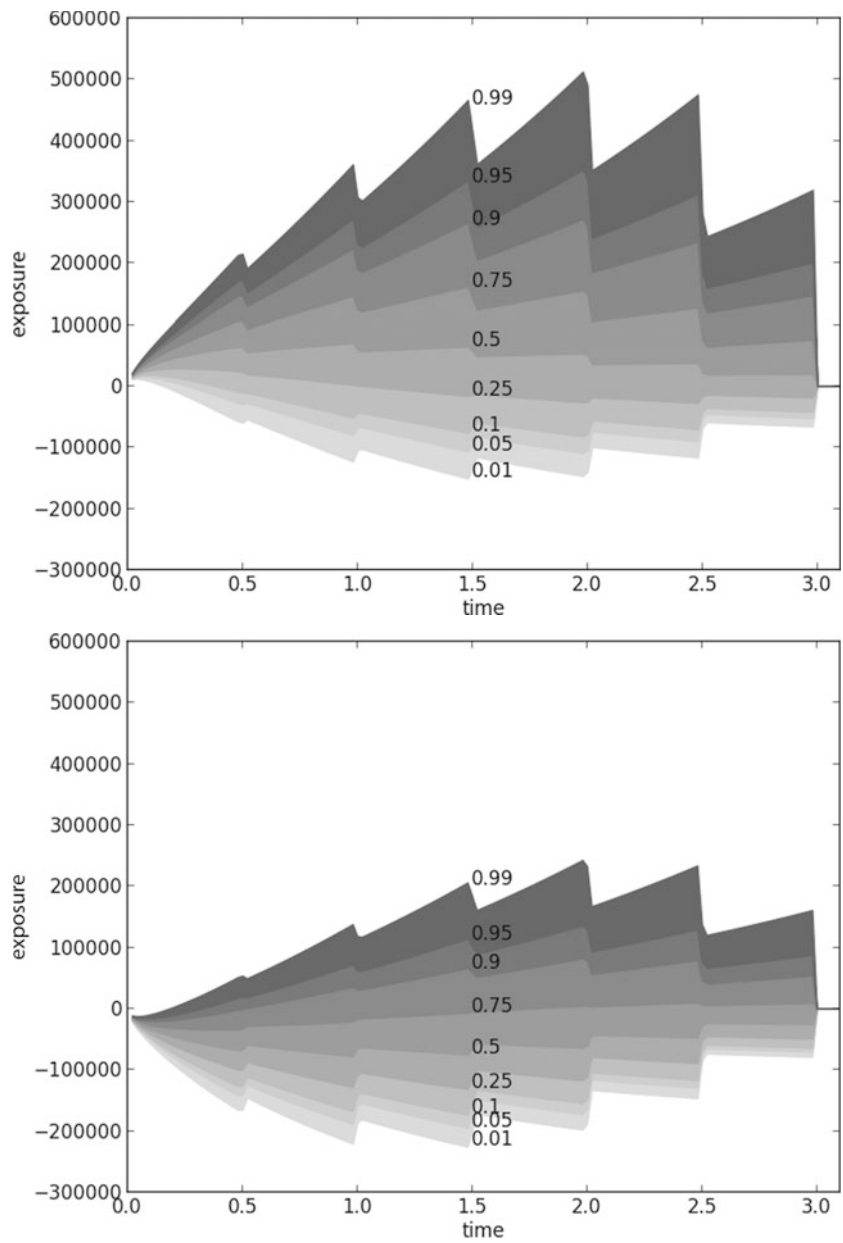


Figure 14.1 CVA quantiles with correlations -25%, 25%

14.4.2 Constant deterministic barrier

In the special case of $B_t = B$ one can derive closed form expressions for CVA for simple cash flows. Expectation

$$\mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}]$$

has been computed in Section 18.4.2, see formulae (18.36) and (18.35) with parameters

$$\begin{aligned} x &= \frac{\log(B/A_0)}{\sigma_A \sqrt{T}}, \quad y = \frac{\log(K/F_T^S) + \sigma_S^2 T/2}{\sigma_S \sqrt{T}}, \\ v &= \frac{\log(F_T^A/A_0)}{T} - \frac{\sigma_A^2}{2}, \quad \mu = \sigma_S \sqrt{T} \end{aligned} \quad (14.5)$$

CVA is thus given as

$$\begin{aligned} \text{CVA} &= \hat{V} - V \\ &= P(T) \left(\mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau < T\}}] + \mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}] - \mathbb{E}[(S_T - K)^+] \right) \\ &= (R - 1)P(T) \left(\mathbb{E}[(S_T - K)^+] - \mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}] \right) \end{aligned}$$

14.4.3 Constant random barrier à la CreditGrades

As discussed in Section 12.3 an asset model with constant barrier does not allow one to calibrate spread term structure. The CreditGrades (2002) approach with log-normal random barrier allows one to achieve a better approximation to CDS data. For options on S , with a counterparty defaulting if a correlated log-normal asset hits a constant log-normal barrier, a closed form expression can be derived. It can be computed from

$$\begin{aligned} &\mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}] \\ &= F(T)^S e^{-\lambda^2/2} \mathbb{E}[\mathbf{1}_{\{X \leq \gamma\}} f(x(X), y, \rho, v, \mu)] - K \mathbb{E}[\mathbf{1}_{\{X < \gamma\}} f(x(X), y, \rho, v, 0)] \end{aligned}$$

with y, v, μ as in (14.5) as well as

$$\begin{aligned} x(X) &= \frac{\log(B/A_0)}{\sigma_A \sqrt{T}} = \frac{\log(\bar{B}/A_0) + \lambda X - \lambda^2/2}{\sigma_A \sqrt{T}} = \alpha X + \beta \\ \gamma &= \frac{\log(A_0/\bar{B}) + \lambda^2/2}{\lambda} \end{aligned}$$

for some constants α and β where f is defined in (18.35). In fact,

$$\begin{aligned} & \mathbb{E} \left[\mathbf{1}_{\{X \leq y\}} f(x(X), y, \rho, v, \mu) \right] \\ &= \mathbb{E} \left[\mathbf{1}_{\{X \leq y\}} e^{\mu^2/2 + 2x(X)(v + \rho\mu)} \Phi_2(x(X) + v + \rho\mu, -y + \mu + 2\rho x(X); \rho) \right] \\ &= e^{\mu^2/2 + 2\beta(v + \rho\mu)} \mathbb{E} \left[\mathbf{1}_{\{X \leq y\}} e^{\delta X} \Phi_2(\alpha_1 X + \beta_1, \alpha_2 X + \beta_2; \rho) \right] \end{aligned}$$

with parameters

$$\begin{aligned} \delta &= 2\alpha(v + \rho\mu) \\ \alpha_1 &= \alpha \\ \beta_1 &= \beta + v + \rho\mu \\ \alpha_2 &= 2\rho\alpha \\ \beta_2 &= -y + \mu + 2\rho\beta \end{aligned}$$

is given as a trivariate function in (18.43). An approximate bivariate formula can be obtained by considering a full barrier, as suggested in CreditGrades (2002), compare formula (12.14). One has

$$\begin{aligned} & \mathbb{E} \left[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}} \right] \\ &= \mathbb{E} \left[(\tilde{S}_{T + \lambda^2/\sigma_A^2} - K)^+ \mathbf{1}_{\left\{ \forall t \in [\lambda^2/\sigma_A^2, T + \lambda^2/\sigma_A^2]: \tilde{A}_t > \tilde{B}e^{-\lambda^2} \right\}} \right] \\ &\approx \mathbb{E} \left[(\tilde{S}_{T + \lambda^2/\sigma_A^2} - K)^+ \mathbf{1}_{\left\{ \tilde{A}_{*, T + \lambda^2/\sigma_A^2} > \tilde{B}e^{-\lambda^2} \right\}} \right] \end{aligned}$$

where the distribution of processes $(\tilde{S}, \tilde{A})$ is defined by

$$\begin{aligned} F^{\tilde{S}}(t) &= S_0 \exp \left(\log \left(\frac{F^S(T)}{S_0} \right) \frac{t - \lambda^2/\sigma_A^2}{T} \right) \\ \tilde{\sigma}^2 &= \sigma^2 \frac{T}{T + \lambda^2/\sigma_A^2} \\ F^{\tilde{A}}(t) &= A_0 \exp \left(\log \left(\frac{F^A(T)}{A_0} \right) \frac{t - \lambda^2/\sigma_A^2}{T} \right) \\ \tilde{\sigma}_A &= \sigma_A \\ \tilde{\rho} &= \frac{\rho}{\sqrt{1 + \lambda^2/(\sigma_A^2 T)}} \end{aligned} \tag{14.6}$$

This leads to

$$\mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}] = F(T)^S e^{-\lambda^2/2} f(\tilde{x}, y, \tilde{\rho}, \nu, \mu) - Kf(\tilde{x}, y, \tilde{\rho}, \nu, 0)$$

with parameters y, μ, ν as in (14.5), $\tilde{\rho}$ as in (14.6), as well as

$$\tilde{x} = \frac{\log(\bar{B}/\tilde{A}_0) - \lambda^2}{\sigma_A \sqrt{T + \lambda^2/\sigma_A^2}} = \frac{\log(\bar{B}/A_0) + \log(F^A(T)/A_0) \lambda^2 / (\sigma_A^2 T) - \lambda^2}{\sigma_A \sqrt{T + \lambda^2/\sigma_A^2}}$$

Thus the adjustment of the standard outside barrier formula (18.36) due to the random barrier consists in lower correlation $\tilde{\rho}$ and modified distance to default $\tilde{x}$.

Figure 14.2 exhibits CVA (factor $R - 1$ is omitted for the remainder of this chapter) of a Vanilla option with different values of the counterparty's asset volatility and barrier volatility (forward 100, volatility 20%, maturity 5y, call, strike 100, interest rate 0%, survival probability 95%). The barrier is calibrated to survival probability. As expected, correlation impact decreases with barrier volatility and increases with asset volatility.

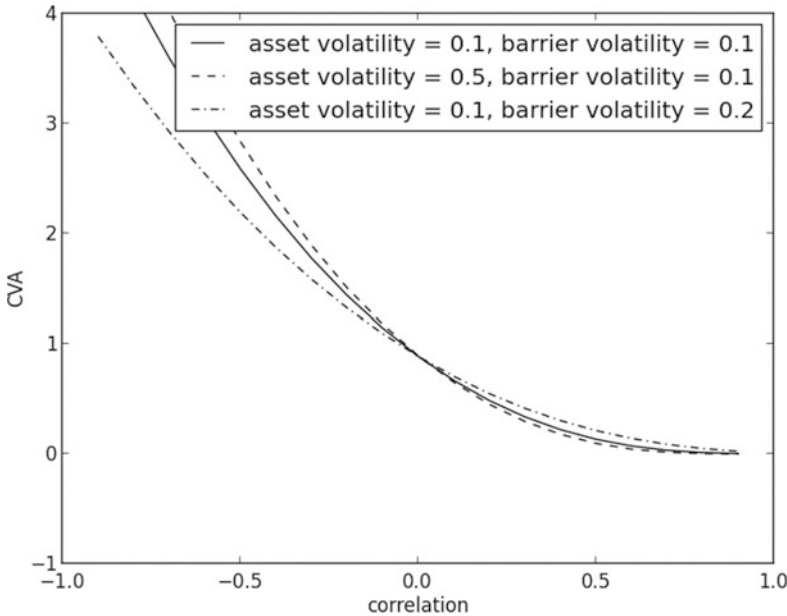


Figure 14.2 CVA with CreditGrades default barrier

14.4.4 Gaussian copula

Counterparty default time τ consistent with survival probabilities $p(t) = 1 - q(t)$ can be defined as

$$\tau = q^{-1}(\Phi(X))$$

where X is a standard Gaussian variate, compare Section 12.3.6, equation (12.16). The price of a call on a log-normal stock S_T with counterparty default time joined via a Gaussian copula can be expressed as a European outside barrier, namely

$$\mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}] = \mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{A_T > B\}}]$$

The relationship between X, q, A_T and B , such that equalities

$$\{\tau > T\} = \{\Phi(X) > q(T)\} = \{A_T > B\}$$

hold, is given as

$$A_T = F^A(T) \exp\left(\sigma_A \sqrt{T} X - \frac{\sigma_A^2 T}{2}\right)$$

$$B = F^A(T) \exp\left(\sigma_A \sqrt{T} \Phi^{-1}(q(T)) - \frac{\sigma_A^2 T}{2}\right)$$

Application of formula (18.37) yields

$$\begin{aligned} & \mathbb{E}[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}}] \\ &= F(T) \Phi_2\left(d_2^-(B) + \rho \sigma \sqrt{T}, d^+(K); \rho\right) - K \Phi_2\left(d_2^-(B), d^-(K); \rho\right) \\ &= F(T) \Phi_2\left(\Phi^{-1}(p(t)) + \rho \sigma \sqrt{T}, d^+(K); \rho\right) - K \Phi_2\left(\Phi^{-1}(p(t)), d^-(K); \rho\right) \end{aligned}$$

With zero correlation the Black-Scholes formula multiplied with $p(T)$ is recovered.

14.5 Reduced form models

14.5.1 Black-Scholes and Gaussian hazard rate

This section applies the setup developed in Section 13.1.2 in the context of defaultable equity to the case of counterparty default. Let

$$S_T = F(T) e^{\sigma_S X - \sigma_S^2 T/2}$$

with Gaussian variate X and denote integrated hazard rate by

$$H_T = \int_0^T h_t dt$$

For a positive payoff $f(S_T)$ owed by a counterparty the price can be obtained as

$$\mathbb{E} [f(S_T) \mathbf{1}_{\{\tau > T\}}] = \mathbb{E} [f(S_T) \mathbf{1}_{\{U < e^{-H_T}\}}] = \mathbb{E} [f(S_T) e^{-H_T}]$$

In the case of H_T being independent of S_T the price is given by risky discounting. Assume the Gaussian approximation

$$H_T \approx (a_T + \sigma_H Y)^+ = G_T^+$$

where Y is another Gaussian correlated with X with coefficient ρ . Given survival probability $p(T)$ and volatility σ_H one can solve for a_T using (13.4). The call option price with counterparty credit risk can be computed from

$$\begin{aligned} \mathbb{E} [(S_T - K)^+] &= \mathbb{E} [S_T \mathbf{1}_{\{G_T \leq 0, S_T > K\}}] + \mathbb{E} [S_T e^{-G_T} \mathbf{1}_{\{G_T > 0, S_T > K\}}] \\ &\quad - K \mathbb{P} [G_T \leq 0, S_T > K] - K \mathbb{E} [e^{-G_T} \mathbf{1}_{\{G_T > 0, S_T > K\}}] \end{aligned}$$

The first term yields

$$\begin{aligned} \mathbb{E} [S_T \mathbf{1}_{\{S_T > K, G_T \leq 0\}}] &= F(T) \mathbb{P} [-X < d_+, a_T + \sigma_H(Y + \rho\sigma_S) \leq 0] \\ &= F(T) \Phi_2 \left(d_+, -\frac{a_T}{\sigma_H} - \rho\sigma_S; -\rho \right) \end{aligned}$$

where

$$d_{\pm} = \frac{1}{\sigma_S} \left(\log \frac{F(T)}{K} \pm \frac{1}{2} \sigma_S^2 \right)$$

The second term is

$$\begin{aligned} &\mathbb{E} [S_T \mathbf{1}_{\{S_T > K, G_T > 0\}} e^{-G_T}] \\ &= F(T) \mathbb{E} [\mathbf{1}_{\{-X < d_+, a_T + \sigma_H(Y + \rho\sigma_S) > 0\}} e^{-a_T - \sigma_H(Y + \rho\sigma_S)}] \\ &= F(T) e^{-a_T - \rho\sigma_S\sigma_H + \sigma_H^2/2} \Phi_2 \left(d_+ - \rho\sigma_H, \frac{a_T}{\sigma_H} + \rho\sigma_S - \sigma_H; \rho \right) \end{aligned}$$

Similarly, the third term is

$$\begin{aligned} \text{KIP} [S_T > K, G_T \leq 0] &= \text{KIP} [-X < d_-, a_T + \sigma_H Y \leq 0] \\ &= K \Phi_2 \left(d_-, -\frac{a_T}{\sigma_H}; -\rho \right) \end{aligned}$$

and the fourth term is

$$\begin{aligned} \text{KIE} \left[\mathbf{1}_{\{S_T > K\}} e^{-H_T} \right] &= \text{KIE} \left[\mathbf{1}_{\{-X < d_-, a_T + \sigma_H Y > 0\}} e^{-a_T - \sigma_H Y} \right] \\ &= K e^{-a_T + \sigma_H^2/2} \Phi_2 \left(d_- - \rho \sigma_H, \frac{a_T}{\sigma_H} - \sigma_H; \rho \right) \end{aligned}$$

In summary

$$\begin{aligned} &\text{IE} \left[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}} \right] \\ &= F(T) \Phi_2 \left(d_+, -\frac{a_T}{\sigma_H} - \rho \sigma_S; -\rho \right) \\ &\quad + F(T) e^{-a_T + \sigma_H^2/2 - \rho \sigma_S \sigma_H} \Phi_2 \left(d_+ - \rho \sigma_H, \frac{a_T}{\sigma_H} + \rho \sigma_S - \sigma_H; \rho \right) \\ &\quad - K \Phi_2 \left(d_-, -\frac{a_T}{\sigma_H}; -\rho \right) \\ &\quad - K e^{-a_T + \sigma_H^2/2} \Phi_2 \left(d_- - \rho \sigma_H, \frac{a_T}{\sigma_H} - \sigma_H; \rho \right) \end{aligned} \quad (14.7)$$

Note that if $\sigma_H = 0$ then one obtains the Black-Scholes price with risky discount factor $p(T)P(T)$. In practice one often has $a_T \gg \sigma_H$ leading to the approximation

$$\begin{aligned} &\text{IE} \left[(S_T - K)^+ \mathbf{1}_{\{\tau > T\}} \right] \\ &\approx p(T) \left\{ F(T) e^{-\rho \sigma_S \sigma_H} \Phi(d_+ - \rho \sigma_H) - K \Phi(d_- - \rho \sigma_H) \right\} \end{aligned} \quad (14.8)$$

This approximation is valid if

$$\text{IP} [H_T < 0] \approx 0.$$

Hence it will deteriorate with increasing hazard rate volatility σ_H and decreasing expected hazard rate.

Figure 14.3 displays the CVA of a Vanilla option with different values of the counterparty's hazard rate volatility (forward 100, volatility 20%, maturity 5y, call, strike 100, interest rate 0%, survival probability 95%). Hazard rate mean a_T

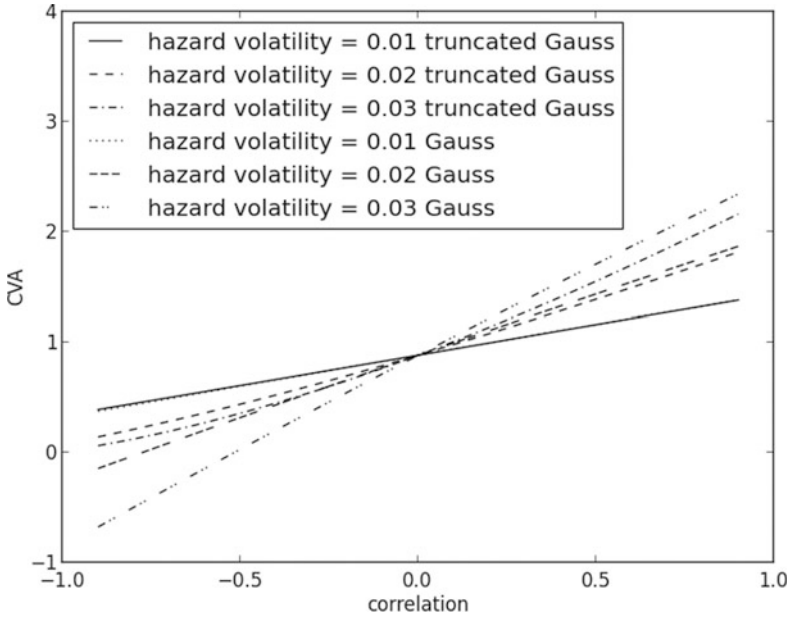


Figure 14.3 CVA with Gaussian hazard (call and put)

is calibrated to survival probability. The truncated Gauss formula (14.7) ensures positive CVA, while the approximating Gauss formula (14.8) fails for negative correlation and large hazard volatility, in the sense that CVA attains negative values.

Figure 14.4 exhibits CVA of Vanilla options (left graph: call, right graph: put) with different values of correlation of hazard rate increments and asset returns where the hazard process is assumed to be CIR (forward 100, volatility 20%, maturity ten years, interest rate 0%, short and long hazard 5%, mean reversion and vol 100%). This is compared with a closed form CVA formula (14.7) and (14.8) with σ_H obtained from (6.12), a_T calibrated to CIR survival probability p_T via (13.4) and ρ computed numerically from the CIR case.

14.5.2 Stochastic hazard rate

Stochastic hazard rates within a defaultable equity model have been discussed in Section 13.1.4. A corresponding non defaulting stochastic volatility stock process with counterparty default driven by stochastic hazard rate is discussed below. The dynamics is again given by

$$\begin{aligned} d\sigma_t^2 &= (\theta_\sigma^2 - \kappa_\sigma \sigma_t^2) dt + \alpha_\sigma \sigma_t dW_t^\sigma \\ h_t &= \beta \sigma_t^2 + z_t \\ dz_t &= (\theta_z - \kappa_z z_t) dt + \alpha_z \sqrt{z_t} dW_t^z \end{aligned}$$

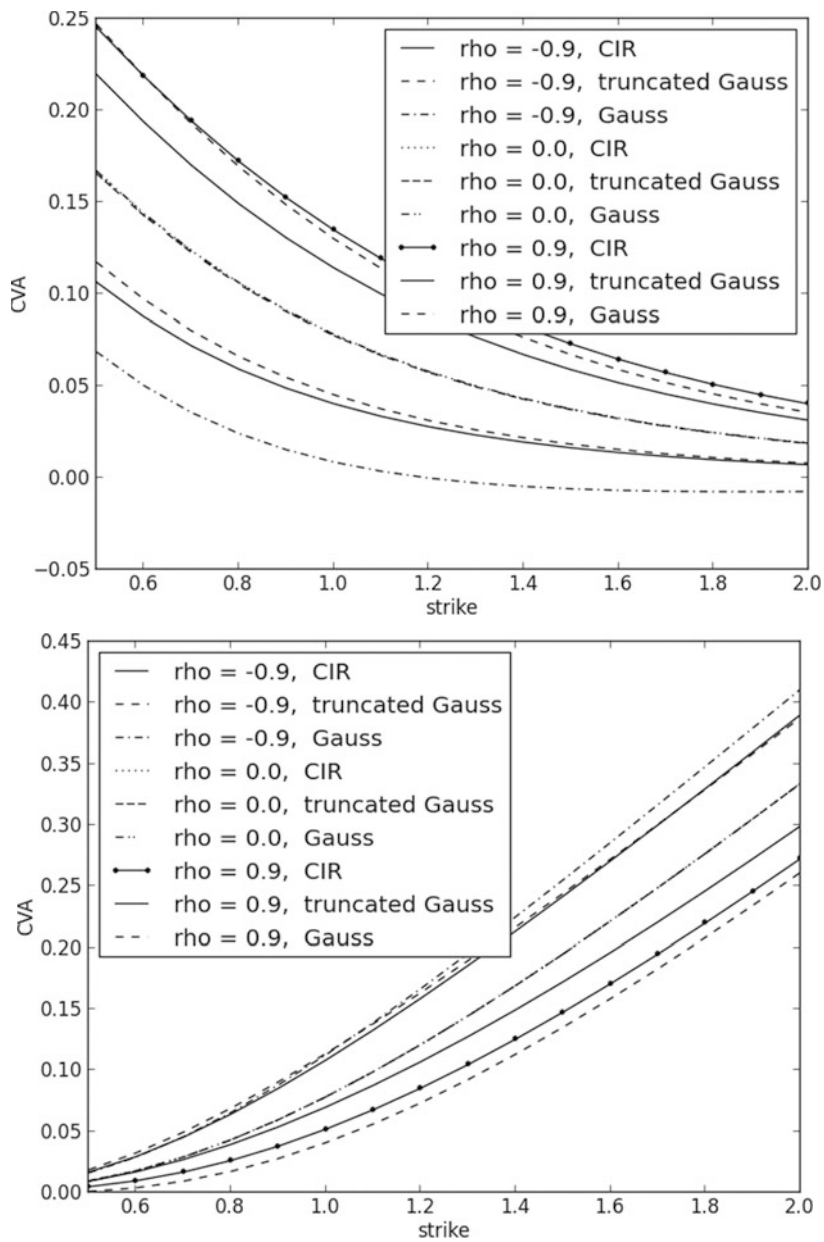


Figure 14.4 CVA with CIR hazard versus Gaussian hazard

with W^z independent of the correlated pair (W, W^σ) . Using notation H_T for $\int_0^T h_t dt$ and Z_T for $\int_0^T z_t dt$ the undiscounted complex payoff

$$\mathbf{1}_{\{\tau > T\}} \exp\left(iu \log \frac{S_T}{F(T)}\right)$$

has value

$$\begin{aligned} \phi(u) &= \mathbb{E} \left[\exp \left(-H_T + iu \left(\int_0^T \sigma_t dW_t - \frac{1}{2} \int_0^T \sigma_t^2 dt \right) \right) \right] \\ &= \mathbb{E}_{iu\sigma dW} \left[\exp \left(-H_T - \frac{iu + u^2}{2} \int_0^T \sigma_t^2 dt \right) \right] \\ &= \mathbb{E}_{iu\sigma dW} \left[\exp \left(-Z_T - \left(\beta + \frac{iu + u^2}{2} \right) \int_0^T \sigma_t^2 dt \right) \right] \\ &= \mathbb{E} [\exp(-Z_T)] \mathbb{E}_{iu\sigma dW} \left[\exp \left(- \left(\beta + \frac{iu + u^2}{2} \right) \int_0^T \sigma_t^2 dt \right) \right] \end{aligned}$$

where the density of $\mathbb{P}_{iu\sigma dW}$ with respect to $\mathbb{P}$ is given by the complex martingale M with $M_0 = 1$ and

$$\frac{dM_t}{M_t} = iu\sigma_t dW_t$$

While Z is independent of M the dynamics of σ^2 with respect to $\mathbb{P}_{iu\sigma dW}$ has a modified CIR parameter

$$\tilde{\kappa}_\sigma = \kappa_\sigma - iu\rho\alpha_\sigma$$

The transform of integrated CIR processes Z_T and $\int_0^T \sigma_t^2 dt$ follows from equation (10.21).

One may consider defaultable equity with hazard rate h^S combined with counterparty default with hazard rate h^C of the form

$$\begin{aligned} h_t^S &= \beta_S \sigma_t^2 + z_t^1 \\ dz_t^S &= (\theta_S - \kappa_S z_t^S) dt + \alpha_S \sqrt{z_t^S} dW_t^S \\ h_t^C &= \beta_C \sigma_t^2 + z_t^2 \\ dz_t^C &= (\theta_C - \kappa_C z_t^C) dt + \alpha_C \sqrt{z_t^C} dW_t^C \end{aligned}$$

where W^S , W^C and the pair (W, W^σ) are independent. Using the same reasoning one obtains

$$\begin{aligned}
 \phi(u) &= \mathbb{E} \left[\exp \left(-H_T^C - H_T^S + iu \left(H_T^S + \int_0^T \sigma_t dW_t - \frac{1}{2} \int_0^T \sigma_t^2 dt \right) \right) \right] \\
 &= \mathbb{E}_{iu\sigma dW} \left[\exp \left(-H_T^C - (1-iu)H_T^S - \frac{iu+u^2}{2} \int_0^T \sigma_t^2 dt \right) \right] \\
 &= \mathbb{E}_{iu\sigma dW} \left[\exp \left(-Z_T^C - (1-iu)Z_T^S - \left(\beta_C + (1-iu)\beta_S + \frac{iu+u^2}{2} \right) \int_0^T \sigma_t^2 dt \right) \right] \\
 &= \mathbb{E} \left[\exp \left(-Z_T^C \right) \right] \mathbb{E} \left[\exp \left(-(1-iu)Z_T^S \right) \right] \\
 &\quad \mathbb{E}_{iu\sigma dW} \left[\exp \left(- \left(\beta_C + (1-iu)\beta_S + \frac{iu+u^2}{2} \right) \int_0^T \sigma_t^2 dt \right) \right]
 \end{aligned}$$

Note that correlation between stock volatility, stock default and counterparty default is only due to correlation of volatility and hazard rate processes. Assuming deterministic hazard rates will result in independence between stock process as well as default times of stock and counterparty. For methods incorporating correlation between default events see Section 12.4. Assuming Gaussian or truncated Gaussian approximations for Z^C, Z^S default event correlation can be incorporated.

14.6 Funding valuation adjustment

There is an ongoing debate on whether an adjustment other than CVA and DVA for the purpose of reflecting funding requirements should be applied. Trading operations are exposed to funding cost by their treasury department and tend to be in favour of adding a funding adjustment value FVA. Both theoretical considerations and regulatory requirements, namely exit cost pricing, support the position of not including funding.

Hull and White (2012) argue that if the derivative is perfectly hedged then the discount rate should be the riskless rate, as in Black and Scholes (1973). If it is not hedged then a rate commensurate with the risk should be charged, as in Merton (1973). Both approaches can be extended to the case of risky assets and counterparty risk leading to the incorporation of bilateral CVA and DVA. Incorporating funding implies that a hedged derivative should earn the bank's cost of funding, which reflects the average risk of the bank's existing portfolio. However, cost, of equity capital are higher than those of debt and it is not clear which rate of return should be applied. In short, there is no economic principle supporting FVA.

Burgard and Kjaer (2011) argue that in case the derivative cannot be used as collateral and cash needs to be borrowed to fund the portfolio then an additional funding spread s is payable. They suggest hedging own credit risk by repurchasing own bonds with cash generated from the proceeds of the derivative (i.e. portfolio). Their key result is the FVA expression

$$\text{FVA}^{[s,T]} = -N_0 \mathbb{E} \left[\int_s^{\tau \wedge \tilde{\tau} \wedge T} \frac{1}{N_t} s_t V_t^+ dt \right] \quad (14.9)$$

If the funding spread is derived from own credit spreads, namely

$$(1 - \tilde{R}) \mathbb{P}[\tilde{\tau} \in dt] = \mathbb{E} [s_t dt \mathbf{1}_{\{\tilde{\tau} > t\}}]$$

then (14.9) implies

$$\text{DVA}^{[s,T]} + \text{FVA}^{[s,T]} = -(1 - \tilde{R}) N_0 \mathbb{E} \left[\frac{1}{N_{\tilde{\tau}}} \mathbf{1}_{\{s < \tilde{\tau} \leq T, \tilde{\tau} < \tau\}} V_{\tilde{\tau}} \right]$$

The reasoning for (14.9) proceeds as follows. As above, it is assumed that upon default of either counterparty the recovery value of the portfolio is given as a fraction of a similar default free portfolio. Let r denote risk free rate and P be the value process of a risky bond issued by the counterparty. An own bond is denoted $\tilde{P}$. The dynamics of P is

$$\frac{dP_t}{P_t} = (r_t + h_t) dt - dN_t$$

where N is a Poisson jump process moving from 0 to 1 with intensity h , with corresponding notation for the dynamics of $\tilde{P}$. For ease of notation the bond expiries are not specified here. Also, risky bonds are assumed to have zero recovery. Consider the value $\hat{V}$ of a portfolio containing a derivative on a risk free asset S with adjustments

$$\hat{V}_t = V_t + \text{CVA}_t + \text{DVA}_t + \text{FVA}_t \quad (14.10)$$

where V_t is the value excluding any adjustments. A self-financing hedge portfolio offsetting the position shall be set up as

$$-\hat{V}_t = \Delta_t S_t + \alpha_t P_t + \tilde{\alpha}_t \tilde{P}_t + \beta_t$$

One expects $\alpha_t \leq 0$ in the case of $V_t \geq 0$ since counterparty default will lead to a loss. Windfall gains due to own default in the case of $V_t \leq 0$ can be hedged with

$\tilde{\alpha}_t \leq 0$. For the hedge to be self-financing the cash account has to satisfy

$$d\beta_t = -r_t \hat{V}_t dt + \Delta_t (d_t - r_t) S_t dt - \alpha_t r_t P_t dt - \tilde{\alpha}_t r_t \tilde{P}_t dt + s_t (-\hat{V}_t - \tilde{\alpha}_t \tilde{P}_t)^- dt$$

The second term is income on stock where parameter d_t is dividend yield minus repo rate. For positions in own and counterparty bond a repo agreement incurs financing cost at the risk free rate as the bonds can be pledged as collateral giving the third and fourth term. If the portfolio excluding own bonds needs to be financed an additional funding spread s_F has to be paid. One has

$$\begin{aligned} -d\hat{V}_t &= \Delta_t dS_t + \alpha_t dP_t + \tilde{\alpha}_t d\tilde{P}_t + d\beta_t \\ &= \left(-r_t \hat{V}_t + \Delta_t (d_t - r_t) S_t dt + \alpha_t h_t P_t + \tilde{\alpha}_t \tilde{h}_t \tilde{P}_t + s_t (-\hat{V}_t - \tilde{\alpha}_t \tilde{P}_t)^- \right) dt \\ &\quad + \Delta_t dS_t - \alpha_t P_t dN_t - \tilde{\alpha}_t \tilde{P}_t d\tilde{N}_t \end{aligned}$$

On the other hand, Itô's formula applied to $\hat{V}_t = \hat{V}_t(S_t, N_t, \tilde{N}_t)$ yields

$$d\hat{V}_t = \frac{\partial \hat{V}_t}{\partial t} dt + \frac{\partial \hat{V}_t}{\partial S} dS_t + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 \hat{V}_t}{\partial S^2} dt + \Delta \hat{V}_t dN_t + \tilde{\Delta} \hat{V}_t d\tilde{N}_t$$

where

$$\Delta \hat{V}_t = \hat{V}_t(S_t, 1, 0) - \hat{V}_t(S_t, 0, 0) \quad \text{and} \quad \tilde{\Delta} \hat{V}_t = \hat{V}_t(S_t, 0, 1) - \hat{V}_t(S_t, 0, 0)$$

It follows that all risks can be hedged with hedge ratios

$$\Delta_t = -\frac{\partial \hat{V}_t}{\partial S}, \quad \alpha_t = \frac{\Delta \hat{V}_t}{P_t}, \quad \tilde{\alpha}_t = \frac{\tilde{\Delta} \hat{V}_t}{\tilde{P}_t}$$

Since a riskless portfolio grows at a risk free rate r one obtains equation

$$\frac{\partial \hat{V}_t}{\partial t} + (r_t - d_t) S_t \frac{\partial \hat{V}_t}{\partial S} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 \hat{V}_t}{\partial S^2} - r_t \hat{V}_t + h_t \Delta \hat{V}_t + \tilde{h}_t \tilde{\Delta} \hat{V}_t - s_t (\hat{V}_t + \tilde{\Delta} \hat{V}_t)^+ = 0$$

with suitable boundary condition, for example $\hat{V}_T(S) = (S_T - K)^+$ for a call option. With the assumption

$$\begin{aligned} \hat{V}_t(S, 1, 0) &= RV_t^+(S, 0, 0) + V^-(S, 0, 0) \\ \hat{V}_t(S, 0, 1) &= V_t^+(S, 0, 0) + \tilde{R}V^-(S, 0, 0) \end{aligned}$$

one obtains

$$\begin{aligned} & \frac{\partial \hat{V}_t}{\partial t} + (r_t - d_t) S_t \frac{\partial \hat{V}_t}{\partial S} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 \hat{V}_t}{\partial S^2} - (r_t + h_t + \tilde{h}_t) \hat{V}_t \\ & = -h_t (V_t^+ + R V_t^-) - \tilde{h} (\tilde{R} V_t^+ + V_t^-) + s_t V_t^+ \end{aligned}$$

This is the special case with deterministic time-dependent parameters $r, q, \sigma, h, \tilde{h}$ of (14.10) with (14.2), (14.3) and (14.9) via Feynman-Kac theorem.

15 Foreign Exchange

Foreign exchange rates are defined by both domestic and foreign currency. These concepts are symmetric in the sense that for a foreign investor my domestic currency is her foreign currency. For example, a European investor may buy 1\$ at today's exchange rate of 0.75€. A call option with strike 0.70€ may cost 0.06€. Expressed in USD one has

$$\frac{1}{X_0} P^{\text{€}}(T) \text{IE}_{\text{€}} \left[(X_T - K)^+ \right] = \frac{0.06}{0.75} = 0.08 \quad (15.1)$$

where $\text{IE}_{\text{€}}$ denotes the risk free measure of the European investor. For the foreign investor the value of 1€ is $1/X_T$ whereas 1\$ has constant value. Thus, the same product has value

$$P^{\text{\$}}(T) \text{IE}_{\text{\$}} \left[\left(1 - \frac{K}{X_T} \right)^+ \right] = K P^{\text{\$}}(T) \text{IE}_{\text{\$}} \left[\left(\frac{1}{K} - \frac{1}{X_T} \right)^+ \right] \quad (15.2)$$

Those prices need to coincide, thus defining a relationship between the riskless measures of domestic and foreign investor. Moreover $K = 0$ implies the foreign exchange forward

$$F^X(T) = X_0 \frac{P^{\text{\$}}(T)}{P^{\text{€}}(T)} \quad (15.3)$$

15.1 Cross currency basis swap

A cross currency basis swap may have the following cash flows:

- Pay 1€ and receive X_0 \$ at inception
- Receive periodic payments of €-Libor plus spread and pay X_0 times \$-Libor
- Receive 1€ and pay X_0 \$ at maturity

Cross currency basis spread s is thus defined through

$$\begin{aligned}
 & -1^{\text{€}} + \sum_{i=0}^{n-1} \alpha_i (L^{\text{€},i} + s) P^{\text{€}}(T_{i+1}) + P^{\text{€}}(T_n) \\
 & = X_0 \left(-1^{\text{\$}} + \sum_{i=0}^{n-1} \alpha_i L^{\text{\$},i} P^{\text{\$}}(T_{i+1}) + P^{\text{\$}}(T_n) \right) \quad (15.4)
 \end{aligned}$$

Given $L^{\text{\$}}$ and $P^{\text{\$}}$ and standard swap rates for $L^{\text{€}}$ allows the bootstrapping of $P^{\text{€}}$, compare sections 10.1.4, 10.1.5 and 10.1.6. Forward F^X can be computed as

$$F^X(T) = X_0 \frac{P^{f,\text{\$}}(T)}{P^{f,\text{€}}(T)} \left(1 + \frac{s}{r} \left(1 - P^{f,\text{€}}(T) \right) \right)$$

where r is the rate of a standard swap in € and P^f refers to the forwarding curve defined by

$$P^f(T_n)^{-1} = \prod_{i=0}^{n-1} (1 + \alpha_i L^i)$$

Resettable cross currency swaps use the prevailing exchange rate X_t for payments at time t , rather than today's rate X_0 .

Since 2007 traded forward levels deviate from forwards implied by cross currency swaps, as well as from the ratio of discount factors (15.3). Giménez et al. (2014) find that cross currency swaps are best aligned with the forward market. Discount factor ratios from OIS curves (see Section 10.1.5) are worse than those from Libor curves.

15.2 Market smile

Foreign exchange option prices are quoted through Black-Scholes implied volatilities $\hat{\sigma}$. Unlike in equities markets the implied volatility smile is defined as function of delta rather than strike, compare Section 4.2.6. For a given term the smile is typically defined through a function interpolating three (atm, p25, c25) or five (atm, p25, c25, p10, c10) deltas. Here c/p stands for call/put and 10 and 25 for 10% and 25% (absolute) delta.

- Note that delta $\Delta(K)$ and hence volatility $\hat{\sigma}(\Delta(K))$ for a given strike K can only be computed iteratively.

- Strike $K(\Delta) = K(\Delta, \hat{\sigma}(\Delta))$ can be computed directly given delta by inverting the function $\Delta(K)$ with given volatility $\hat{\sigma} = \hat{\sigma}(\Delta)$.
- There are various definitions of delta, such as (but not limited to) spot Δ_X , forward Δ_F and driftless delta Δ_0 given as

$$\Delta_X = P^f(T)\Phi(d_+)$$

$$\Delta_F = P^d(T)\Phi(d_+)$$

$$\Delta_0 = \Phi(d_+)$$

15.3 Vanna-Volga approach

The Vanna-Volga approach has been developed by practitioners as an extension of the Black-Scholes model with the aim of allowing first-generation exotics to be priced with smile data included. First-generation exotics include barrier products, in particular single- and double-touch and single- and double-barrier options. Following Castagna and Mercurio (2007) the market price of the derivative is assumed to be given through a Black-Scholes formula for the payoff applied to the market implied volatility:

$$V_t = V_t(X_t, \sigma_t)$$

Price changes are due to changes in implied volatility σ and spot X . One intends to hedge the derivative with respect to changes in volatility and spot. The derivative including hedge

$$\Pi_t = V_t - \Delta X_t - \sum_i x_i \text{Call}_t^i$$

can be expanded up to first order in dt as

$$d\Pi = \frac{\partial \Pi}{\partial t} dt + \frac{\partial \Pi}{\partial X} dX + \frac{\partial \Pi}{\partial \sigma} d\sigma + \frac{1}{2} \frac{\partial^2 \Pi}{\partial \sigma^2} d\langle \sigma \rangle + \frac{\partial^2 \Pi}{\partial \sigma \partial X} d\langle \sigma, X \rangle + \frac{1}{2} \frac{\partial^2 \Pi}{\partial X^2} d\langle X \rangle$$

With three options $\text{Call}^1, \text{Call}^2, \text{Call}^3$ with different strikes one can find weights Δ, x_1, x_2, x_3 such that delta, vega, vanna $\frac{\partial^2 \Pi}{\partial \sigma \partial X}$ and volga $\frac{\partial^2 \Pi}{\partial \sigma^2}$ vanish. Gamma will also vanish due to the Black-Scholes equation. Thus one ends up with a locally riskless portfolio. The Vanilla options will typically trade at different Black-Scholes implied volatilities. Adding the price discrepancy for setting up the hedge to the Black-Scholes price V^{BS} takes into account the market smile. Thus the Vanna-Volga price V^{VV} is given as

$$V^{VV} = V^{BS} + \sum_{i=1}^3 x_i \left(\text{Call}^{i, mkt} - \text{Call}^{i, BS} \right)$$

It turns out that the Vanna-Volga price of Vanilla options is consistent in the following sense; if market prices $Call^{i,mkt}$, $i = 1, 2, 3$ are given at three liquid strikes (such as atm, p25, c25) then Vanna-Volga prices of options with the same maturity and arbitrary strike interpolate those market prices. Thus Vanna-Volga can be used to define the implied volatility smile function.

In the case of barrier products the hedge adjustment is found to be too large. It is often multiplied with a quantity related to a barrier hitting probability p since the hedge adjustment is no longer needed once the barrier is hit. For a given product there may be different conventions depending on the trader's views on how to compute p . Details can be found in Wystup (2006), see the chapter on trader's rule of thumb.

15.4 Models

For Vanilla options the Black-Scholes model with smile interpolation, following Section 15.2, is commonly used. Vanna-Volga may be used in the context of first-generation exotics, namely single-asset options with one or two constant barriers.

Market standard for both investment banks and software providers, such as Murex, Numerix, VolMaster and Fincad, is the use of local stochastic volatility models as described in Section 6.3. These models allow them to take into account, to some extent, market prices of first-generation exotics. The autonomous volatility process is typically chosen to be analytically tractable, such as Heston, possibly extended to allow for piecewise constant parameters or jumps. The local factor may be nonparametric, requiring numerical schemes for calibration, or parametric, allowing for analytic approximations.

In the presence of several exchange rates correlation can be implied through option prices since the ratio of two exchange rates is again an exchange rate if they share one currency. This also introduces restrictions on foreign exchange modelling since asset ratios are required to be processes within the same model class. A consistent framework can be defined by only modelling exchange rates relative to USD. Another consistent framework can be formulated in terms of multivariate Black-Scholes processes, possibly extended by Hull-White interest rates, as discussed in Section 11.1.

15.5 Quanto options

A quanto product converts underlying asset prices into units of the payoff currency by applying a fixed exchange rate. A common example is a basket option on stocks

or indices denominated in different currencies with payoff

$$\left(\sum_k w_k \frac{S_t^k}{S_0^k} - K \right)^+$$

The foreign currency amount S_t^k is converted into a domestic currency amount using exchange rate one. Let S be a foreign asset denoted in currency f . If interest rates are deterministic then the relation between domestic and foreign measure is given by

$$\frac{\mathbb{P}^d}{\mathbb{P}^f} \Big| \mathcal{F}_t = \frac{X_t}{F_0^X(t)}$$

where X denotes the price of one unit of foreign currency and F^X denotes corresponding forward prices. As discussed in the introduction of this chapter one has

$$F_0^X(T) = X_0 \frac{D^f(T)}{D^d(T)}$$

This allows one to compute quanto options in closed form in the log-normal case, see Section 18.2.1.

15.5.1 Quanto forward

In the case of models with stochastic volatility and dividend yield the quanto forward is given by

$$F^q(T) = \mathbb{E} \left[S_T \frac{X_T}{F_0^X(T)} \right] = F_0^S(T) \frac{X_0}{F_0^X(T)} \mathbb{E} \left[\exp \left(\int_0^T \rho_t \sigma_t^X \sigma_t^S dt \right) \right]$$

Giese (2012) computes the quanto forward in closed form for the Schöbel-Zhu model, introduced in Section 6.2.2.

15.5.2 Quanto volatility

The naive approach of choosing quanto implied volatility $\hat{\sigma}^q(K, T)$ from the corresponding domestic implied volatility surface at the same strike

$$\hat{\sigma}^q(K, T) = \hat{\sigma}(K, T)$$

ignores the effect of the quanto forward being different from domestic forward. Arguably, a better approach using the appropriate moneyiness is

$$\hat{\sigma}^q(K, T) = \hat{\sigma} \left(K \frac{F_0^S(T)}{F^q(T)}, T \right)$$

Giese (2012) finds that both approaches deviate from theoretical prices obtained by calibrating X and S to Vanilla options as well as (stock/foreign exchange and stock volatility/foreign exchange) correlations to market quotes for F^q .

15.6 Government intervention

Governments have the tendency to interfere through central banks with exchange rates for the purpose of improving the competitiveness of the represented economy or reducing the pain of debt servicing. Currencies are often pegged to another currency, which leads to unofficial exchange rates deviating from official quotes. Alternatively, central banks may change interest rates or trade in assets of a given currency to achieve a desired exchange rate. Intervention often fails due to market pressure resulting in market turbulence.

- The German Democratic Republic (DDR) forced visitors to buy their currency Mark (M) at parity to Deutsche Mark (DM) while the unofficial price for one DM in 1989 was up to 20 M.
- The European exchange rate mechanism (ERM) was introduced on March 13, 1979, ahead of the launch of the Euro on January 1, 1999, to keep participating currencies quoted in reference currency ECU within defined price bands.
- Parity between the Irish Pound (IEP) and Pound Sterling (GBP) was abandoned with the introduction of an exchange rate on March 30, 1979 as GBP (non ERM) appreciated against all ERM member currencies, including the IEP.
- GBP joined the ERM in 1990 but was forced to exit in 1992 due to investors, including George Soros, taking a position against GBP. This led to the crash on September 16, 1992 (Black Wednesday).
- On September 6, 2011 the Swiss National Bank (SNB) introduced a lower bound on the price of one EUR of 1.20 CHF. This bound was achieved by SNB buying EUR. The intervention was abandoned on January 15, 2015 leading to the price dropping to a low of 0.9866 CHF.

To model government intervention on EUR/CHF one may assume a latent (unobservable) exchange rate X^l , which would prevail without intervention, see

Hanke et al. (2014). The observed price X subject to intervention is given as

$$X_t = X_t^l + Put(X_t^l, K^*, \tau_t)$$

where $Put(X_t^l, K^*, \tau_t)$ refers to the value of an American put at time t . The (unobservable) maturity τ_t refers to a market consensus on the time when SNB will give up intervention. Hanke et al. argue that due to EUR interest rates being higher than CHF rates the American put is in fact European, see Section 2.5.6 and the introduction to this chapter.

The (observable) value $Put(X_t, K, T)$ of a put on X with strike $K \leq K^* = 1.20$ and maturity T can only be positive if the event $\{\tau_t < T\}$ has non-zero probability. Define

$$p_t(T) = \mathbb{IP}[\tau_t > T] = \mathbb{IP}_t[\tau > t]$$

Assuming independence of τ and X^l processes¹ one has

$$Put(X_t, K, T) = p_t(T)Put(X_t^l + Put(X_t^l, K^*, \tau_t), K, T) + (1 - p_t(T))Put(X_t^l, K, T)$$

If X^l is log-normal then this option can be valued using a compound option formula, such as (12.12). Hanke et al. suggest fitting model parameters to observed data separately for every trading day. In view of X^l being revealed on $\tau = \text{January 15, 2015}$ research in this area will have to take this information into account.

¹ Note that intuitively the hazard rate $h_t(T)$ of τ is more likely to be negatively correlated with X_t^l moves.

16] Affine Processes

16.1 General framework

Closed form pricing via Fourier transform methods is the basis for many stochastic volatility models. A general framework has been formulated by Duffie, Pan and Singleton (2000). They define an n -dimensional jump diffusion process X with time-dependent coefficients through

$$dX_t = \mu_t(X_t)dt + \sigma_t(X_t)dW_t + dZ_t \quad (16.1)$$

where W is an n -dimensional Brownian motion with independent components and Z is a pure jump process with intensity $\lambda_t(X_t)$ and jumps distributed according to measure ν on $\mathbb{R}^n$ with Laplace transform

$$\theta(c) = \int e^{cz} d\nu(z)$$

Such a model is called affine if coefficients satisfy

$$\begin{aligned} \mu_t(x) &= m_t^0 + (m_t^1)^T x \\ \sigma_t(x)\sigma(x)^T &= s_t^0 + (s_t^1)^T x \\ \lambda_t(x) &= l_t^0 + (l_t^1)^T x \\ r_t(x) &= r_t^0 + (r_t^1)^T x \end{aligned} \quad (16.2)$$

for all $x \in \mathbb{R}^n$ and functions m^i, s^i, l^i, r^i . Here r refers to an interest rate process. A key result in Duffie et al. (2000) states that

$$\psi(u, X_t, t, T) = \mathbb{E} \left[\exp \left(- \int_t^T r_s(X_s) ds + u^T X_T \right) \middle| \mathcal{F}_t \right] = e^{A(u,t,T) + B^T(u,t,T)X_t}$$

holds with functions A, B satisfying

$$\begin{aligned}\frac{\partial B(u, t, T)}{\partial t} &= r_t^1 - (m_t^T)^1 B(u, t, T) - \frac{1}{2} B(u, t)^T s_t^1 B(u, t, T) \\ &\quad - l_t^1 (\theta(B(u, t, T)) - 1) \\ \frac{\partial A(u, t, T)}{\partial t} &= r_t^0 - m_t^0 B(u, t, T) - \frac{1}{2} B(u, t)^T s_t^0 B(u, t, T) \\ &\quad - l_t^0 (\theta(B(u, t, T)) - 1)\end{aligned}\tag{16.3}$$

with boundary conditions $B(u, T, T) = u$ and $A(u, T, T) = 0$.

- The proof of this result relies on the fact that given certain technical conditions the process M with

$$M_t = \exp \left(- \int_0^t r_s(X_s) ds + A(u, t, T) + B^T(u, t, T) X_t \right)$$

is a martingale due to Itô's lemma.

- Duffie, Pan and Singleton focus on the time-homogenous case where parameters in (16.2) do not depend on t . In that case one can consider functions $A(t), B(t)$ and $A(u, t, T) = A(u, T - t)$ and define $B(u, t, T) = B(u, T - t)$ instead. Those functions will satisfy

$$\begin{aligned}\frac{\partial B(u, t)}{\partial t} &= -r^1 + (m^T)^1 B(u, t) + \frac{1}{2} B(u, t)^T s^1 B(u, t) + l^1 (\theta(B(u, t)) - 1) \\ \frac{\partial A(u, t)}{\partial t} &= -r^0 + m^0 B(u, t) + \frac{1}{2} B(u, t)^T s^0 B(u, t) + l^0 (\theta(B(u, t)) - 1)\end{aligned}\tag{16.4}$$

with boundary conditions $B(u, 0) = u$ and $A(u, 0) = 0$.

- Equations (16.3) can be solved using the Runge-Kutta method. In the time-homogenous case explicit solutions are known for some parameter combinations.

16.2 European options and Fourier transform

Assuming that the stock process S is a deterministic function of X of the form

$$S_t = e^{a_t + b_t X_t} + c_t$$

option valuation requires computation

$$G_{a,b}(k, X_0, t) = \mathbb{E} \left[e^{-\int_0^t r(X_s) ds} e^{a X_t} \mathbf{1}_{\{b X_t \leq k\}} \right]$$

The Fourier transform of this quantity is given in terms of ψ since

$$\begin{aligned} \int_{-\infty}^{\infty} e^{iux} dG_{a,b}(x, X_0, t) &= \mathbb{E} \left[e^{-\int_0^t r(X_s) ds} e^{(a+iub)X_t} \right] \\ &= \psi(a + iub, X_0, t) \end{aligned}$$

Inverting this transform leads to

$$G_{a,b}(k, X_0, t) = \frac{\psi(a, X_0, t)}{2} - \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{\psi(a + iub, X_0, t) e^{-iuk}}{iu} \right] du$$

see Duffie et al. (2000). Thus, option valuation is reduced to a one-dimensional integration.

In the case $r = 0$ this is equivalent to the representation

$$G_{a,b}(k, X_0, t) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{\psi(u - ia, X_0, t) e^{-iuk}}{iu\psi(-ia)} \right] du$$

namely the representation suggested by Heston (1993), Kahl (2004) and Gatheral (2006).

To avoid the subtleties due to complex logarithms in some applications, as well as numerical issues related to Fourier inversion, the alternative representation

$$\mathbb{E} \left[\left(e^{X_t} - K \right)^+ \right] = e^{X(0)} - \frac{K}{2\pi} \int_{-\infty}^{\infty} \frac{\psi^1(v, t) e^{(-iv+1/2)(X_0 - \log(K))}}{v^2 + 1/4} dv$$

with

$$\psi^1(u, t) = \mathbb{E} \left[e^{\left(-iu + \frac{1}{2}\right)(X_t - X_0)} \right]$$

has been suggested by Lewis (2000) and Lipton (2012).

16.2.1 Time-dependent parameters

Given a time grid $0 = t_0 < t_1 < t_2 < \dots < t_n = T$ and an affine process X with constant parameters on intervals $[t_{i-1}, t_i]$ the characteristic function of X_T can be

computed iteratively as

$$\begin{aligned}
 & \mathbb{E} \left[e^{-\int_0^T r(X_s) ds + u^T X_T} \right] \\
 &= \mathbb{E} \left[e^{-\int_0^{t_{n-1}} r(X_s) ds + A(u, \Delta_n) + B^T(u, \Delta_n) X_{t_{n-1}}} \right] \\
 &= e^{A(u, \Delta_n)} \mathbb{E} \left[e^{-\int_0^{t_{n-1}} r(X_s) ds + B^T(u, \Delta_n) X_{t_{n-1}}} \right] \\
 &= e^{A(u, \Delta_n) + A(B(u, \Delta_n), \Delta_{n-1})} \mathbb{E} \left[e^{-\int_0^{t_{n-2}} r(X_s) ds + B^T(u, \Delta_{n-1}) X_{t_{n-2}}} \right] \\
 &= e^{A(u, \Delta_n) + A(B(u, \Delta_n), \Delta_{n-1}) + A(B(u, \Delta_{n-1}), \Delta_{n-2})} \mathbb{E} \left[e^{-\int_0^{t_{n-2}} r(X_s) ds + B^T(u, \Delta_{n-1}) X_{t_{n-2}}} \right] \\
 &= e^{\sum_{i=1}^n A(B(u, \Delta_{i+1}), \Delta_i) + X_0 B(u, \Delta_2), \Delta_1)}
 \end{aligned}$$

where $\Delta_i = t_i - t_{i-1}$ as well as $B(u, \Delta_{n+1}) = u$, compare Kahl (2004). Note that $A(\cdot, \Delta_i), B(\cdot, \Delta_i)$ refer to the solution of equations (16.3) with parameters valid on interval $[t_{i-1}, t_i]$. In the special case of Heston's model this result is due to Mikhailov and Nögel (2003).

16.2.2 Forward-starting options

To price forward-starting European options for stock processes $S_t = e^{a_t + b_t X_t^1}$ where X is an affine process, quantities of type

$$\mathbb{E} \left[e^{-\int_0^T r(X_s) ds} \left(e^{a_T X_T - a_t X_t} - k \right)^+ \right]$$

are required, with one-dimensional constants a_t, a_T . One obtains the characteristic function of $a_T X_T - a_t X_t$ following Kahl (2004):

$$\begin{aligned}
 & \mathbb{E} \left[e^{-\int_0^T r(X_s) ds + u^T (a_T X_T - a_t X_t)} \right] \\
 &= \mathbb{E} \left[\phi(a_T u, X_t, T - t) e^{-\int_0^t r(X_s) ds - a_t u^T X_t} \right] \\
 &= e^{A(a_T u, T - t)} \mathbb{E} \left[e^{-\int_0^t r(X_s) ds + (B^T(a_T u, T - t) - a_t u^T) X_t} \right] \\
 &= e^{A(a_T u, T - t)} \phi(B(a_T u, T - t) - a_t u, X_0, t) \\
 &= e^{A(a_T u, T - t) + A(B(a_T u, T - t) - a_t u, t) + B^T(B(a_T u, T - t) - a_t u, t) X_0}
 \end{aligned}$$

Note that piecewise constant parameters on $[0, t]$ and $[t, T]$ can be incorporated by using corresponding functions $A(\cdot, t), B(\cdot, t)$ and $A(\cdot, T - t), B(\cdot, T - t)$.

17] Monte Carlo

Equity derivatives contracts can be valued using closed form solutions, finite difference methods and Monte Carlo simulation. The former two approaches are limited to specific products and models. Closed form solutions are available primarily for European options and affine models as discussed in Chapters 11 and 16. For log-normal assets (approximate) formulae are also known for barrier products, as well as basket and Asian products, see Sections 3.1, 3.3 and 3.8. Finite difference, including tree methods, are important for products with early exercise features, such as American options and convertible bonds. For path-dependent and multi-asset products these methods are of limited use due to the high dimensionality of the state space.

Monte Carlo simulation is a flexible approach which is not limited to specific products and models and is therefore an indispensable numerical method for pricing and risk management of equity derivatives beyond Vanilla options. There are many comprehensive resources on Monte Carlo methods applied to financial derivatives, including the books by Jäckel (2002) and Glasserman (2003). This section summarizes the approach and adds recent and less known aspects.

17.1 Method

Monte Carlo simulation allows one to perform high-dimensional integration. Let

$$S = \left(S_{t_j}^l \right)$$

denote the value of one or more assets at one or more future dates. Given market data and a model calibrated to those market data S has a distribution

$$F_S(x) = \mathbb{P} \left[\forall l, j : S_{t_j}^l < x_{jl} \right]$$

In order to compute the expected value of a payoff $f(S)$ one has to value the integral

$$\mathbb{E}[f(S)] = \int f(x) dF_S(x)$$

In case the amount $f(S)$ is payable at time T the distribution F_S may be defined with respect to the T -forward measure $\mathbb{IP} = \mathbb{IP}_T$ and the present value of the expectation is obtained by multiplication with the current discount factor $P(T)$. An estimation $\mathbb{E}_n[f(S)]$ of $\mathbb{E}[f(S)]$ is obtained based on n independent draws $S(i), i = 1, 2, \dots, n$ from the distribution F_S as

$$\mathbb{E}[f(S)] \approx \mathbb{E}_n[f(S)] = \frac{1}{n} \sum_{i=1}^n f(S(i))$$

Each draw $S(i)$ is computed from d independent uniform random numbers $(U(k, i), k = 1, 2, \dots, d)$. The $U(k, i)$ are computed from a random number generator. The quality of the estimation $\mathbb{E}_n[f(S)]$ is governed by the law of large numbers, namely

$$\mathbb{IP} \left[|\mathbb{E}_n[f(S)] - \mathbb{E}[f(S)]| \geq \varepsilon \right] \leq \frac{\text{Var}[f(S)]}{n\varepsilon^2} \quad (17.1)$$

for $\varepsilon > 0$. The variance of $f(S)$ on the right hand side can be estimated as

$$\text{Var}[f(S)] \approx \text{Var}_n[f(S)] = \frac{1}{n-1} \sum_{i=1}^n (f(S(i)) - \mathbb{E}_n[f(S)])^2$$

These theoretical results lead to practical considerations.

- For Monte Carlo simulation to be efficient it is required that mean and variance are bounded. Payoffs such as $f(S) = S_t^2$ or $f(S) = \log S_t$ may not converge if the distribution of S_t has heavy tails.
- Computation time of a payoff $f(S) = f(S_{t_j}^l)$ will increase with an increasing number of assets l and time points j .
- The speed of a Monte Carlo simulation for a given payoff $f(S)$, asset distribution F_S and convergence requirement depends on the following factors:

paths: The number of paths n required in order to achieve convergence. This number depends on the payoff $f(S)$ and the distribution of S .

rng: The time required to generate random numbers U as well as their quality

- steps: The number m of random numbers U required to generate a draw $S(i)$. This number depends on the model choice (number of factors) as well as the number of time steps.
- process: The complexity of the dependence of $S(i)$ on $U(k, i), k = 1, 2, \dots, d$. This is related to the number of time steps: a more complicated dependence may allow one to use fewer time steps and thus fewer random numbers U .
- memory: The memory required. This is related to the previous points. For example, path storage reduces computation cost for random number and process generation while increasing memory footprint.
- payoff: The complexity of evaluating the payoff $f(S)$ from S . When using payoff languages rather than hard coded payoffs this will be important.

17.2 Random numbers

Random numbers U are typically generated as sequences

$$(u(i)) = (u_k(i) \in [0, 1], i \in \mathbb{N}, k = 1, 2, \dots, d)$$

with the aim of having independent uniform variates. Those sequences are then used to generate other sequences, such as Gaussian vectors or Poisson jump processes. On a computer randomness can be represented via pseudo and low discrepancy sequences.

17.2.1 Pseudo random sequences

Pseudo random numbers are random in the sense that (17.1) holds. This allows one to define confidence intervals. Implementations for standard algorithms named Ran0, Ran1, Ran2, Ran3 can be found in Press et al. (2007). A popular algorithm for pseudo random numbers is the Mersenne Twister. An implementation has been given by Matsumoto and Nishimura (1998). Generation of pseudo random numbers is slower than generation of low discrepancy numbers.

17.2.2 Low discrepancy sequence

A low discrepancy sequence $(u(i)) = (u_k(i), i \in \mathbb{N}, k = 1, 2, \dots, d)$ of dimension d satisfies

$$\sup_{y \in [0, 1]^d} \left| \frac{1}{n} \sum_{i=1}^n \prod_{k=1}^d 1_{\{u_k(i) \leq y_k\}} - \prod_{k=1}^d y_k \right| \leq c_d \frac{(\ln n)^d}{n}$$

for a constant c_d depending on d only. This inequality specifies the rate of convergence as worst absolute Monte Carlo error

$$|\mathbb{E}_n[f(U)] - \mathbb{E}[f(U)]|$$

where f is of type

$$f(u) = \mathbf{1}_{\{u \leq y\}} = \prod_{k=1}^d \mathbf{1}_{\{u_k \leq y_k\}}$$

Examples are the sequences named after Halton, Sobol and Niederreiter, see Jäckel (2002). An implementation of Sobol up to dimension 32 can be found in Quantlib. For higher dimensions see Jäckel (2002). Although (17.1) is not guaranteed to hold, convergence with low discrepancy sequences is typically faster than with pseudo random numbers. This, in connection with high generation speed, explains why low discrepancy numbers are popular among practitioners.

For high dimensions d one observes undesirable regularities. This can lead to spikes in the convergence function $n \rightarrow \mathbb{E}_n[f(U)]$ and thus mispricing if f depends strongly on U_k with $k \gg 1$. Suitable constructions of asset paths S from uniforms U with the aim of limiting dependency on higher dimensions of U are discussed in the next section.

17.3 Path construction for Brownian motion

Brownian motion $(W_t, t \in [0, T])$ can be constructed from independent Gaussian variates (X_i) using basis functions. The first variate represents the final value, i.e. $W_T = \sqrt{T}X_1$. The next variate X_2 determines $W_{T/2}$ etc. In practice, one generates a Brownian motion at $2^N + 1$ equidistant times using 2^N independent Gaussians. The impact of the Gaussians on the path decreases with index i . Thus this method is particularly suitable in connection with low discrepancy sequences, such as Sobol. The path thus constructed minus the first basis function yields a Brownian bridge $W^{T,0}$.

When generating d correlated Brownian motions one can use spectral decomposition in connection with this path construction. Thus the first d Gaussian variates $X_1^k, k = 1, 2, \dots, d$ are used to generate the terminal level of d independent Brownian motions B_T^I , the next batch $X_2^k, k = 1, 2, \dots, d$ for the levels $B_{T/2}^I$ etc. In order to spatially correlate the Brownian increments $B_{t_i}^I - B_{t_{i-1}}^I$ one employs spectral decomposition U_i of covariance increments $C_i = \text{Cov}(t_{i-1}) - \text{Cov}(t_i)$, namely $C_i = U_i U_i^t$ and

$$B_{t_i} - B_{t_{i-1}} = U_i(B_{t_i}^I - B_{t_{i-1}}^I)$$

since

$$\begin{aligned} & \mathbb{E} \left[(B_{t_i} - B_{t_{i-1}})(B_{t_i} - B_{t_{i-1}})^t \right] \\ &= U_i \mathbb{E} \left[(B_{t_i}^I - B_{t_{i-1}}^I)(B_{t_i}^I - B_{t_{i-1}}^I)^t \right] U_i^t = C_i \end{aligned}$$

An alternative method involving Cholesky decomposition ensures that adding new processes (or factors) does not modify existing processes (or factors).

17.4 Discretization

Monte Carlo simulation of a path-dependent payoff $f(S) = f(S_{t_i})$ requires time discretization of the process in continuous time $S = (S_t, t \in [0, T])$. A general purpose approach is the Euler scheme.

Better convergence (or the same convergence with fewer time steps) can be achieved with higher order schemes at the cost of more complex terms in the approximating equation. In practice, the added value of higher order schemes does generally not compensate for the complexity of deriving, implementing and simulating additional terms. An exception are processes S_t with stochastic volatility σ_t , where efficient schemes evolving integrated variance $V_t(t)$ in addition to stock and volatility processes are available.

Barrier and lookback products can be efficiently priced by simulating continuous extrema of discretely sampled processes.

17.4.1 Euler

When simulating processes with dynamics

$$dX_t = \mu_t(X_t)dt + \sigma_t(X_t)dW_t \quad (17.2)$$

for the purpose of path-dependent product valuation the standard method is a linearization of the stochastic differential equation. One step from s to t of an Euler scheme X^e approximating X is defined as

$$X_t^e = X_s^e + \mu_s(X_s^e)(t-s) + \sigma_s(X_s^e)Z\sqrt{t-s}$$

where $X_0^e = X_0$ and Z denotes a standard Gaussian random variate. Assuming the growth condition

$$|\mu_t(x) - \mu_s(x)| + |\sigma_t(x) - \sigma_s(x)| \leq c_1(1 + |x|)\sqrt{t-s}$$

for some c_1 the Euler scheme has a strong order of convergence $1/2$ in the sense that if $t_i = ih$ and $T = t_n$ for some $n = n(h)$ then

$$\mathbb{E} \left[|X_T^e - X_T| \right] \leq c_2 h^p$$

for a constant c_2 and all h with $p = 1/2$. This result extends to the case of multivariate processes X and W in (17.2).

17.4.2 Broadie-Kaya

For asset processes with stochastic volatility

$$\begin{aligned} \frac{dS_t}{S_t} &= (r_t - d_t)dt + \sigma_t dW_t^S \\ d\sigma_t &= \mu_t^\sigma(\sigma_t)dt + \sigma_t^\sigma(\sigma_t)dW_t^\sigma \end{aligned} \quad (17.3)$$

an alternative to Euler has been proposed by Broadie and Kaya (2006). Assuming constant correlation ρ between Brownian motions W^σ and W^S , driving volatility and returns respectively, one can define

$$W^S = \rho W^\sigma + \sqrt{1 - \rho^2} W$$

with W and W^σ denoting independent Brownian motions. The solution to (17.3) can be written as

$$\begin{aligned} &\log\left(\frac{S_t}{F(t)}\right) - \log\left(\frac{S_s}{F(s)}\right) \\ &= \rho \int_s^t \sigma_u dW_u^\sigma + \sqrt{1 - \rho^2} \int_s^t \sigma_u dW_u - \frac{1}{2} \int_s^t \sigma_u^2 du \\ &\stackrel{(d)}{=} \rho \int_s^t \sigma_u dW_u^\sigma + \sqrt{1 - \rho^2} \sqrt{\frac{V(s, t)}{t - s}} (W_t - W_s) - \frac{1}{2} V(s, t) \end{aligned} \quad (17.4)$$

where variance $V(s, t)$, defined as

$$V(s, t) = \int_s^t \sigma_u^2 du \quad (17.5)$$

is independent of W . This scheme allows one to simulate returns using a single random number for each time step once the volatility process has been simulated.

Conditional on $V(s, t)$ and $\int_s^t \sigma_u dW_u^\sigma$ the ratio S_t/S_s is log-normally distributed. This can be used within a time discretization scheme with $0 = t_0 < t_1 < t_2 < \dots, t_n = T$ and $s = t_{i-1}, t = t_i$. Note that the single step case $n = 1$ generates the

distribution of S_T . In connection with closed form formulae this approach allows one to price some products efficiently, such as Vanilla cliquets and Asian options. McGhee (2014) also covers barrier products.

The integral $V(s, t)$ from (17.5) can be approximated as $\sigma_s^2(t-s)$ or $(\sigma_s^2 + \sigma_t^2)(t-s)/2$. The Euler variant using the left point σ_s^2 only leads to decorrelation between stock and volatility relative to the true model. The second variant has improved correlation dynamics but requires a drift correction to ensure the martingale condition for $S_t/F(t)$.

17.4.3 Heston

The CIR process assumed in the Heston model (see Section 6.2.1) with constant mean reversion κ and volatility of volatility α can be integrated as

$$\begin{aligned}\sigma_t^2 &= \sigma_s^2 + \int_s^t \kappa (\theta_u - \sigma_u^2) du + \int_s^t \alpha \sigma_u dW_u^\sigma \\ &= \sigma_s^2 + \kappa \int_s^t \theta_u du - \kappa V(s, t) + \alpha \int_s^t \sigma_u dW_u^\sigma\end{aligned}$$

Substituting term $\int_s^t \sigma_u dW_u^\sigma$ in (17.4) leads to the exact scheme

$$\begin{aligned}\log\left(\frac{S_t}{F(t)}\right) - \log\left(\frac{S_s}{F(s)}\right) \\ \stackrel{(d)}{=} \frac{\rho}{\alpha} \left(\sigma_t^2 - \sigma_s^2 - \kappa \int_s^t \theta_u du \right) + \left(\frac{\kappa \rho}{\alpha} - \frac{1}{2} \right) V(s, t) + \sqrt{1 - \rho^2} \sqrt{V(s, t)} \frac{W_t - W_s}{\sqrt{t - s}}\end{aligned}$$

This scheme requires simulation of the CIR process $\nu_t = \sigma_t^2$ at time t given ν_s which is known to have non-central χ^2 -distribution. Euler schemes fail if the probability of ν being near the origin is too high. An alternative efficient method, due to Andersen (2008), is quadratic exponential (QE). Quadratic and exponential refers to

$$\begin{aligned}\nu_t &= \nu_s + a(b + X)^2 & \text{and} \\ \nu_t &= \nu_s + \beta^{-1} \ln\left(\frac{1-p}{1-U} \vee 1\right)\end{aligned}$$

respectively, where U and X denote uniform and standard Gaussian variates. Parameters a, b, β, p are computed such that the first two moments of ν_t given ν_s are matched. The quadratic scheme can only be moment matched for conditional variance over conditional mean squared $\Psi > 1$. The exponential scheme requires $\Psi < 2$. Andersen suggests a switching rule at $\Psi = 1.5$. The drift correction for this

scheme with approximation $V(s, t) = (\sigma_s^2 + \sigma_t^2)(t - s)/2$ is also given in Andersen (2008).

17.4.4 Schöbel-Zhu

Mean reverting Gaussian volatility

$$d\sigma_t = \kappa(\theta - \sigma_t)dt + \alpha dW_t$$

as introduced by Schöbel and Zhu (1999) and discussed in Section 6.2.2 can be rewritten in terms of variance

$$d\sigma_t^2 = (\alpha^2 + 2\kappa\theta\sigma_t - 2\kappa\sigma_t^2)dt + 2\alpha\sigma_t dW_t$$

leading to

$$2\alpha \int_s^t \sigma_u dW_u^\sigma = \sigma_t^2 - \sigma_s^2 - \alpha^2(t - s) - 2\kappa\theta \int_s^t \sigma_u du - 2\kappa V(s, t)$$

With constant parameters α, κ, θ and ρ in period $[s, t]$ one thus has the exact scheme

$$\begin{aligned} \log\left(\frac{S_t}{F(t)}\right) - \log\left(\frac{S_s}{F(s)}\right) &= \frac{\rho}{2\alpha} \left(\sigma_t^2 - \sigma_s^2 - \alpha^2(t - s) - 2\kappa\theta \int_s^t \sigma_u du \right) \\ &\quad + \left(\frac{\kappa\rho}{\alpha} - \frac{1}{2} \right) V(s, t) + \sqrt{1 - \rho^2} \sqrt{V(s, t)} \frac{W_t - W_s}{\sqrt{t - s}} \end{aligned}$$

The OUP process σ can be simulated at discrete times from one Gaussian variate per time step, as discussed in Section 10.4, see equation (10.17). When using the approximation $V(s, t) \approx (\sigma_s^2 + \sigma_t^2)(t - s)/2$ a martingale correction can be derived, see Haastrecht, Lord and Pelsser (2009). Another contribution of (2009) is an extension of the simulation scheme to Schöbel and Zhu with Hull-White interest rates.

17.4.5 Log-normal volatility

In the case of log-normal volatility, also known as SABR with exponent $\beta = 1$ (see Section 6.3.1),

$$\frac{d\sigma_t}{\sigma_t} = \alpha dW_t^\sigma$$

one has

$$\int_s^t \sigma_u dW_u^\sigma = \frac{1}{\alpha} (\sigma_t - \sigma_s)$$

Inserting into (17.4) yields the exact scheme

$$\begin{aligned} & \log\left(\frac{S_t}{F(t)}\right) - \log\left(\frac{S_s}{F(s)}\right) \\ &= \frac{\rho}{\alpha}(\sigma_t - \sigma_s) + \sqrt{1 - \rho^2} \sqrt{V(s, t)} \frac{W_t - W_s}{\sqrt{t - s}} - \frac{1}{2} V(s, t) \end{aligned}$$

Thus the distribution of S_T is given in terms of the joint distribution of $(V(T), \sigma_T)$. This joint distribution can be approximated using a log-normal approximation of $V(T)$ conditional on σ_T leading to an efficient numerical scheme to compute European option prices for the SABR model, see McGhee and Tralalini (2014). A closed form expression for the joint distribution is also available, see Islah (2009). Integrating the Scott-Chesney model (see Section 6.2.4)

$$d\log\sigma_t = \kappa(\theta - \log\sigma_t)dt + \alpha dW_t^\sigma$$

leads to

$$\sigma_t = \sigma_s + \int_s^t \sigma_u \left(\kappa(\theta - \log\sigma_u) + \frac{1}{2}\alpha^2 \right) du + \alpha \int_s^t \sigma_u dW_u^\sigma$$

and hence

$$\begin{aligned} & \log\left(\frac{S_t}{F(t)}\right) - \log\left(\frac{S_s}{F(s)}\right) \\ &= \frac{\rho}{\alpha} \left(\sigma_t - \sigma_s - \left(\kappa\theta + \frac{1}{2}\alpha^2 \right) \int_s^t \sigma_u du + \kappa \int_s^t \sigma_u \log\sigma_u du \right) \\ & \quad + \sqrt{1 - \rho^2} \sqrt{V(s, t)} \frac{W_t - W_s}{\sqrt{t - s}} - \frac{1}{2} V(s, t) \end{aligned}$$

The SABR model is not a special case of Scott-Chesney but can be recovered with $\theta = -\alpha^2/(2\kappa)$ and $\kappa \searrow 0$.

17.4.6 Sampling continuous extrema

Brownian bridge $W^{s,a,t,b}$ is the distribution of Brownian motion $(W_u, u \in [s, t])$ conditioned on start and end points a at time s and b at time t respectively, see Section 18.1.3. One can sample from the maximum and minimum of Brownian

bridge distributions derived in Section 18.1.4

$$\begin{aligned}\mathbb{P}\left[W_*^{s,a,t,b} < x\right] &= \exp\left(-2\frac{(a-x)(b-x)}{t-s}\right) & x < a \wedge b \\ \mathbb{P}\left[W_*^{s,a,t,b,*} > x\right] &= \exp\left(-2\frac{(a-x)(b-x)}{t-s}\right) & x > a \vee b\end{aligned}$$

using a standard uniform variate U as

$$\begin{aligned}W^{s,a,t,b,*} &\stackrel{(d)}{=} \frac{1}{2} \left(a + b + \sqrt{(a-b)^2 - 2(t-s)\log U} \right) \\ W_*^{s,a,t,b} &\stackrel{(d)}{=} \frac{1}{2} \left(a + b - \sqrt{(a-b)^2 - 2(t-s)\log U} \right)\end{aligned}$$

Thus, given a discrete path $((S_{t_i}, \sigma_{t_i}), i = 1, 2, \dots, n)$ of spot and instantaneous volatility the continuous maximum of S in period $[t_{i-1}, t_i]$ can be sampled as

$$S_{[t_{i-1}, t_i]}^* = \sqrt{S_{t_{i-1}} S_{t_i}} \exp \sqrt{\log \left(\frac{S_{t_{i-1}}}{S_{t_i}} \right)^2 - 2\sigma_{t_{i-1}}^2 (t_i - t_{i-1}) \log U}$$

Note that maximum $W^{s,a,t,b,*}$ and minimum $W_*^{s,a,t,b}$ are not independent. The above simulation scheme will not generate the correct dependence. This can be an issue for double-barrier products if upper and lower barriers are both close to the end points a and b .

17.5 Greeks

Risk management relies on computing sensitivities of prices to changes in market data. Let f denote a payoff contingent on the value of one or more assets at one or more future points in time S with value π . The distribution of S depends on one or more model parameters θ , namely $S = S(\theta)$. The simplest example is a Vanilla call with maturity t on a Black-Scholes stock S , namely

$$\begin{aligned}S_t &= S_0 e^{(r-d)t + \sigma W_t - \frac{1}{2}\sigma^2 t} \\ f(S) &= (S_t - K)^+\end{aligned}$$

as well as θ being one of the model parameters S_0, σ, r and d for sensitivities delta, vega, rho and dividend delta. The naive approach

$$\frac{\partial}{\partial \theta} \pi(\theta) = \frac{1}{\varepsilon} (\mathbb{E}[f(S(\theta + \varepsilon))] - \mathbb{E}[f(S(\theta))])$$

requires one extra simulation per sensitivity required. Convergence is slow if the payoff f has discontinuities.

More efficient methods can be based on differentiation of density (likelihood ratio), payoff (pathwise method) or discretization scheme (adjoint method). Differentiation can now be performed algorithmically.

Computation of second-order sensitivities, such as gamma or correlation delta, is challenging due to the amount of sensitivities as well as slow convergence. Path storage (or path recycling) in combination with specific discretization schemes can massively reduce computation time for path generation. This will not help if payoff evaluation dominates computation time. However, it may solve the convergence issue.

17.5.1 Likelihood ratio

The likelihood ratio method is based on the observation

$$\frac{\partial}{\partial \theta} \int f(x) p(x, \theta) dx = \int f(x) \frac{\partial}{\partial \theta} \log p(x, \theta) p(x) dx$$

which can be rewritten as

$$\frac{\partial}{\partial \theta} \pi(\theta) = \mathbb{E} \left[f(S) \frac{\partial \log p}{\partial \theta}(S) \right]$$

This method requires differentiation of the density rather than the payoff. First- and second-order likelihood ratios for the Black-Scholes and some extensions, including CDO sensitivities with respect to the Gauss copula model, can be found in Jäckel's book (2002) and article (2004). Likelihood ratios for stochastic volatility models are discussed in Glasserman (2003). The latter effectively computes the likelihood ratio for the Euler scheme approximating the process with stochastic volatility rather than for the process itself. The main limitation is the requirement to know and implement analytical expressions for likelihood ratios. Some likelihood ratios lead to expressions with slow convergence.

17.5.2 Pathwise method

The pathwise method consists of applying the chain rule path by path, namely

$$\frac{\partial}{\partial \theta} \pi(\theta) = \mathbb{E} \left[\frac{\partial f}{\partial S}(S) \frac{\partial S}{\partial \theta}(\theta) \right]$$

For this method to be applicable Lipschitz conditions for both payoff and model parameters have to be satisfied. The Black-Scholes call example yields for

delta (i.e. $\theta = S_0$)

$$\begin{aligned}\frac{\partial f}{\partial S}(S) &= \mathbf{1}_{\{S_t > K\}} \\ \frac{\partial S}{\partial \theta}(\theta) &= \frac{S_t}{S_0}\end{aligned}$$

and the sensitivity can be computed as

$$\frac{\partial}{\partial \theta} \pi(\theta) = \mathbb{E} \left[\frac{S_t \mathbf{1}_{\{S_t > K\}}}{S_0} \right]$$

A discretization scheme for an equation

$$dS_t = a_t(S_t)dt + b_t(S_t)dW_t$$

describing the dynamics of one or more assets (including possibly other factors such as stochastic volatility, interest and hazard rates) through time can be formalized as

$$S_{i+1} = F_i(S_i)$$

where $S_i = S_{\tau_i}$ and $0 = \tau_0 < \tau_1 < \tau_2 < \dots < \tau_n = T$. A given sensitivity parameter θ can impact F_i directly or through S_i , namely

$$S_{i+1}(\theta) = F_i(\theta, S_i(\theta)) \quad (17.6)$$

Going forward arguments of F_i will be suppressed. With the forward matrix recursion

$$\Delta(i+1) = \frac{\partial F_i}{\partial \theta} + \frac{\partial F_i}{\partial S_i} \Delta(i) \quad (17.7)$$

with initial value $\Delta(0)$ being the identity matrix I with entries $I_{ij} = \mathbf{1}_{\{i=j\}}$ one obtains the sensitivity for a European payoff $f(S_T)$ as

$$\begin{aligned}\frac{\partial f(S_n)}{\partial \theta} &= \frac{\partial f(S_n)}{\partial S_n} \frac{\partial S_n}{\partial \theta} \\ &= \frac{\partial f(S_n)}{\partial S_n} \left(\frac{\partial F_{n-1}}{\partial \theta} + \frac{\partial F_{n-1}}{\partial S_{n-1}} \frac{\partial S_{n-1}}{\partial \theta} \right) \\ &= \dots = \frac{\partial f(S_n)}{\partial S_n} \Delta(n)\end{aligned}$$

Once the $\Delta(i)$ are computed one obtains the sensitivity with respect to θ for path-dependent payoffs $f(S_1, S_2, \dots, S_n)$ as

$$\frac{\partial f(S_1, S_2, \dots, S_n)}{\partial \theta} = \sum_{i=1}^n \frac{\partial f(S_1, S_2, \dots, S_n)}{\partial S_i} \Delta(i)$$

17.5.3 Adjoint method

This method was introduced to financial engineering by Giles and Glasserman (2006). Given a European payoff $f(S_T)$ one has

$$\frac{\partial f(S_n)}{\partial \theta} = \frac{\partial f(S_n)}{\partial S_n} \frac{\partial S_n}{\partial \theta}$$

as well as the recursive equation

$$\frac{\partial S_{i+1}}{\partial \theta} = \frac{\partial F_i}{\partial \theta} + \frac{\partial F_i}{\partial S_i} \frac{\partial S_i}{\partial \theta}$$

with F_i defined in (17.6). Defining the backward vector recursion

$$V(i)^t = V(i+1)^t \frac{\partial F_i}{\partial S_i} \quad (17.8)$$

with initial value

$$V(n)^t = \frac{\partial f(S_n)}{\partial S_n}$$

one obtains

$$\begin{aligned} \frac{\partial f(S_n)}{\partial \theta} &= V(n)^t \frac{\partial S_n}{\partial \theta} = V(n)^t \left(\frac{\partial F_{n-1}}{\partial \theta} + \frac{\partial F_{n-1}}{\partial S_{n-1}} \frac{\partial S_{n-1}}{\partial \theta} \right) \\ &= V(n)^t \frac{\partial F_{n-1}}{\partial \theta} + V(n-1)^t \left(\frac{\partial F_{n-2}}{\partial \theta} + \frac{\partial F_{n-2}}{\partial S_{n-2}} \frac{\partial S_{n-2}}{\partial \theta} \right) \\ &= \dots = \sum_{i=1}^n V(i)^t \frac{\partial F_{i-1}}{\partial \theta} \end{aligned}$$

Note that equation (17.8) is a vector recursion in contrast to the computationally more involved matrix recursion (17.7). The vector recursion relies on a specific payoff but allows one to compute several sensitivities with little overhead.

17.5.4 Algorithmic differentiation

Implementation of pathwise and adjoint methods require differentiation of payoff and discretization schemes with respect to their arguments. For that purpose one can rely on software tools performing algorithmic differentiation, see Bücker and Hovland. Those tools either generate code implementing differentiation or allow one to access differentials through operator overloading. For a description of how this works see Capriotti (2011). Given a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ algorithmic differentiation computes the Jacobian J with $J_{ij} = \frac{\partial F_i(x)}{\partial x_j}$. Assuming some computational complexity model one can show that

$$\frac{\text{Cost}[F\&J]}{\text{Cost}[F]} \leq \frac{5}{2}n$$

One thus has for a single linear combination of columns

$$\frac{\text{Cost}[F\&L_c(J)]}{\text{Cost}[F]} \leq \frac{5}{2}$$

This is intuitive as it involves a single bump into a specified direction. Using the adjoint method one has

$$\frac{\text{Cost}[F\&J]}{\text{Cost}[F]} \leq 4m$$

For a scalar function this reads

$$\frac{\text{Cost}[F\&L_c(J)]}{\text{Cost}[F]} \leq 4$$

Hence the gradient of a scalar function can be computed at four times the cost of evaluating the function, irrespective of the number n of arguments (i.e. sensitivities).

Pricing may require calibration in addition to Monte Carlo simulation. In that case differentiation of the calibration routine can be avoided, as pointed out by Henrard (2011). A pricing routine $z = f(a)$ with calibration may be summarized through functions g_1, g_2, g_3 such that

$$\begin{aligned} b &= g_1(a) \\ c \text{ s.t. } g_2(b, c) &= 0 \\ z &= g_3(c) \end{aligned}$$

Here g_1 maps market data a to calibration instruments b , g_2 is the calibration routine represented as a root search where model parameters c have to be found.

Thus g_2 is the gradient of the sum of squares of differences between market prices b and model prices obtained with model parameters c . Function g_3 maps calibrated model parameters c to exotics prices using Monte Carlo simulation. Recursion (17.8) leads to the differentiation of a function of type g_3 , namely the sensitivities to model parameters. The implicit function theorem states that if there is a solution $g_2(b_0, c_0) = 0$ and $\frac{\partial g_2}{\partial c}$ is invertible then there is an implicit function g_4 such that $g_2(b, g_4(b)) = 0$ near b as well as

$$\frac{\partial g_4}{\partial b}(b) = - \left(\frac{\partial g_2}{\partial c}(b, c) \right)^{-1} \frac{\partial g_2}{\partial b}(b, c)$$

The adjoint version of the algorithm reads

$$\begin{aligned} \bar{z} &= I \\ \bar{c} &= \left(\frac{\partial g_3}{\partial c}(c) \right)^t \bar{z} \\ \bar{b} &= \left(\frac{\partial g_4}{\partial b}(b) \right)^t \bar{c} = - \left(\left(\frac{\partial g_2}{\partial c}(b, c) \right)^{-1} \frac{\partial g_2}{\partial b}(b, c) \right)^t \bar{c} \\ \bar{a} &= \left(\frac{\partial g_1}{\partial a}(a) \right)^t \bar{b} \end{aligned}$$

compare (17.8). Note that the same observation has already been made by Knudsen (2002), see Section 6.6. In short, market sensitivities computation only requires the differentiation of model prices.

17.5.5 Path recycling

To compute sensitivities for a d -asset derivative one may, for each simulation i , store the base path

$$S(i, \theta) = (S^1(i, \theta), \dots, S^d(i, \theta))$$

alongside the paths $S(i, \theta + \varepsilon)$ and $S(i, \theta - \varepsilon)$ with parameter shift. This allows one to simulate individual delta and gamma for payoff $f(S)$ as

$$\begin{aligned} \Delta^f(k) &= \frac{1}{2\varepsilon n} \sum_{i=1}^n \left(f(S^1(i, \theta), \dots, S^k(i, \theta + \varepsilon), \dots, S^d(i, \theta)) \right. \\ &\quad \left. - f(S^1(i, \theta), \dots, S^k(i, \theta - \varepsilon), \dots, S^d(i, \theta)) \right) \end{aligned}$$

$$\Gamma^f(k) = \frac{1}{\varepsilon^2 n} \sum_{i=1}^n \left(f(S^1(i, \theta), \dots, S^k(i, \theta + \varepsilon), \dots, S^d(i, \theta)) \right. \\ \left. - 2f(S(i, \theta)) + f(S^1(i, \theta), \dots, S^k(i, \theta - \varepsilon), \dots, S^d(i, \theta)) \right)$$

as well as cross gamma

$$\Gamma^f(k, l) = \frac{1}{4\varepsilon^2 n} \sum_{i=1}^n \left(f(\dots, S^k(i, \theta + \varepsilon), \dots, S^l(i, \theta + \varepsilon), \dots) \right. \\ \left. - f(\dots, S^k(i, \theta - \varepsilon), \dots, S^l(i, \theta + \varepsilon), \dots) \right. \\ \left. - f(\dots, S^k(i, \theta + \varepsilon), \dots, S^l(i, \theta - \varepsilon), \dots) \right. \\ \left. + f(\dots, S^k(i, \theta - \varepsilon), \dots, S^l(i, \theta - \varepsilon), \dots) \right)$$

Thus one price, d deltas, d gammas and $d(d-1)/2$ cross gammas, can be simulated with three sets of paths. This may sound better than it is as efficiency saving is limited to path generation and does not extend to payoff evaluation.

17.5.6 Correlation sensitivity

Computation of correlation sensitivities can be combined with path recycling. Indeed, given correlation matrix ρ one has Cholesky decomposition c with

$$c_{ij}c_{jj} = \rho_{ij} - \sum_{k=1}^{j-1} c_{ik}c_{jk} \\ c_{ii}^2 = 1 - \sum_{k=1}^{i-1} c_{ik}^2$$

for $j < i$ and $c_{ij} = 0$ for $j > i$. This can be written in terms of angles $\theta_{ij}, j < i$ as

$$c_{ij} = \cos(\theta_{ij}) \prod_{k=1}^{j-1} \sin(\theta_{ik}) \\ c_{ii} = \prod_{k=1}^{i-1} \sin(\theta_{ik})$$

One has

$$\frac{\partial V}{\partial \rho_{kl}} = \sum_{j < i} \frac{\partial V}{\partial \theta_{ij}} \frac{\partial \theta_{ij}}{\partial \rho_{kl}}$$

Sensitivity with respect to θ_{ij} requires bump of asset S^i only. Hence $i - 1$ bumps of asset S^i are required in order to compute all correlation sensitivities.

17.6 Variance reduction

Depending on the problem at hand there are various methods which can lead to efficiency gains.

17.6.1 Antithetic

It can be useful to use antithetic numbers to reduce computation time for the generation of random numbers. For example, one may use the sequence of independent standard Gaussian variates

$$X_1 = N^{-1}(U_1), -X_1, X_2 = N^{-1}(U_2), -X_2, \dots$$

obtained from the Mersenne Twister sequence (U_i). This does not apply to quasi random numbers where antithetic is implicit and random number generation is fast.

17.6.2 Moment matching

This method exploits knowledge of the expectation of a random variable. Let $S_i(k)$ denote the k -th simulation of stock S at time t_i with sample mean

$$F(i, n) = \frac{1}{n} \sum_{k=1}^n S_i(k)$$

If the forward is given as $F(t_i)$ then matching the first moment consists in using the adjusted variate

$$S_i^1(k) = S_i(k) \frac{F(t_i)}{F(i, n)}$$

If log-volatility $\hat{\sigma}(t_i)$ is also to be matched one may use sample log volatility

$$\hat{\sigma}(i, n)^2 = \frac{1}{n-1} \sum_{k=1}^n \left(\log \frac{S_i(k)}{F(i, n)} \right)^2$$

as well as adjusted sample forward

$$F^1(i, n) = \frac{1}{n} \sum_{k=1}^n (S_i(k))^{\frac{\hat{\sigma}(i)}{\hat{\sigma}(i, n)}}$$

Matching log volatility is achieved with

$$S_i^2(k) = (S_i(k))^{\frac{\hat{\sigma}(i)}{\hat{\sigma}(i,n)}} \frac{F(t_i)}{F^1(i, n)}$$

One can also enforce exact matching of covariances and higher order moments. This method hides weaknesses, such as bias, outliers, weak convergence, present in the simulation scheme and is thus not recommended.

17.6.3 Control variate

Let $f(S)$ denote a payoff with expected value $\mathbb{E}[f(S)]$. If the value

$$\pi(g) = \mathbb{E}[g(S)]$$

of payoff $g(S)$ is known in closed form then the control variate technique consists of simulating

$$f^g(S(k)) = f(S(k)) + \beta (\pi^g - g(S(k)))$$

where $S(k)$ denotes the k -th simulated path. The optimal parameter β minimizing the sample variance of $f^g(S)$ is

$$\beta = \frac{\text{Cov}[f(S), g(S)]}{\text{Var}[g(S)]}$$

leading to variance

$$\text{Var}[f^g(S)] = (1 - \text{Corr}[f(S), g(S)]^2) \text{Var}[f(S)]$$

This optimal parameter can be computed alongside the simulation at limited extra cost. In many cases setting $\beta = 1$ is sufficient. The control variate method can also be applied with the same payoff but different process $\tilde{S}$, namely

$$\tilde{f}(S(k)) = f(S(k)) + \beta (\tilde{\pi}^f - f(\tilde{S}(k)))$$

where

$$\tilde{\pi}^f = \mathbb{E}[f(\tilde{S})]$$

Let, for example, S be a multi-asset hybrid model with stochastic interest rates and stochastic volatility for S . In that case the process $\tilde{S}$ may be:

- a Black-Scholes process,
- a stochastic volatility process without stochastic rates,
- a process with stochastic rates and deterministic volatility,
- the original process S approximated with fewer time steps,
- the original process S based on approximate multivariate Brownian drivers constructed with fewer Gaussian variates or
- the original process S computed with single precision rather than double precision.

To achieve high correlation the faster scheme $\tilde{S}$ should be implemented, such that $\tilde{S}(i) = S(i)$ holds for all i if model parameters for S are set such that S and $\tilde{S}$ have the same distribution.

The key for performance gain is that $\tilde{\pi}^f$ (or π^g) is faster to compute than π^f for the same accuracy and that $f(S)$ and $f(\tilde{S})$ (or $g(S)$) are highly correlated. One may also consider a combined payoff and process control variate $g(\tilde{S})$ leading to the scheme

$$\tilde{f}^g(S(k)) = f(S(k)) + \beta \left(\tilde{\pi}^g - g(\tilde{S}(k)) \right)$$

17.6.4 Multilevel

Multilevel Monte Carlo methods iterate the control variate method at the process level. Let $S^0, S^1, S^2, \dots$ denote a sequence of processes approximating S , such that S^l is slower to simulate than S^{l-1} but is a better approximation of S . Independent simulations of process S^l are enumerated $S^l(1), S^l(2), \dots$. The simulation proceeds as

$$\pi^L = \frac{1}{n_0} \sum_{k=1}^{n_0} f(S^0(k)) + \sum_{l=1}^L \left(\frac{1}{n_l} \sum_{k=1}^{n_l} \left(f(S^l(k)) - f(S^{l-1}(k)) \right) \right)$$

Cost and variance of this approach are

$$\begin{aligned} \text{Cost}[\pi^L] &= n_0 \text{Cost}[f(S^0)] + \sum_{l=1}^L n_l \text{Cost}[f(S^l) - f(S^{l-1})] \\ \text{Var}[\pi^L] &= \frac{1}{n_0} \text{Var}[f(S^0)] + \sum_{l=1}^L \frac{1}{n_l} \text{Var}[f(S^l) - f(S^{l-1})] \end{aligned}$$

In the case of standard control variate method $L = 1$ for given cost the variance is minimized with simulation ratio

$$\left(\frac{n_1}{n_0}\right)^2 = \frac{\text{Var}[f(S^1) - f(S^0)] \text{Cost}[f(S^0)]}{\text{Cost}[f(S^1) - f(S^0)] \text{Var}[f(S^0)]}$$

In other words, an optimal combination of the $n_l, l = 0, 1, \dots, L$, is given as

$$n_l = c \sqrt{\frac{\text{Var}[f(S^l) - f(S^{l-1})]}{\text{Cost}[f(S^l) - f(S^{l-1})]}} \quad (17.9)$$

where c does not depend on l and $f(S^{-1}) = 0$. This statement also holds for $L > 1$. In order to bound variance

$$\text{Var}[\pi^L] \leq \varepsilon^2$$

Giles (2008) suggests first choosing L such that

$$\left(\mathbb{E}[f(S^L)] - \mathbb{E}[f(S)]\right)^2 \leq \frac{\varepsilon^2}{2}$$

and then determining c in equation (17.9) such that

$$\text{Var}[\pi^L] \leq \frac{\varepsilon^2}{2}$$

holds.

A standard Monte Carlo estimation of $\mathbb{E}[f(S)]$ based on n paths has variance $\text{Var}[f(S)]/n$. Assuming a Lipschitz payoff f the discretization error for a standard Euler scheme, as given in (17.3), can be estimated as

$$\mathbb{E}\left[\left(f(S^h) - f(S)\right)^2\right] = \mathcal{O}(h^2)$$

Here $h = T/m$ is the step size obtained with maturity T and m number of steps. Thus, computational cost for a root mean squared error of ε is $\mathcal{O}(\varepsilon^{-3})$, assuming cost to be proportional to time steps and number of iterations. The efficiency gain of multilevel MC follows from the multilevel Monte Carlo theorem, see Giles (2008). In short, if variance decays no slower than cost increases with level l (as is the case for S^l referring to Euler discretizations with 2^l steps) then computational cost is reduced to $\mathcal{O}(\varepsilon^{-2} \log(\varepsilon)^2)$. Recent research on multilevel Monte Carlo can be found on Giles' website, *Multilevel Monte Carlo research*.

18] Gauss

This chapter collects (and derives) closed form formulae for integrals related to Brownian motion for reference elsewhere in the book. It includes Black-Scholes and Bachelier formulae, as well as a range of barrier products.

18.1 Brownian motion

18.1.1 Gaussian random variable

Gaussian random variables play an important role in finance. In option pricing they are used to describe asset returns in the famous Black-Scholes model. Risk management relies on them to assess value at risk. In this book they are denoted with capital letters such as X, Y or Z . Distribution and density of normalized Gaussian random variables are denoted by Φ and ϕ , respectively.

18.1.2 Brownian motion

Brownian motion W is a continuous Gaussian process with mean 0 and covariance

$$\mathbb{E} [W_s W_t] = s \wedge t$$

It is both a Markov process and a martingale with respect to its own filtration

$$\mathcal{F}_t = \sigma (W_s, s \leq t)$$

The first statement refers to the property

$$\mathbb{E} [f(W_t, t \geq s) | \mathcal{F}_s] = \mathbb{E} [f(W_t, t \geq s) | W_s] \quad (18.1)$$

for all functionals f and $s \geq 0$, while the second states

$$\mathbb{E} [W_t | W_s] = W_s \quad (18.2)$$

for all s, t with $0 \leq s \leq t$.

18.1.3 Brownian bridge

Brownian bridge $W^{x,T}$ is Brownian motion conditioned on a final value x at time T , namely

$$\mathbb{E}[f(W_t, t \in [0, T]) \mid W_T = x] = \mathbb{E}\left[f\left(W_t^{x,T}, t \in [0, T]\right)\right]$$

This implies the equality of distributions

$$W_t^{x,T} \stackrel{(d)}{=} W_t - \frac{t}{T}(W_T - x)$$

It is a continuous Gaussian process with mean

$$\mathbb{E}\left[W_t^{x,T}\right] = \frac{t}{T}x$$

and covariance

$$\text{Cov}\left[W_s^{x,T}, W_t^{x,T}\right] = \frac{s(T-t)}{T}$$

if $0 \leq s \leq t \leq T$.

Brownian bridge $W^{s,x,t,y}$ is Brownian motion conditioned on start and end value x at time s and y at time t , namely

$$\mathbb{E}[f(W_u, u \in [s, t]) \mid W_s = x, W_t = y] = \mathbb{E}\left[f\left(W_u^{s,x,t,y}, u \in [s, t]\right)\right]$$

It has mean

$$\mathbb{E}\left[W_u^{s,x,t,y}\right] = \frac{t-u}{t-s}x + \frac{u-s}{t-s}y$$

and covariance

$$\text{Cov}\left[W_u^{s,x,t,y}, W_v^{s,x,t,y}\right] = \frac{(u-s)(t-v)}{t-s}$$

if $u \leq v$. The distribution of the maximum $W^{*,s,x,t,y} = \max\{W_u^{s,x,t,y}, u \in [s, t]\}$ of a Brownian bridge is derived below, see equation (18.4).

18.1.4 Reflection principle

Let W_{*t} denote the running minimum of a Brownian motion, namely

$$W_{*t} = \min\{W_s, s \in [0, t]\}$$

The corresponding running maximum process is denoted W^* . The reflection principle states that, for all $x < 0$ and $x < y$,

$$\mathbb{P}[W_{*1} < x, W_1 > y] = \mathbb{P}[W_1 < 2x - y] \quad (18.3)$$

Deriving with respect to y one obtains the distribution of the minimum of a Brownian bridge as

$$\mathbb{P}[W_{*1} < x | W_1 = y] = \frac{\varphi(2x - y)}{\varphi(y)} = e^{-2x(x-y)}$$

Repeating the argument with time $t - s$ instead of 1 and $W + z$ instead of W leads to

$$\mathbb{P}\left[W_*^{s,z,t,y} < x\right] = \exp\left(-2\frac{(x-z)(x-y)}{t-s}\right) \quad (18.4)$$

if $x \leq y \wedge z$.

18.1.5 Discrete monitoring

This section follows Broadie et al. (1997). In order to cover the case of discretely monitored extrema let W^m denote the random walk defined as

$$W_t^m = \frac{1}{\sqrt{m}} \sum_{1 \leq i \leq tm} X_i$$

where the X_i are independent Gaussian variates. The corresponding running maximum and hitting time at level x are denoted W^{m*} and τ_x^m respectively. A reflection principle holds for W^m , namely

$$\mathbb{P}[W_1^{m*} > x, W_1^m < y] = \mathbb{P}[W_1^m > 2(x + R_m) - y]$$

if $x > 0$ and $x > y$. Here R_m is the overshoot

$$R_m = W_{\tau_x^m}^m - x$$

One can show that

$$\sqrt{m}\mathbb{E}[R_m] \xrightarrow{m \rightarrow \infty} \beta$$

where

$$\beta = \frac{\zeta(1/2)}{\sqrt{2\pi}} \approx 0.5826$$

holds. Hence

$$\mathbb{P}[W_1^{m*} > x, W_1^m < y] = \mathbb{P}\left[W_1^* > x + \frac{\beta}{\sqrt{m}}, W_1 < y\right] + o\left(\frac{1}{\sqrt{m}}\right)$$

18.1.6 Brownian motion with drift

Let ν be a real number. Denote by W^ν Brownian motion with drift ν if

$$W_t^\nu = W_t + \nu t$$

for all $t \geq 0$ and a Brownian motion W . One has

$$\mathbb{E}\left[f(W_t, t \in [0, T])e^{\nu W_T - \nu^2 T/2}\right] = \mathbb{E}\left[f(W_t^\nu, t \in [0, T])\right]$$

for all functionals f . This is a special case of Girsanov's theorem. The running maximum and minimum of Brownian motion with drift ν are denoted $W^{*\nu}$ and W_*^ν respectively.

18.1.7 Bachelier

With

$$S_t = F(t) + \sigma W_t$$

following an early approach to options pricing pioneered by Bachelier (1900), one obtains the call price

$$\begin{aligned} \mathbb{E}[(S_t - K)^+] &= \mathbb{E}[S_t \mathbf{1}_{\{S_t > K\}}] - K \mathbb{P}[S_t > K] \\ &= (F(t) - K) \mathbb{P}[F(t) + \sigma W_t > K] + \sigma \mathbb{E}[W_t \mathbf{1}_{\{F(t) + \sigma W_t > K\}}] \\ &= (F(t) - K) \Phi\left(\frac{F(t) - K}{\sigma \sqrt{t}}\right) + \sigma \phi\left(\frac{F(t) - K}{\sigma \sqrt{t}}\right) \end{aligned} \quad (18.5)$$

Similarly or via call-put parity

$$(S_t - K)^+ - (K - S_t) = S_t - K \quad (18.6)$$

the put price is given as

$$\mathbb{E}[(K - S_t)^+] = (K - F(t)) \Phi\left(\frac{K - F(t)}{\sigma \sqrt{t}}\right) + \sigma \phi\left(\frac{F(t) - K}{\sigma \sqrt{t}}\right) \quad (18.7)$$

18.2 Black-Scholes

Let

$$S_t = F(t)e^{\sigma W_t - \sigma^2 t/2}$$

Then

$$\begin{aligned} & \mathbb{E}[(S_t - K)^+] \\ &= \mathbb{E}[S_t \mathbf{1}_{\{S_t > K\}}] - K \mathbb{P}[S_t > K] \\ &= F(t) \mathbb{E}\left[e^{\sigma W_t - \sigma^2 t/2} \mathbf{1}_{\{S_t > K\}}\right] - K \mathbb{P}[S_t > K] \\ &= F(t) \mathbb{P}\left[\sigma(W_t + \sigma t) - \frac{\sigma^2 t}{2} > \log \frac{K}{F(t)}\right] - K \mathbb{P}\left[\sigma W_t - \frac{\sigma^2 t}{2} > \log \frac{K}{F(t)}\right] \\ &= F(t) \Phi(d_+) - K \Phi(d_-) \end{aligned} \quad (18.8)$$

where

$$d_{\pm} = \frac{\log \frac{F(t)}{K} \pm \frac{\sigma^2 t}{2}}{\sigma \sqrt{t}} \quad (18.9)$$

If $S_t = S_0 C_t$ and C_t does not depend on S_0 one has

$$\begin{aligned} \frac{\partial}{\partial S_0} (S_t - K)^+ &= C_t \frac{\partial}{\partial S_0} (S_0 - K/C_t)^+ = C_t \mathbf{1}_{\{S_0 > K/C_t\}} \\ &= C_t \mathbf{1}_{\{-W_t/\sqrt{t} < d_-\}} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial^2}{\partial S_0^2} (S_t - K)^+ &= C_t \frac{\partial}{\partial S_0} \mathbf{1}_{\{-W_t/\sqrt{t} < d_-\}} = C_t \delta_{\{-W_t/\sqrt{t} = d_-\}} \frac{\partial}{\partial S_0} d_- \\ &= C_t \delta_{\{-W_t/\sqrt{t} = d_-\}} \frac{1}{S_0 \sigma \sqrt{t}} \end{aligned}$$

By integrating one obtains delta

$$\begin{aligned} \Delta &= \frac{\partial}{\partial S_0} \mathbb{E}[(S_t - K)^+] \\ &= \frac{F(t)}{S_0} \mathbb{E}\left[e^{\sigma W_t - \sigma^2 t/2} \mathbf{1}_{\{-W_t/\sqrt{t} < d_-\}}\right] = \frac{F(t)}{S_0} \Phi(d_+) \end{aligned}$$

and gamma

$$\begin{aligned}\Gamma &= \frac{\partial^2}{\partial S_0^2} \mathbb{E}[(S_t - K)^+] \\ &= \frac{F(t)}{S_0^2 \sigma \sqrt{t}} \mathbb{E}\left[e^{\sigma W_t - \sigma^2 t/2} \delta_{\{-W_t/\sqrt{t}=d_-\}}\right] = \frac{F(t)}{S_0^2 \sigma \sqrt{t}} \phi(d_+)\end{aligned}$$

Vega is given as

$$\Upsilon = \sqrt{t} K \phi(d_-) \quad (18.10)$$

18.2.1 Quanto

Given two log-normal assets $S^k, k = 1, 2$, with means F^k , volatilities σ_k and correlation ρ one may consider the payoff

$$S_t^1 (S_t^2 - K)^+$$

Applications include quanto options, see Section 15.5, and Vanilla options with Gaussian interest rates, see Chapter 11. One has

$$\mathbb{E}[S_t^1 (S_t^2 - K)^+] = \mathbb{E}[S_t^1 S_t^2 \mathbf{1}_{\{S_t^2 > K\}}] - K \mathbb{E}[S_t^1 \mathbf{1}_{\{S_t^2 > K\}}]$$

For the second term one computes

$$\begin{aligned}\mathbb{E}[S_t^1 \mathbf{1}_{\{S_t^2 > K\}}] &= F^1(t) \mathbb{E}\left[e^{\sigma_1 W_t^1 - \sigma_1^2 t/2} \mathbf{1}_{\left\{\sigma_2 W_t^2 - \frac{\sigma_2^2 t}{2} > \log \frac{K}{F^2(t)}\right\}}\right] \\ &= F^1(t) \mathbb{P}\left[\sigma_2 (W_t^2 + \rho \sigma_1 t) - \frac{\sigma_2^2 t}{2} > \log \frac{K}{F^2(t)}\right] \\ &= F^1(t) \Phi(d_-^2 + \rho \sigma_1 \sqrt{t})\end{aligned}$$

with

$$d_{\pm}^k = \frac{1}{\sigma_k \sqrt{t}} \left(\log \frac{F^k(t)}{K} \pm \frac{1}{2} \sigma_k^2 t \right) \quad (18.11)$$

For the first term notice that $S_t^{12} = S_t^1 S_t^2$ has mean

$$F^{12}(t) = F^1(t) F^2(t) e^{\rho \sigma_1 \sigma_2 t}$$

and variance

$$\sigma_{12}^2 = \sigma_1^2 + 2\rho\sigma_1\sigma_2 + \sigma_2^2$$

The correlation with S_t^2 is given as

$$\rho_{12,2} = \frac{\rho\sigma_1 + \sigma_2}{\sigma_{12}}$$

Thus, with S^{12} instead of S^1 , the computation above yields

$$\mathbb{E} \left[S_t^{12} \mathbf{1}_{\{S_t^2 > K\}} \right] = F^{12}(t) \Phi \left(d_-^2 + \rho_{12,2} \sigma_{12} \sqrt{t} \right)$$

Hence

$$\mathbb{E} \left[S_t^1 (S_t^2 - K)^+ \right] = F^1(t) \left\{ F^2(t) e^{\rho\sigma_1\sigma_2 t} \Phi \left(d_+^2 + \rho\sigma_1 \sqrt{t} \right) - K \Phi \left(d_-^2 + \rho\sigma_1 \sqrt{t} \right) \right\} \quad (18.12)$$

with $d_{\pm}^k$ defined in equation (18.11).

18.3 Barrier

18.3.1 Barrier toolbox

Some quantities are derived below for the purpose of barrier option pricing. One has for all λ, v and y

$$\begin{aligned} \mathbb{E} \left[e^{\lambda W_1^v} \mathbf{1}_{\{W_1^v > y\}} \right] &= \mathbb{E} \left[e^{(\lambda+v)W_1 - v^2/2} \mathbf{1}_{\{W_1 > y\}} \right] \\ &= e^{\lambda^2/2 + \lambda v} \mathbb{P} \left[W_1^{\lambda+v} > y \right] \\ &= e^{\lambda^2/2 + \lambda v} \Phi(\lambda + v - y) \\ &= f_0(y, v, \lambda) \end{aligned} \quad (18.13)$$

as well as

$$\begin{aligned} \mathbb{E} \left[e^{\lambda W_1^v} \mathbf{1}_{\{W_1^v < y\}} \right] &= e^{\lambda^2/2 + \lambda v} \Phi(-\lambda - v + y) \\ &= f_0(-y, -v, -\lambda) \end{aligned} \quad (18.14)$$

For all λ, ν and x, y with $x \leq 0$ and $x \leq y$ one has

$$\begin{aligned}
 \mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_{*1}^\nu < x, W_1^\nu > y\}} \right] &= \mathbb{E} \left[e^{(\lambda+\nu)W_1 - \nu^2/2} \mathbf{1}_{\{W_{*1} < x, W_1 > y\}} \right] \\
 &= \mathbb{E} \left[e^{(\lambda+\nu)W_1 - \nu^2/2} \mathbb{P}[W_{*1} < x | W_1] \mathbf{1}_{\{W_1 > y\}} \right] \\
 &= \mathbb{E} \left[e^{(\lambda+\nu)W_1 - \nu^2/2} e^{-2x(x-W_1)} \mathbf{1}_{\{W_1 > y\}} \right] \\
 &= e^{(\lambda+\nu+2x)^2/2 - \nu^2/2 - 2x^2} \mathbb{P}[W_1^{\lambda+\nu+2x} > y] \\
 &= e^{\lambda^2/2 + \lambda\nu + 2(\lambda+\nu)x} \Phi(\lambda + \nu + 2x - y) \\
 &= f_1(x, y, \nu, \lambda)
 \end{aligned} \tag{18.15}$$

This formula can be generalized to all x and y via

$$f_1(x, y, \nu, \lambda) = \begin{cases} f_0(y, \nu, \lambda) & \text{if } x > 0 \\ f_1(x, x, \nu, \lambda) + f_0(y, \nu, \lambda) - f_0(x, \nu, \lambda) & \text{if } x < 0, x > y \end{cases}$$

The other equalities are

$$\begin{aligned}
 \mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_{*1}^\nu > x, W_1^\nu > y\}} \right] &= \mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_1^\nu > y\}} \right] - \mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_{*1}^\nu < x, W_1^\nu > y\}} \right] \\
 &= f_0(y, \nu, \lambda) - f_1(x, y, \nu, \lambda) \\
 &= f_2(x, y, \nu, \lambda)
 \end{aligned} \tag{18.16}$$

as well as

$$\mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_{*1}^\nu > x, W_1^\nu < y\}} \right] = f_2(x, x, \nu, \lambda) - f_2(x, y, \nu, \lambda) \tag{18.17}$$

and

$$\begin{aligned}
 &\mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_{*1}^\nu < x, W_1^\nu < y\}} \right] \\
 &= \mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_1^\nu < y\}} \right] - \mathbb{E} \left[e^{\lambda W_1^\nu} \mathbf{1}_{\{W_{*1}^\nu > x, W_1^\nu < y\}} \right] \\
 &= f_0(-y, -\nu, -\lambda) - f_2(x, x, \nu, \lambda) + f_2(x, y, \nu, \lambda)
 \end{aligned} \tag{18.18}$$

Since

$$(W_1^{*\nu}, W_1^\nu) \stackrel{(d)}{=} (-W_{*1}^{-\nu}, -W_1^{-\nu})$$

one obtains corresponding formulae for the maximum, namely

$$\mathbb{E} \left[e^{\lambda W_1^v} \mathbf{1}_{\{W_1^{*v} < x, W_1^v < y\}} \right] = f_2(-x, -y, -v, -\lambda) \quad (18.19)$$

and

$$\mathbb{E} \left[e^{\lambda W_1^v} \mathbf{1}_{\{W_1^{*v} > x, W_1^v < y\}} \right] = f_1(-x, -y, -v, -\lambda) \quad (18.20)$$

Deriving (18.15) with respect to x results in

$$\begin{aligned} & \mathbb{E} \left[e^{\lambda W_1^v} \mathbf{1}_{\{W_{*1}^v \in dx, W_1^v > y\}} \right] \\ &= 2e^{\lambda^2/2 + \lambda v + 2(\lambda + v)x} \left((\lambda + v) \Phi(\lambda + v + 2x - y) + \phi(\lambda + v + 2x - y) \right) dx \end{aligned}$$

Using (18.39) and (18.40) allows one to compute

$$\begin{aligned} & \mathbb{E} \left[e^{\delta W_{*1}^v + \lambda W_1^v} \mathbf{1}_{\{W_{*1}^v < x, W_1^v > W_{*1}^v + y\}} \right] \\ &= \int_{-\infty}^x e^{\delta z} \mathbb{E} \left[e^{\lambda W_1^v} \mathbf{1}_{\{W_{*1}^v \in dz, W_1^v > z + y\}} \right] \\ &= 2e^{\lambda^2/2 + \lambda v} \int_{-\infty}^x e^{\tilde{\delta} z} \left((\lambda + v) \Phi(\lambda + v + z - y) + \phi(\lambda + v + z - y) \right) dz \\ &= \frac{2}{\tilde{\delta}} e^{\lambda^2/2 + \lambda v} (\lambda + v) \left(e^{\tilde{\delta} x} \Phi(x + \lambda + v - y) - e^{\tilde{\delta}(\delta/2 + y)} \Phi(x + \lambda + v - y - \tilde{\delta}) \right) \\ &\quad + 2e^{\lambda^2/2 + \lambda v + \delta^2/2 + (\lambda + v)\delta + \tilde{\delta} y} \Phi(x + \lambda + v - y - \tilde{\delta}) \\ &= e^{\lambda^2/2 + \lambda v} \left(\left(1 - \frac{\delta}{\tilde{\delta}} \right) e^{\tilde{\delta} x} \Phi(x + \lambda + v - y) \right. \\ &\quad \left. + e^{\tilde{\delta}(y + \delta/2)} \left(1 + \frac{\delta}{\tilde{\delta}} \right) \Phi(x + \lambda + v - y - \tilde{\delta}) \right) \quad (18.21) \end{aligned}$$

with

$$\tilde{\delta} = \delta + 2(\lambda + v)$$

The above formulae extend to statements on W_T^v, W_{*T}^v with arbitrary $T > 0$ by replacing x, y, v, λ, δ with $\frac{x}{\sqrt{T}}, \frac{y}{\sqrt{T}}, v\sqrt{T}, \lambda\sqrt{T}, \delta\sqrt{T}$.

18.3.2 Barrier rebate

A contract paying one unit at maturity time T if the barrier B is not hit can be priced using

$$\mathbb{P}[S_{*T} > B] = \mathbb{P}\left[\forall t < T: W_t + \frac{1}{\sigma} \left(\frac{\log(F(T)/S_0)}{T} - \frac{\sigma^2}{2} \right) t > \frac{\log(B/S_0)}{\sigma}\right]$$

Applying equation (18.15) with parameters

$$x = y = \frac{\log(B/S_0)}{\sigma}, \quad \nu = \frac{1}{\sigma} \left(\frac{\log(F(T)/S_0)}{T} - \frac{\sigma^2}{2} \right), \quad \lambda = 0$$

one obtains for $B \leq S_0$

$$\mathbb{P}[S_{*T} > B] = \Phi\left(\nu\sqrt{T} - \frac{x}{\sqrt{T}}\right) - e^{2\nu x} \Phi\left(\nu\sqrt{T} + \frac{x}{\sqrt{T}}\right) \quad (18.22)$$

from (18.15). The corresponding upper barrier value is obtained by changing the sign of both x and ν resulting in

$$\mathbb{P}[S_T^* < B] = \Phi\left(-\nu\sqrt{T} + \frac{x}{\sqrt{T}}\right) - e^{2\nu x} \Phi\left(-\nu\sqrt{T} - \frac{x}{\sqrt{T}}\right) \quad (18.23)$$

for $B \geq S_0$.

18.3.3 Hitting time

Defining the hitting time

$$\tau_x^\nu = \inf\{t: W_t^\nu = x\}$$

one has

$$\mathbb{P}[\tau_x^\nu > T] = \begin{cases} \mathbb{P}[W_{*T}^\nu > x] & \text{if } x < 0 \\ \mathbb{P}[W_T^{\nu*} < x] & \text{if } x > 0 \end{cases}$$

Hence

$$\begin{aligned} \mathbb{P}[\tau_x^\nu > t] &= \Phi\left(-\operatorname{sgn}(x)\nu\sqrt{t} + \frac{|x|}{\sqrt{t}}\right) \\ &\quad - e^{2\nu x} \Phi\left(-\operatorname{sgn}(x)\nu\sqrt{t} - \frac{|x|}{\sqrt{t}}\right) \end{aligned} \quad (18.24)$$

from (18.15). Deriving with respect to t one obtains for all $x \neq 0$ the density

$$\mathbb{P}[\tau_x^v \in dt] = \frac{|x|}{t^{3/2}} \phi\left(v\sqrt{t} - \frac{x}{\sqrt{t}}\right) dt$$

Note that

$$-rt - \frac{1}{2}\left(v\sqrt{t} - \frac{x}{\sqrt{t}}\right)^2 = (v - \tilde{v})x - \frac{1}{2}\left(\tilde{v}\sqrt{t} - \frac{x}{\sqrt{t}}\right)^2$$

with

$$\tilde{v} = \sqrt{2r + v^2} \quad (18.25)$$

This implies

$$e^{-rt} \mathbb{P}[\tau_x^v \in dt] = e^{(v-\tilde{v})x} \mathbb{P}[\tau_x^{\tilde{v}} \in dt]$$

Integrating this equation from 0 to T one obtains from (18.24)

$$\begin{aligned} \mathbb{E}\left[e^{-r\tau_x^v} \mathbf{1}_{\{\tau_x^v < T\}}\right] &= e^{(v-\tilde{v})x} \Phi\left(\operatorname{sgn}(x)\tilde{v}\sqrt{T} - \frac{|x|}{\sqrt{T}}\right) \\ &\quad + e^{(v+\tilde{v})x} \Phi\left(-\operatorname{sgn}(x)\tilde{v}\sqrt{T} - \frac{|x|}{\sqrt{T}}\right) \end{aligned} \quad (18.26)$$

with $\tilde{v}$ defined in (18.25). Deriving with respect to r and evaluating at $r = 0$ yields

$$\begin{aligned} \mathbb{E}\left[\tau_x^v \mathbf{1}_{\{\tau_x^v < T\}}\right] &= \frac{x}{|v|} \left(-\Phi\left(\operatorname{sgn}(x)v\sqrt{T} - \frac{|x|}{\sqrt{T}}\right) \right. \\ &\quad \left. + e^{2vx} \Phi\left(-\operatorname{sgn}(x)v\sqrt{T} - \frac{|x|}{\sqrt{T}}\right) \right) \end{aligned} \quad (18.27)$$

18.3.4 Barrier rebate with payment at hit

Let τ_B denote the hitting time at level B , namely

$$\tau_B = \inf\{t : S_t = B\} \quad (18.28)$$

With Black-Scholes dynamics for S one has $\tau_B = \tau_x^v$ with

$$x = \frac{\log(B/S_0)}{\sigma}, \quad v = \frac{1}{\sigma} \left(\frac{\log(F(T)/S_0)}{T} - \frac{\sigma^2}{2} \right)$$

If a rebate is paid at the first hitting time one can compute $\mathbb{E}[e^{-r\tau_B}\mathbf{1}_{\{\tau_B < T\}}]$ respectively $\mathbb{E}[\tau_B\mathbf{1}_{\{\tau_B < T\}}]$ from equations (18.26) respectively (18.27).

18.3.5 European barrier rebate

The survival probability $p(t)$ in Merton's model (1974) with corresponding τ_M is given as

$$p(t) = \mathbb{E}[A_t > B] = \Phi\left(v\sqrt{t} - \frac{x}{\sqrt{t}}\right) \quad (18.29)$$

Let τ_M be a stopping time satisfying

$$\mathbb{P}[\tau_M > t] = p(t)$$

Deriving with respect to t yields

$$\begin{aligned} \mathbb{P}[\tau_M \in dt] &= -\frac{1}{2t} \left(v\sqrt{t} + \frac{x}{\sqrt{t}}\right) \phi\left(v\sqrt{t} - \frac{x}{\sqrt{t}}\right) dt \\ &= \frac{1}{2} \left(\frac{v}{x}t + 1\right) \mathbb{P}[\tau_v^x \in dt] \end{aligned}$$

Note that in order to ensure positive density of τ_M one should have $v \leq 0$. Otherwise the density is only positive for $t \in [0, -x/v]$. Applying equations (18.26) and (18.27) one obtains

$$\begin{aligned} &\int_0^T e^{-rt} \mathbb{P}[\tau_M \in dt] \\ &= \frac{1}{2} e^{(v-\tilde{v})x} \int_0^T \left(\frac{v}{x}t + 1\right) \mathbb{P}[\tau_v^x \in dt] \\ &= \frac{1}{2} e^{(v-\tilde{v})x} \left(\frac{v}{x} \mathbb{E}[\tau_v^x \mathbf{1}_{\{\tau_v^x < T\}}] + \mathbb{P}[\tau_v^x < T]\right) \\ &= \frac{1}{2} e^{(v-\tilde{v})x} \frac{v}{x} \left(\frac{x}{|\tilde{v}|} \left(-\Phi\left(-\tilde{v}\sqrt{T} + \frac{x}{\sqrt{T}}\right) + e^{2\tilde{v}x} \Phi\left(\tilde{v}\sqrt{T} + \frac{x}{\sqrt{T}}\right)\right)\right) \\ &\quad + \frac{1}{2} e^{(v-\tilde{v})x} \Phi\left(-\tilde{v}\sqrt{T} + \frac{x}{\sqrt{T}}\right) + \frac{1}{2} e^{(v+\tilde{v})x} \Phi\left(\tilde{v}\sqrt{T} + \frac{x}{\sqrt{T}}\right) \\ &= \frac{e^{(v-\tilde{v})x}}{2} \left(1 - \frac{v}{|\tilde{v}|}\right) \Phi\left(-\tilde{v}\sqrt{T} + \frac{x}{\sqrt{T}}\right) \\ &\quad + \frac{e^{(v+\tilde{v})x}}{2} \left(1 + \frac{v}{x}\right) \Phi\left(\tilde{v}\sqrt{T} + \frac{x}{\sqrt{T}}\right) \end{aligned} \quad (18.30)$$

with $\tilde{v}$ defined in (18.25).

18.3.6 Partial barrier

Consider the partial hitting time

$$\tau_x^{\nu,t} = \inf\{s > t : W_s^\nu \leq x\} = t + \inf\{s > 0 : W_t^\nu + \tilde{W}_s^\nu \leq x\}$$

where W^ν and $\tilde{W}^\nu$ are independent Brownian motions with drift ν . Conditioning on W_t^ν one has from (18.26)

$$\begin{aligned} & \mathbb{E} \left[e^{-rt} \tau_x^{\nu,t} \mathbf{1}_{\{\tau_x^{\nu,t} < T\}} \right] \\ &= e^{-rt} \mathbb{P} \left[x - W_t^\nu > 0 \right] \\ &+ e^{-rt} \mathbb{E} \left[e^{(\nu-\tilde{\nu})(x-W_t^\nu)} \Phi \left(-\tilde{\nu}\sqrt{T-t} + \frac{x-W_t^\nu}{\sqrt{T-t}} \right) \mathbf{1}_{\{x-W_t^\nu < 0\}} \right] \\ &+ e^{-rt} \mathbb{E} \left[e^{(\nu+\tilde{\nu})(x-W_t^\nu)} \Phi \left(\tilde{\nu}\sqrt{T-t} + \frac{x-W_t^\nu}{\sqrt{T-t}} \right) \mathbf{1}_{\{x-W_t^\nu < 0\}} \right] \end{aligned}$$

Applying equation (18.40) and simplifying one obtains

$$\begin{aligned} & \mathbb{E} \left[e^{-rt} \tau_x^{\nu,t} \mathbf{1}_{\{\tau_x^{\nu,t} < T\}} \right] \\ &= e^{-rt} \Phi \left(-\nu\sqrt{t} + \frac{x}{\sqrt{t}} \right) + e^{(\nu-\tilde{\nu})x} \Phi_2 \left(\tilde{\nu}\sqrt{t} - \frac{x}{\sqrt{t}}, -\tilde{\nu}\sqrt{T} + \frac{x}{\sqrt{T}}; -\sqrt{\frac{t}{T}} \right) \\ &+ e^{(\nu+\tilde{\nu})x} \Phi_2 \left(-\tilde{\nu}\sqrt{t} - \frac{x}{\sqrt{t}}, \tilde{\nu}\sqrt{T} + \frac{x}{\sqrt{T}}; -\sqrt{\frac{t}{T}} \right) \end{aligned} \quad (18.31)$$

This implies

$$\begin{aligned} \mathbb{P} \left[\tau_x^{\nu,t} > T \right] &= \Phi_2 \left(\nu\sqrt{t} - \frac{x}{\sqrt{t}}, \nu\sqrt{T} - \frac{x}{\sqrt{T}}; \sqrt{\frac{t}{T}} \right) \\ &- e^{2\nu x} \Phi_2 \left(-\nu\sqrt{t} - \frac{x}{\sqrt{t}}, \nu\sqrt{T} + \frac{x}{\sqrt{T}}; -\sqrt{\frac{t}{T}} \right) \end{aligned} \quad (18.32)$$

In particular

$$\mathbb{P} \left[\tau_x^{\nu,t} > t \right] = \Phi \left(\nu\sqrt{t} - \frac{x}{\sqrt{t}} \right) \quad (18.33)$$

18.3.7 Barrier option

With asset dynamics

$$S_t = F(t)e^{\sigma W_t - \sigma^2 t/2}$$

a down and out call is given as

$$\begin{aligned} & \mathbb{E} \left[(S_T - K)^+ \mathbf{1}_{\{S_{*T} > B\}} \right] \\ &= \mathbb{E} \left[S_T \mathbf{1}_{\{S_{*T} > B, S_T > K\}} \right] - K \mathbb{P} \left[S_{*T} > B, S_T > K \right] \\ &= S_0 \mathbb{E} \left[e^{\lambda W_T^\nu} \mathbf{1}_{\{W_{*T}^\nu > x, W_T^\nu > y\}} \right] - K \mathbb{P} \left[W_{*T}^\nu > x, W_T^\nu > y \right] \quad (18.34) \end{aligned}$$

with parameters

$$\begin{aligned} x &= \frac{\log(B/S_0)}{\sigma}, \quad y = x \vee \frac{\log(K/S_0)}{\sigma}, \\ \nu &= \frac{1}{\sigma} \left(\frac{\log(F(T)/S_0)}{T} - \frac{\sigma^2}{2} \right), \quad \lambda = \sigma \end{aligned}$$

if $B \leq S_0$. The expectations have been computed in (18.16), see also the statement at the end of Section 18.3.1. The corresponding put and upper barrier options are given as

$$\begin{aligned} & \mathbb{E} \left[(K - S_T)^+ \mathbf{1}_{\{S_{*T} > B\}} \right] \\ &= S_0 \mathbb{E} \left[e^{\lambda W_T^\nu} \mathbf{1}_{\{W_{*T}^\nu > x, W_T^\nu < y\}} \right] - K \mathbb{P} \left[W_{*T}^\nu > x, W_T^\nu < y \right] \end{aligned}$$

as well as

$$\begin{aligned} & \mathbb{E} \left[(S_T - K)^+ \mathbf{1}_{\{S_T^* < B\}} \right] \\ &= S_0 \mathbb{E} \left[e^{\lambda W_T^\nu} \mathbf{1}_{\{W_T^{*\nu} < x, W_T^\nu > y\}} \right] - K \mathbb{P} \left[W_T^{*\nu} < x, W_T^\nu > y \right] \\ &= S_0 \mathbb{E} \left[e^{-\lambda W_T^{-\nu}} \mathbf{1}_{\{W_{*T}^{-\nu} > -x, W_T^{-\nu} > -y\}} \right] - K \mathbb{P} \left[W_{*T}^{-\nu} > -x, W_T^{-\nu} > -y \right] \end{aligned}$$

and

$$\begin{aligned} & \mathbb{E} \left[(K - S_T)^+ \mathbf{1}_{\{S_T^* < B\}} \right] \\ &= S_0 \mathbb{E} \left[e^{-\lambda W_T^{-\nu}} \mathbf{1}_{\{W_{*T}^{-\nu} > -x, W_T^{-\nu} > -y\}} \right] - K \mathbb{P} \left[W_{*T}^{-\nu} > -x, W_T^{-\nu} > -y \right] \end{aligned}$$

18.4 Outside barrier

18.4.1 Outside barrier toolbox

To price two-asset products distributions related to the extremum of a Brownian motion, W_{*1} and the terminal value of another Brownian $\tilde{W}_1$ are useful. One has, for $x \leq 0$,

$$\begin{aligned} \mathbb{P} \left[W_{*1} \geq x, \tilde{W}_1 \geq y \right] &= \mathbb{E} \left[\mathbb{P} [W_{*1} \geq x \mid W_1] \mathbb{E} \left[\tilde{W}_1 \geq y \mid W_1 \right] \right] \\ &= \mathbb{E} \left[\left(1 - e^{-2x(x-W_1)} \right) \mathbf{1}_{\{W_1 > x\}} \Phi \left(\frac{\rho}{\bar{\rho}} W_1 - \frac{y}{\bar{\rho}} \right) \right] \end{aligned}$$

Applying equation (18.38) with $X = -W_1$, $\alpha = -\frac{\rho}{\bar{\rho}}$, $\beta = -\frac{y}{\bar{\rho}}$, $\gamma = -x$, $\delta = 0$ and $\delta = -2x$ one obtains

$$\mathbb{P} \left[W_{*1} \geq x, \tilde{W}_1 \geq y \right] = \Phi_2(-x, -y; \rho) - \Phi_2(x, -y + 2x\rho; \rho)$$

Introducing drift ν one has

$$\begin{aligned} &\mathbb{P} \left[W_{*1}^\nu \geq x, \tilde{W}_1 \geq y \right] \\ &= \mathbb{E} \left[e^{\nu W_1 - \nu^2/2} \mathbf{1}_{\{W_{*1} \geq x, \tilde{W}_1 \geq y + \rho\nu\}} \right] \\ &= \mathbb{E} \left[e^{\nu W_1 - \nu^2/2} \mathbb{P} [W_{*1} \geq x \mid W_1] \mathbb{E} \left[\tilde{W}_1 \geq y + \rho\nu \mid W_1 \right] \right] \\ &= \mathbb{E} \left[e^{\nu W_1 - \nu^2/2} \left(1 - e^{-2x(x-W_1)} \right) \mathbf{1}_{\{W_1 > x\}} \Phi \left(\frac{\rho}{\bar{\rho}} W_1 - \frac{y + \rho\nu}{\bar{\rho}} \right) \right] \end{aligned}$$

Applying again equation (18.38) with $X = -W_1$, $\alpha = -\frac{\rho}{\bar{\rho}}$, $\beta = -\frac{y + \rho\nu}{\bar{\rho}}$, $\gamma = -x$, $\delta = -\nu$ and $\delta = -2x - \nu$ one obtains the corresponding equation with drift

$$\mathbb{P} \left[W_{*1}^\nu \geq x, \tilde{W}_1 \geq y \right] = \Phi_2(-x + \nu, -y; \rho) - e^{2x\nu} \Phi_2(x + \nu, -y + 2\rho x; \rho)$$

Note that

$$\mathbb{E} \left[e^{\lambda \tilde{W}_1} \mathbf{1}_{\{W_{*1}^\nu \geq x, \tilde{W}_1 \geq y\}} \right] = e^{\lambda^2/2} \mathbb{P} \left[W_{*1}^{\nu + \rho\lambda} \geq x, \tilde{W}_1 \geq y - \lambda \right]$$

Hence, for $x \leq 0$,

$$\begin{aligned} f(x, y, \rho, v, \lambda) &= \mathbb{E} \left[e^{\lambda \tilde{W}_1} \mathbf{1}_{\{W_{*1}^v \geq x, \tilde{W}_1 \geq y\}} \right] \\ &= e^{\lambda^2/2} \Phi_2(-x + v + \rho\lambda, -y + \lambda; \rho) \\ &\quad - e^{\lambda^2/2 + 2x(v + \rho\lambda)} \Phi_2(x + v + \rho\lambda, -y + \lambda + 2\rho x; \rho) \quad (18.35) \end{aligned}$$

Opposite inequalities follow with (18.35) from the symmetry of Brownian motion with respect to 0 as

$$\begin{aligned} \mathbb{E} \left[e^{\lambda \tilde{W}_1} \mathbf{1}_{\{W_{*1}^v \geq x, \tilde{W}_1 \leq y\}} \right] &= f(x, -y, -\rho, v, -\lambda) \\ \mathbb{E} \left[e^{\lambda \tilde{W}_1} \mathbf{1}_{\{W_1^{*v} \leq x, \tilde{W}_1 \geq y\}} \right] &= f(-x, y, -\rho, -v, \lambda) \\ \mathbb{E} \left[e^{\lambda \tilde{W}_1} \mathbf{1}_{\{W_1^{*v} \leq x, \tilde{W}_1 \leq y\}} \right] &= f(-x, -y, \rho, -v, -\lambda) \end{aligned}$$

Expectations with knock-in events such as $\{W_{*1}^v < x\}$ and $\{W_1^{*v} > x\}$ can be written as the difference between corresponding no barrier expectations and knock-out expectations with events $\{W_{*1}^v \geq x\}$ and $\{W_1^{*v} \leq x\}$.

18.4.2 Outside barrier

With asset dynamics

$$S_t^i = F^i(t) e^{\sigma_i W_t^i - \sigma_i^2 t/2}$$

one has

$$\begin{aligned} &\mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_{*T}^2 > B\}} \right] \\ &= \mathbb{E} \left[S_T^1 \mathbf{1}_{\{S_T^1 > K, S_{*T}^2 > B\}} \right] - K \mathbb{P} \left[S_T^1 > K, S_{*T}^2 > B \right] \\ &= F^1(T) \mathbb{E} \left[e^{\sigma_1 \tilde{W}_T - \sigma_1^2 T/2} \mathbf{1}_{\left\{ \sigma_1 \tilde{W}_T > \log(K/F^1(T)) + \sigma_1^2 T/2, \sigma_2 W_{*T}^{\log(F^2(T)/S_0^2)/T - \sigma_2^2/2} > \log(B/S_0^2) \right\}} \right] \\ &\quad - K \mathbb{P} \left[\sigma_1 \tilde{W}_T > \log(K/F^1(T)) + \sigma_1^2 T/2, \sigma_2 W_{*T}^{\log(F^2(T)/S_0^2)/T - \sigma_2^2/2} > \log(B/S_0^2) \right] \end{aligned}$$

Applying equation (18.35) with parameters

$$x = \frac{\log(B/S_0^2)}{\sigma_2\sqrt{T}}, \quad y = \frac{\log(K/F^1(T)) + \sigma_1^2 T/2}{\sigma_1\sqrt{T}},$$

$$v = \frac{\log(F^2(T)/S_0^2) - \sigma_2^2 T/2}{\sigma_2\sqrt{T}}, \quad \lambda = \sigma_1\sqrt{T}$$

one obtains, for $B \leq S_0^2$,

$$\begin{aligned} \mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_{*T}^2 > B\}} \right] &= F^1(T) e^{-\lambda^2/2} f(x, y, \rho, v, \lambda) \\ &\quad - K f(x, y, \rho, v, 0) \end{aligned} \quad (18.36)$$

The corresponding put and upper barrier options are given as

$$\begin{aligned} \mathbb{E} \left[(K - S_T^1)^+ \mathbf{1}_{\{S_{*T}^2 > B\}} \right] &= F^1(T) e^{-\lambda^2/2} f(x, -y, -\rho, v, -\lambda) \\ &\quad - K f(x, -y, -\rho, v, 0) \\ \mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_T^{2*} < B\}} \right] &= F^1(T) e^{-\lambda^2/2} f(-x, y, -\rho, -v, \lambda) \\ &\quad - K f(-x, y, -\rho, -v, 0) \\ \mathbb{E} \left[(K - S_T^1)^+ \mathbf{1}_{\{S_T^{2*} < B\}} \right] &= F^1(T) e^{-\lambda^2/2} f(-x, -y, \rho, -v, -\lambda) \\ &\quad - K f(-x, -y, \rho, -v, 0) \end{aligned}$$

In barriers are obtained as the difference between corresponding no-barriers and out-barriers.

18.4.3 European outside barrier

A European down and out outside barrier call has payoff

$$(S_T^1 - K)^+ \mathbf{1}_{\{S_T^2 > B\}}$$

The price can be computed as

$$\begin{aligned} &\mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_T^2 > B\}} \right] \\ &= \mathbb{E} \left[\left(F^1(T) e^{\sigma_1(\rho X + \bar{\rho} Y)\sqrt{T} - \sigma_1^2 T/2} - K \right)^+ \mathbf{1}_{\{-X < d_2^-(B)\}} \right] \\ &= \mathbb{E} \left[BSCall \left(F^1(T) e^{\sigma_1 \rho X \sqrt{T} - \sigma_1^2 \rho^2 T/2}, K, \sigma_1 \sqrt{1 - \rho^2}, T \right) \mathbf{1}_{\{-X < d_2^-(B)\}} \right] \end{aligned}$$

with

$$d_i^\pm(K) = \frac{\log(F_T^i/K) \pm \sigma_i^2 T/2}{\sigma_i \sqrt{T}}$$

The Black-Scholes formula leads to two integrals, namely

$$\begin{aligned} & \mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_T^2 > B\}} \right] \\ &= F^1(T) e^{-\delta^2/2} \mathbb{E} \left[e^{\delta X} \mathbf{1}_{\{X \leq \gamma\}} \Phi(\alpha X + \beta_+) \right] - K \mathbb{E} \left[\mathbf{1}_{\{X \leq \gamma\}} \Phi(\alpha X + \beta_-) \right] \end{aligned}$$

with parameters

$$\begin{aligned} \alpha &= \frac{-\rho}{\sqrt{1-\rho^2}} \\ \beta_\pm &= \frac{\log(F^1(T)/K) + (-\rho^2 \pm (1-\rho^2))\sigma_1^2 T/2}{\sigma_1 \sqrt{1-\rho^2} \sqrt{T}} \\ \delta &= -\rho \sigma_1 \sqrt{T} \\ \gamma &= d_2^-(B) \end{aligned}$$

Application of (18.38) yields

$$\begin{aligned} & \mathbb{E} \left[(S_T^1 - K)^+ \mathbf{1}_{\{S_T^2 > B\}} \right] \\ &= F^1(T) \Phi_2 \left(\gamma - \delta, \frac{\alpha \delta + \beta_+}{\sqrt{1+\alpha^2}}, \frac{-\alpha}{\sqrt{1+\alpha^2}} \right) - K \Phi_2 \left(\gamma, \frac{\beta_-}{\sqrt{1+\alpha^2}}, \frac{-\alpha}{\sqrt{1+\alpha^2}} \right) \\ &= F^1(T) \Phi_2 \left(d_2^-(B) + \rho \sigma_1 \sqrt{T}, d_1^+(K); \rho \right) - K \Phi_2 \left(d_2^-(B), d_1^-(K); \rho \right) \quad (18.37) \end{aligned}$$

18.5 Useful integrals

This section collects basic integrals for use within this section.

$$\begin{aligned}
 \mathbb{E} \left[e^{\delta X} \mathbf{1}_{\{X \leq \gamma\}} \Phi(\alpha X + \beta) \right] &= \mathbb{E} \left[e^{\delta X} \mathbf{1}_{\{X \leq \gamma, Y \leq \alpha X + \beta\}} \right] \\
 &= e^{\delta^2/2} \mathbb{P} \left[X + \delta \leq \gamma, Y \leq \alpha(X + \delta) + \beta \right] \\
 &= e^{\delta^2/2} \mathbb{P} \left[X \leq \gamma - \delta, \frac{Y - \alpha X}{\sqrt{1 + \alpha^2}} \leq \frac{\alpha\delta + \beta}{\sqrt{1 + \alpha^2}} \right] \\
 &= e^{\delta^2/2} \Phi_2 \left(\gamma - \delta, \frac{\alpha\delta + \beta}{\sqrt{1 + \alpha^2}}; \frac{-\alpha}{\sqrt{1 + \alpha^2}} \right) \quad (18.38)
 \end{aligned}$$

This integral is useful for outside barrier options. The following two equations are referred to in the derivation of barrier option prices.

$$\begin{aligned}
 \int_{-\infty}^{\gamma} e^{\delta x} \phi(\alpha x + \beta) dx &= \mathbb{E} \left[e^{\delta \frac{X - \beta}{\alpha}} \mathbf{1}_{\left\{ \frac{X - \beta}{\alpha} < \gamma \right\}} \right] \\
 &= e^{\frac{\delta^2}{2\alpha^2} - \frac{\delta\beta}{\alpha}} \Phi \left(\alpha\gamma + \beta - \frac{\delta}{\alpha} \right) \quad (18.39)
 \end{aligned}$$

Partial integration and (18.39) yields

$$\begin{aligned}
 \int_{-\infty}^{\gamma} e^{\delta x} \Phi(\alpha x + \beta) dx \\
 = \frac{1}{\delta} e^{\delta\gamma} \Phi(\alpha\gamma + \beta) - \frac{\alpha}{\delta} e^{\frac{\delta^2}{2\alpha^2} - \frac{\delta\beta}{\alpha}} \Phi \left(\alpha\gamma + \beta - \frac{\delta}{\alpha} \right) \quad (18.40)
 \end{aligned}$$

The next integral can be applied to outside partial barrier options, compare (18.36).

$$\begin{aligned}
 \mathbb{E} \left[e^{\delta X} \Phi_2(\alpha_1 X + \beta_1, \alpha_2 X + \beta_2; \rho) \right] \\
 = e^{\delta^2/2} \mathbb{P} \left[Y \leq \alpha_1(X + \delta) + \beta_1, \rho Y + \sqrt{1 - \rho^2} \rho Z \leq \alpha_2(X + \delta) + \beta_2 \right] \\
 = e^{\delta^2/2} \Phi_2 \left(\frac{\alpha_1\delta + \beta_1}{\sqrt{1 + \alpha_1^2}}, \frac{\alpha_2\delta + \beta_2}{\sqrt{1 + \alpha_2^2}}; \frac{\rho + \alpha_1\alpha_2}{\sqrt{1 + \alpha_1^2}\sqrt{1 + \alpha_2^2}} \right) \quad (18.41)
 \end{aligned}$$

Counterparty credit risk for default driven by a random barrier à la CreditGrades requires a trivariate formula, namely

$$\begin{aligned}
 & \mathbb{E} \left[\mathbf{1}_{\{X \leq \gamma\}} e^{\delta X} \Phi_2(\alpha_1 X + \beta_1, \alpha_2 X + \beta_2; \rho) \right] \\
 &= e^{\delta^2/2} \mathbb{P} \left[X + \delta \leq \gamma, Y \leq \alpha_1(X + \delta) + \beta_1, \rho Y + \sqrt{1 - \rho^2} Z \leq \alpha_2(X + \delta) + \beta_2 \right] \\
 &= e^{\delta^2/2} \Phi_3 \left(\gamma - \delta, \frac{\alpha_1 \delta + \beta_1}{\sqrt{1 + \alpha_1^2}}, \frac{\alpha_2 \delta + \beta_2}{\sqrt{1 + \alpha_2^2}}; \frac{-\alpha_1}{\sqrt{1 + \alpha_1^2}}, \frac{-\alpha_2}{\sqrt{1 + \alpha_2^2}}, \frac{\rho + \alpha_1 \alpha_2}{\sqrt{1 + \alpha_1^2} \sqrt{1 + \alpha_2^2}} \right)
 \end{aligned}
 \tag{18.42}$$

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